



PhotoMesh™

Version 8.0.4

User Guide

Skyline Software Systems Inc.

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CHAPTER 1 INTRODUCTION

About PhotoMesh

Skyline's PhotoMesh fully automates the generation of high-resolution, textured, 3D mesh models from standard 2D photographs, LiDAR and Full Motion Video (FMV), offering a significant reduction in cost and time compared to traditional modeling methods. PhotoMesh's breakthrough technology is based on the highest-performance photogrammetry, computer vision, and computational geometry algorithms. Combining any number of photographs, in a wide range of formats and resolutions, PhotoMesh generates highly-detailed 3D models that can be viewed and queried using TerraExplorer or other 3D and GIS products.

PhotoMesh employs elaborate tiling mechanisms to efficiently handle projects with even hundreds of thousands of photos. Running on standard hardware, PhotoMesh can also exploit computer clusters and cloud computing to run a single project simultaneously on hundreds of fuser machines. Sharing the workload involved in generating PhotoMesh output vastly accelerates all steps of the build.

PhotoMesh Components

The PhotoMesh application is composed of three components:

- **PhotoMesh Editor** – The main PhotoMesh working area for various workflow actions such as preparing, submitting for processing, monitoring, and reviewing the build.
 - **PhotoMesh Fuser** – Worker component of PhotoMesh which allows you to share the workload between several computers on the same network. The master computer, equipped with PhotoMesh fuser, utilizes the computing power of the network's client computers by connecting to PhotoMesh fusers on the client machines.
- Note:** Fuser pools can be used to share fusers between several PhotoMesh master computers in your organization. PhotoMesh can also work with Amazon Web Services (AWS), adding and terminating virtual fuser machines, based on resource needs.
- **PhotoMesh Build Manager** – Manager application that is responsible for managing the build process and PhotoMesh fusers. PhotoMesh Build Manager keeps track of all build tasks and fusers, distributing tasks among the fusers and assigning new tasks as fusers complete their tasks and become available. During the build process, PhotoMesh Build Manager monitors the build and provides the user with detailed information about the build's progress.

PhotoMesh Resources

Resources that provide information about the application and about your application license are easily accessible from the File menu.

To open PhotoMesh resources:

- From the **File** menu, select **Resources** and then select one of the following resources:

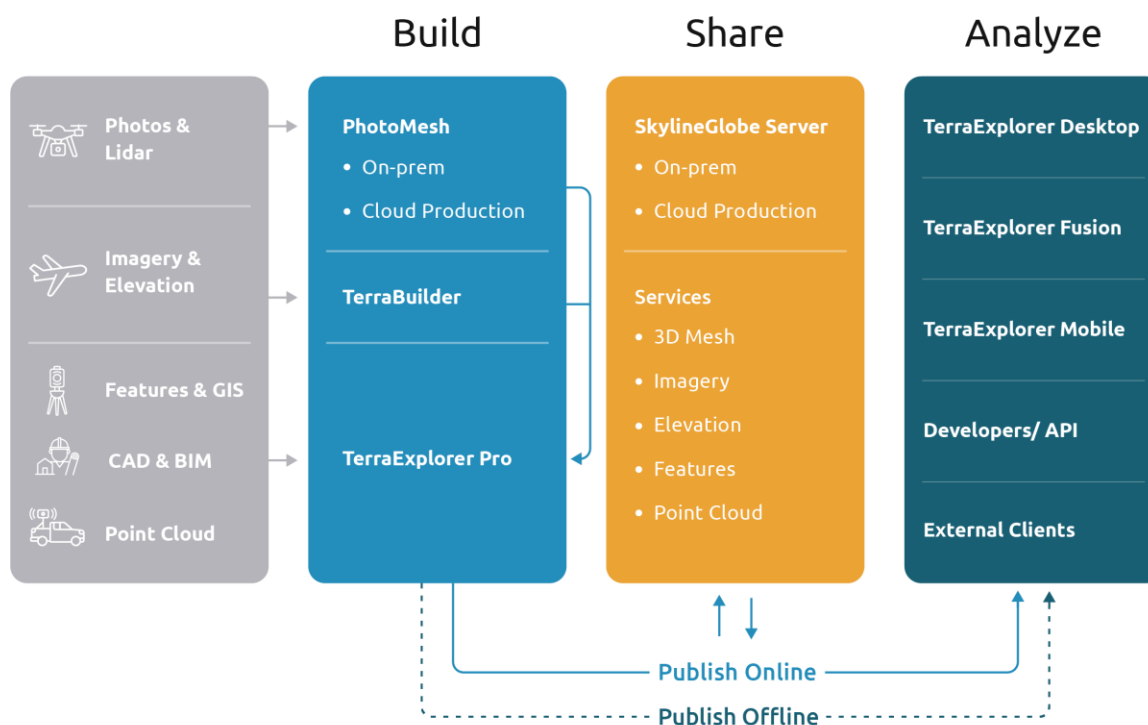
Resource	Description
PhotoMesh Help (F1)	Provides access to PhotoMesh Help files.
License Manager	Opens the License Manager tool.

PhotoMesh API	Provides access to interactive documentation of the PhotoMesh REST API to automate the production phases, e.g., load photo files or list, set AT and reconstruction areas, begin and manage a build, and create new build versions.
PhotoMesh Start Page	Opens the Start page from which you can create a new project, open an existing one, open the automation dashboard, and more.
About PhotoMesh	Access basic application information, including version number, contact information, and license information.

SkylineGlobe Product Suite

The SkylineGlobe software suite sets the standard for 3D desktop and web-based applications, enabling an enterprise to build and efficiently publish realistic 3D environments that can be explored, queried, analyzed and edited by desktop, web, and mobile clients. The suite includes:

- **Editors** – PhotoMesh, TerraBuilder, and TerraExplorer Pro
- **Cloud Services** – Streaming, Storing, and Cataloging
- **Clients** – Desktop, Mobile, and Web applications



SkylineGlobe Product Suite

The SkylineGlobe Enterprise solution is scalable, with licenses to support from ten to tens of thousands of concurrent users.

Whether deploying a desktop, mobile or web-based application, SkylineGlobe Enterprise provides a full Application Programming Interface (API) allowing you to customize it according to your requirements.

Because SkylineGlobe Enterprise is based on open OGC standards such as WFS and WMS, it can operate as a seamless 3D interface with other existing, legacy systems within the organization.

"On the fly" data fusion from disparate and distributed sources without data pre-processing allows you to keep your 3D environment current and relevant.

Skyline's range of products allows users to design an implementation customized to meet their requirements. Deployment options include the ability to work in a networked or disconnected (off-line) mode and make content available to the public or keep it restricted to secure networks and authorized users. Highly efficient streaming server technology supports large enterprise deployments with minimal hardware requirements.

Editors

Skyline offers powerful applications for generating, editing and publishing 3D terrain and urban model databases from a wide variety of data types.

TerraBuilder

TerraBuilder is a 3D terrain database creator with professional-grade tools for manipulating and merging orthophotos, DEM and DSM raster files of different sizes and resolutions. The resulting photo-realistic, geographically accurate terrain database (MPT) can be made available to local TerraExplorer clients or published to remote clients using the SkylineGlobe Server.

PhotoMesh

Skyline's breakthrough PhotoMesh application fully automates the building of high-resolution, textured, 3D mesh models from oblique and nadir photographs and LiDAR captured from street view, drones, UAV and airplanes. PhotoMesh results can be generated in standard 3D model (3DML, OSGB, DAE, OBJ), raster (Orthophoto, DSM, DTM) and point cloud (LAS) formats.

TerraExplorer Pro

TerraExplorer Pro is a powerful GIS tool for merging unlimited, geo-referenced 3D terrain, 3D mesh, LiDAR, BIM, raster and feature layers into a 3D high-resolution, geospatial environment. The project and all its individual layers can be directly uploaded and published to SkylineGlobe cloud server from TerraExplorer. Through a single publishing operation, the project and its individual layers are made available to all TerraExplorer clients: Desktop, Mobile, and TerraExplorer Fusion, as well as standard geospatial clients. TerraExplorer Pro also provides powerful tools and a high resolution 3D environment in which to view, query, analyze and present geospatial data.

Cloud Services

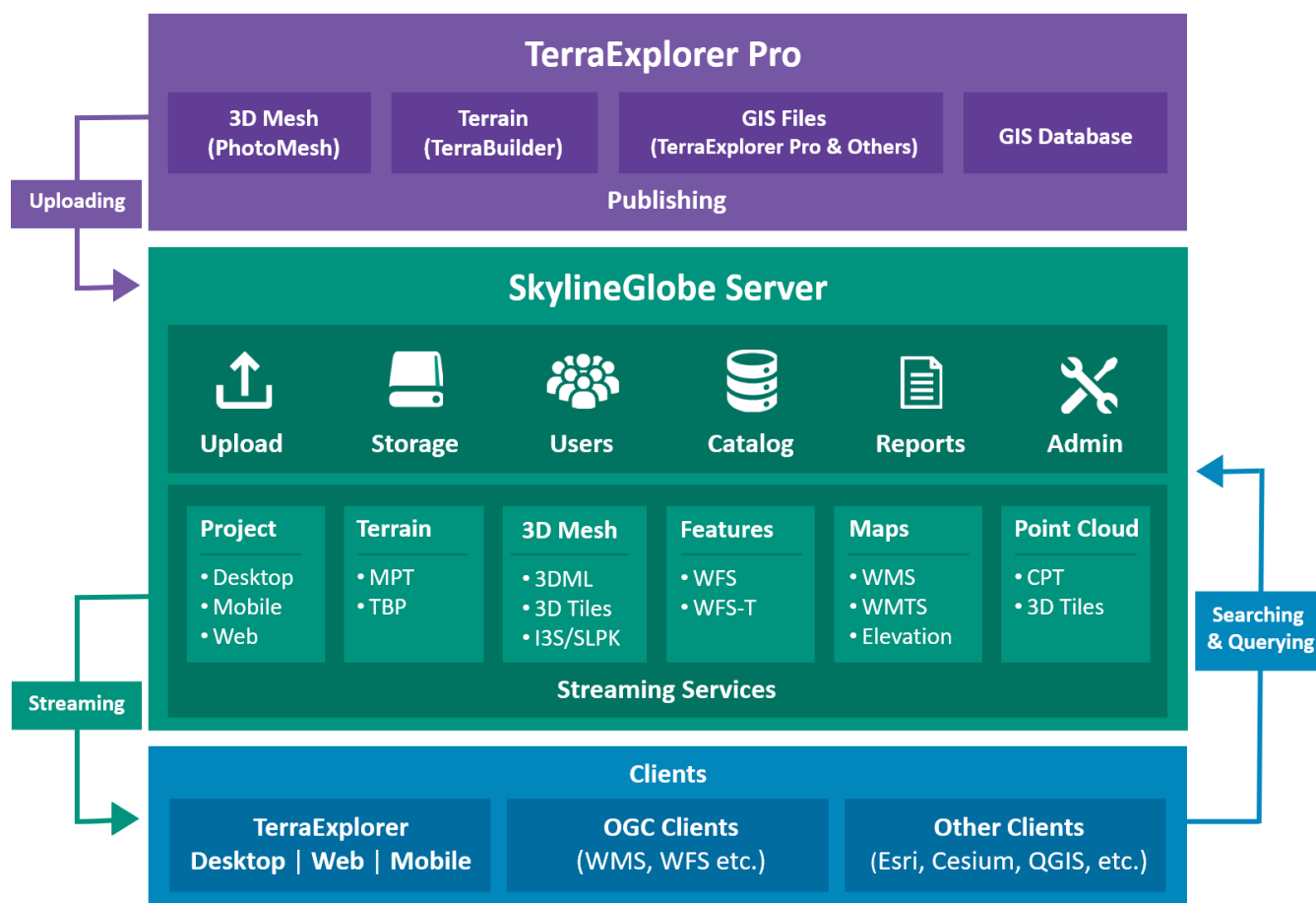
SkylineGlobe Server (SGS) is a private cloud solution that provides a comprehensive set of web services for publishing, storing, managing and streaming 3D spatial data. SkylineGlobe Server provides streaming services for all your spatial data types including terrain (MPT/TBP), map (raster-WMS/WMTS), feature (WFS/WFS-T), 3D Mesh (3DML, OGC 3D Tiles, SLPK (I3S)), and point cloud (CPT, OGC 3D Tiles). All data on your server is cataloged and referenced so clients can easily find the specific published layer required. Your data is kept safe by multiple security layers, including restriction of each user group to predefined data folders and a robust user authentication mechanism.

The server's built-in complete user access control system allows easy management of users, groups, and administrative roles controlling the server-side storage and client-side view/edit permissions.

SkylineGlobe Server is compliant with the following OpenGIS Implementation Specifications:

- OpenGIS® Web Feature Service (WFS) Implementation Specification

- OpenGIS® Web Feature Service Transactions (WFS-T) Implementation Specification
- OpenGIS® Web Map Service (WMS) Implementation Specification
- OpenGIS® Web Map Tiling Service (WMTS) Implementation Specification



SkylineGlobe Terrain Service

Powerful terrain service technology for streaming terrain data stored either in Skyline optimized terrain databases (MPT) or in terrain databases defined by the TerraBuilder project file (TBP) to remote TerraExplorer clients and any application that reads the standard OGC WMS/WMTS protocol.

SkylineGlobe Feature Service

Powerful feature service technology for streaming point, polyline, and polygon feature layers from files and from the Oracle, SQL server, PostGIS and ArcSDE databases to remote TerraExplorer clients and any application that reads the standard OGC WFS protocol. The Feature Service can stream most TerraExplorer supported feature layer formats. It also supports advanced spatial and attribute queries from TerraExplorer and other WFS clients. With SGS's support for WFS-T (Web Feature Service-Transactional), remote clients can edit (create, delete, and update features) the feature layer and save changes to the data source.

SkylineGlobe Maps Service

Powerful maps service technology for streaming most TerraExplorer supported raster layer formats to remote TerraExplorer clients and any application that reads the standard OGC WMS/WMTS protocol. The layers are streamed as WMS/WMTS tiles in JPEG and PNG formats.

SkylineGlobe 3D Mesh Service

Powerful 3D city model service technology for streaming 3D Mesh Layer (3DML) databases from 3DML files to remote TerraExplorer clients. 3D mesh layers (3DML) are high-resolution, textured, 3D mesh models that have been compressed and stream-optimized to accelerate loading speed and efficiently handle memory. The 3D Mesh Service also exposes the data as 3D tiles for Cesium based clients.

SkylineGlobe Point Cloud Service

Powerful point cloud service technology for streaming multi-resolution and stream-optimized point cloud data from CPT files to remote TerraExplorer clients. The Point Cloud Service also exposes the data as 3D tiles for Cesium based clients.

Clients

Skyline offers a range of 3D visualization clients that enable users to interact with, edit and annotate a geo-referenced 3D environment. Users can customize the database with data from a network, local drive or the Internet. Overlaying data specific to the user's requirements onto a 3D map creates a targeted, interactive picture that can meet the needs of a diverse user base and the specific requirements of each individual user. Skyline's client applications can be used in a local desktop environment, accessing terrain and content data directly, or in a network environment via the SkylineGlobe server.

TerraExplorer Desktop

A cutting-edge 3D GIS desktop viewer and creator that provides powerful tools and a high resolution 3D environment in which to view, query, analyze and present geospatial data.

With TerraExplorer's robust and extensive capabilities and ever-increasing interoperability, stunningly realistic 3D visualizations can be created by overlaying the terrain with unlimited data layers, 3D models, virtual objects and more.

TerraExplorer Fusion

TerraExplorer Fusion is a powerful 3D geospatial viewer that enables you to view and analyze high-resolution, stunningly realistic 3D content right in your web browser with no download or installation required. TerraExplorer Fusion enables viewing of most online layers and objects published to SkylineGlobe Server, performing powerful measurements and analysis operations, and dynamically exploring the 3D World via presentations which merge a custom flight path with distinct displays of the project. Built on HTML5/ WebGL standards, TerraExplorer Fusion ensures compatibility with multiple platforms and browsers.

TerraExplorer Fusion is based on TerraExplorer Desktop source code converted to WebAssembly. This conversion technology provides exceptional compatibility, enabling most of the original TerraExplorer Desktop code to run as-is in web browsers with numerous advantages, including support for nearly all the layers and objects supported by the Desktop version, consistent API for code sharing, and powerful performance.

TerraExplorer for Mobile

TerraExplorer for Mobile is a customizable, 3D GIS viewer for Android and Apple iOS devices that enables you to view high-resolution SkylineGlobe 3D content. The TerraExplorer for Mobile app can display most of your spatial databases including 3D city layers and feature layers, and it provides

powerful tools to query, analyze and present your data on the go. TerraExplorer for Mobile seamlessly accesses online data from Skyline's SkylineGlobe server and other OGC-compliant servers, and quickly loads online TerraExplorer projects.

CHAPTER 2 GETTING STARTED

Software and Hardware Requirements (Master/Fuser Computer)

Operating System	Windows® 8 / 10 / 11 Windows® Server 2012 R2 / 2016 / 2019 – 64bit required. Ubuntu 18.04 and Debian 10.0 – 64-bit required (Only for PhotoMesh fusers).
System Memory	16 GB RAM (32 GB recommended) per fuser. On a computer with 16 or more logical processors, 25 GB RAM is required per fuser. Note: RAM requirements are mostly affected by the reconstruction tile size and complexity. See "Setting Advanced Reconstruction Parameters" in the "Building" chapter for more information. 50 GB disk space or more per fuser is recommended for temporary files (%TEMP%) on each computer.
Graphics Processing Unit (GPU)	Dedicated or onboard GPU (dedicated recommended, especially when running multiple fusers on a single machine).
Processor	2 logical processors (8 recommended). PhotoMesh works best in a multi-core environment and can utilize multiple CPU's and hyper-threaded processors. In the latest PhotoMesh performance tests, using a single computer, 16 GB RAM and 8 logical processors, output was between 10-15 gigapixels/day, depending on the machine and resource availability.
Additional Software	.Net Framework 4.6.1 required.

Note: These requirements apply to both the master and fuser computers. [See the "Fusers" chapter](#) for more information on working with fusers.

License Mechanism for PhotoMesh

PhotoMesh is protected by a licensing mechanism that controls its operation, license term, and available modules. A license file issued by Skyline determines the licensing mechanism used.

The following licensing mechanisms are available for PhotoMesh:

- [Local Hardware Protection \(USB Key\)](#)
- [Local Hardware Protection \(HASP Key\)](#)
- [Workstation Dependent Software License](#)
- [Skyline Floating License Server](#)
- [HASP Floating License](#)

Local Hardware Protection (USB Key)

A USB key is a physical hardware key delivered with the PhotoMesh software package that plugs into the USB port of the computer – to allow the operation of the product. Verify that the USB key is plugged into the USB port before starting PhotoMesh.

Local Hardware Protection (HASP Key)

In some cases, Gemalto's HASP key solution is used instead of a USB key. The HASP key is a physical hardware key delivered with the PhotoMesh software package that plugs into the USB port of the computer – to allow the operation of the product. Verify that the HASP key is plugged into the USB port before starting PhotoMesh. To use a HASP key, the necessary driver must be downloaded and installed on your computer. **See** ["Manual Installation of the HASP Driver"](#) in this chapter for information.

Workstation Dependent Software License

Another type of license is the workstation dependent software license. This license does not require any hardware protection but is limited to a specific computer. Before working with the workstation dependent software license, you must first register to get a license locked to your computer. The license cannot be transferred without repeating the registration process.

Skyline Floating License Server

The license information can also be obtained using a floating license server which enables a number of licenses to be shared among a larger number of users. When an authorized user wants to run PhotoMesh, the user requests a license from the license server, and if one is available, the license server allows PhotoMesh to run. When the user finishes running PhotoMesh, the license is returned to the license server for use by other users. **See** ["Configuring PhotoMesh to Use the Skyline Floating License Server \(SFLS\)"](#) and ["Checking Out Licenses on the Client Side"](#) in this chapter for information. The floating license server also supports the PhotoMesh GX mechanism, which allows you to pre-purchase a bank of gigapixel credits, load them to your local Skyline Floating License Server (SFLS), and automatically apply them towards different PhotoMesh projects. **See** ["Applying Gigapixel Credits"](#) in the "Building" chapter for more information.

HASP Floating License

In some cases, Gemalto's NetHASP solution is used instead of Skyline's Floating License Server. The NetHASP key is a physical hardware key delivered with the PhotoMesh software package that plugs into the USB port of the server computer along with installation of the NetHASP License Manager software. It allows a predetermined number of concurrent users on the same network to use PhotoMesh, according to the license level defined by Skyline, without an individual hardware key on each of their computers.

License Manager

The License Manager allows the user to view and update the current license level.

To activate the License Manager:

- From the **File** menu, point to **Resources**, and then click **License Manager**. The License Manager window is displayed.

PhotoMesh License Options

The following PhotoMesh license options are available:

- **PhotoMesh Standard** – Pay per fuser. No gigapixel limit.
- **PhotoMesh GX** – Pay per gigapixel in your project. Unlimited managers/fusers. This option is generally implemented with Skyline Floating License Server (SFLS), or using an online gigapixel

credit form or request file. Gigapixel credits are loaded to SFLS, and then applied to a project's photos when building the project. See the SFLS Setup Guide for more information.

Installing PhotoMesh

The PhotoMesh application installation includes fuser installation. A standalone fuser installation is also available. See ["Installing PhotoMesh Fusers"](#) in this chapter for information.

Note: If your project will include a WKT of a coordinate system that references a grid shift file (GSB, GTX) that was not bundled with the PhotoMesh installation, the grid shift file can be automatically installed during the PhotoMesh installation process by placing a zip file that includes the required grid shift files next to the PhotoMesh.exe file in the installation folder. See ["Grid Shift Files"](#) in the "Basic Concepts" chapter for information on grid shift files and how to install them after PhotoMesh was installed.

To install PhotoMesh:

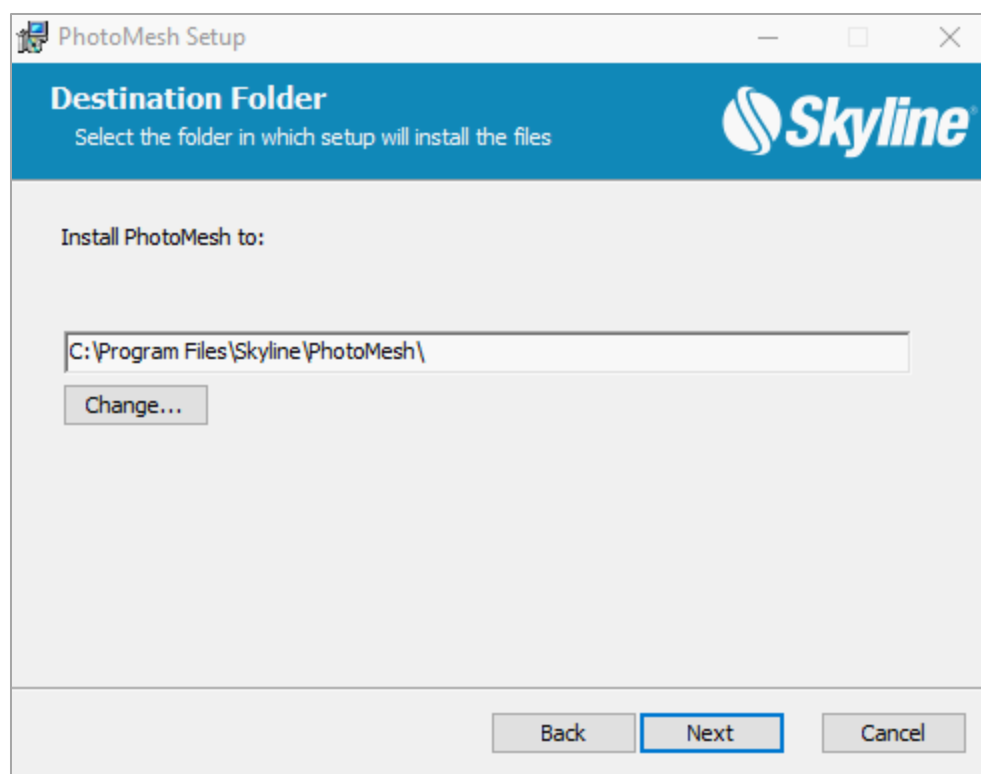
Note: On some operating systems, administrative privileges may be required.

1. If you have an old version of PhotoMesh running, close it before continuing with the installation.
2. Place the PhotoMesh license file (SLPhotoMesh.lic) next to the PhotoMesh.exe file in the installation folder.
3. On your computer, browse to the **Photomesh.exe** file and double-click. The Skyline Software End User License Agreement dialog is displayed.



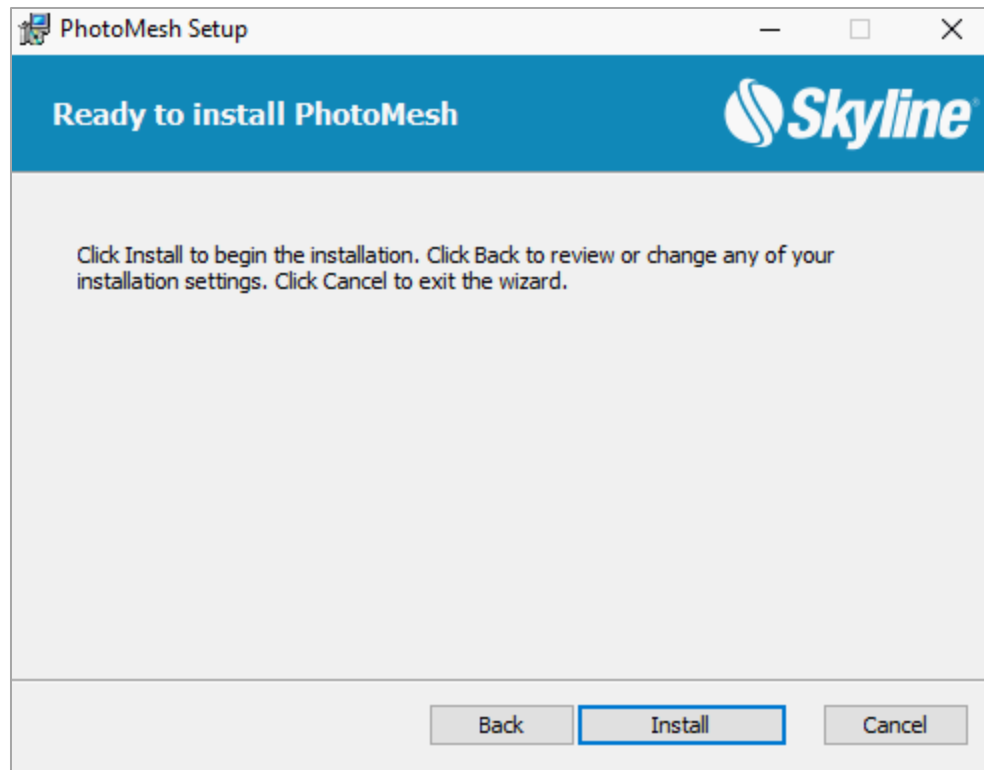
End User License Agreement

4. Select the check box to agree to the license agreement terms and then click **Install** to continue the installation.



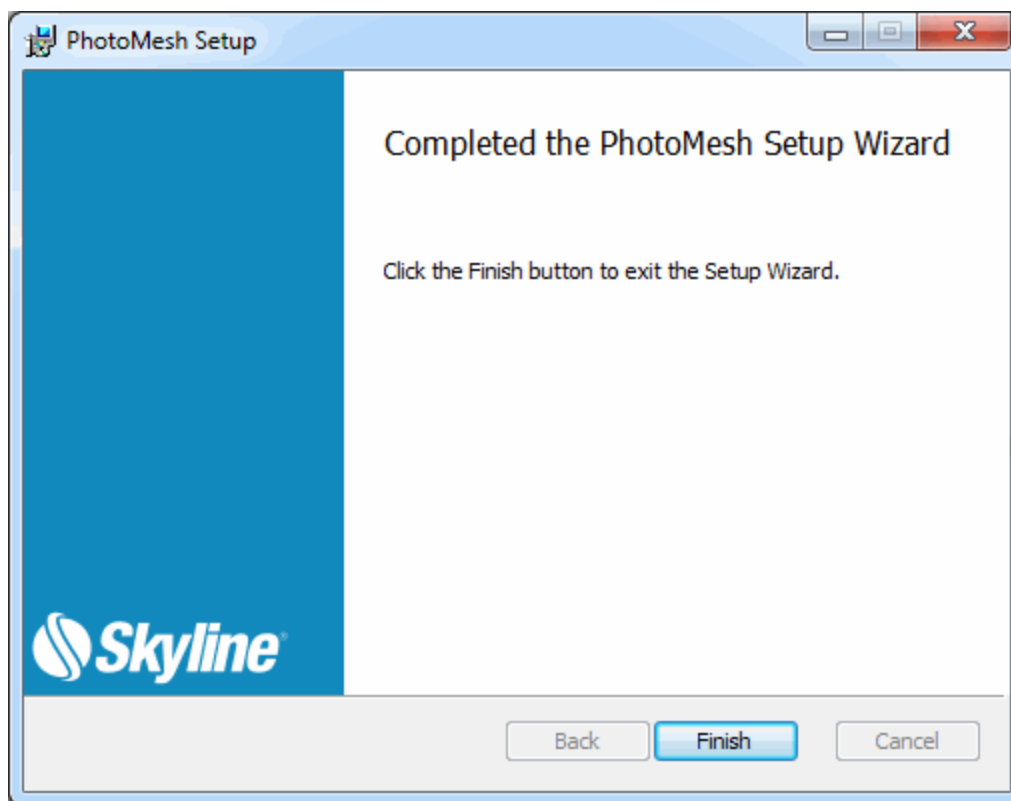
Destination Folder

5. Select the folder in which setup should install the files, and then click **Next**. The Ready to Install PhotoMesh dialog is displayed.



Ready to Install PhotoMesh Dialog

6. Click **Install**. A dialog is displayed when the installation process is complete.



PhotoMesh Setup Complete

7. Click **Finish** to exit the Setup Wizard.

Manual Installation of the HASP Driver

If your software was delivered with a HASP license key, you can manually install the required HASP driver.

To install the HASP driver:

1. [Download](#) the HASP driver zip file: Sentinel_LDK_Run-time_setup.
2. Unzip the file to a temporary directory, and run the HASPUserSetup.exe file.
3. Follow the installation wizard instructions to complete the installation.

Silent Installation of PhotoMesh

To install PhotoMesh:

1. Open a command prompt and use the "cd" command to change directories to where PhotoMesh.exe is located.
2. Run the following command to install PhotoMesh to a particular folder. If no installation folder is included in the command, PhotoMesh is installed to the default folder (c:\Program Files\Skyline\PhotoMesh).

```
PhotoMesh.exe -s INSTALLFOLDER=[Path to installation folder]
```

Note: Place the license file in the same directory as the installation executable (.exe) file before calling this command.

To install PhotoMesh fuser:

1. Open a command prompt and use the "cd" command to change directories to where PhotoMeshFuser.exe is located.
2. Run the following command to install PhotoMeshFuser to a particular folder. If no installation folder is included in the command, PhotoMeshFuser is installed to the default folder (c:\Program Files\Skyline\PhotoMeshFuser).

```
PhotoMeshFuser.exe -s INSTALLFOLDER=[Path to installation folder]
```

Note: Place the license file in the same directory as the installation executable (.exe) file before calling this command.

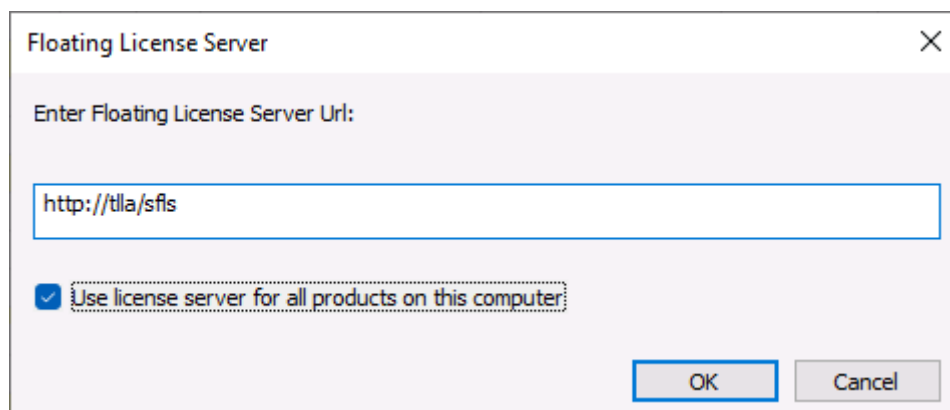
Configuring PhotoMesh to Use the Skyline Floating License Server (SFLS)

Each machine that is sharing the floating license for PhotoMesh must be configured with the address of the Floating License Server.

Note: The Skyline Floating License should be installed on the server before configuring it on each of the machines that is sharing the license.

To configure PhotoMesh to the SFLS address:

1. Open the License Manager. See "[License Manager](#)" in this chapter for information.
2. If the Skyline Floating License received from Skyline was not used in the [installation](#) (i.e., placed next to the **PhotoMesh.exe** file in the installation folder), from the **File** menu, select **Load License File**, and browse to the license file.
3. To set the floating license server URL, from the **File** menu, select **Set Floating License Server**. The Floating License Server dialog box is displayed.



Floating License Server

4. Enter the floating license server URL, e.g., `http://<server_name>/SFLS`.
5. If you want to override the protection type set in the currently loaded license file for each Skyline product on the computer, select **Use license server for all products on this computer**.

Checking Out Licenses on the Client Side

To check out a license:

1. Close the application for which you want to check out a license.

2. Open the **License Manager**, and under the section for the desired application select **Check Out**. The Application Protection Type field will refresh and read Floating (Checked Out). When the License Server Products page is refreshed, the Checked out count will have increased by the number of licenses checked out.

To check a license back in:

- Open the License Manager and select Check In.

See the [SFLS Setup Guide](#) for more information.

Uninstalling PhotoMesh

The Uninstall feature lets you quickly remove all folders and files that you installed to your hard disk during the installation procedure.

To uninstall PhotoMesh (including PhotoMesh fuser):

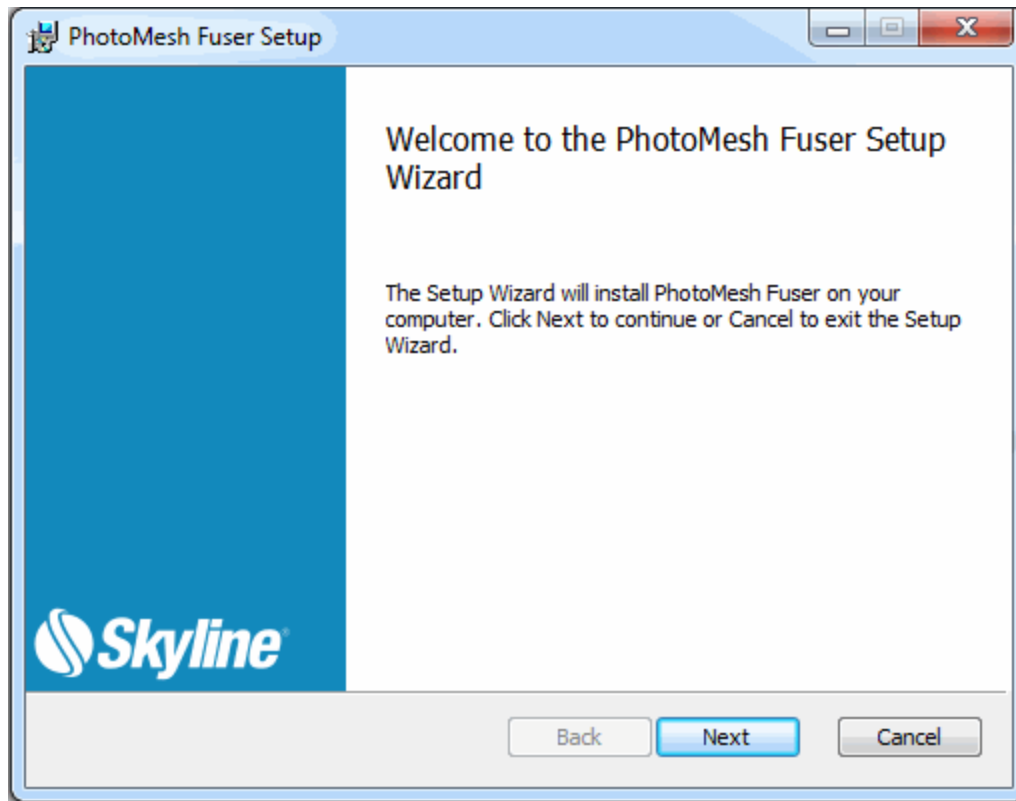
1. Exit the PhotoMesh and PhotoMesh fuser applications.
2. From the Windows **Start** menu, click **All Programs**, then click the **Skyline PhotoMesh** folder, and select **Uninstall**. The Windows Installer dialog is displayed.
3. Click **Yes** to uninstall.

Installing PhotoMesh Fusers

PhotoMesh fuser can be installed as a standalone installation on any client computer on the same network. The number of fusers that can run simultaneously is determined according to the PhotoMesh license. See "[PhotoMesh License Options](#)" in the "Getting Started" chapter. See the "[Fusers](#)" chapter for information on installing fusers for a fusers pool and for general fuser information.

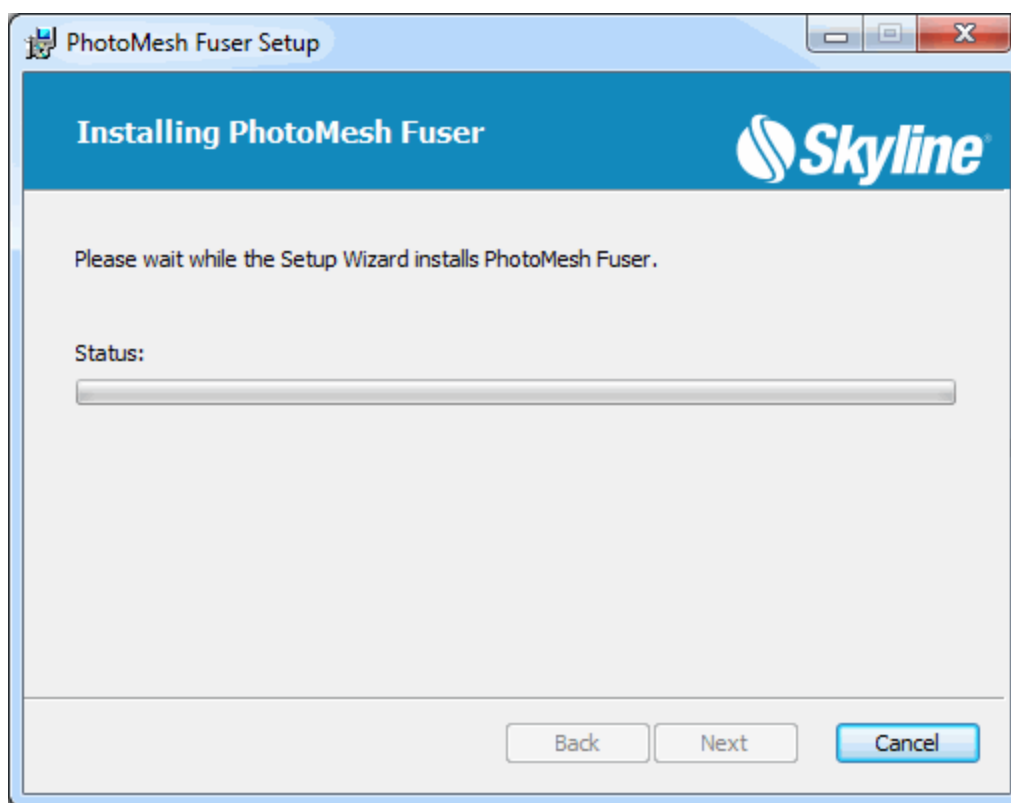
To install a PhotoMesh fuser:

1. Log on as a user with administration privileges.
2. On your computer, browse to the PhotoMesh Fuser.exe file and double-click. The Welcome dialog is displayed.



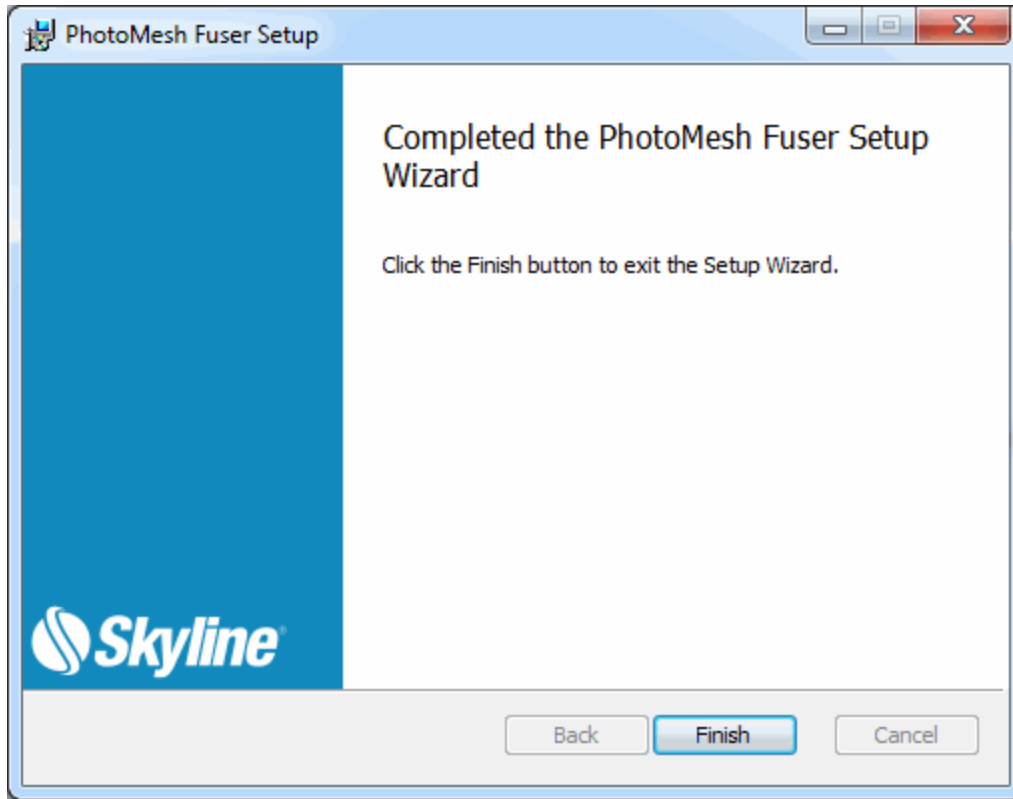
PhotoMesh Fuser Welcome Dialog

3. Click **Next** to view the Software License Agreement dialog. If you agree to the licensing terms, click **Yes** to continue the installation. The Installing PhotoMesh Fuser dialog is displayed.



Installing PhotoMesh Fuser Dialog

4. Click **Next**. The Completed PhotoMesh Fuser Setup dialog is displayed.



Completed Fuser Setup Dialog

5. Click **Finish**.

Installing Linux Fusers

To install a Linux fuser on **Ubuntu 24.04**:

See: ["Customizing the Instance"](#) in the "Getting Started with AWS" chapter for more information.

1. Make sure you have the latest graphics drivers for your graphics card and X Window System configured.

Note: To verify that your graphics card is configured properly, type the following in the terminal window: `glxinfo -B` and see if the right graphics card info is displayed.

2. Install and configure [Wine](#) by doing the following:

- a. Add 32-bit architecture, and then install Wine with Winetricks:

```
sudo dpkg --add-architecture i386
sudo apt-get -y update
sudo apt-get -y install wine winetricks
```

- b. Click [here](#) to download the configuration file for AWS. For running a local fuser, only the installation package `"wineinst.tar"` is required. Copy this file to your Linux machines, under `"/home/[your Linux user name]/"`.

Note: If you don't know your user name, you can run the command `"whoami"` in the Linux command line to retrieve it.

- c. Extract the contents of the `"wineinst.tar"` file and run the fuser setup script::

```
tar -xvf wineinst.tar
sh fuser.sh
```

Note: This process can take a few minutes to complete.

3. Run PhotoMeshFuser:

- a. In the terminal window, mount all the relevant shared drives on the Linux machine. (Working folder, Project, Input data)

```
mount -t cifs //<IP>/<SharedDrive> /mnt/mount1 -o
username=<username>,password=<password>,vers=2.0,uid=<user ID, either a number or name>
```

Note: If you don't know your user ID, you can run the command "*id*" in the Linux command line to retrieve it.

- b. In the terminal window, mount and hard link the Wine DOS devices:

```
mkdir ~/.wine/dosdevices/unc
mkdir ~/.wine/dosdevices/unc/<IP>
ln -s /mnt/mount1 ~/.wine/dosdevices/unc/<IP>/<SharedDrive>
```

- c. In the terminal window, navigate to the PhotoMesh fuser folder and run PhotoMeshFuser.exe with Wine. The fuser files can be on your Linux machine or on a shared drive.

```
wine photomeshfuser.exe
```

Uninstalling PhotoMesh Fusers

The uninstall feature lets you quickly remove all folders and files that were installed on the hard disk during the installation procedure.

To uninstall a standalone PhotoMesh fuser:

1. Exit PhotoMesh application.
2. From the Windows **Start** menu, click **All Programs**, then click the **Skyline PhotoMesh Fuser** folder, and select **Uninstall PhotoMesh Fuser**.
3. The Windows Installer dialog is displayed. Click **Yes** to uninstall.

Starting PhotoMesh

Start PhotoMesh by doing one of the following:

- Click Start > Skyline PhotoMesh> PhotoMesh.
- Click the **PhotoMesh** icon on your desktop.
- Double-click any .PhotoMeshXML file on your file system.

CHAPTER 3 NEW FEATURES

What's New in PhotoMesh 8.0.4

Stability Improvements and Bug Fixes

- **Fixed Fuser Regression:** Resolved an issue where a local or remote fuser mistakenly reset its working folder and was removed from the project.
- **Improved LAS/LAZ Accuracy:** Enhanced accuracy when generating LAS/LAZ files from DSM and ortho images.
- **Added Preset for Enhanced Bundle Adjustment:** Introduced a new preset that improves the robustness of the bundle adjustment process when working with multiple collections that are poorly aligned or have significant offsets from a reference source. This preset is now part of the "Close Inspection" build option.
- **Fixed Inconsistent Fuser Status Reporting:** Resolved an issue where some fusers were shown as "Not Available" in the Build Manager at the start of a build—despite being listed as "Allocated" in the Pool Manager. This mismatch, seen occasionally in large projects, no longer occurs.
- **AT Pyramid Step Optimization:** Resolved excessive processing time for machines with many cores (255+), allowing the AT Pyramid step to complete reliably.
- **Orientation Handling in AT:** PhotoMesh now consistently honors the "Ignore Orientation" setting when splitting images during AT.
- **Fixed Reduced CPU Utilization on E-core Machines:** Resolved an issue where fuser processes running on machines with Intel 13th/14th gen CPUs (with E-cores) showed significantly reduced CPU usage (~60%) when minimized or running in the background. Fusers now maintain full CPU performance regardless of window focus or visibility.

What's New in PhotoMesh 8.0.3

Output Quality Improvements

- **Enhanced True Orthophoto Quality:** Improved photo selection process for True Orthophotos provides more consistent texturing in dynamic scenarios (such as changing shadows, moving vehicles, and vegetation variations). This update also features sharper edges and enhanced clarity.
- **Enhanced Quick Orthophoto Quality:** Improved algorithms for better handling of break lines enhance the quality of Quick Orthophotos.
- **Improved Support for Close-Range Modeling:** Leverages the [Depth Anything AI algorithm](#) to focus more precisely on target elements. This release also features improved handling of close-range data collections in limited coverage areas and other challenging scenarios, boosting aerotriangulation success rates.
- **Water Polygon Improvements:** New options ensure better handling of water polygons, even in challenging environments such as whitewater areas.



Increased Compatibility

- **Reference Project Mechanism Enhancements:** Improved performance when utilizing reference data in projects without GPS information. New Reference Project report graphs enhance visualization of referencing adjustments.
- **Enhanced Vertical Datum Transformation Support:** The PhotoMesh 8.0.3 installation kit now comes equipped with an extensive collection of Grid Shift Files (GTX, GSB, TIFF). These files enhance the software's ability to perform precise vertical datum transformations by providing elevation correction values, ensuring more accurate height measurements across various geographic coordinate systems.
- **Fixes for 3D Tiles Output Format:** Enhanced support for the 3D Tiles 1.0 format, ensuring greater compatibility across various 3D viewers.

Stability and Performance Enhancements

- **Stability Improvements for Large Project Production:** Enhanced stability boosts processing efficiency for larger projects.
- **Optimized Production Time:** Reconstruction algorithm optimizations reduce production times across various scenarios by up to 20%.
- **Fusers Pool Fixes:** Multiple fixes and optimizations implemented in the fusers pool ensure reliability and smoother performance.

What's New in PhotoMesh 8.0.2

Performance Enhancements

- **Project Reporting:** Improved reference project reports and graphs. Reference ("base") projects are often added to a project when building a project that overlaps partially or entirely with a different project that will be overlaid on the same map in a client viewer. When building the new project, PhotoMesh establishes a correspondence between points in the reference project and target project, so that the positions of identified features in the target project match the geographic positions of the

same features in the reference project.

- **Data Preparation:** Added a failure threshold setting, configurable via the PhotoMesh API or the PhotoMesh.exe.config file, to better manage data preparation failures.
- **Improvement in PM 802:** Enhanced tile generation by including vertexes in the bounding box (BBox) and splitting large tiles to improve processing efficiency.

Output Quality Improvements

- **Orthophoto Quality:** Significant improvements in the quality of orthophotos and True orthophotos.
- **DSM Output:** Enhanced detail in Digital Surface Model (DSM) outputs for more precise topographic data.

New Flags

New flags available through Skyline support to do the following:

- Generate compressed 3D Tiles (3TZ format).
- Generate compressed OSGB files (zip format).
- Generate Normalized Difference Vegetation Index (NDVI) orthophoto outputs

Technical Advances

- **Containerization with Docker:** In PM 8.0.2, PhotoMesh fusers have been successfully containerized with Docker and deployed in both AWS EKS and an on-prem Kubernetes environment. With PhotoMesh Fuser's Dockerfile, you can now easily create your own Docker image and automate management of your fuser resources.

Preset Updates

- **Enhanced True Ortho and DSM Presets:** Improved noise reduction and point accuracy in the DPC point cloud, increased tile consistency, and reduced processing times with the SmoothSurface setting disabled.
- **Quick Ortho Enhancements:** Adjusted DPC overlap mode to -dpc_om 3 (smaller triangles) for better texture preservation on building tops and reduced ground noise.

Bug Fixes

- **Smeared or Black Textures (Holes):** Fixed an issue causing smeared or black textures (holes) in the rapid orthophoto.
- **Double Image:** Resolved a double image issue that occurred when building a project using the "Rapid Ortho Composite" preset.
- **Project Opening Error in PM 801:** Corrected an error that prevented projects from opening.
- **3DML Output Limitation:** Addressed a bug in PM 801 where generating more than ten 3DML outputs resulted in new outputs incorrectly overwriting the tenth model instead of creating subsequent ones (e.g., #11, 12, 13).
- **PM GCP Editor Viewer Limitation:** Improved the PM GCP editor viewer's capability to display large

images.

- **Fixed Match2.unt Crash:** Resolved an issue where having a space before the file extension caused the Match2.unt process to crash.
- **Build Deletion Issue Resolved:** Fixed a problem in the Build Version Manager where builds could not be deleted.
- **Missing GCP Notification Fix:** Resolved an issue where PhotoMesh would run indefinitely without alerting the user when no ground control points were present in a project. The system now provides a prompt notification.
- **AWS Fuser Resource Management:** Fixed a bug that caused an infinite round-trip between the manager and AWS by introducing a two-minute timeout to prevent this issue.

What's New in PhotoMesh 8.0.1

Model Quality and Output Improvements

New Presets

- **Tile Generation for Close Range Presets:** Eliminates tile gaps that create a "floating" effect on upper segments of vertical structures, such as towers, ensuring they appear connected to lower portions.
- **New Enhanced True Orthophoto Preset:** Introduces an alternative 3D correlation method focused on horizontal surfaces to better represent vertical structures and complex geometries. Best used with nadir collections that have high overlap.
- **New Camera Calibration Refinement for Non-Metric Sensors and Lenses AT Preset:** Adds an extra Bundle Block Adjustment iteration using a unique camera calibration method to address image geometry inconsistencies in non-metric sensors and lenses. Recommended for cameras with issues such as large distortions, rolling shutter, auto-zoom, auto-stabilization, and motion-based distortion.
- **Modified the default build setting to set the origin of Ortho tiles at the pixel's corner instead of the center, aligning with the accepted standard.**

New Settings

Introduced two new settings, adjustable via configuration files:

- **ImagesPerTask:** Configures the number of images allocated per task in the Image Processing stage, allowing for more flexible task management.
- **ImageProcessingMaxFailThreshold:** Establishes a failure threshold for the Image Processing task, whereby exceeding this limit will result in the task being marked as failed.

NIR Photo Calibration

PhotoMesh 8.0.1 includes an optional flag that provides advanced calibration of Near-Infrared (NIR) photos during the photo preparation step. Using DJI Exif tags (drone-dji:BlackLevel and drone-dji:SensorGainAdjustment) embedded within the captured photos, PhotoMesh normalizes the light sensitivity and establishes consistent baseline black levels across all NIR photos. This calibration process is critical for ensuring uniform reflectance measurements throughout the imagery, thereby effectively

neutralizing the impact of variable sunlight conditions which can otherwise compromise the quality and analytical value of NIR data.

Stability Improvements and Bug Fixes

Project Build

- Fixed an issue where the number of photos participating in the build was not updated after changes to the effective range value.
- Resolved an issue where some build stages failed or entered a loop on German locale machines due to decimal separator issues.
- Fix to the "More Aggressive Match" and "Reconstruction Close Range with Long Lens" presets

Fuser Management

- Conducted a comprehensive refactor and stability enhancement of the fusers pool.
- Corrected a display issue in fuser manager window that incorrectly showed zero active fusers.
- Fixed inaccurate reporting in the Build manager regarding the number of required, allocated, and maximum fusers.
- Prevent job release failures caused by fuser machine reboots during builds.
- Addressed an issue where fusers would get stuck and fail to display completion percentage.
- Enhanced user data management by ensuring the deletion of the fuser's share folder alongside its intermediate files.

AWS Stability and Automation Enhancements

- Resolved issues with fusers running on AWS cloud, including one related to a malfunctioning .dll file that caused installation problems.
- Fixed issue where builds in AWS would fail due to the use of reserved values as tag keys.
- Addressed InvalidArgument error when using AWS fusers in regions without specified subnet.
- Resolved a potential issue where duplicate keys on AWS fusers could disrupt operation.
- Enabled the "Build and Rerun Failed Tasks" functionality using the PhotoMesh Project Queue REST API.

Usability

- Introduced a new logging system where a separate out.log file is created for every task, addressing the issue of concurrent log access in multi-machine environments.
- Fixed Photos List "Exclude" check box issue where exclusion mark was not initially visible.
- Allow selection of reconstruction tiles after creating a new modified terrain object in the area, improving user experience.

Reports

- Fixed error affecting reporting when one of the AT tiles has no calculated photos.

- Addressed a bug that caused projects with reference images to be inaccurately represented in the connection and projection error maps.

What's New in PhotoMesh 8.0

Quality and Performance

Aerotriangulation

- Improved performance for projects with unreliable/No GPS
- Improved accuracy for projects with unreliable/No GPS
- New option to define custom position and orientation accuracy settings for each collection

Improved mesh mode quality

- Close range, sub-centimeter collections
- Multiple collections with large resolution variance
- Clean noise
- Remove gaps in low-resolution pyramid levels

Accelerated production time

- Optimization for close range and vertical collections
- Optimization for wooded area

Automatic Ground Control Points

The latest version of PhotoMesh has been enhanced with an automatic control point template detection feature, which optimizes efficiency by enabling fully automated control point marking. After selecting a template that matches your marker type, such as a checkerboard or cross, PhotoMesh can automatically detect the control point markers in your photos. Any auto-marked control points can also be manually adjusted or entirely overridden, if needed.



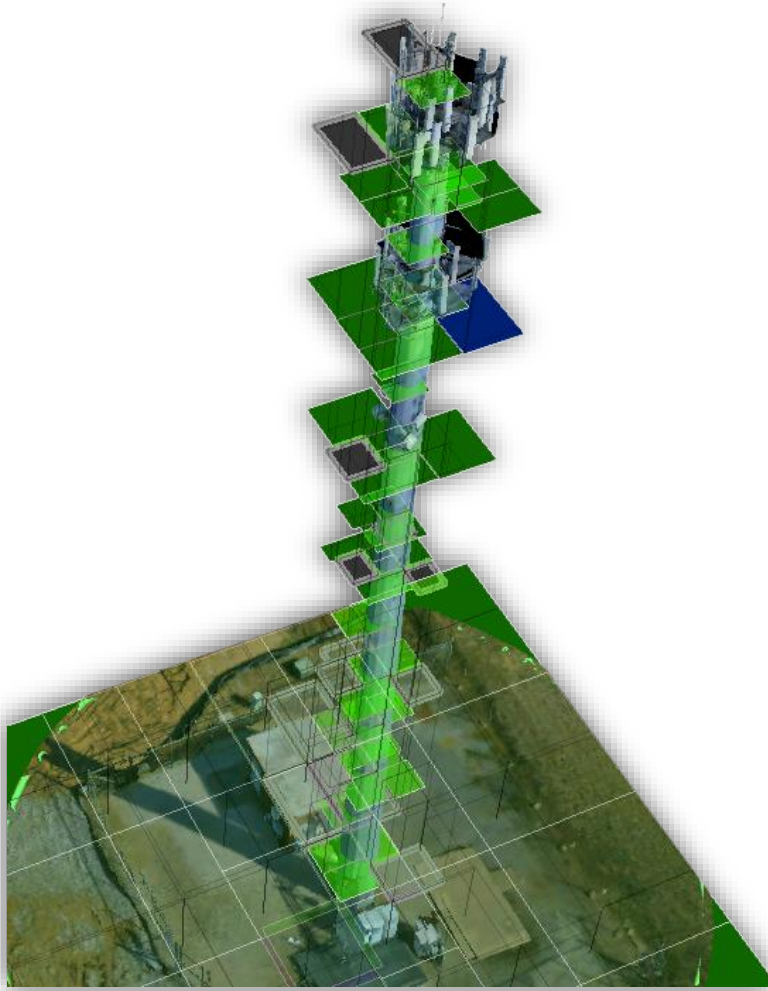
Control Point Markers

Data Validation

The latest version of PhotoMesh includes a comprehensive data validation process for your input data, immediately after loading photos. This process checks for potential issues and displays a detailed dialog with warnings and errors specific to each collection and photo, as well as providing suggestions on how to fix them, ensuring that your project is free of errors and runs smoothly.

Dynamic Tiles

New adaptive 3D tiling algorithm optimizes production time and quality for vertical close-range collections, such as towers, bridges and tall buildings. When reconstruction tiles are created as adaptive 3D boxes, the display style option is automatically set to "3D Dynamic". In this display mode, the tiles and their boxes are displayed in absolute heights. Tiles that were created one on top of each other will appear as if they are floating in this mode.



Manual Retouch Improvements

We've added a new "Remove moving cars" texturing option, specifically designed to tackle reconstruction issues caused by moving vehicles in the photographed area, such as cars that are partially sunk in the road since only some of the photos captured the car in a particular spot. This option ensures that the polygon area is textured only with photos that do not include moving cars. Users can now also fill retouch polygon areas with a single color or a specific texture or use photos from the area.

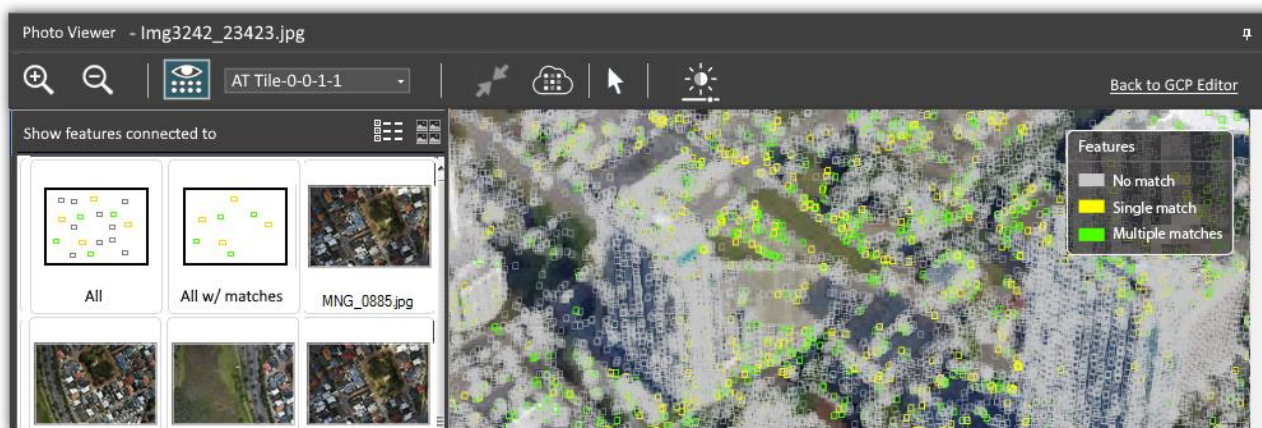
The manual retouch tool has been further enhanced with the ability to perform basic retouch operations on orthophoto format output, in addition to the previously supported mesh layer format, for greater flexibility and precision in your retouch projects.



Photo Viewer Changes

PhotoMesh 8.0 introduces a host of new and exciting improvements to the Photo Viewer, that make it more powerful and user-friendly than ever before. Open the viewer from multiple locations, including the photos list, photo ribbon, control point editor, or when showing photos in point. With the new option to open multiple viewer popups, you can now easily analyze the strength of the features in multiple photos at once.

We've also added new capabilities to the viewer, such as the "Dim" mode which darkens the photo, highlighting feature markers and making them easier to see. The "Show features" command now offers the choice to show matched features identified in the preliminary match phase of the AT step or from the sparse point cloud whose match quality was validated during the AT's bundle adjustment process. In feature display mode, you can select which features to display in the photo viewing window, whether it's all features, only features found in at least one more photo, or only matched features between the photo selected in the photo panel and the main photo. And you can even select a specific feature and identify all photos that contain that feature.



GUI Changes

In the 8.0 release, a new Photos Wizard and a redesigned user interface facilitate a more intuitive user engagement and increase your situational awareness throughout the mesh model preparation and construction process. PhotoMesh 8.0 also introduces a new layout for the Project Panel, with a one-level index control and the transfer of control points output, object/layer, and water polygon information from

the Tree to dedicated item lists. This change streamlines the GUI and makes for a more intuitive workflow experience.

The screenshot displays the PhotoMesh 8.0 interface, highlighting the streamlined workflow for managing water polygons. The 'Control and Tie Points' panel is visible at the top, showing a table with columns for Point Type, Auto-GCP, Exclude, Visible, AT Tile, Sampled Photos, Calculated Photos, Horizontal Accuracy (m), and Vertical Accuracy (m). Below it, the 'Water Polygon' panel is shown, featuring a 'Polygons' tab and a table with columns for Name, Layer, Visible, Excluded, Leveled, and Average Elevation (Source CS). The 'Project' panel at the bottom lists various project items, including Photos, LiDAR, Control and Tie Points, AT Tiles, Reconstruction Tiles, Manual Retouch, Water Polygons, Reference Projects, Build Outputs, Objects and Layers, and Messages.

Point Type	Auto-GCP	Exclude	Visible	AT Tile	Sampled Photos	Calculated Photos	Horizontal Accuracy (m)	Vertical Accuracy (m)
Control	☐	☐	☒	-	0	0	0.10	0.20

Name	Layer	Visible	Excluded	Leveled	Average Elevation (Source CS)
1	water	☒	☒	☒	24.36
2	water	☒	☐	☐	40.00
3	water	☒	☐	☒	25.00
4	water	☒	☒	☒	25.70
5	water	☒	☒	☒	27.00
6	water	☒	☒	☒	25.22
7	water	☒	☐	☒	32.91

Project Item	Count	Status
Photos	2673	☒
LiDAR	0	☐
Control and Tie Points	0	☒
AT Tiles	4	☐
Reconstruction Tiles	0	☐
Manual Retouch	0	☐
Water Polygons	7	☐
Reference Projects	0	☐
Build Outputs	62	
Objects and Layers	15	
Messages	0	

Water Polygons

In PhotoMesh 8.0, we've streamlined the water polygon creation and editing process with the introduction of a new Water Polygon list. This list consolidates all water polygon properties and settings, such as visibility, inclusion in the project, and water level constancy, in one convenient location, eliminating the need for a separate property sheet. The Water Polygon list has two tabs: Polygons and Layers, providing an easy way to view and manage all individual polygons in the project and polygon layers respectively.

Water Polygon

Polygons Layers

Enter text to search...

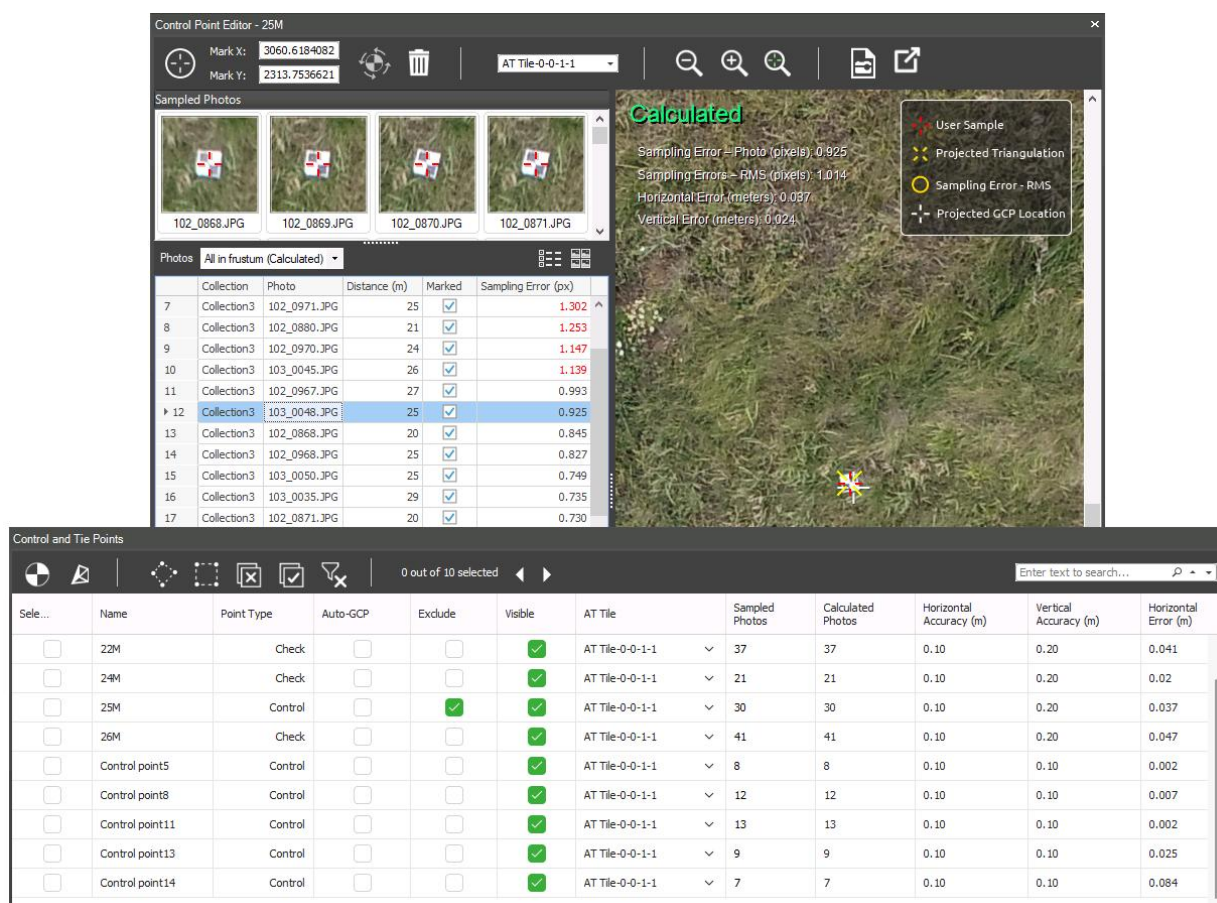
Selection	Name	Layer	Visible	Excluded	Leveled	Average Elevation (Source CS)
☐	wp_6	Manually Created	☒	☐	☒	208.60
☐	wp_7	Manually Created	☒	☐	☒	226.74
☐	wp_8	Manually Created	☒	☐	☒	240.01
☐	wp_39	666	☒	☐	☐	228.26
☐	wp_40	666	☒	☐	☐	328.26
☐	wp_41	666	☒	☒	☐	328.72
☐	wp_42	666	☒	☐	☐	493.87
☐	wp_43	666	☒	☐	☐	479.90
☐	wp_44	666	☒	☐	☐	379.90
☐	wp_45	666	☒	☐	☐	241.77

AWS Improvements: Fuser Throttling

New Launch Rate setting lets you control the maximum number of fuser instances launched per minute, e.g., if Maximum Instances was set to 200, and the Launch Rate to 50, the Build Manager will spread the launch of the fusers over 4 minutes. This prevents a potential bottleneck that could be caused by a large number of fusers attempting to read the build data at the same time.

Control Point Editor

In PhotoMesh 8.0, we've streamlined the control point editing process by introducing a new control point list that consolidates control point properties and their editing in one place, while reserving the control point editor for marking control points in photos. You can seamlessly navigate between the editor and list using either the "Open GCP Editor" button in the list's toolbox or by selecting a control point row in the list, when the GCP editor is open.



Multi-Fuser Support

PhotoMesh now formally supports the launching of multiple fusers on powerful machines to improve overall production performance and fully utilize the computer's power. In addition to standard fuser use, PhotoMesh also offers two options for auto-scaling of fusers to enable dynamic allocation of fusers based on project requirements:

- **Amazon Web Services (AWS) Fusers** – AWS fuser instances can be dynamically launched whenever there are unassigned pending fuser tasks, to further accelerate model creation.
- **Fusers Pool** – Fusers can be shared between several PhotoMesh Build Managers in your organization, enabling each machine running PhotoMesh to use the combined resources of all the machines in the pool.

New Output Options

- PhotoMesh supports the latest 3D Tiles, SLPK, and o3DML output formats
- Point cloud output colorized by the number of correlated photos
- Watermark option has been removed. In PhotoMesh evaluation version, output is generated without a watermark, but it is mandatory to include a Skyline copyright notice and logo.
- Optimized mesh model to reduce size without compromising on quality

- New option to export the selected photos' aerotriangulation information to a CSV file.

Quality Report

New Quality Report, containing statistical and graphical information about the build and its aerotriangulation and reconstruction results, highlights key information and alerts you to potential problems such as missing information, triangulation errors, or failed tiles so you can address and resolve the necessary issues. It includes four maps that graphically depict information about photo matches, level of detail, position accuracy and more.

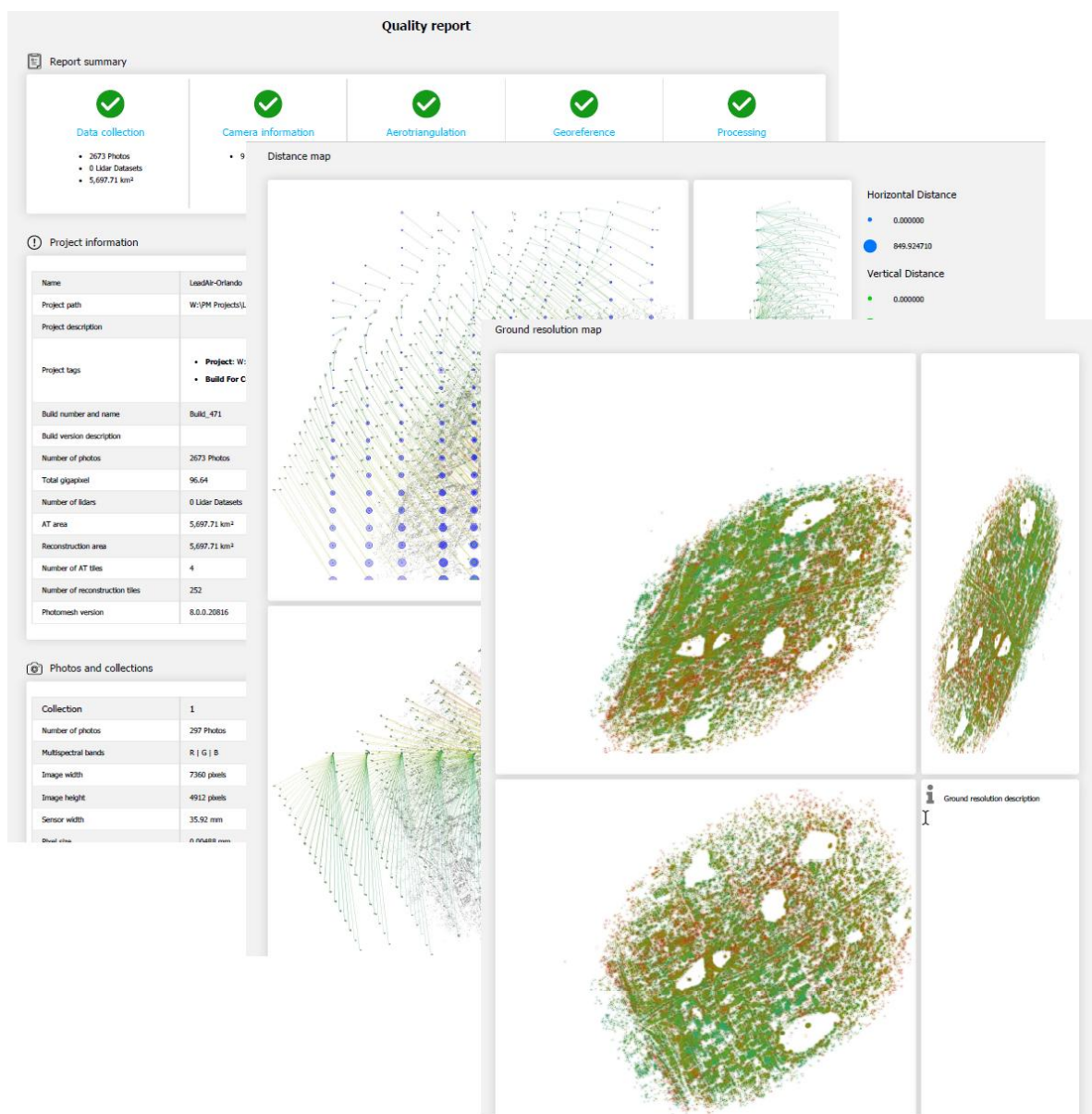
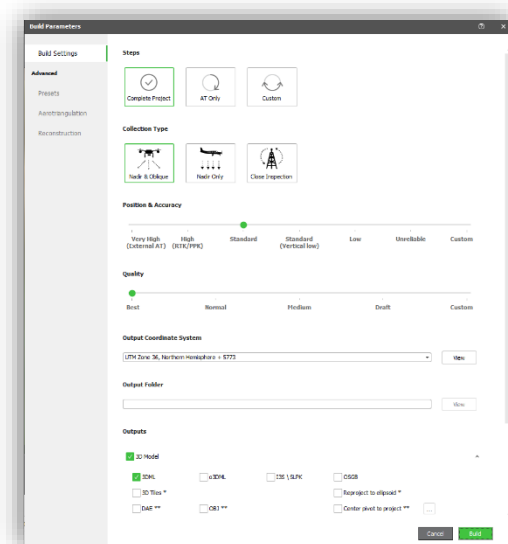
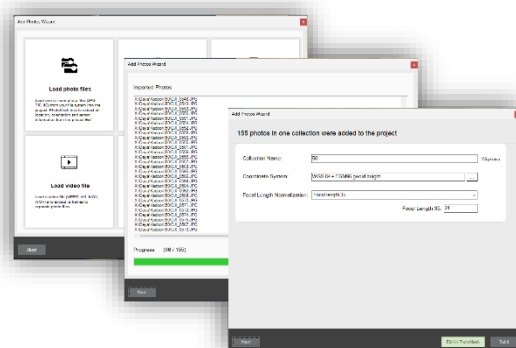


Photo Wizard

The new PhotoMesh Wizard streamlines the process of adding photos to a project by consolidating all load and import options, including photo files/folders, Excel/XML photo list file, video, and photos with

metadata into a single, user-friendly dialog. This eliminates the need to navigate multiple menus and simplifies the process of bringing photos and metadata into your project. In addition to this general revamping, the following specific improvements and capabilities have been introduced:

- **CSV Import** – PhotoMesh now remembers and retains last mapping of CSV columns to corresponding photo properties. This means that for subsequent imports with the same CSV file, you won't need to re-map the columns.



Build Wizard

The Build dialog now includes a simplified view that makes it easier to use. In this view, all the main settings for a standard build are consolidated in one tab labeled "Build Settings." Along with this change, some styling enhancements have also been made to the dialog.

Several new presets have been added to make it easier that make it easy to automatically set the parameters that will generate the optimum output for your project's data :

- **Close Inspection** - For projects with a single object captured from all perspectives. PhotoMesh focuses on the foreground object to achieve best image correlation relative to the close-range object and optimizes 3D reconstruction for modeling of thin features and complex geometry.

- **Skeletal Objects** – Aggressive cleaning of spaces within skeletal objects, resulting in a cleaner, more visually accurate 3D model. This is particularly effective for objects like telecommunications towers and other mostly hollow objects.
- **Nadir Only Dataset** – Improves edge sharpness and overall orthophoto quality.

We've also incorporated two new sliders:

- **Position Accuracy slider** - Allows you to determine what weight should be assigned to provided camera positioning when performing aerotriangulation.
- **Output Quality slider** - Includes a Draft mode option for faster processing.

Build Parameters

Build Settings

Advanced

Presets

Aerotriangulation

Reconstruction

Steps

Complete Project AT Only Custom

Collection Type

Nadir & Oblique Nadir Only Close Inspection

Position & Accuracy

Very High (External AT) High (RTK/PPK) Standard Standard (Vertical low) Low Unreliable Custom

Quality

Best Normal Medium Draft Custom

Output Coordinate System

WGS84 / UTM zone 17N View

Output Folder

View

Outputs

☒ 3D Model

Cancel Build

CHAPTER 4 THE PHOTOMESH WORKFLOW

About PhotoMesh Workflow

PhotoMesh converts standard 2D photographs into high-resolution, textured, 3D mesh models in a streamlined, step-by-step process:

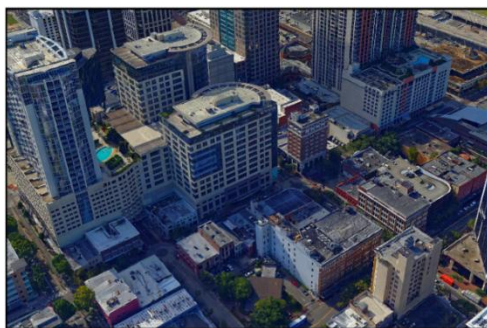
- [Step 1: Imagery Acquisition](#)
- [Step 2: Creating a PhotoMesh Project](#)
- [Step 3: Loading Photos and Other Input Sources and Preparing the Project](#)
- [Step 4: Managing Local, Pool, and AWS Fusers](#)
- [Step 5: Building the Model](#)
- [Step 6: Managing and Reviewing the Build](#)

Step 1: Imagery Acquisition

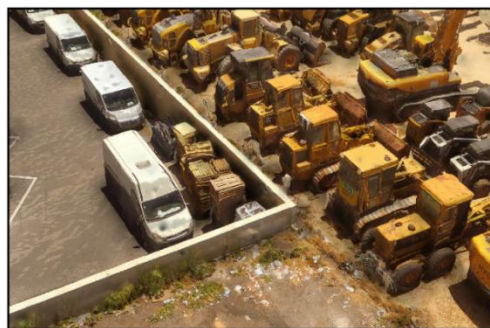
The first step to creating a 3D model is imagery acquisition. All visible surfaces of the intended object or scene must be photographed from multiple overlapping viewpoints.

The accuracy of the resulting 3D mesh model is affected by several factors including: input image resolution and quality, lens distortion, overlapping ratio, directionality, and lighting. While PhotoMesh optimizes the accuracy of the 3D mesh model by selecting the best photos for an area, each factor can affect the overall output or cause varying levels of accuracy and quality in different areas of a project.

See the "[Imagery Acquisition](#)" chapter for more information.



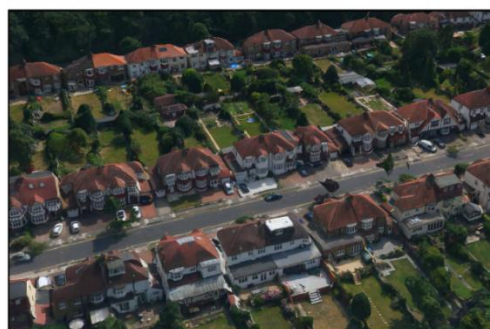
Ground resolution = 5cm per pixel – Orlando USA



Ground resolution = 1 cm per pixel – Tel Aviv (drone)



Ground resolution = 3 cm per pixel - Frederick



Ground resolution = 7cm per pixel – Enfield England (Midas)

Step 2: Creating a PhotoMesh Project

Before beginning to work in PhotoMesh, you must create a new project or open an existing project. See ["Creating and Opening a PhotoMesh Project"](#) in the "Project Management" chapter for more information.

Step 3: Loading Photos and Other Input Sources and Preparing the Project

Photos and other input sources are loaded into a project using any of the following methods:

- Load photo files from disk (JPG, TIF, IIQ).
- Load an Excel/XML file with a list of photos, their file paths, all photo information (e.g., XYZ, OPK/YPR) and photo collection and camera information (e.g., focal length, sensor width, pixel size).
- Load a CSV file with a list of photos, their file paths and XYZ and OPK/YPR photo information.
- Load a PhotoMesh project. When a PhotoMesh project (.PhotoMeshXML) is loaded into a second project, only the project's photos and photo and collection parameters are loaded.
- Load a video from which PhotoMesh extracts individual image frames.
- Import photos of different data types (e.g., Inpho, Bingo, Stellacore) into the project with their metadata (e.g., sensor width, focal length, and aerotriangulation data). When AT results are imported, the AT step of the build process can be skipped (by selecting "Do not calculate AT (Fully trusted)" in the Calculation Mode AT parameter).
- Load LiDAR data in .las, .laz or .e57 formats into a PhotoMesh project, integrating it with project photos to supplement project data and increase model accuracy.

See ["Loading Photos"](#) and ["Importing Photos with Metadata"](#) in the "Photo Management" chapter, and ["Loading LiDAR Data"](#) in the "LiDAR Management" chapter for information.

After photos have been loaded, do any of the following to prepare the project for the build:

- Modify photo and photo collection properties
- Verify that the camera symbols generate the expected cluster form
- Project photos on the terrain to see if they fit the terrain imagery or to better understand their position
- Show LiDARs on the terrain to see if they fit the terrain imagery or to better understand their position
- Associate a LiDAR with its trajectory data that provides information about the scanner's position while capturing the point cloud
- Improve the accuracy of the model's position by creating ground control points that associate real-world X, Y, Z coordinates with corresponding locations (in pixels) in photos
- Establish a correspondence between project photos and improve aerotriangulation results by creating tie points in the same physical point in three or more photos
- View photos in the photo viewer
- Import or draw water body polygons to generate a flat surface with correct texturing over user-defined water bodies
- Display the directions map to identify which areas of the project have sufficient coverage and which require more

- Display the overlap map
- Create a subset AT area in which to validate camera parameters and build settings
- Set a reconstruction area

See ["Modifying Photo and Photo Collection Properties"](#) in the "Photo Management" chapter, ["Working with Water Body Polygons"](#) in the "Reviewing your Build" chapter and the ["LiDAR Management"](#), ["Preparing the Project"](#), and ["Control Points"](#) chapters for information.

Step 4: Managing Local, Pool, and AWS Fusers

Fusers allow you to share the demanding workload involved in processing the different build steps (data preparation, AT, point cloud, mesh model, texture, 3DML/other outputs) between several computers on the same local network. The master computer utilizes the computing power of the network's client computers by connecting to PhotoMesh fusers on the client machines.

Fusers can be shared between several PhotoMesh Build Managers in your organization, enabling each machine running PhotoMesh to use the combined resources of all the machines in the pool. PhotoMesh can also work with Amazon Web Services (AWS), adding and terminating virtual fuser machines, to scale a project up and down based on resource needs.

See ["Installing PhotoMesh Fusers"](#) and ["Uninstalling PhotoMesh Fusers"](#) in the "Getting Started" chapter for information on installing and starting the fusers. See ["Fuser Management \(on the Master computer\)"](#) and ["Fuser Auto-Scaling"](#) in the "Fusers" chapter for information on managing the fusers, and the ["Getting Started with AWS"](#) chapter for information about AWS fusers.

Step 5: Building the Model

After all photos are added to the project and all properties are set as required, you can launch PhotoMesh Build Manager to begin the build process. The build process provides flexibility both with regard to the area of the project being built (entire or selected area) and the build steps that are to be performed. To achieve optimal results and avoid unnecessary rebuilding, it is recommended to initially perform only the aerotriangulation step ("AT Only"), and then review the results, to ensure a precise AT which includes all project's photos that intersect with the AT area, before proceeding with the rest of the build. When an AT Only build is repeated following project adjustments, PhotoMesh determines precisely which tiles and which specific AT sub-processes were affected by the modification of properties or control points, enabling a much lighter and faster AT process.

The build process involves the following steps:

- Data Preparation
 - Conversion of photos to PhotoMesh's internal format (SMPT)
 - Conversion of LiDAR to PhotoMesh's internal format (CPT)
 - Color correction of photos
- Aerotriangulation
- Reconstruction
 - Point Cloud Creation
 - Model Creation
 - Model Texturing

- Project Output - Select all required output formats: 3DML (for loading directly into TerraExplorer and streaming with SkylineGlobe Server), DAE, OBJ, 3D Tiles, SLPK (I3S), OSGB, Orthophoto, DSM, DTM/DEM, and LAS (Point cloud public file format). **See** ["Setting Build Steps, Parameters, and Outputs"](#) in the "Building" chapter for information on the different file formats.

See the ["Building"](#) chapter for information.

Step 6: Managing and Reviewing the Build

Throughout the build process, you can monitor build progress from PhotoMesh Build Manager and view the build data already created in the PhotoMesh 3D Window. As soon as a build step is completed for a tile, its build data is viewable in the 3D Window, e.g. if aerotriangulation was performed, PhotoMesh can display the calculated camera positions in the 3D Window, and if reconstruction tiles were calculated, they can be displayed in the 3D Window, with each tile colorized according to the build step completed. You can also view point cloud, 3D model, and texture data for one or more tiles once this build step for the selected tile is completed. Build quality and production reports are available, as well, with statistical and graphical information about the build process. **See** ["Reviewing Aerotriangulation and Reconstruction Tiles"](#) in the "Reviewing Your Build" chapter for information.

After a build is complete, water bodies can be defined by drawing or importing a water body polygon, and mesh imperfections, such as bumps, irregular surfaces, or floating artifacts, can be marked for flattening, filling, and clean up using the Manual Retouch tool. **See** ["Working with Water Body Polygons"](#) and ["Marking a Mesh Layer for Retouching"](#) in the "Reviewing Your Build" chapter for information.

After reviewing build data, you can easily rebuild any of the following: the entire project, error tiles, or selected tiles. **See** ["Monitoring a Build"](#) in the "Building" chapter and the ["Reviewing Your Build"](#) chapter for information. Alternatively, you can create a new build version with different project steps and different build parameters. You have the option to create a completely new build version or one which copies the AT information from the previous build, thereby accelerating the build process. **See** ["Creating a New Build Version"](#) and ["Creating a New Build Version with a Copy of the Current Aerotriangulation Results"](#) in the "Building" chapter for information.

Before rebuilding, 3D models can be exported when necessary to external software, such as Autodesk® MeshMixer, to perform certain edits, e.g. to remove a particular building that is being torn down, flatten surfaces and correct certain imperfections such as bumps or irregular surfaces, or edit textures, and then reimported to PhotoMesh. **See** the ["Importing and Exporting Models"](#) chapter for more information.

CHAPTER 5 BASIC CONCEPTS

Overview

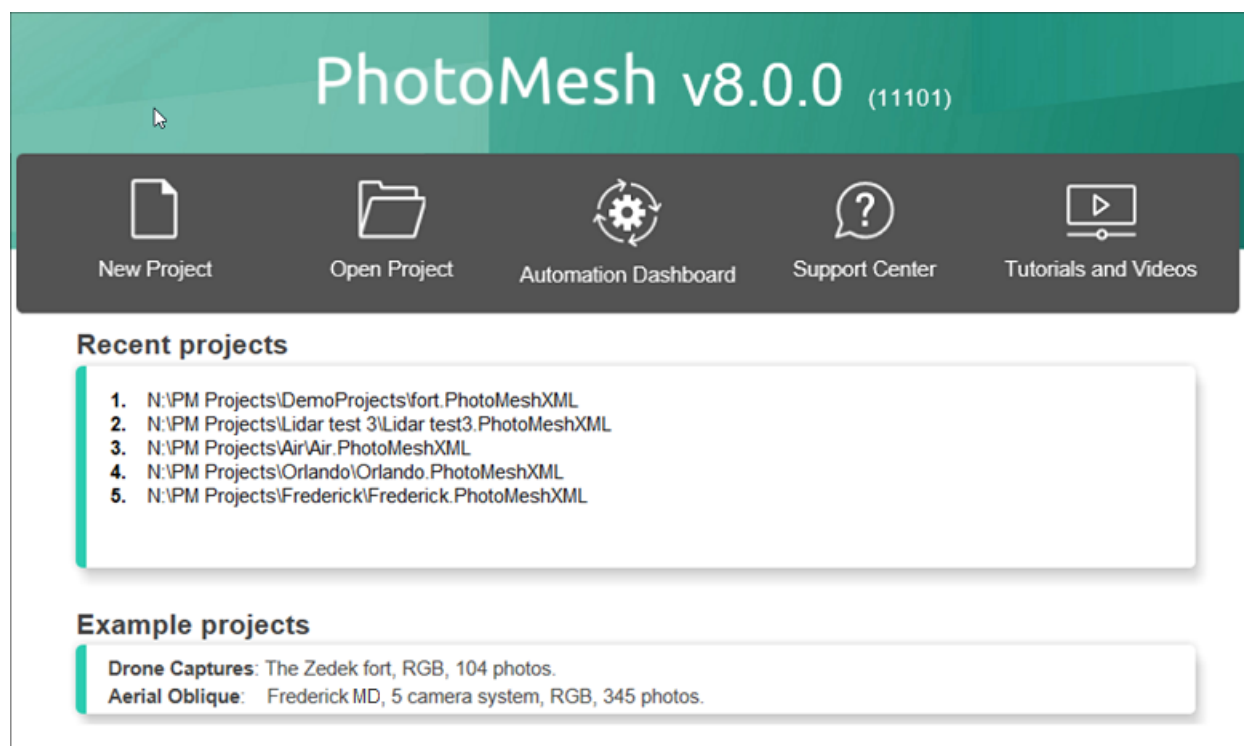
This chapter addresses the following topics:

- [Start Page](#)
- [Application Layout](#)
- [Error/Message Log](#)
- [PhotoMesh Ribbon](#)
- [Setting the Display of the 3D Window](#)
- [Saving a Project](#)
- [Selecting Objects in the 3D Window](#)
- [Dragging and Dropping Files in PhotoMesh](#)
- [Setting PhotoMesh Options](#)
- [Coordinate System Advanced Capabilities](#)

Start Page

The Start Page is the first page you see after opening PhotoMesh from the **Program Files** menu or from the desktop icon. The Start Page also appears after you close your current project.

When you start PhotoMesh by selecting a *.PhotoMeshXML file or link, PhotoMesh opens the project directly and does not show the Start Page.



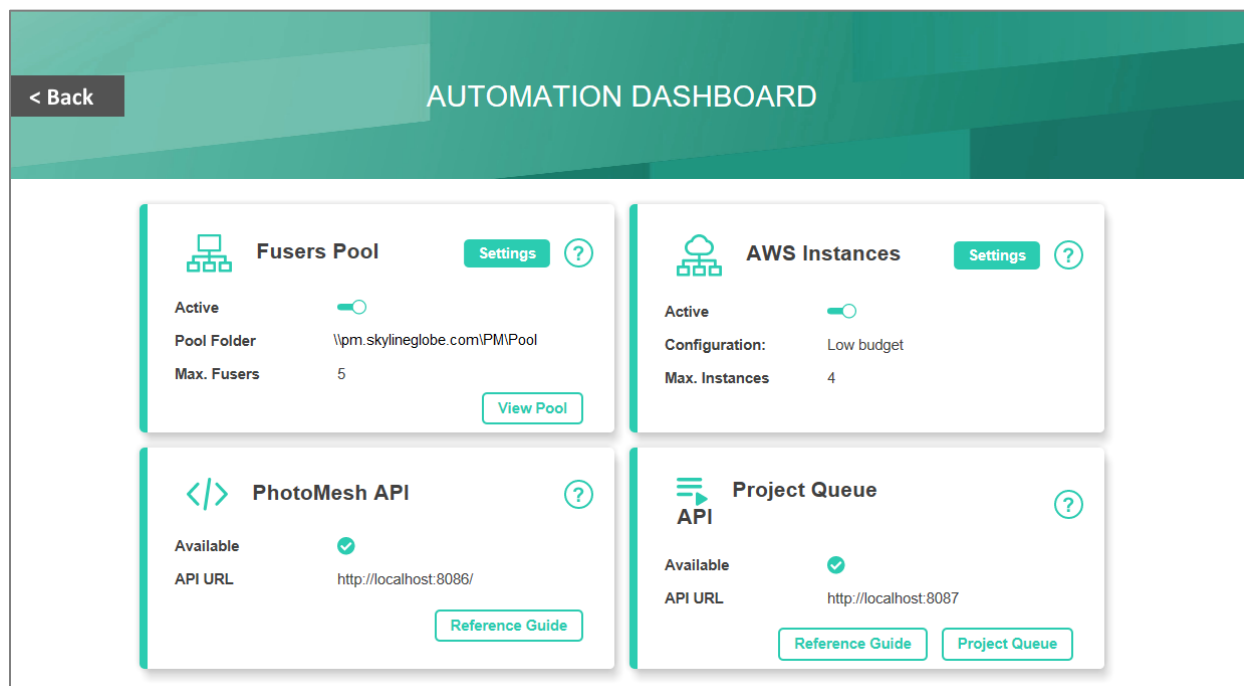
Start Page

Start page options:

- **New Project** – Opens a dialog in which you can create a new project.
- **Open Project** – Opens a dialog in which you can browse to a project file.
- **Automation Dashboard** – Opens PhotoMesh's automation center, from where you can do all the following:
 - [Fusers Pool](#) – Monitor and manage fusers in your fusers pool.
 - [AWS Instances](#) – Monitor and manage your AWS instances.
 - [PhotoMesh API](#) – View interactive documentation of the REST API for PM Production.
 - [Project Queue API](#) – View technical and interactive documentation of the REST API for project queues and open the Project Queue Manager.
- **Support Center** – Opens Skyline's [knowledge base](#), a centralized repository of information about PhotoMesh.
- **Videos of major PhotoMesh Features** – Opens [Skyline's YouTube channel](#) with videos demonstrating PhotoMesh key features.
- **Recent Projects** – Opens one of your recent PhotoMesh projects.
- **Example Projects** – Download one of the ready-to-build example projects in PhotoMesh, and follow the instructions to get started building.
- **Contact Business and Sales** – Opens the Contact Skyline page on Skyline's website.

- **Technical Support** – Opens the Skyline Technical Support page on Skyline's website.
- **License Agreement** – Displays the PhotoMesh software license agreement on the main window.

Automation Dashboard



From PhotoMesh's [Start](#) page, you can access PhotoMesh's Automation Dashboard, from which you can control all the following:

- **Fusers Pool** – Monitor and manage fusers in your fusers pool:
 - Change pool settings: Active / Not active, and maximum number of fusers. Click **Settings** to open the **Auto Scaling Fusers Settings** dialog to change the pool status, pool folder, or maximum fusers. See ["Using a Fusers Pool"](#) in the "Fusers" chapter for more information.
 - Manage the fusers pool. Click **View Pool** to open the Pool Manager with information on all the fusers in the pool. See ["Managing a Fusers Pool"](#) in the "Fusers" chapter for more information.
 - Click **Help** (?) to open the relevant section in Skyline's online knowledge base.
- **AWS Instances** – Monitor and manage your AWS instances:
 - Change AWS fuser settings: Active / Not active, and maximum number of instances. Click **Settings** to open the **Auto Scaling Fusers Settings** dialog to use/stop using AWS Instances, change maximum instances or other AWS settings. See ["Using Amazon Web Services \(AWS\) Fusers"](#) in the "Fusers" chapter for more information.
 - Click **Help** (?) to open the relevant section in Skyline's online knowledge base.
- **PhotoMesh API** – View interactive documentation of the REST API for PM Production. See ["PhotoMesh API"](#) in the "PhotoMesh Automation" chapter for more information. You can do all the following:
 - Click **Reference Guide** to view interactive documentation of the REST API for PM Production.

- Click **Help**  to open the relevant section in Skyline's online knowledge base.
- **Project Queue API** – View technical and interactive documentation of the REST API for project queues and open the Project Queue Manager. See "[Queue API](#)" in the "PhotoMesh Automation" chapter for more information. You can do all the following:
 - Click **Reference Guide** to view technical and interactive documentation of the REST API for project queues.
 - Click **Project Queue** to open the Project Queue Manager.
 - Click **Help**  to open the relevant section in Skyline's online knowledge base.

Application Layout

The PhotoMesh display includes the following panels:

Panel	Description
3D Window	The main working area for various actions including preparing, monitoring, and reviewing the build.
Project Tree	Provides access to the items in the project. When you select a group in the Project Tree, the corresponding "list" is displayed with the group's items. See "Using the Project Tree and Item List" chapter for further information. The Project Tree has the following groups: ▪ Photos ▪ LiDARs ▪ Control and Tie Points ▪ AT Tiles ▪ Reconstruction Tiles ▪ Manual Retouch ▪ Water Polygons (manually created and imported) ▪ Reference Projects ▪ Build Outputs ▪ Objects and Layers ▪ Messages
List	The Item List displays a list of all the items in the currently selected Project Tree group, and information about each item.
Properties	Displays data relating to the currently selected photo collection or photo. See "Modifying Photo and Photo Collection Properties" in the "Photo Management" chapter. Note: LiDAR, build outputs, and terrain object/ layer properties are not displayed in the property sheet panel. You can view LiDAR point cloud properties by right-clicking the LiDAR in the LiDAR List, and selecting Style Point Cloud. You can view terrain object/layer properties and build outputs by right-clicking in the respective List, and selecting Display Properties.

Error/Message Log	Records and displays warnings, messages, and errors to help you troubleshoot any PhotoMesh issue. See " Error/Message Log " in this chapter for more information.
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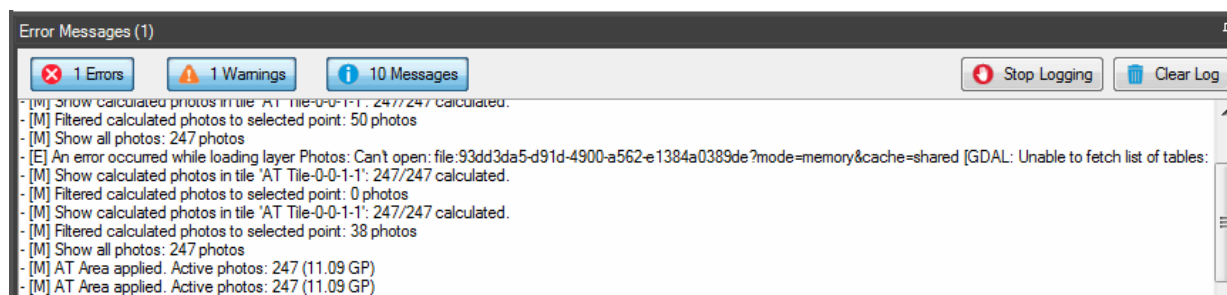
Error/Message Log

The PhotoMesh log records and displays warnings, messages, and errors to help you troubleshoot any PhotoMesh issue. Logs can be exported and sent to Skyline support:

- **Errors** – Alerts users of critical problems.
- **Warnings** – Alerts users of conditions they should be aware of, which might cause problems in the future.
- **Messages** – Records various activities.

Viewing the Log

To view the log:



Error, Warning, and Message Log

- In the **Project Panel**, select **Messages**, and then in the **Messages** list, click the **Errors**, **Warnings**, and **Messages** buttons to toggle the display of their respective messages on and off.

Clearing the Log

To clear the log:

- At the top of the log, click **Clear Log**.

Stop Logging

To stop logging:

- At the top of the log, click **Stop Logging**.

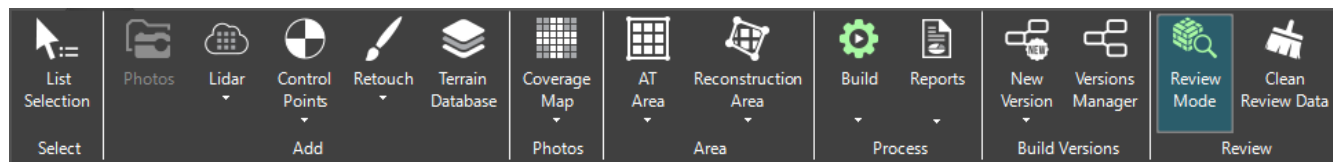
PhotoMesh Ribbon

The following commands are available from the File menu located at the upper left hand corner of the application:

Commands	Shortcut Key	Activity
New	CTRL+N	Creates a new PhotoMesh project. See "Creating and Opening a PhotoMesh Project" in this chapter for information.
Open	CTRL+O	Opens an existing PhotoMesh project. See "Creating and Opening a PhotoMesh Project" in the "Project Management" chapter for information.
Close		Closes the open PhotoMesh project.
Save	CTRL+S	Saves the current project.
Resources		Opens resources that display license information, basic application information, or help files. See "PhotoMesh Resources" in the "Introduction" chapter for information.
Options		Set default system settings for PhotoMesh. See "Setting PhotoMesh Options" in this chapter for information.

Home Tab

The Home tab contains commands for setting up your project (setting build settings, adding photos to a project, setting photo display on the terrain, and setting the aerotriangulation and reconstruction areas), building the 3D model, and reviewing your project.



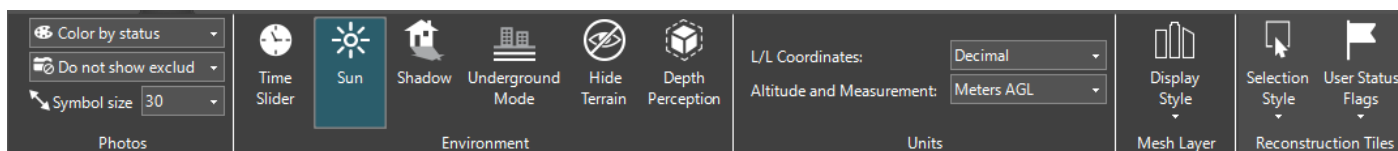
Home Tab

Group	Description
Select	See "Selecting Objects in the 3D Window" in the "Basic Concepts" chapter.
Add	See "Loading Photos" in the "Photo Management" chapter, "Loading LiDAR Data" in the "LiDAR Management" chapter, "Setting the Terrain Database" in the "Project Management" chapter, "Working with Water Body Polygons" and "Retouching a Mesh Layer" in the "Reviewing your Build" chapter, "Importing a Model" in the "Importing and Exporting Models" chapter and the "Control Points" chapter.
Photos	See "Showing the Directions Map" , "Showing the Connection Map" , and "Showing the Overlap Map" in the "Preparing the Project" chapter.
Area	See "Setting the AT Area" and "Setting the Reconstruction Area" in the "Preparing the Project" chapter.

Process	See the "Building" chapter" and " Viewing a Production Report ", and " Viewing a Quality Report " in the "Reviewing Your Build" chapter.
Build Versions	See " Creating a New Build Version " in the "Building" chapter and " Managing Build Versions " in the "Reviewing Your Build" chapter.
Review	See the " Reviewing Your Build " chapter.

View Tab

The View tab contains commands for changing your units and view in the 3D Window.

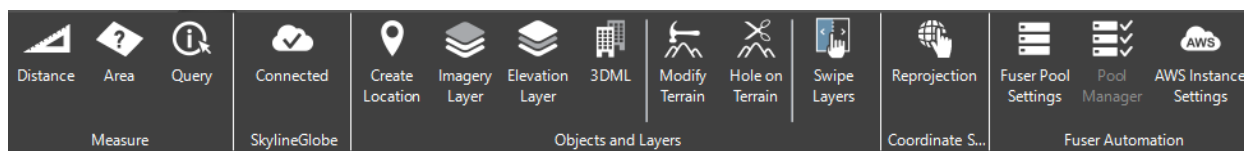


View Tab

Group	Description
Photos	See " Displaying Camera Symbols " and " Selecting Camera Symbol Size and Color " in the "Preparing the Project" chapter.
Environment	See " Setting the Display of the 3D Window " in this chapter.
Units	See " Units " in this chapter.
Mesh Layer	See " 3DML Display Style Options " in the "Reviewing Your Build" chapter.
Reconstruction Tiles	See " Setting a Reconstruction Tile's Selection Style " and " Displaying User Status Flags and Legend " in the "Reviewing Your Build" chapter.

Tools Tab

The Tools tab provides a set of TerraExplorer analysis tools that help you validate photo parameters. It also contains terrain objects and layers that can be added to a PhotoMesh project and a command for connecting to SkylineGlobe Server.



Tools Tab

Group	Description
Measure	See the " Analysis Tools " chapter.
SkylineGlobe	See " Connecting to a SkylineGlobe Server " in the "Working with SkylineGlobe Server" chapter.

Objects and Layers	See the "Adding Terrain Objects and Layers" chapter and "Swipe Mesh, Point Cloud, and Imagery Layers" in the "Analysis Tools" chapter.
Coordinate System	See "Coordinate System" in the "Basic Concepts" chapter.
Fuser Automation	See "Fuser Auto-Scaling" in the "Fusers" chapter and the "PhotoMesh Automation" chapter.

Photo Tab

The Photo (contextual) tab contains tools for viewing the project's photos. This tab is displayed when a photo is selected in the photo list.

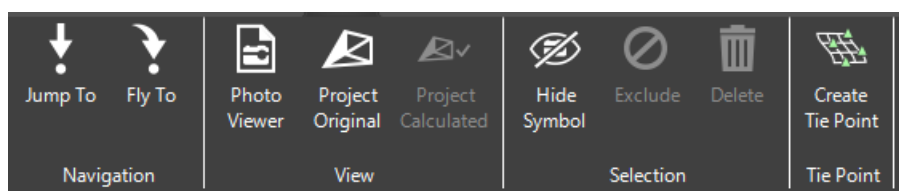
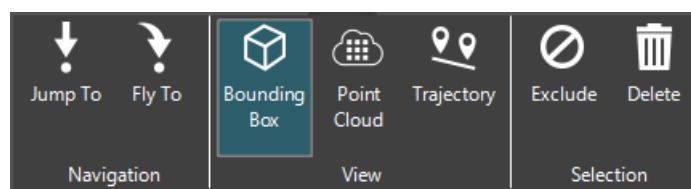


Photo Tab

Group	Description
Navigation	See "Navigating to a Photo" in the "Preparing the Project" chapter.
View	See "Viewing Photos in the Photo Viewer" and "Projecting a Photo on the Terrain" in the "Preparing the Project" chapter.
Selection	See "Displaying Camera Symbols" in the "Preparing the Project" chapter and "Excluding a Photo" , and "Deleting a Photo" in the "Photo Management" chapter.
Tie Point	See "Creating Tie Points" in the "Control Points" chapter.

LiDAR Tab

The LiDAR (contextual) tab is displayed when a LiDAR file is selected and it provides tools for viewing the LiDAR data.



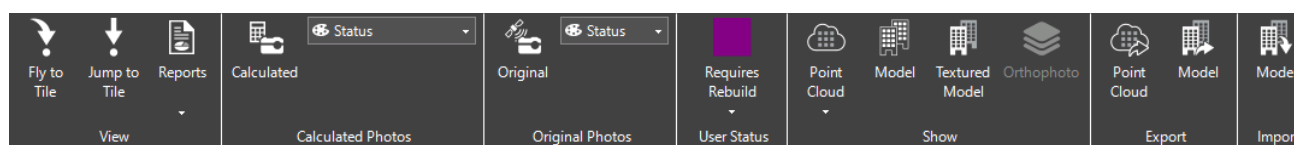
LiDAR Tab

Group	Description
Navigation	See "Navigating to a LiDAR" in the "LiDAR Management" chapter.

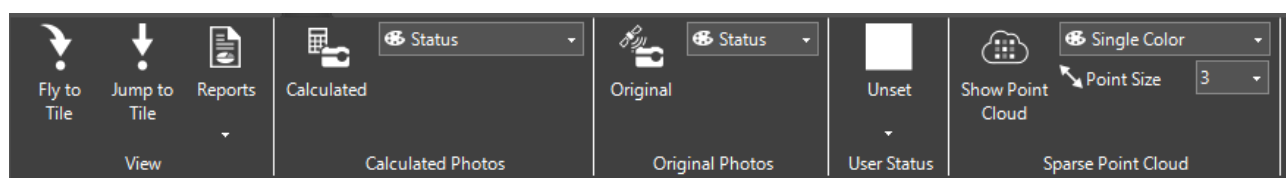
View	See "Showing a LiDAR Point Cloud on the Terrain" and "Associating a LiDAR Point Cloud with its Trajectory Data" in the "LiDAR Management" chapter.
Selection	See "Excluding a LiDAR" and "Deleting a LiDAR" in the "LiDAR Management" chapter.

Tile Tab

The Tile (contextual) tab is displayed when either an AT or reconstruction tile is selected and it provides tools for displaying and exporting the different steps of the build output. The commands included in the tab depend on whether an AT or reconstruction tile is selected.



Reconstruction Tile Tab



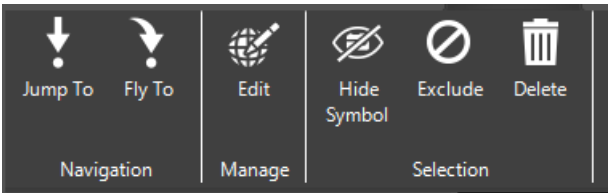
AT Tile Tab

Group	Description
View	See "Navigating to a Tile" , "Viewing a Production Report" , and "Viewing a Quality Report" in the "Reviewing Your Build" chapter.
Calculated Photos	See "Displaying Camera Positions for Tiles" in the "Reviewing Your Build" chapter.
Original Photos	See "Displaying Camera Positions for Tiles" in the "Reviewing Your Build" chapter.
User Status	See "Selecting Tiles" and "Viewing Tile Information in the Tile List" in the "Reviewing Your Build" chapter for information.
Sparse Point Cloud	See "Viewing an Aerotriangulation Tile's Sparse Point Cloud" in the "Reviewing Your Build" chapter. This group only displays when an AT tile is selected.
Show	See "Displaying Reconstruction Tile Results" in the "Reviewing Your Build" chapter. This group only displays when a reconstruction tile is selected.
Export	See "Exporting a Point Cloud" and "Exporting a Model" in the "Importing and Exporting Models" chapter. This group only displays when a reconstruction tile is selected.

Import	See "Importing a Model" in the "Importing and Exporting Models" chapter. This group only displays when a reconstruction tile is selected.
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Control Point Tab

The Control Point (contextual) tab provides commands for working with control points and marking the position of control points in photos.



Control Point Tab

Group	Description
Navigation	See "Navigating to a Control Point" in the "Control Points" chapter.
Manage	See "Editing Control Points" and "Marking Control Points in Photos" in the "Control Points" chapter.
Selection	See "Hiding Control Point Symbols" , "Excluding Control Points" , and "Deleting Control Points" in the "Control Points" chapter.

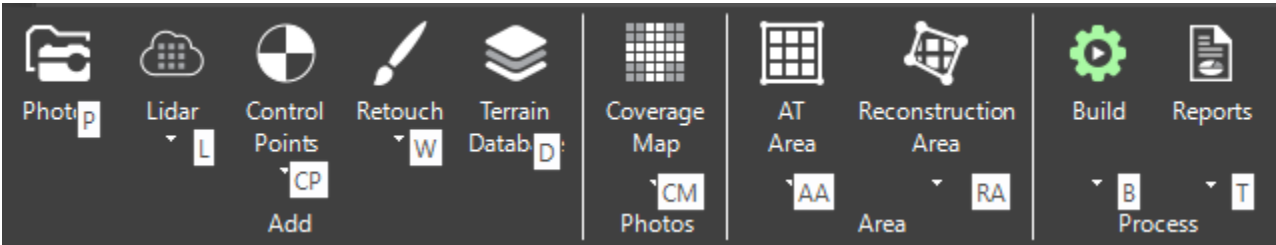
Shortcut Keys

Most commands can be activated through a shortcut key combination (KeyTip) that appears on the Ribbon when you press the ALT key.

Note: Some alternate shortcut keys appear in this documentation in parentheses in the section where the command is explained.

To use KeyTips:

1. Press **ALT**. The KeyTips are displayed over each main tab and over each command on the Quick Access toolbar.



PhotoMesh KeyTips

2. Press the letter or number shown in the KeyTip over the tab/command that you want to use.
3. If you pressed a letter over one of the tabs, that tab is displayed showing KeyTips over the enabled commands. Press the letter of the required command.

Note: Pressing ALT a second time hides the KeyTips.

Quick Access Toolbar

The Quick Access toolbar is a customizable toolbar that contains a set of commands that are independent of the tab that is currently displayed.

Commands can be added or deleted from the Quick Access toolbar, according to each user's needs and convenience.

Adding/Deleting Commands from the Quick Access Toolbar

The Quick Access toolbar can be customized in two ways:

- By right-clicking the required commands on each tab.
Note: This method is recommended for addition/deletion of just a few commands.
- By opening a dialog in which you select the commands to include in the Quick Access toolbar.
Note: This method is recommended for extensive modification of the Quick Access toolbar.

Adding/Deleting Commands by Right-Clicking

To add or delete a command:

1. If you want to add a command to the Quick Access toolbar, do the following:
 - a. On the required tab, right-click the command you want added to the Quick Access toolbar.
 - b. From the shortcut menu, select **Add to Quick Access Toolbar**. The command is added to the Quick Access toolbar.
2. If you want to delete a command from the Quick Access toolbar, do the following:
 - a. On the **Quick Access** toolbar, right-click the command you want to delete.
 - b. From the shortcut menu, select **Remove from Quick Access Toolbar**. The command is deleted from the Quick Access toolbar.

Adding/Deleting Commands Using the Customize Dialog

To add or delete commands from the Customize dialog:

1. Right-click anywhere on the Ribbon.
2. From the shortcut menu, select **Customize Quick Access Toolbar**. The Customize Quick Access Toolbar dialog is displayed.
3. From the **Choose commands from** dropdown list, select the Ribbon group whose commands you want to add or remove from the Quick Access toolbar.
The Commands section displays all of the commands that are available on the selected option. The section on the right side of the dialog displays all the commands that are presently included on your Quick Access toolbar.
4. If you want to add a command, select it in the **Commands** list and click **Add**.
5. If you want to remove a command, select it in the section on the right side of the dialog, and click **Remove**.
6. If you want to change the order in which the commands are displayed on the Quick Access toolbar, in the section on the right side of the dialog, drag the command to the desired position.

Note: The default command included in the Quick Access toolbar is Save.

7. Click **OK**.

Setting the Display of the 3D Window

You can control the view, layout, and overlay of the 3D Window. This includes the display of several controls and indicators to orient you, options for creating a more realistic 3D World environment, and options for displaying the 3D model:

- [Navigation Controls](#)
- [Status Bar](#)
- [Sun](#)
- [Date and Time Slider](#)
- [Underground Mode](#)
- [Units](#)
- [Hide Terrain](#)
- [Mesh Model Display Style](#)
- [Highlight Depth Changes](#)

Navigation Controls

The navigation controls and virtual joystick enable you to zoom in or out or turn the plane to face north.



Navigation Controls

To navigate in the 3D Window using the navigation control:

- Click the navigation controls to turn north or zoom in or out of the terrain.

Navigation Control	Description
	Turn the plane to face north.
	Zoom in.
	Zoom out.

Status Bar

The status bar displays:

- The longitude, latitude, altitude, and direction of the plane/camera. Clicking on the status bar opens the Jump to Location dialog. See the "Navigating in the 3D World" chapter in the TerraExplorer manual for further information.
- The current terrain buffering level in a graphical format. The display quality indicator represents the current terrain resolution for the area shown in the 3D Window, compared to the estimated maximum resolution available in the terrain database. The graphic bar indicates values from 0%-100% of quality.



Status Bar

Sun

You can turn the sunlight source on or off. When turned on, the sun is used as the project's light source and affects the lighting on the terrain and on building objects.

To toggle the sunlight source on/off:

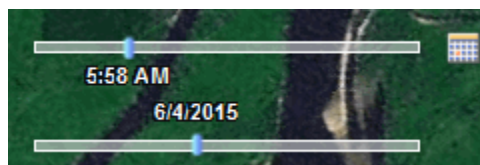
- On the **View** tab, in the **Environment** group, click **Sun**.

Date and Time Slider

The Date and Time Slider enables you to modify the current date and time so you can view each area of terrain in the date and time that you require. If the sun is used as the project's light source, the displayed light on the terrain corresponds to the sunlight there is on the terrain at the hour and date indicated by the blue marker on the slider.

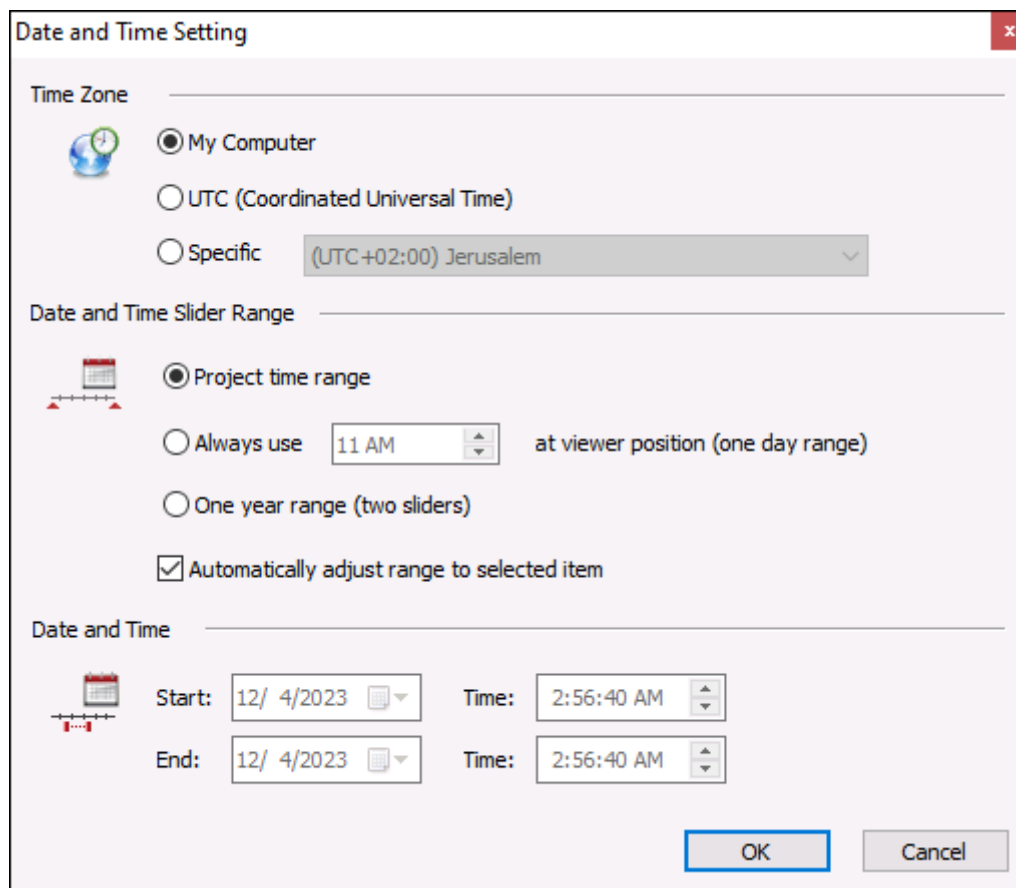
To use the Date and Time slider:

1. On the **View** tab, in the **Environment** group, click **Time Slider**. The Date and Time slider is displayed on the upper left corner of the 3D Window.



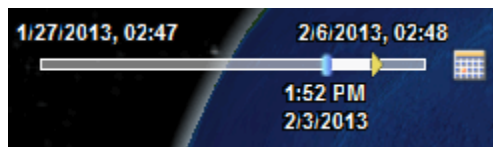
Date and Time Slider

2. Click the **Date and Time Settings**  icon to the right of the slider to select a mode. The Date and Time Setting dialog is displayed.



Date and Time Setting Dialog

3. Select one of the following time zones. The time zone you select determines which time zone is used for the time you select in the Date and Time Setting section in the same dialog.
 - **My computer** – Your local time zone.
 - **UTC** (coordinated Universal Time) – UTC time (GMT + 0:00).
 - **Specific** – The time zone you select from the dropdown list.
4. Control the interval of time for which you can view objects and features with timespans by selecting a date and time range for the date/time slider:
 - Project time range



The slider's time range corresponds to the project's time range. Date and hour adjustments are both made on the same slider.

- Always use <> at viewer position



The slider's time range is 24 hours.

This option also requires you to select a specific time that will be the default current time (until the time is changed on the slider).

- One year range (two sliders)



The time slider's range is 24 hours and the date slider's range is one year.

5. If you want the date and time sliders ranges to automatically readjust to correspond to the timespan of the selected group, select the **Automatically adjust range to selected group** check box.

Note: This option does not apply in PhotoMesh.

6. Set the project's **current** time and date or time range.

Note: The current date and time can only be set when Project time range is selected above and there are objects or groups in the project for which timespans were set.

- If you want to set a specific current time and date, set the **same** Start and End Date and Time.

The blue marker and yellow arrow are displayed together on the slider .

- If you want to set a current time **range** so that all objects and features whose timespans fall anywhere in the range display, set a Start Date and Time, and then set an End Date and Time. The blue marker on the slider indicates the Start Date/Time and the yellow arrow indicates the End Date/Time .

Note: Start and end date and time must be within the project range.

Underground Mode

The underground navigation mode allows you to explore the subsurface of the terrain. You can navigate under the terrain's surface and through buildings. A subsurface grid navigation aid appears when you navigate below the terrain, allowing you to navigate the same way as above ground.

To fly in underground mode:

- On the **View** tab, in the **Environment** group, click **Underground Mode**. You can freely go through the terrain surface in any direction.

Units

You can change the units used throughout PhotoMesh. Changing the unit settings substantially changes the display of your project: inputs and outputs and property sheets are all affected.

To change units:

- On the **View** tab, in the **Units** group, select the coordinate units and altitude:

Parameter	Description
Lat-Long Coordinates	
Decimal	Presents coordinate values in a decimal degrees format. This option is only available when the terrain is defined in the Lat-Long coordinate system.
Degrees; Minutes; Seconds	Presents coordinate values in a degrees, minutes and seconds format. This option is only available when the terrain is defined in the Lat-Long coordinate system.
Altitude Units	
Above Ground Level (AGL)	Displays altitude values as meters or feet above ground level.
Absolute Values	Displays altitude values as meters or feet above the terrain database vertical datum base ellipsoid.

Hide Terrain

You can hide the terrain texture (the terrain appears as a black image).

To toggle the display of the terrain on/off:

- On the **View** tab, in the **Environment** group, click **Hide Terrain**.

Display Style

There are several options for displaying the mesh model once it is built. The mesh model can include a wireframe and be textured or solid color. to view the interior of buildings or expose obscured sections of the 3D View.

To select a display style:

- On the **View** tab, in the **Mesh Layer** group, click **Display Style**, and select a style:
 - Texture
 - Texture + wireframe
 - Solid color
 - Solid color + wireframe

Depth Perception

You can highlight depth changes in the 3D model to make it easier to recognize the model's contours and shape. This can be particularly useful for an untextured model or point cloud without colors.

To highlight depth changes:

- On the **View** tab, toggle on **Depth Perception**.

Saving a Project

You can save a project and its system information.

To save a project:

- From the **File** menu, select **Save** (Ctrl + S).

Closing a Project

To close a project:

- From the **File** menu, select **Close**. If you haven't saved changes, a dialog is displayed asking whether you want to save your changes.

Selecting Objects in the 3D Window

Two selection modes are available for selecting objects in the 3D Window: Global Selection and List Selection. Use Global Selection to select whichever PhotoMesh item you point to in the 3D Window. Use List Selection to limit your selection to only the item type of the list currently open, e.g., if the tile list is open, and you click in the 3D Window on a camera icon, the tile below it is selected.

To change the selection mode:

- On the **Home** tab, in the **Select** group, toggle on **Global Selection** or **List Selection**.

Dragging and Dropping Files in PhotoMesh

PhotoMesh's drag and drop capability supports project files, all photo and LiDAR formats, control point lists, and video files. A single or multiple files can be dragged and dropped anywhere on the 3D Window.

File Type	How Loaded
.photmeshxml (project already open)	Open project file
.jpg .jpeg .jp2 .j2k .jpc .tif .tiff, folder	Load photo or all photos in folder
.las .laz e57, folder	Load LiDAR or all LiDAR in folder
.gcp	Import control point data
.asf, .wmv, .mp4, .avi, .fly, .mpg, .ts, .mov (only a single file at a time)	Load video

Setting PhotoMesh Options

Default system settings for the current PhotoMesh project and all future projects are set from the Options dialog.

To modify PhotoMesh Options:

1. From the **File** menu, select **Options (F9)**. Then select one of the following from the menu in the left pane:
 - **Project** – Show properties related to the current project
 - **Preferences** – Default values for the current project and new projects
 - **AT Area & Tiles** – AT/Reconstruction tiles
 - **View** – View options

Options

Project

Preferences

AT Area & Tiles

View

Project

Project Description

Build Version Name

Build Version Description

Project Vertical Datum

VERT_CS["EGM96 geoid height",VERT_DATUM["EGM96 geoid",2005,AUTHORITY["E

Project Tags

Project

C:\Users\Administrator\Documents\yyyy\yyyy.Phot...

Build For Customer

Add

Note: this information is also added as AWS tags to all fuser instances

Cancel

OK

Options Dialog

2. Modify the following settings as required:

Parameter	Description
Project	
Note: Project settings only apply to the current project.	
Project Description	Free text describing the build. This field is initially populated with the description from the previous build.
Build Version Name	Name of the build version. This field is initially populated with the name of the previous build.
Build Version Description	Free text describing the build version.
Project Vertical Datum	Vertical datum for the current project (Read-only). Note: The Output Vertical Datum field below is only applied to future projects. It is not applied to the currently open project.
Project Tags	Key words or terms (and their values) associated with the project file for advanced searches. There are two built in tags: "Project Name" with the path to the project file, and "Build For Customer" for information about the project customer.

	Tags can help you identify the fuser instance in the Amazon EC2 console after it is launched. Click Add to add more tags.
Preferences	
Note: All properties in the Preferences section are applied to the current project and future projects.	
API Binding Address	Determines the network accessibility of your PhotoMesh and ProjectQueue API for fuser pool management: ▪ Local Only - Binds the PM API exclusively the local machine (http://localhost:8086). ▪ Network Access - Allows the PM API to bind to all network interfaces (http://*:8086), making it accessible from other machines on the network. Select this setting when multiple PM managers need to communicate and query each other's statuses. To enable network access, run PhotoMesh with administrator privileges and open ports [8086-8095] to allow inbound connections.
Working Folder	The working folder that PhotoMesh Build Manager (on the master computer) and PhotoMesh fusers use to communicate: to deposit and collect pending tasks, and share information about fuser availability, status and progress. The working folder should be set to a network folder with read-write access permissions, for both the master and client computers. See "Setting the Working Folder (on the Master Computer)" in the "Fusers" chapter for information. Click Browse to browse to a different working folder, and then click Select Folder.
Preset Manager	Click Open to open the Preset Manager to load or manage preset files. See "Working with Presets" in the "Building" chapter for information.
Output Vertical Datum	Well-Known Text of the output's vertical datum. By default, this is set to EGM96. Note: You can use the Coordinate System Generator to generate a WKT string defining the required coordinate system. Only enter the Vert_CS entity of the WKT in this field. See "Coordinate System" in the "Basic Concepts" chapter for information. Select Auto to automatically set the output vertical datum based on the first one of the following that has a defined vertical datum: 1. Ground control points. 2. LiDARs 3. Photo collections. If none of these include a vertical datum, the output vertical datum is set to the default EGM96.
Maximum Local Fusers	Maximum number of local fusers to run. Select Auto to automatically set the number of local fusers based on the amount of memory on the machine and its processing power. See "Launching Multiple Fusers" in this chapter for more information.

DJI Altitude	Select the DJI altitude information to read from the metadata encoded in the photo files: ▪ Load Absolute – Load the absolute altitude values. ▪ Load Relative – Load the relative altitude values, and define altitude offset based on the terrain elevation at the position of the first photo in each collection.
Load Multispectral Bands	Select the check box to load multispectral bands for photos.
Terrain Database	Default MPT (local or remote) or .FLY to use as the base project when creating new project. Click Browse to browse to a different terrain database, and then click Open. Note: If a project is open, changes to the terrain database will take effect only after saving the project file and reopening it.
Terrain Vertical Datum	Well-Known Text of vertical datum of terrain database. Note: You can use the Coordinate System Generator to generate a WKT string defining the required coordinate system. Only enter the Vert_CS entity of the WKT in this field. See "Coordinate System" in the "Basic Concepts" chapter for information.
Areas and Tiles	
Set AT area to include	PhotoMesh can automatically calculate the AT area based on all the project's photos plus a buffer. The automatic AT area's buffer can be based on the collections' largest effective range, i.e., the maximum distance from the camera for which photo information is considered valid for aerotriangulation, or on a buffer you set. Select one of the following options: ▪ Photo collections' effective range - Calculate the AT area based on the positions of all the project's photos plus a buffer that is based on the collections' largest effective range. ▪ Use buffer of - Calculate the AT area based on the positions of all the project's photos plus the buffer that you set.
Tile Bottom Buffer	When splitting photos into AT tiles, photos whose frustum intersects with a box that defines the AT tile are included in the tile. This box's base is defined by the AT area's minimum terrain elevation minus the Tile Bottom Buffer value (in meters).
Tile Top Buffer	When splitting photos into AT tiles, photos whose frustum intersects with a box that defines the AT tile are included in the tile. This box's top is defined by the AT area's maximum terrain elevation plus the Tile Top Buffer value (in meters).
Maximum Photos Per Tile	Maximum number of photos that can be included in an AT tile, even if the Do Not Split If Sum Exceed or Do Not Split If Child Exceed properties (below) are met. When the Maximum Photos Per Tile is reached, tiles are only split if the Do Not Split If Sum Exceed and Do Not Split If Child Exceed properties (below) are both met. Note: When a tile is split, some photos will be included in both sub-tiles, based on frustum area.

Do Not Split If Sum Exceed	In cases where splitting a tile into sub-tiles, based on Maximum Photos Per Tile, results in a significant number of photos that are included in both tiles, it may not be desirable to split the tile. This property sets the maximum percent the sum of the photos in the sub-tiles can exceed the photos in the original tile. If the percent increase exceeds this value, the tile will not be split into sub-tiles even though Maximum Photos Per Tile was exceeded.
Do Not Split If Child Exceed	In cases where splitting a tile into sub-tiles, based on Maximum Photos Per Tile, results in sub-tiles that are nearly as large as the original tile, i.e., a significant number of photos were included in both tiles, it may not be desirable to split the tile. This property sets the maximum percentage (of the original tile) for either sub-tile after the split. If either of the sub-tiles exceeds this value, the tile will not be split into sub-tiles even though Maximum Photos Per Tile was exceeded.
Photo Frustum Buffer	When splitting photos into AT tiles, photos whose frustum volume intersects with a box that defines the AT tile are included in the tile. This property lets you increase or decrease the frustum volume by a set percent.
Fully Trusted Max GP	Maximum number of gigapixels per AT tile for fully trusted input data when aerotriangulation is not performed. See "Importing Photos with Metadata" in the "Photo Management" chapter for more information on fully trusted data.

View

Point Cloud Point Size	Enter the maximum size, in pixels, to display each point.
Show Legend in Review Mode	Select the check box to display AT Tile and Reconstruction Tile legends in the 3D Window when in review mode, which indicate what color is associated with each AT and Build status.
Original Position	The camera symbol's color for photos in the original position. Click Change to select a different color, and then select the color.
Calculated Position	The camera symbol's color for photos in the calculated position. Click Change to select a different color, and then select the color.
Failed to Calculate	The camera symbol's color for photos whose position PhotoMesh was unable to calculate. Click Change to select a different color, and then select the color.
Excluded	The camera symbol's color for photos manually excluded by the user or outside the project crop area. Click Change to select a different color, and then select the color.
AT Tile	Select the label and color to represent each of the user status options in the AT Tiles list. The user status indicates what actions still need to be performed on the tile, e.g., rebuild or control points. See "Selecting Tiles" and "Viewing Tile Information in the Tile List" in the "Reviewing Your Build" chapter for information.
Reconstruction Tile	Select the label and color to represent each of the user status options in the Reconstruction Tiles list. The user status indicates what actions still need to be performed on the tile, e.g., rebuild, editing or external retouch. See "Selecting Tiles" and "Viewing Tile Information in the Tile List" in the "Reviewing Your Build" chapter for information.

Coordinate System

A coordinate system is a reference system used to represent the locations of project data within a common coordinate framework. A coordinate system can be projected (X, Y), geographic (latitude-longitude), or geocentric (X, Y, Z). Each coordinate system is associated with its Well-Known Text (WKT) description, a simple structured, text-based format for spatial reference system information that is easy to store and share between systems (**See** <https://www.ogc.org/standard/wkt-crs/>). A coordinate system can optionally include a vertical datum that defines the origin for height values.

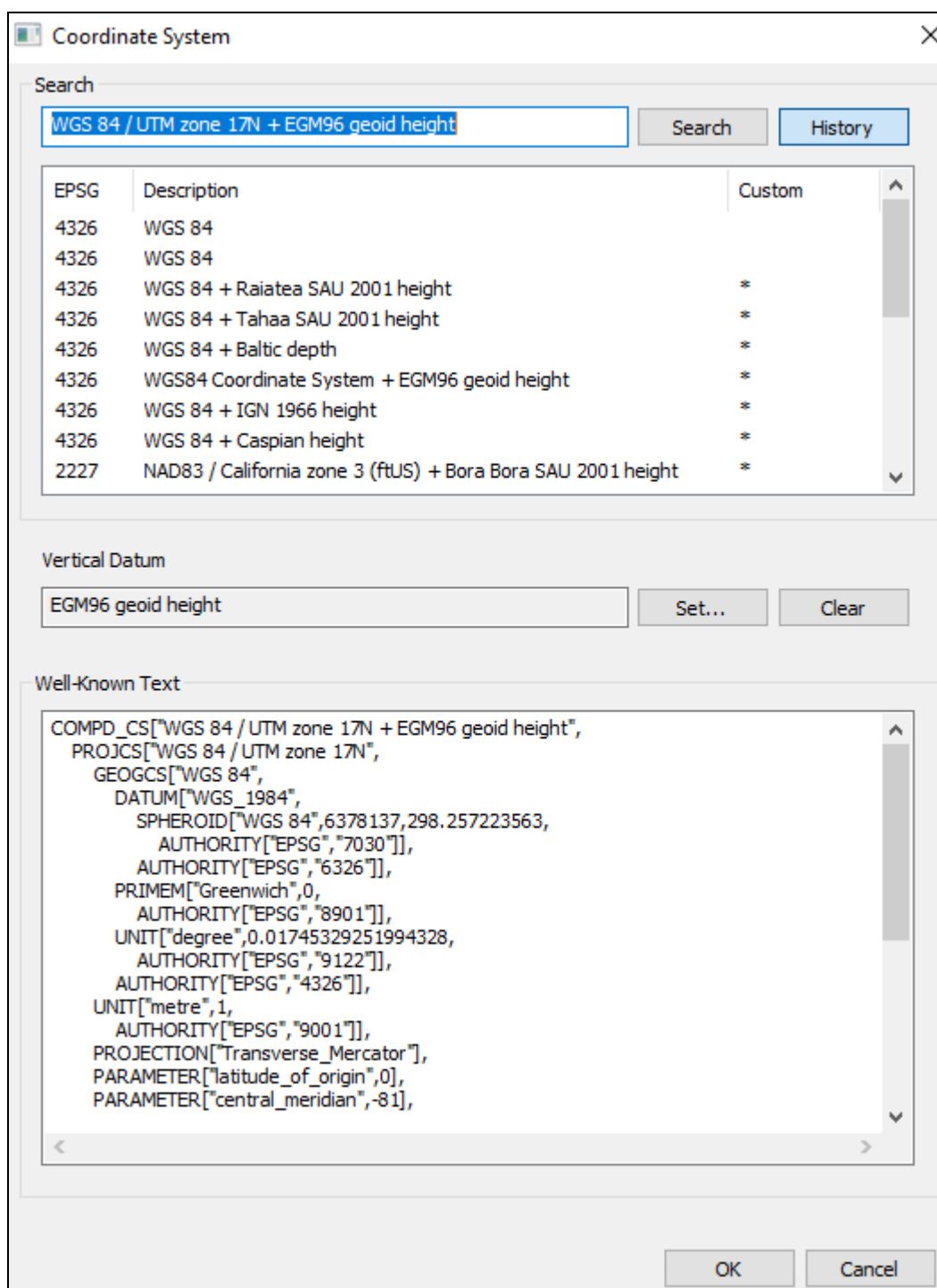
When you add input data to a project, PhotoMesh reprojects the various inputs to the [AT's project coordinate system](#) and eventually to the [output's coordinate system](#). The inputs and other project information (e.g., calculated photos and output previews) are also reprojected to the 3D Window's Lat-Long WGS84 coordinate system (with [vertical datum](#) if provided). If an input source does not include coordinate system information or PhotoMesh is unable to read this information, you must supply it so that PhotoMesh can reproject the source. Usually, the data supplier provides the information about the coordinate system used for each source.

The Coordinate System dialog is used to define the coordinate system for all project data.

When data is in Lat-Long WGS84 or ECEF coordinate system and no other coordinate system was used for photos and ground control points, etc., the default system used is the Universal Transverse Mercator (UTM) grid system with the UTM zone based on where the center of the data falls.

To set the coordinate system:

1. Open the **Coordinate System** dialog.



Coordinate System Dialog

- Enter the coordinate system information using one of the following methods:
 - Click **History** and select one of the recently used coordinate systems. The last 20 coordinate systems used are saved in the History for future use.

- In the **Search** field, type the EPSG code or a phrase from the Coordinate system Description, as indicated by the supplier, and click **Search**. Then select the required result from the search results displayed directly below.
 - In the **Well-Known Text** section, type the WKT. The Well-Known Text representation of spatial reference systems, regulated by the Open Geospatial Consortium (OGC), provides a standard textual representation for coordinate system information. You can edit an existing WKT.

Note: A custom coordinate system is listed with all the other coordinate systems, with an asterisk in the Custom column.
3. If your coordinate system includes vertical datum, click **Set** and select a vertical datum from a list of supported datums.

Note: If the WKT of your coordinate system references a particular grid shift file that was not bundled with the PhotoMesh installation, you can add the file by downloading it and placing it in the correct directory. See ["Adding Grid Shift Files"](#) in this chapter for information.
 4. Click **OK**.

The following advanced coordinate system capabilities are available:

- [Generate WKT strings for coordinate systems](#)
- [Add missing grid shift files \(GSB, GTX\)](#)

Coordinate System Reprojection Tool

The Reprojection Tool reprojects data from one coordinate system to another:

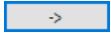
- Generate a WKT string that defines a coordinate system (optionally, with vertical datum) for inclusion in Excel/XML photo list files, control point files, and trajectory files.
- Convert coordinate, OPK, and YPR values from one coordinate system to another.
- Convert the rotation format from a rotation matrix to OPK/YPR. The rotation matrix transforms coordinates in the specified coordinate system into camera coordinates using: a 3x3 matrix with M_{ij} tags, where M_{ij} denotes the coefficient of the $(i+1)$ row and the $(j+1)$ column of the rotation matrix.
- Convert the rotation format from OPK/YPR to a rotation matrix

To generate a WKT string for a coordinate system:

1. On the **Tools** tab, in the **Coordinate System** group, click **Reprojection**. The Reprojection Tool dialog is displayed.
2. Click **Open List**, and select a coordinate system.
3. Click **Set** and select a vertical datum from a list of supported datums.
4. Click **OK**. The generated WKT string can be copied into your files as required.

To convert from one coordinate system to another:

1. On the **Tools** tab, in the **Coordinate System** group, click **Reprojection**. The Reprojection Tool dialog is displayed.
2. On the left side, do the following:
 - a. Click **Open List**, and select a coordinate system.
 - b. Click **Set** and select a vertical datum from a list of supported datums.

- c. Click **OK**.
 - d. Enter the XYZ values, and optionally, the OPK or YPR values.
3. On the right side, do the following:
 - a. Click **Open List**, and select the coordinate system you want to project to.
 - b. Click **Set** and select a vertical datum from a list of supported datums.
 - c. Click **OK**.
4. Click **Project** . The XYZ, OPK, and YPR fields on the right side are populated with the projected values.

To convert a rotation matrix to OPK and YPR:

1. On the **Tools** tab, in the **Coordinate System** group, click **Reprojection**. The Reprojection Tool dialog is displayed.
2. On either of the sides, do the following:
 - a. Click **Open List**, and select a coordinate system.
 - b. Click **Set** and select a vertical datum from a list of supported datums.
 - c. Click **OK**.
 - d. Enter the XYZ and OPK or YPR values.

3. Click **Convert** .

To convert OPK and YPR to a rotation matrix:

1. On the **Tools** tab, in the **Coordinate System** group, click **Reprojection**. The Reprojection Tool dialog is displayed.
2. Enter a rotation matrix.
3. Click **Convert** .

Adding Grid Shift (GTX, GSB) Files

PhotoMesh uses grid shift files (GTX, GSB) for shifting and transforming between some coordinate systems that have different datums. Sometimes the Well-Known Text (WKT) of one of the coordinate systems in your project references a particular grid shift file that was not bundled with your installation, or the reference itself is missing from the WKT. The steps below describe how to add the missing grid shift information in both of these cases.

Note: Grid shift files can be automatically installed during the PhotoMesh installation process by placing a zip file that includes the required files next to the PhotoMesh.exe file in the installation folder. See ["Installing PhotoMesh"](#) in the "Getting Started" chapter for more information.

To add a grid shift file that was not found on your computer, find and install the missing file:

1. Generally, when one of the coordinate systems in your project references a particular grid shift file that was not found on your computer, a notification will display in the coordinate system dialog or in the Error/Message Log with a link to a page with the required file, packaged as a zip file. Download the required zip file.
2. If no zip was made available to you with the missing grid shift file, you can download the required file from one of the following websites with download links to many common grid shift files:

- <https://github.com/OSGeo/proj-datumgrid/tree/master/>
 - <https://vdatum.noaa.gov/download.php>
 - <https://raw.githubusercontent.com/OSGeo/proj.4/gh-pages/proj.pdf>
 - https://proj.org/resource_files.html#transformation-grids
3. Extract the grid shift file to both the following directories for computers with full PhotoMesh installation, and to only the second directory for computers with only fuser installation:
 - The \PhotoMesh\proj subdirectory of PhotoMesh's installation directory.
 - The \PhotoMesh\Fuser\proj subdirectory of PhotoMesh's installation directory.
 4. If a reference to the grid shift file is missing in the WKT, add the following reference within the WKT's VERT_DATUM object that is under the VERT_CS object:

```
EXTENSION["PROJ4_GRIDS","[Name of Grid Shift File]"
```

E.g.,

```
COMPD_CS["RGF93 / Lambert-93 + NGF-IGN69 height",
  PROJCS["RGF93 / Lambert-93",
    ...,
    AUTHORITY["EPSG","2154"]],
  VERT_CS["NGF-IGN69 height",
    VERT_DATUM["Nivellement General de la France - IGN69",2005,
      EXTENSION["PROJ4_GRIDS","RAF18.gtx"],
    AUTHORITY["EPSG","5119"]],
    UNIT["metre",1,AUTHORITY["EPSG","9001"]],
    AXIS["Up",UP],AUTHORITY["EPSG","5720"]]
]
```

CHAPTER 6 WORKING WITH SKYLINEGLOBE SERVER

About Working with SkylineGlobe Server

SkylineGlobe Server (SGS) is a private cloud solution that provides a comprehensive set of web services for publishing, storing, managing and streaming 3D spatial data. Remote PhotoMesh clients can connect to a SkylineGlobe Server to load imagery, elevation and 3DML layers into their projects. All data on your server is cataloged and referenced so clients can easily find the specific published layer required.

For more information about SkylineGlobe Server, see the [SkylineGlobe Server User Guide](#).

Connecting to a SkylineGlobe Server

To connect to a SkylineGlobe Server:

1. On the **Tools** tab, in the **SkylineGlobe** group, click **Disconnect**. The SkylineGlobe Server dialog is displayed.

Note: If you are already connected to a server and want to connect to a different one, on the **Tools** tab, in the **SkylineGlobe** group, click **Connect**. In the Connect to SkylineGlobe Server dialog that is displayed, first click **Disconnect**, and then enter the new server connection information.

Connect to SkylineGlobe Server Dialog

2. Enter the connection information and click **Connect**:

Parameter	Activity
Server	The Server Host Name, which is the URL link of the server.
User Name	User name to access the server (mandatory).
Password	Enter the password.

Remember Password	Select this option to save the password. The password will be saved and will be displayed using asterisks the next time the SkylineGlobe Server dialog is displayed.
-------------------	--

The values that you enter for Server and User Name are saved. The next time you load this window, this information is automatically filled in.

Loading SkylineGlobe Layers

To add a layer from a SkylineGlobe server:

- 1. On the **Tools** tab, in the **Objects and Layers** group, click the required layer type (Imagery Layer, Elevation Layer, or 3DML). The Browse dialog is displayed.
- 2. In the **From Server** section, select **SkylineGlobe Server** as the **Server Type**, and click **Select**. The SkylineGlobe Layers dialog is displayed, showing the selected layer type.

SkylineGlobe Layers

Search String:

Advanced

Search

Search Results [1-100 of 129. Previous / Next]

Layer Name	Type	Name	Data Source	Published at
	Mesh3DML		prmdemos_data	28/11/2023 0
	Mesh3DML		demossiteuploads	06/10/2023 0
	Mesh3DML		demossiteuploads	22/09/2023 0
	Mesh3DML		demossiteuploads	30/06/2023 0
	Mesh3DML		demossiteuploads	30/06/2023 0
	Mesh3DML		demossiteuploads	27/06/2023 0
	Mesh3DML		demossiteuploads	27/06/2023 0
	Mesh3DML		demossiteuploads	27/06/2023 0
	Mesh3DML		demossiteuploads	27/06/2023 0

SkylineGlobe Layers Dialog

- 3. If you aren't already connected to the required SkylineGlobe Server, the Connect to SkylineGlobe Server dialog is first displayed to enable you to connect. See "Connecting to a SkylineGlobe Server" in this chapter for information. After connecting, the SkylineGlobe Layers dialog is displayed showing all layer types.
- 4. Search for the required layer. See "Searching for SkylineGlobe Layers" in this chapter.
- 5. Select the layer you want to load, and click **Load Layer**. Use CTRL-click or SHIFT-click to multi-select layers.

Searching for SkylineGlobe Layers

To search for a layer in the SkylineGlobe Layers dialog:

Note: If only a basic search field is displayed, and you want more advanced search options, click the **Advanced** link next to the **Search String** field.

1. If you want to search for layers whose property information contains a specific string (e.g., ID, Name, Data Source), do the following:
 - a. In the **Search String** field, type a search string or enter an attribute, condition, and value and click **Insert String** . The search expression is displayed in the Search string field.
 - b. If you want to include additional strings in your search expression, select a connector:

And

▼

 and OR or, and repeat step a for the next condition.
 - c. If you want to filter by group or owner, select the required value in the respective field.
2. If you want to search according to geographic location, do the following:
 - a. Click **Set Bounding Box**, and select the required region by drawing a rectangle on the map on the right side of the page around the area to which the search is confined
 - b. From the **Map** field, select one of the following:
 - Do Not Use – To ignore the bounding box.
 - Contained – If you want to search only for layers whose entire geometry fall within the bounding box.
 - Intersect – If you want to search for all layers whose geometry intersect at any point with the bounding box.
 - c. If you want to clear the bounding box, click **Clear**.
3. Click **Search**. Your search results are displayed in the search results list at the bottom of the page.

CHAPTER 7 IMAGERY ACQUISITION

About Imagery Acquisition

The first step in the PhotoMesh process entails photographing all visible surfaces of the intended object or scene from multiple overlapping viewpoints. PhotoMesh supports standard image formats and large frame sensors.

The accuracy of the resulting 3D mesh model is affected by several factors including: input photo resolution and quality, lens distortion, overlapping ratio, directionality, and lighting. While PhotoMesh optimizes the accuracy of the 3D mesh model by selecting the best photos for an area, each factor can affect the overall output or cause varying levels of accuracy and quality in different areas of a project.

In most projects with good quality input, the resolution of the 3D model has been found to be approximately 1-2 times the ground sample distance of the source imagery.

Input Format and Resolution

Input Format/Data	
Image Format	Standard image formats (JPG, TIF, IIQ, etc.).
Positioning Information	XY coordinates and altitude of each photo.
Orientation Information	Omega, Phi, Kappa values (Recommended for large scale projects).
Camera	
Large Frame Sensors (100+ Mega Pixel)	Supported.
Focal Length	Photos with different focal lengths are supported.
Digital Zoom	No.

Directionality

Since PhotoMesh models can be viewed from any angle, it is important to capture nearly all visible surfaces, including walls and facades. Any object in the area of interest should be photographed multiple times from the top and at least four other directions.

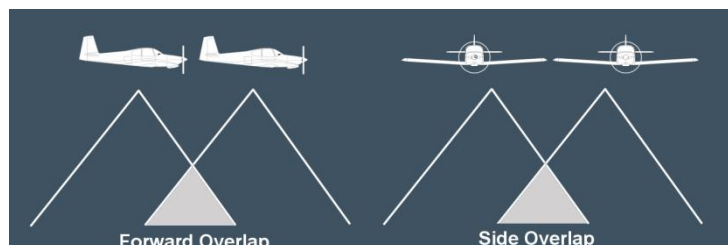


Overlap

To create an accurate 3D model, each point in the area of interest should be visible in multiple photos. A >66% overlap between consecutive photos is recommended. Each point in the area of interest should be visible by, at least, 3 cameras (min 10° offset). PhotoMesh can combine photos with different ground resolutions.

For **structured** aerial photography

- Overlap in direction of flight (forward overlap) – 75 - 90% (recommended - 80%)
- Overlap between neighboring rows (side overlap) – 70 - 90% (recommended - 80%)



Structured Aerial Photography Recommendations

The following guidelines are recommended for optimal model accuracy and texture quality:

- **Capture time** – Avoid long shadows and sun reflections
- **Capture date** – Capture all photos in the same season
- **Collection system** – Use stable platforms with minimal smearing

Data Collection Recommendations for Photos

This section outlines Skyline's recommendations for optimal 3D PhotoMesh production from collected data:

- [Camera and lens combinations \(hardware preferences\)](#)
- [Camera settings and image format](#)
- [Flight planning and execution](#)
- [Mobile photography](#)

Camera and Lens Combinations (Hardware Preferences)

The following factors in sensor and lens selection are critical for 3D model geometry and quality:

- A high quality lens, with minimal distortion.
- Fixed focal length - to avoid alternating focal lengths in photos which are meant to be processed in the same dataset.
- For a standard grid collection pattern, it is important to have a lens with a field of view (FOV) at least 90 degrees wide in the long axis (60 degrees in the short axis, ~108 degrees in diagonal).
- A FOV < 120 degrees in the long axis, since these lenses tend to be fish-eye and too distorted for high-quality aerotriangulation.
- A lens which can maintain sharpness in the edges of the image - for proper aerotriangulation and texturing of facades in the 3D model.
- A sensor which can collect images rapidly - to maintain high overlap collection in low altitude flights

while achieving proper overlap.

Camera Settings and Image Format

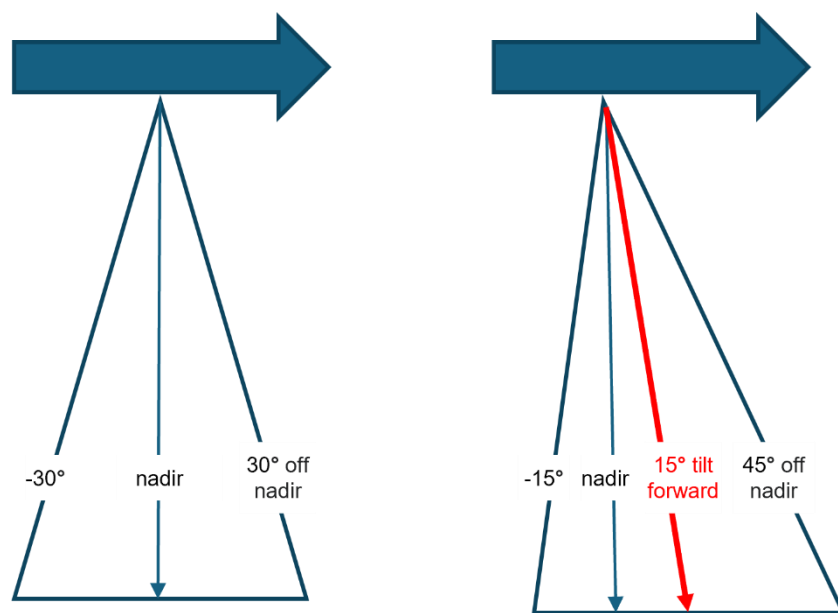
The following camera settings are recommended:

- Disable any "auto-rotate" options, to ensure that all of the images maintain a uniform orientation during the flight.
- Avoid slow shutter speed (generally $< 1/1000$) since it will cause significant motion blur, resulting in photos that cannot be used in the PhotoMesh production process. Instead of allowing a shutter speed drop to $1/640$ or $1/500$, it is preferable to open up the aperture value as wide as possible, even at risk of slight softness at the corners of the image. It is also possible to increase the ISO value in high quality cameras, and still maintain non-grainy images with high dynamic range in bright and dark areas.
- Disable any settings which might affect the relative location of the lens to the sensor, such as anti-shock or vibration features, since this will create a random principal point for the camera model. Vibration or motion in the UAV stabilization platform should be addressed with the camera and lens combination as a static object.
- Avoid auto-focus or automatic zoom. Focus the lens (usually to infinity) and lock it in that position.
- Use full resolution format, so that it is not down-sampled from the sensor pixels.
- Collect the photos in full uncompressed format. If JPEG format is necessary, use minimum compression to avoid artifacts or missing information in the images.

Flight Planning and Execution

The following flight patterns and execution are recommended:

- For single building or small area collection, use a circular flight pattern, with the target area in the center of the image from all perspectives.
- For large scale collection, use a grid pattern which allows for efficient area collection with the goal of creating 3D products.
- For nadir cameras with a field of view of approximately 60 degrees, tilt the camera 10-15 degrees forward to achieve the necessary 40-45 degree oblique perspective in all directions.



- Maintain a frontal (vertical) overlap of 75 - 90% (recommended - 80%).
- Maintain a side (lateral) overlap of 70 - 90% (recommended - 80%).

Mobile Photography

The following perspective and overlap are recommended for optimum reconstruction results:

- **Perspective** – Capture required objects and facades at a perpendicular angle and at a 30 degree oblique variance (60 and 120 degrees) to maximize triangulation quality, while minimizing perspective distortion. In close-range data collection, any perspective which captures the object or façade at more than a 45 degree oblique will most likely be at a considerably lower resolution to provide effective input for the reconstruction.
- **Sky** – Limit the sky component in the photo to no more than 30%. This can be achieved by setting the tripod to a **higher** altitude when possible, as well as by slightly angling the camera down (instead of pointing the camera directly at the horizon, 90 degrees from the ground, it can be angled 70-80 degrees from the ground).
- **Overlap** – Maintain at least an 80% overlap, so that each object will appear in at least five photos. This provides PhotoMesh with the data to fill in holes in planes and facades that are partially obstructed by objects (such as trees and street furniture) which are closer to the camera.

Data Collection Recommendations for Full Motion Video (FMV)

This section outlines Skyline's recommendations for optimal 3D PhotoMesh reconstruction from full motion video. The recommendations below are for oblique data. It is generally recommended to use nadir video datasets only for production of orthophotos or point clouds, and not for full 3D Mesh models.

Video Camera – Hardware and Settings

- Use video setting most suitable for sharp video frames in motion.
- Use gyroscope or accelerometer-based gimbal or other 3-axis stabilization.

Flight Planning and Execution

- Slow flight to prevent motion blur.
- Minimum field of view - 45 degrees (optimal: 60-100 degrees), depending on the lens quality.
- Minimum lateral (side) overlap - 40% (optimal: 60-80%).
- Minimum overlap between consecutive photos - 60%. This can be achieved by a combination of flight speed and frame rate.
- It is recommended to use consistent zoom, especially in random flight patterns.
- For large area collection, use a criss-cross grid pattern.
- For single building or small area collection, video the target feature from different altitudes, and different pointing angles for complete coverage.

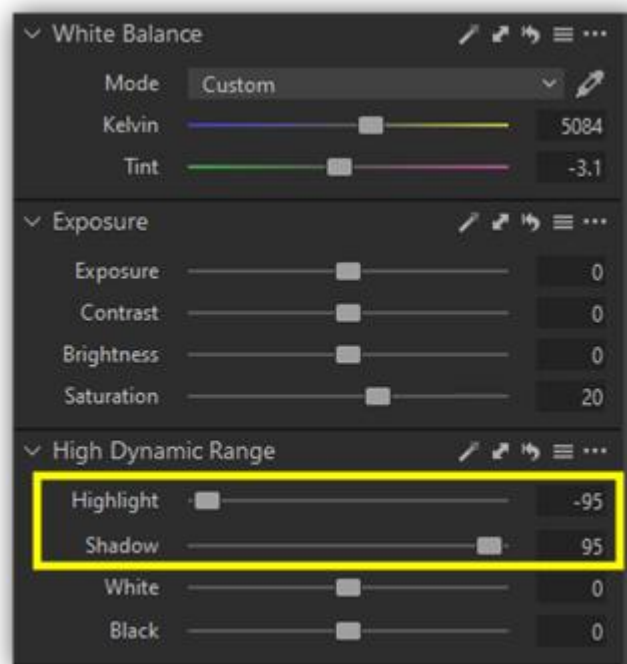
Photo Pre-Processing Recommendations for Non-Optimal Illumination Conditions

When dealing with sub-optimal lighting or camera settings unsuited to the data's light level, it is advisable to pre-process the photos before importing them into PhotoMesh for final processing. This step is especially critical when converting high-bit depth data (such as 12-bit or 16-bit) to a lower bit depth format (like 8-bit TIFF or minimally compressed JPG).

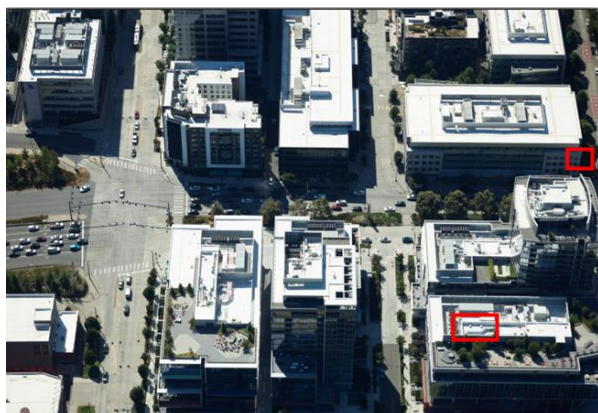
Unprocessed photos often have a high-contrast histogram, accentuating brightness variations, as seen in brightly lit areas next to shadows or dark roads beside light sidewalks. While visually striking, these photos may not be suitable for detailed analysis or processing.

Pixel-based correlation, which requires high detail at the image's native resolution, becomes particularly challenging for photos captured in environments with extreme lighting conditions. In such settings, disparities in light and shadow are more pronounced, negatively impacting the accuracy of 3D modeling.

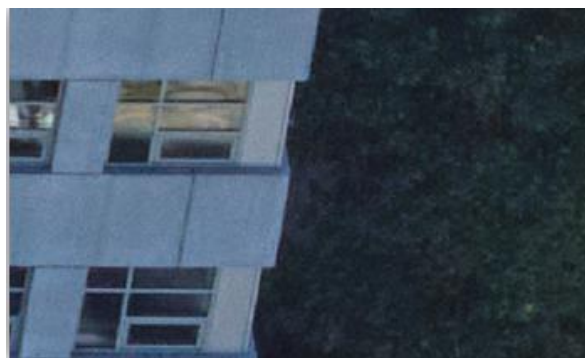
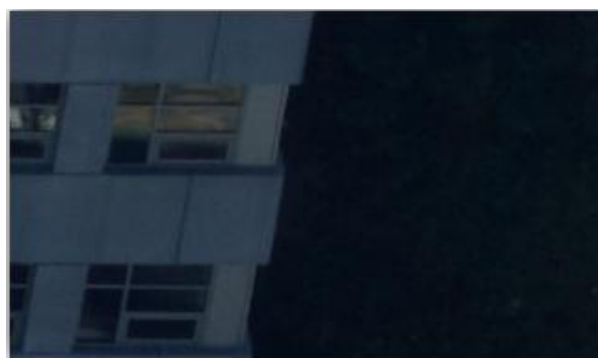
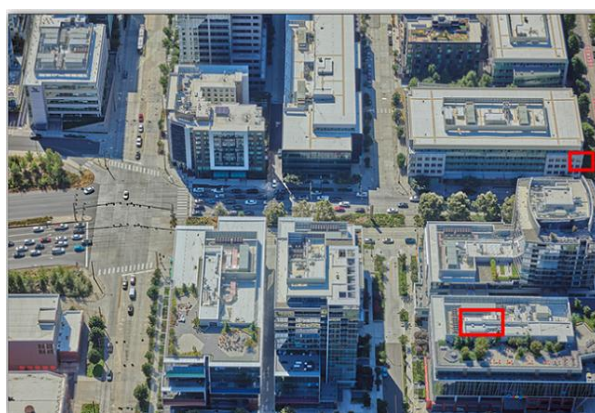
To address these issues, adjustments on the histogram's edges are recommended, particularly shadow and highlight corrections (see below), along with white balance adjustments to establish a neutral gray color scheme. A slight increase in saturation can also enhance non-gray elements, achieving a more even pixel value distribution.



Before



After



These adjustments effectively reduce the contrast between shadowed and brightly lit areas, resulting in a more balanced histogram. This balance is critical for two reasons: it improves 3D correlation by ensuring consistent feature detection across images, and it aids in smoother texture blending, thus enhancing the visual coherence of the 3D model.

Further refinements include dehaze and minor color balance adjustments in the shadows and highlights to compensate for [Rayleigh Scattering](#). The extent of dehazing required depends on the visibility conditions during data collection. Adjustments to shadows and highlights should be tailored to match the light intensity and the contrast between illuminated and shaded areas.



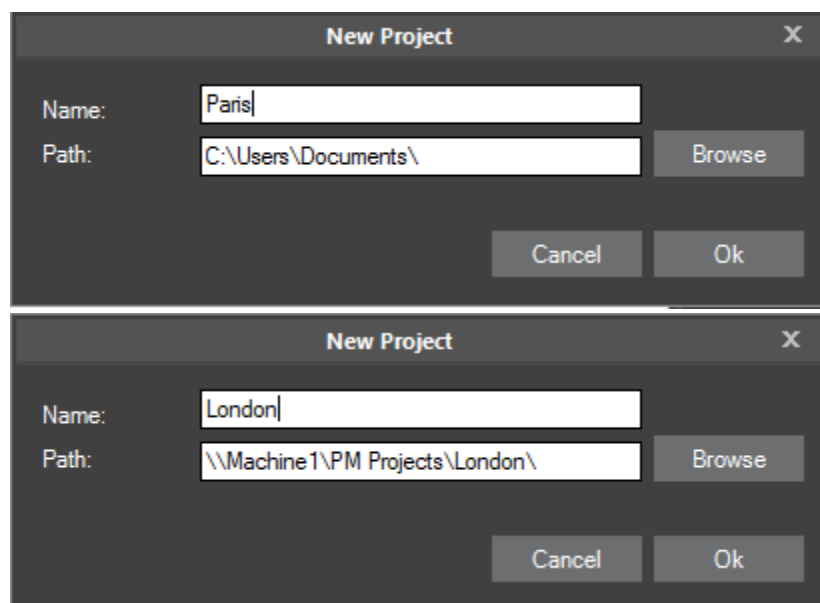
CHAPTER 8 PROJECT MANAGEMENT

Creating and Opening a PhotoMesh Project

When you create a new project, PhotoMesh creates an output folder with the same name as the project file for the project file and output files.

To start a new project:

1. From the **File** menu, select **New** (CTRL+N). The New Project dialog is displayed.



New Project on Local Drive / New Project on Network Drive

2. Browse to the required path for the project folder.

Note: PhotoMesh can take advantage of computer clusters to accelerate database creation. See the "[Fusers](#)" chapter for more information. To enable this, both the project folder and the working folder should be set to network folders with read-write access permissions, for both the master and client computers. It is recommended to use UNC format (`\\UNC Master] \folder\ProjectFolder`, e.g.: `\\Machine1\c\PhotoMesh\ProjectFolder` and `\\UNC Master] \folder\WorkingFolder`, e.g.: `\\Machine1\c\PhotoMesh7.0\WorkingFolder`), though a mapped network drive can also be used as long as the same letter is mapped to the same share on all computers. The shared project folder and working folder must both be on a device that supports the number of concurrent network connections you require, e.g., Windows 10 limits the number of supported concurrent connections to 20.

3. In the **Name** field, type a project name, and then click **Ok**. PhotoMesh creates an output folder (with the same name as the project file) for the project file and output files. The Add Photos Wizard is displayed. See the "[Photo Management](#)" chapter for more information.

Note: If PhotoMesh cannot load the default base terrain (generally because you are not connected currently to the Internet), the Terrain Database dialog is displayed. See "[Setting the Terrain Database](#)" in this chapter for more information.

To open an existing project:

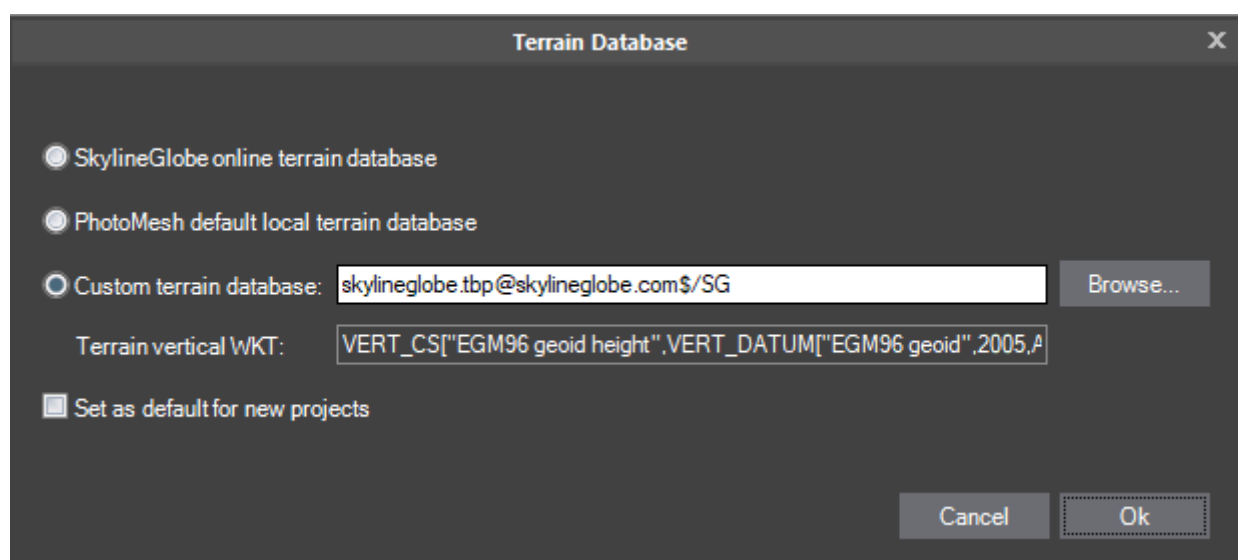
1. From the **File** menu, select **Open**. The Open dialog is displayed.
2. Browse to the required file and click **Open**.
3. If PhotoMesh cannot load the default base terrain (generally because you are not connected currently to the Internet), the Terrain Database dialog is displayed. See "[Setting the Terrain Database](#)" in this chapter for more information.

Setting the Terrain Database

You can set the MPT or .FLY to use as the base terrain for your project.

To set the terrain database:

1. On the **Home** tab, in the **Add** group, click **Terrain Database**. The Terrain Database dialog is displayed.



Terrain Database Dialog

2. Select one of the terrain database options:
 - SkylineGlobe online terrain database – To use the default terrain database (skylineglobe.tbp@tglobe.skylinesoft.com).
 - PhotoMesh default local terrain database – To use the base terrain file included with the PhotoMesh installation (Default_Local_Terrain.mpt).
 - Custom terrain database – To use a custom terrain database. Browse to the required file.
3. If you want to set your selected option as the default for new projects, select the check box. The default terrain database for new projects is listed in the PhotoMesh Options dialog. See "[Setting PhotoMesh Options](#)" in the "Basic Concepts" chapter for more information.
4. Click **OK**.

Viewing Build Settings

The Build Settings dialog is used to view project parameters, such as AT and reconstruction settings, output coordinate system, and output formats. Build settings can only be **edited** when beginning a build

or rebuild process. See "[Setting Build Steps, Parameters, and Outputs](#)" in the "Building" chapter for information.

To view Build Settings:

1. On the **Home** tab, in the **Process** group, click the arrow next to **Build**, and select **View Build Settings**. The Build Settings dialog is displayed, showing the simplified view that consolidates all the main settings under a single tab ("Build Settings"). See "[Setting Build Steps, Parameters and Outputs](#)" in the "Building" chapter for more information.
2. Click the tabs to view preset information and advanced AT and reconstruction parameters. See "[Setting Build Parameters – Advanced Settings](#)" in the "Building" chapter for more information.

Setting the Coordinate System

The Coordinate System Settings dialog is used to set and view the project's internal coordinate system, in which aerotriangulation and reconstruction are performed. The coordinate system is also relevant for calculating the horizontal and vertical accuracy of the control points.

Note: When Review mode is on, you can only **view** the coordinate system.

To view Coordinate System Settings:

1. On the **Home** tab, in the **Process** group, click the arrow next to **Build**, and select **Coordinate System Settings**. The Coordinate System Settings dialog is displayed, showing the name of the project's coordinate system.

Note: When Review mode is off, you can click the arrow to select a different coordinate system from the dropdown list. Select the **Custom** option to create a new custom coordinate system. See "[Coordinate System](#)" in the "Basic Concepts" chapter for more information.

2. Click **View** to view the full Well-Known Text (WKT) for the coordinate system.

CHAPTER 9 USING THE PROJECT PANEL AND ITEM LIST

About the Project Panel

The Project Panel provides quick access to PhotoMesh items. The Project Panel organizes all the project's items by type:

- Photos
- LiDARs
- Control and tie points
- AT tiles
- Reconstruction tiles
- Manual retouch layer
- Water polygons (manually created and imported)
- Reference projects
- Build outputs
- Objects and layers (locations, modify terrain objects, and imagery, elevation and mesh layers)
- Error messages

Note: The number in brackets next to the name of each group indicates the number of items in this group. When **all** the items in a group are set to "visible", a check box is displayed . When only some of the items in the Photos, Water Polygons or Control Points groups were set to "visible" (via the list or ribbon commands), the mixed selection check box is displayed  by the Project Panel's respective group.

	Photos [107]	
	LiDAR [2]	
	Control and Tie Points [12]	
	AT Tiles [0]	☐
	Reconstruction Tiles [0]	☐
	Manual Retouch [1]	
	Water Polygons [2]	
	Reference Projects [0]	☐
	Build Outputs [18]	
	Objects and Layers [1]	
	Messages [3]	

Project Panel

About the Item List

Located under the 3D Window, the Item List lists all of the items in the currently selected Project Panel group, providing information about each item. The columns in each item list vary depending on the item type. Many actions can be performed on each list's items, either from the commands on the list's top bar or from its right-click menu options. See ["Using the Item List"](#) and ["Right-Click Shortcut Menus"](#) in this chapter for details.

Photos: Photos Collections

Selection	Collection	Name	Visible	Exclude	Project Original	X	Y	Altitude	Order
☐	1	006263-1210151633189-Cam1.jpg	☒	☐	☐	463977.757692	3158673.086929	646.249920	▲
☐	1	006264-1210151633203-Cam1.jpg	☒	☐	☐	463974.646977	3158569.077678	645.698227	
☐	1	006266-1210151633231-Cam1.jpg	☒	☐	☐	463971.841819	3158360.213313	643.737209	
☐	1	006265-1210151633217-Cam1.jpg	☒	☐	☐	463972.822301	3158465.081835	644.939634	
☐	1	006267-1210151633245-Cam1.jpg	☒	☐	☐	463970.901787	3158256.186094	642.384891	
☐	1	006268-1210151633259-Cam1.jpg	☒	☐	☐	463969.668025	3158151.892208	641.086461	
☐	1	006269-1210151633273-Cam1.jpg	☒	☐	☐	463968.144824	3158047.676008	639.348194	
☐	1	006271-1210151633301-Cam1.jpg	☒	☐	☐	463964.747482	3157840.310925	635.001262	
☐	1	006270-1210151633287-Cam1.jpg	☒	☐	☐	463966.509303	3157943.998846	637.179005	
☐	1	006272-1210151633315-Cam1.jpg	☒	☐	☐	463962.757763	3157734.399007	634.030675	
☐	1	006273-1210151633329-Cam1.jpg	☒	☐	☐	463960.712863	3157631.626202	633.713767	
☐	1	006275-1210151633357-Cam1.jpg	☒	☐	☐	463957.702798	3157423.427612	635.232562	
☐	1	006274-1210151633343-Cam1.jpg	☒	☐	☐	463958.963749	3157527.838217	634.149695	
☐	1	006276-1210151633371-Cam1.jpg	☒	☐	☐	463956.873505	3157319.243237	637.087808	
☐	1	006277-1210151633385-Cam1.jpg	☒	☐	☐	463956.718409	3157215.621008	639.549388	▼

Photo List

Showing and Hiding Group Items

To toggle the visibility in the 3D Window of all items within a group, click the group's show/hide icon. This action affects only those items set as "visible" through list or ribbon commands. Items specifically set to "hide" using these controls will remain hidden, even when the group's show icon is toggled on in the Project Panel.

Using the Item List

You can search for and select items, e.g., photos, LiDARs, and tiles, and perform many other commands using the Item List. All selected items are selected both in the list (a check is displayed in the Selection column and their row is highlighted) and in the 3D Window. Different commands are available in the various lists.



Selection Point – Select all list items that intersect with the point you draw. This option is only available for photos and LiDAR.



Selection Polygon – Select all list items that intersect with the polygon you draw.



Selection Rectangle – Select all list items that intersect with the rectangle you draw.



Show Original Photos in Selected Point – Display all photos that intersect with a selected point based on original photo positions. This option is only available for photos. See ["Displaying Camera Symbols"](#) in the "Preparing the Project" chapter.



Show Calculated Photos in Selected Point – Display all photos that intersect with a selected point based on aerotriangulation calculations. This option is only available for photos. See ["Displaying Camera Symbols"](#) in the "Preparing the Project" chapter.



Show All Photos – Display camera symbols for all photos. This option is only available for photos. See ["Displaying Camera Symbols"](#) in the "Preparing the Project" chapter.



Load Photo File – Load one or more photo files to the selected collection. This option is only available for photos – Collections tab.



Select All – Select all list items.



Clear Selection – Clear all selected list items.



Clear Columns Filter – Clear any active filter that was applied. See ["Filtering Data"](#) in this chapter for information.



Manual Retouch – Open Manual Retouch tool. This option is only available for manual retouch polygons. See ["Retouching a Mesh Layer"](#) in the "Reviewing Your Build" chapter for information.



Import – Import manual retouch or water polygon layer. This option is only available for manual retouch and water polygon layers. See ["Importing Water Body Polygons"](#) and ["Exporting and Importing a Retouch Layer"](#) in the "Reviewing Your Build" chapter for more information.



Export – Export manual retouch or water polygon layer. This option is only available for manual retouch polygon and water polygon layers. See ["Exporting Water Body Polygons"](#) and ["Exporting and Importing a Retouch Layer"](#) in the "Reviewing Your Build" chapter for more information.



Filter by Reconstruction Area – Select all tiles in the current reconstruction area. This option is only available for reconstruction tiles. See ["Setting the Reconstruction Area"](#) in the "Preparing the Project" chapter for information on defining the area of the project on which PhotoMesh should perform the reconstruction steps of the build process.



Draw Water Body Polygon – Create a new water body. This option is only available for water polygons. See ["Creating Water Body Polygons"](#) in the "Reviewing Your Build" chapter.



Draw Leveled Polygon – Create a new leveled water body. This option is only available for water polygons. See ["Creating Water Body Polygons"](#) in the "Reviewing Your Build" chapter.



Edit Water Body – Edit a water body. This option is only available for water polygons. See ["Editing a Water Body Polygon"](#) in the "Reviewing Your Build" chapter.



Jump to Object – Jump to the object or layer. This option is only available for objects and layers.



Open Object's Properties – Open the object or layer's properties sheet. This option is only available for objects and layers.



Open in TerraExplorer – Open the build outputs in TerraExplorer. This option is only available for build outputs.



Open Project's Output Folder – Open the folder to access any of the project's outputs. This option is only available for build outputs.



Expand Only Current Build's Outputs – Expand only the outputs generated in the current build. All other builds' output information will be collapsed. This option is only available for build outputs.



Hide All Layers – Hide all build output layers. This option is only available for build outputs.



Open Control Point Editor – Open the selected control point in the [Control Point Editor](#). This option is only available for control points.



Display Rays – Display in the 3D Window only the photos that were sampled for the selected control point and project rays from each control point back to the photos it is marked in. This option is only available for control points.



Add Reference Project – Add a project that will be used as a reference, to adjust the geographic location of photos in your project.



Toggle Tile Base – Toggle the display of the tile bases of tiles created as adaptive 3D boxes. This option is only available for reconstruction tiles.



Flatten Terrain – Flatten terrain to height of five meters below lowest tile. This option is only available for aerotriangulation and reconstruction tiles.



Hide Terrain – Hide the terrain texture in the area of the project (the terrain appears as a black image). This option is only available for aerotriangulation and reconstruction tiles.

Note: You can hide the entire terrain texture from the **View** tab. See "[Hide Terrain](#)" in the "Basic Concepts" chapter for more information.

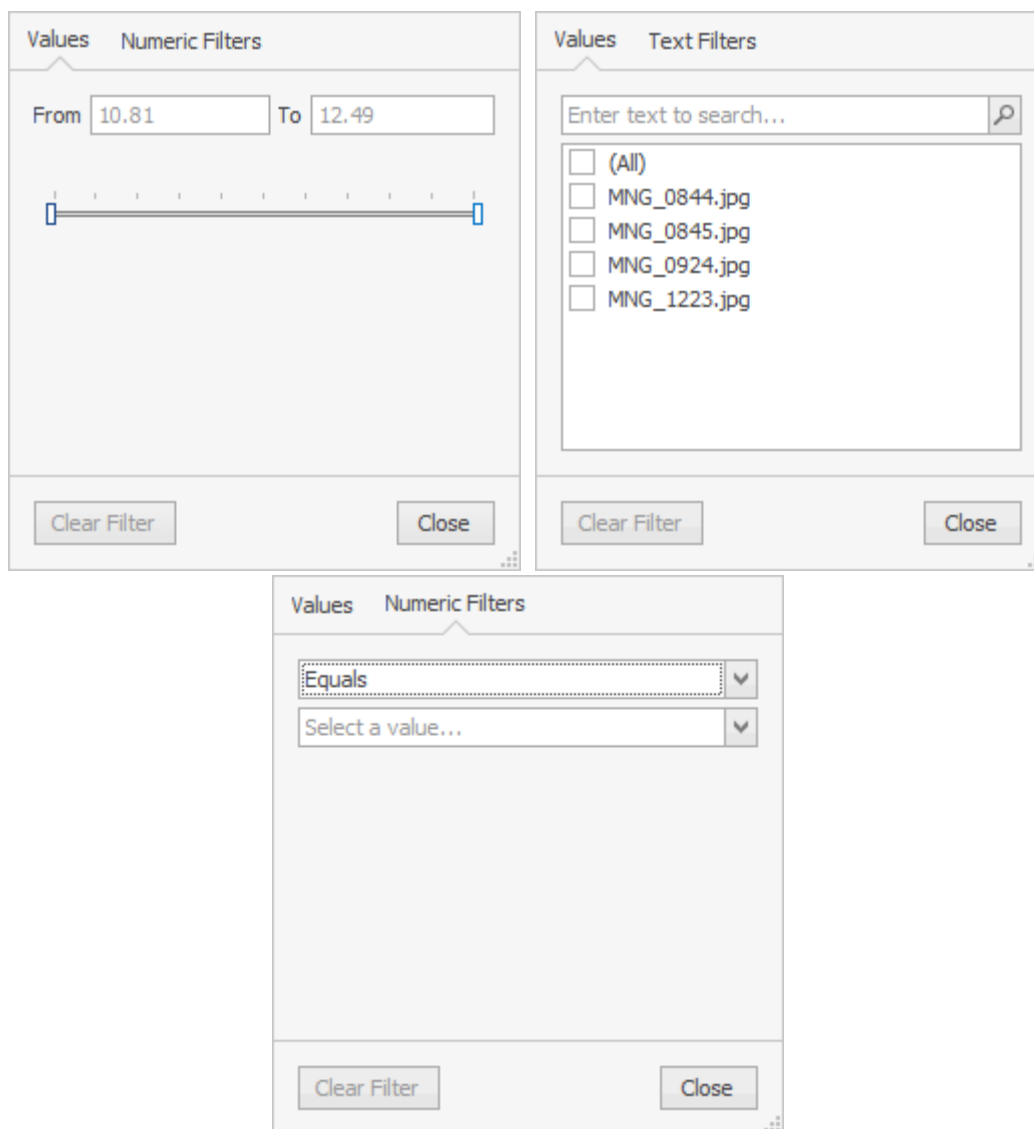
To select items using an item list:

1. If you want to search for items according to geographic location, use any of the relevant selection buttons.
2. If you want to search for items whose property information contains a specific string (e.g., Name, Altitude or X/Y value), in the list's search field, type a search term, and press **ENTER**. The specified text is highlighted wherever it appears in the list. You can use the **Previous** and **Next** buttons to jump in the list from one search result to the next.

Filtering Data

To filter data:

1. Point to the column by which you want to filter, and click the **Filter** icon . The Filter dialog is displayed.



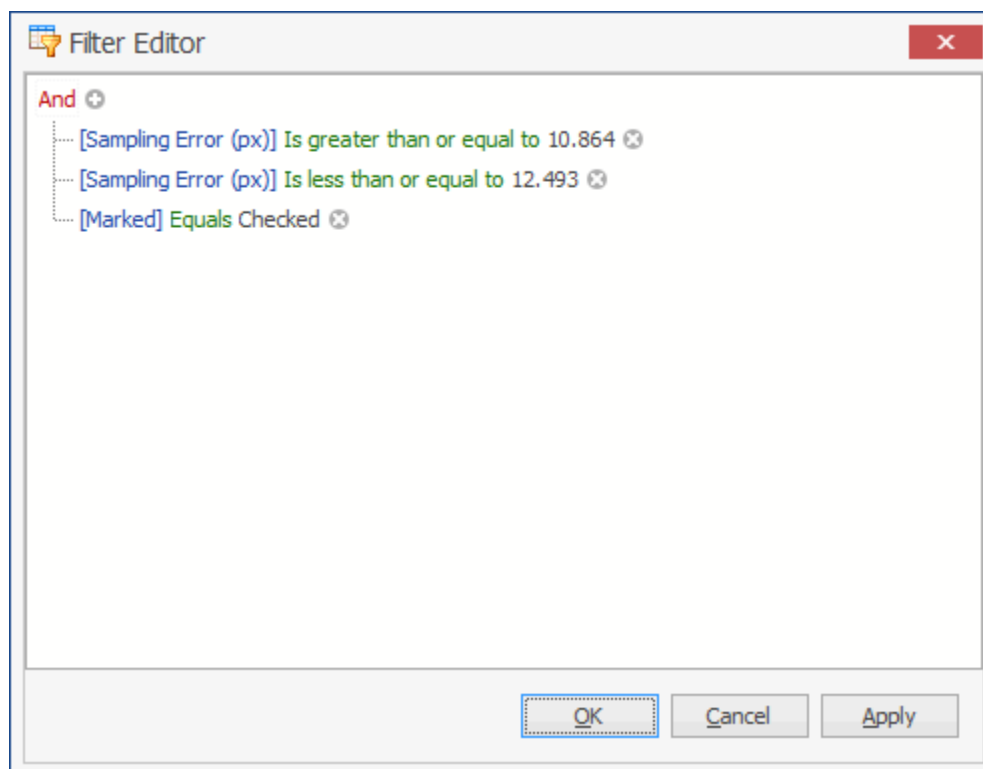
Numeric/Text Filters

2. Do any of the following:
 - To select a numeric/text filter, select the **Numeric Filters/Text Filters** tab. A filter is comprised of the (already) selected property, an operator, and a property value. Each property type has its corresponding operators. Select an **operator** from a list of operators, and a **property value**.
 - To filter by values, select the **Values** tab. For numeric values, enter the **From/To** values of the data you want to view. For text values, select the check boxes of the data you want to view. All other data will be filtered, i.e., temporarily hidden.
 - If you want to clear the filter, click **Clear Filter**.
3. Click **Close**. The applied filter(s) are listed at the top of the panel, e.g.,

[visible] = False And [projected] = False
4. If you want to clear all filters that were applied, click **Clear Columns Filter** .

Filter Editor

You can use the Filter Editor to add and remove conditions. A search expression can consist of more than one condition. In this case, you need to insert a connector between the conditions to specify their relations. In addition to AND, you can use the OR, NOT and [] connectors to define the relations between conditions.



To access additional sorting commands, right-click any column header. Select any of the following commands:

Command	Description
Sort Ascending	Sort all data in the table by this column in ascending order.
Sort Descending	Sort all data in the table by this column in ascending order.
Group by This Column	Group the data by this column. Note: If you want to clear the grouping, right click above the column headers, and select Clear Grouping.
Show Group by Box	Display boxes, above the photo table, indicating how the data is grouped in the table. Note: If you want to clear the Group by boxes, right click above the column headers, and select Hide Group by Box.
Hide This Column	Hide the selected column

Column Chooser	Customize which columns show in the table. Drag and drop columns to and from the Customization dialog.
Best Fit	Adjust the selected column's width for the best fit for its contents.
Best Fit (all columns)	Adjust the all the table's column widths for the overall best fit.
Filter Editor	Open the Filter Editor to edit the table filter.
Show Auto Filter Row	Display a row with a filter option for each of the columns.

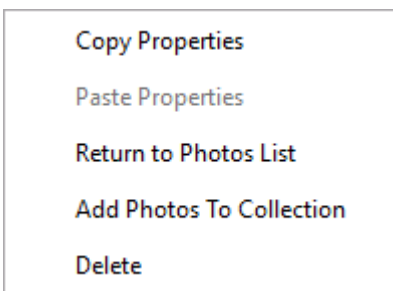
Right-Click Shortcut Menus

Each project item type displays a unique menu of options when you right-click it in its respective list.

Photo Collection Menu

To open a photo collection's shortcut menu:

- In the **Project Panel**, select the **Photos** group, and then in the **Photos** list, click the **Collections** tab, and right-click the required collection.



Collection Shortcut Menu

The Collection menu has the following options:

Note: Only the Copy Properties and Return to Photos List options are available when in Review mode.

Option	Description
Copy Properties	Copy the collection's properties.
Paste Properties	Paste the previously copied collection properties to this collection.
Return to Photos List	Open the photos tab so you can view/edit the properties of the photos in the selected collection.
Add Photos To Collection	Load photos into the project and add them to the selected collection.
Delete	Delete the collection from the project.

Photo Menu

To open a Photo's shortcut menu:

- In the **Project Panel**, select the **Photos** group, and then in the **Photos** list, click the **Photos** tab, and right-click the required photo.

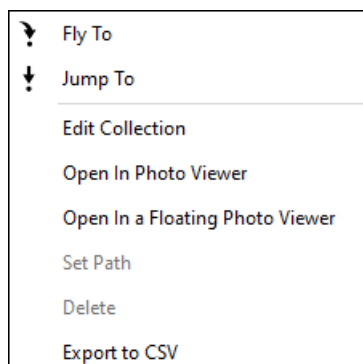


Photo Shortcut Menu

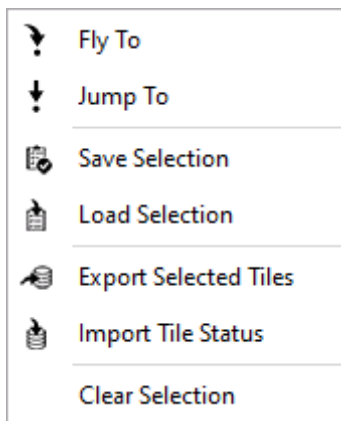
The Photo menu has the following options:

Option	Description
Fly To	Fly to the photo. Note: This option is only available when all position parameters (X, Y, and Altitude) were provided.
Jump To	Jump to the photo.
Edit Collection	Open the collection tab so you can view/edit collection properties.
Open in Photo Viewer	Open the photo in a Photo Viewer .
Open in a Floating Photo Viewer	Open the photo in an additional floating Photo Viewer without closing or replacing the photos in currently open viewers.
Set Path	Set the photo's directory. Note: This option is not available in Review mode.
Delete	Delete the photo from the project. Note: This option is not available in Review mode.
Export to CSV	Export the selected photos' information to a CSV file.

Tile (AT or Reconstruction) Menu

To open a Tile's shortcut menu:

- In the **Project Panel**, select the **AT Tiles/Reconstruction Tiles** group, and then in the **AT Tiles** or **Reconstruction Tiles** list, right-click the required tile.



Tile Shortcut Menu

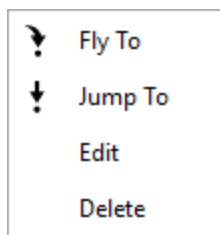
The Tile menu has the following options:

Option	Description
Fly To	Fly to the selected tile.
Jump To	Jump to the selected tile.
Save Selection	Save the selection information (which tiles were selected) to a .select text file.
Load Selection	Load a tile's .select text file with selection information (which tiles should be selected).
Export Selected Tiles	Export the selected tiles with all their information (e.g., status, photo count) to an SQLite file. Note: The exported tiles can be reviewed in TerraExplorer or other standard geospatial applications, and user status and comments added, and then reimported to PhotoMesh.
Import Tile Status	Reimport a tile from an SQLite file (generally after the tile was reviewed in TerraExplorer or another standard geospatial application, and user status and comments were added).
Clear Selection	Clear all selections.

Control Point Menu

To open a control point's shortcut menu:

- In the **Project Panel**, select the **Control and Tie Points** group, and then in the **Control and Tie Points** list, right-click the required control point.



Control Point Shortcut Menu

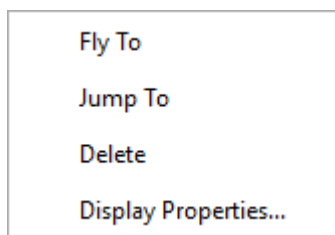
The Control Point menu has the following options:

Option	Description
Fly To	Fly to the control point.
Jump To	Jump to the control point.
Edit	Opens the Control Point Editor to mark control points in the photos. (See " Marking Control Points in Photos " in the "Control Points" chapter for information).
Delete	Delete the control point from the project.

Objects and Layers Menu

To open an [object or layer's](#) shortcut menu:

- In the **Project Panel**, select the **Objects and Layers** group, and then in the **Objects and Layers** list, right-click the required terrain object or layer.



Objects and Layers Shortcut Menu

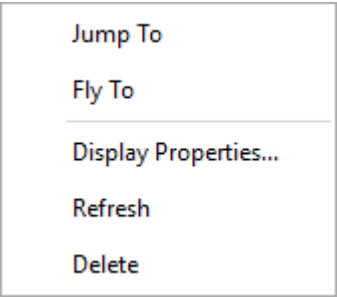
The Objects and Layers menu has the following options for objects and layers:

Option	Description
Fly To	Fly to the terrain object.
Jump To	Jump to the terrain object.
Refresh	Refresh layer. This option is only available for mesh layers.
Delete	Delete the terrain object from the project.
Display Properties	Open the object or layer's property sheet to edit or view its properties.

Build Outputs Menu

To open a Build Outputs shortcut menu:

- In the **Project Panel**, select the **Build Outputs** group, and then in the **Build Outputs** list, right-click the required build output.



Build Outputs Shortcut Menu

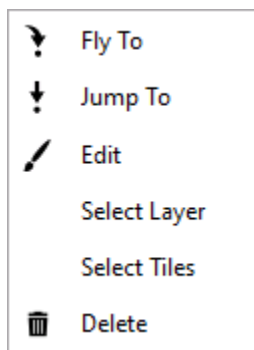
The Build Outputs menu has the following options:

Option	Description
Jump To	Jump to the output.
Fly To	Fly to the output.
Display Properties	Open the output’s property sheet to edit or view its properties.
Refresh	Refresh the output.
Delete	Delete the output from the project.
Export to LAS	Export DSM output to LAS. This option is only available for DSM output.
Export to LAZ	Export DSM output to LAZ. This option is only available for DSM output.

Water Polygon Menu

To open a specific [water polygon’s](#) shortcut menu:

- In the **Project Panel**, select the **Water Polygons** group, and then in the **Water Polygon** list, click the **Polygons** tab, and right-click the required polygon.



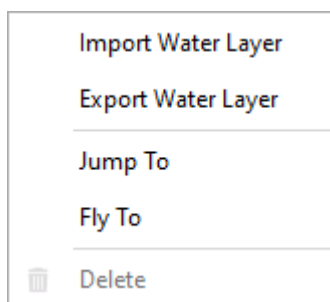
Water Polygon Shortcut Menu

The Water Polygon menu has the following options:

Option	Description
Fly To	Fly to the water polygon.
Jump To	Jump to the water polygon.
Edit	Edit the polygon's shape.
Select Layer	Open the Water Polygon list showing the Layers tab, so you can view/edit layer properties
Select Tiles	Select the intersecting tiles.
Delete	Delete the water polygon from the project.

To open a water polygon layer's shortcut menu:

- In the **Project Panel**, select the **Water Polygons** group, and then in the **Water Polygon** list, click the **Layers** tab, and then right-click the required layer.



Water Polygon Layer Shortcut Menu

The Water Polygon Layers menu has the following options:

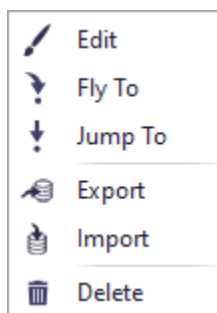
Option	Description
Import Water Layer	Import a water polygon shapefile or KML layer.
Export Water Layer	Export a water polygon layer to a shapefile.
Jump To	Jump to the water polygon layer.

Fly To	Fly to the water polygon layer.
Delete	Delete the water polygon layer

Retouch Layers Menu

To open a specific [manual retouch](#) operation's shortcut menu:

- In the **Project Panel**, select the **Manual Retouch** group, and then in the **Manual Retouch** list, right-click the required retouch polygon (i.e., row).



Manual Retouch Shortcut Menu

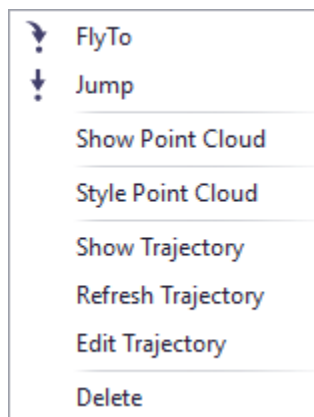
The Manual Retouch menu has the following options:

Option	Description
Edit	Edit the polygon.
Fly To	Fly to the polygon.
Jump To	Jump to the polygon.
Export	Export the retouch polygon layer with the selected retouch polygons to an SQLite file. The layer can be modified in external software and reimported to PhotoMesh.
Import	Reimport a retouch layer from an SQLite file (generally after modification in external software).
Delete	Delete the selected retouch polygons from the project.

LiDAR Menu

To open a [LiDAR](#) shortcut menu:

- In the **Project Panel**, select the **LiDARs** group, and then in the **LiDARs** list, right-click the required LiDAR.



LiDAR Shortcut Menu

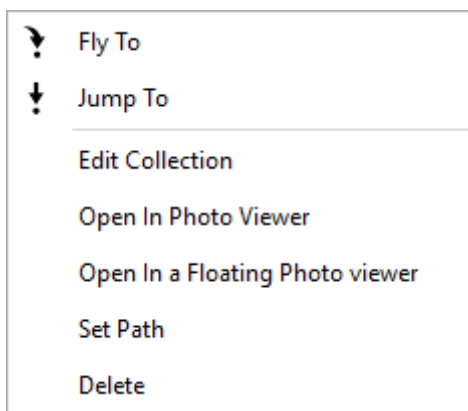
The LiDAR menu has the following options:

Option	Description
Fly To	Fly to the LiDAR point cloud.
Jump	Jump to the LiDAR point cloud.
Show Point Cloud	Show the LiDAR point cloud on the terrain.
Style Point Cloud	Open the LiDAR point cloud's property sheet to set point size, color, visibility, etc. See "LiDAR Point Cloud Properties Dialog" in the "LiDAR Management" chapter.
Show Trajectory	Show the LiDAR point cloud's trajectory. The trajectory is the path of the scanner's position while capturing the LiDAR point cloud. See "Associating a LiDAR Point Cloud with its Trajectory Data" in the "LiDAR Management" chapter for information.
Refresh Trajectory	Refresh trajectory data.
Edit Trajectory	Open the trajectory data file in a text editor, so you can edit its data. See "Associating a LiDAR Point Cloud with its Trajectory Data" in the "LiDAR Management" chapter for information.
Delete	Delete the LiDAR point cloud from the project.

Reference Project Menu

To open a reference photo's shortcut menu:

- In the **Project Panel**, select the **Reference Project** group, and then in the **Reference Photos** list, click the **Photos** tab, and right-click the required photo.



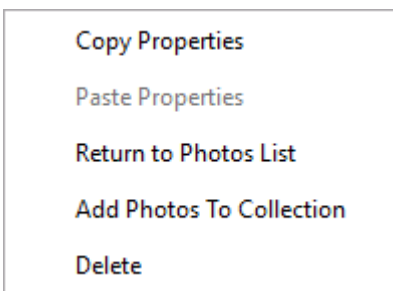
Reference Photos Shortcut Menu

The Reference Project menu has the following options:

Option	Description
Fly To	Fly to the photo. Note: This option is only available when all position parameters (X, Y, and Altitude) were provided.
Jump To	Jump to the photo.
Edit Collection	Open the Collections list so you can view/edit collection properties.
Open in Photo Viewer	Open the photo in the Photo Viewer.
Open in a Floating Photo Viewer	Open the photo in an additional floating Photo Viewer without closing or replacing the photos in currently open viewers.
Set Path	Set the photo's directory. Note: This option is not available in Review mode.
Delete	Delete the photo from the project. Note: This option is not available in Review mode.

To open a reference photo's collection shortcut menu:

- In the **Project Panel**, select the **Reference Project** group, and then in the **Reference Photos** list, click the **Collections** tab, and right-click the required collection.



Collection Shortcut Menu

The Collection menu has the following options:

Note: Only the Copy Properties and Return to Photos List options are available when in Review mode.

Option	Description
Copy Properties	Copy the collection's properties.
Paste Properties	Paste the previously copied collection properties to this collection.
Return to Photos List	Open the photos tab so you can view/edit the properties of the photos in the selected collection.
Add Photos To Collection	Load photos into the project and add them to the selected collection.
Delete	Delete the collection from the project.

CHAPTER 10 PHOTO MANAGEMENT

About Photo Management

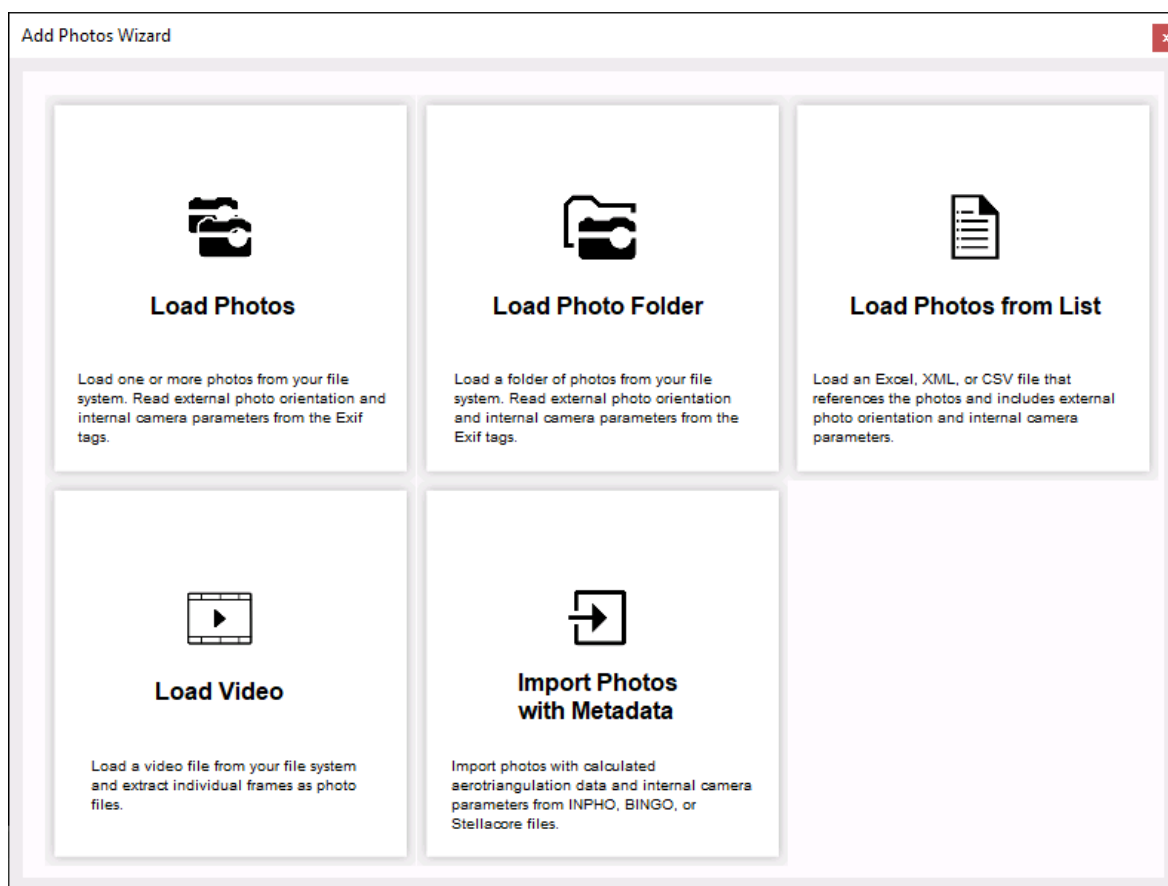
This chapter deals with basic photo management tasks:

- [Loading of new photos in a project](#)
- [Importing photos with metadata](#)
- [Excluding/removing a photo from the project](#)
- [Modifying photo and photo collection properties](#)

Photo Requirements

See "[About Imagery Acquisition](#)" in the "Imagery Acquisition" chapter for a list of requirements and recommendations for the photos from which the model will be generated.

Adding Photos



Add Photos Wizard

Photos are added to the project using the Photos Wizard. The following methods are available:

- Load one or more photos or a folder of photos from your file system. Read external photo orientation and internal camera parameters from the Exif tags.
- Load an Excel, XML, or CSV file that references the photos and includes external photo orientation and internal camera parameters.
- Extract individual frames from video and save them as image files.
- Load a PhotoMesh project. When a PhotoMesh project (.PhotoMeshXML) is loaded into a second project, only the project's photos and photo and collection parameters are loaded.
- Import photos from INPHO, BINGO, or Stellacore files with calculated aerotriangulation data and internal camera parameters. See ["Importing Photos with Metadata"](#) in this chapter.

Photos can only be loaded when Review mode is off. See ["About Reviewing Your Build"](#) in the "Reviewing Your Build" chapter for more information about Review mode.

After photos are added, they are automatically validated. This process checks for any missing data, such as focal length, sensor width/pixel size, or coordinate system information. It also verifies the validity of photo properties, such as photo dimensions, focal length value, and principal point (not too far from photo center). If discrepancies are detected, a warning or error symbol is displayed in the Project Tree to alert you to view and correct the issues. For more information on the specific problem, you can click the warning/error symbol to open a comprehensive Quality Report. See ["Viewing a Quality Report"](#) in the "Reviewing Your Build" chapter for more information. If only photos were added and no build was yet performed, the report will only contain the relevant sections.

Loading Photo Files from Disk

For optimal performance, photographs should be divided into one or more **collections**, homogeneous groups of photographs, all taken by the same physical camera with identical focal length and principal point.

To select photo files from your local disk:

1. On the **Home** tab, in the **Add** group, click **Photos**, and then select one of the following from the **Add Photos Wizard**:
 - **Load Photos** – To add selected files. The Load Photo dialog is displayed.
 - **Load Photo Folder** – To add an entire folder. The Load Photo Folder dialog is displayed.
2. Browse to the required folder/file(s) and click **Open**.
3. If you are adding a folder of photos, PhotoMesh automatically creates a group (collection) and places all photos under the folder in this group. You can see the list of photo collections in the collections list under the Collections tab. If you are adding a folder with sub-folders, a dialog is displayed asking if you want to create a separate collection for each sub-folder. Click **Yes** to create.

Note: If a folder is added with photos that were taken by different cameras or with different focal lengths and principal points, PhotoMesh will automatically place the photos in different collections as long as the required metadata (e.g., Exif) is encoded in the photo files.
4. After loading the photos, the photo collections' basic properties are displayed in the Add Photos Wizard. Edit the properties as required:

Photo Wizard Properties

Collection Name	Name of the collection in the Photos list.
Coordinate System	Coordinate system of the collection. Click Select  to set a different coordinate system. See "Coordinate System" in the "Basic Concepts" chapter for more information.
Focal Length Normalization	Select which camera properties should be used to set the focal length, and then set the values of the selected properties: ▪ Focal length and sensor width ▪ Focal length and pixel size ▪ Focal length 35

5. If you loaded more than one photo collection, an **Apply to all**  button is displayed next to the Coordinate System and Focal Length Normalization fields in each collection. Click **Apply to all**  to apply the property value set for that collection to all the collections that were loaded.
6. Click **Edit in PhotoMesh** to open the Collections list and property sheet so that you can edit additional properties and make other preparations for the build (e.g., add control points or set AT and reconstruction areas) before beginning a build. See ["Modifying Photo and Photo Collection Properties"](#) in this chapter and the "Preparing the Project" chapter for more information.
7. Click **Build** to directly open the Build Settings dialog. See ["Setting Build Steps, Parameters and Outputs"](#) in the "Building" chapter for information.

Defining Collection Properties with collectionProperties.xml

You can define collection properties and override those included in the metadata (e.g., Exif) encoded in photo files by creating a file named collectionProperties.xml with the required collection properties. This saves you the trouble of creating a full Excel or XML file that contains a list of all the photos. The collectionProperties.xml file is commonly used to add band information, focal length, or other collection level properties. Each collectionProperties.xml file can only contain data for a single collection, so a separate file must be created for each collection for which you want to provide supplementary information and placed in the corresponding collection folder.

There are several mandatory properties in the collectionProperties.xml file. If you want to use the values in the metadata encoded in the photo files, for these properties, set the required property in the collectionProperties.xml file equal to 0, e.g., "Width = 0".

A sample collectionProperties.xml follows below. Property descriptions are in the comments `<!-- ....>`.

```
<?xml version="1.0" encoding="UTF-8"?>
<BlocksExchange version="2.2">

  <!-- Coordinate system definition -->
  <SpatialReferenceSystems>
    <SRS>
      <Id>0</Id>
      <Name>WGS 84</Name>
      <Definition>GEOGCS["WGS 84",
        DATUM["WGS_1984",
          SPHEROID["WGS 84",6378137,298.257223563,
            AUTHORITY["EPSG","7030"]],
          AUTHORITY["EPSG","6326"]],
        PRIMEM["Greenwich",0,
          AUTHORITY["EPSG","8901"]],
        UNIT["degree",0.0174532925199433,
          AUTHORITY["EPSG","9122"]],
```

```

        AUTHORITY["EPSG","4326"]]</Definition>
    </SRS>
</SpatialReferenceSystems>

<!-- Block definition -->
<Block>
    <Photogroups>
        <!-- A photogroup is a homogeneous group of photographs, all taken by the
same physical camera with identical focal length and principal point. -->
        <Photogroup>
            <!-- Name of the photogroup -->
            <Name>NIR</Name>
            <!-- (Mandatory) Width and height of the photos, in pixels. If you do
not want the collectionProperties.xml to override the width or height information in
the metadata encoded in the photo files, set their values equal to 0. -->
            <ImageDimensions>
                <Width>0</Width>
                <Height>0</Height>
            </ImageDimensions>
            <Accuracy>
                <!-- (Optional) Weight to assign to the photos' provided accuracy
values. See "Collection Property Sheet" in the "Photo Management" chapter for
information. -->
                <HorizontalAccuracyFactor>30</HorizontalAccuracyFactor>
                <!-- (Optional) Weight to assign to the photos' provided XY
coordinates in determining the calculated XY values. This ranges from 0-100, where 0
is Unreliable and 100 is Very High - External AT. -->
                <VerticalAccuracyFactor>40</VerticalAccuracyFactor>
                <!-- (Optional) Weight to assign to the photos' provided altitude
in determining the calculated altitude value. This ranges from 0-100, where 0 is
Unreliable and 100 is Very High - External AT. -->
                <FocalLengthAccuracy>25</FocalLengthAccuracy>
                <!-- (Optional) A percentage that indicates the allowable
deviation from the provided principal point when calculating the AT. This represents
the accuracy of the provided principal point. This ranges from 0-100, where 0 is Use
Input Value and 100 is Unreliable. -->
                <PrincipalPointAccuracy>15</PrincipalPointAccuracy>
                <!-- (Optional) Radial accuracy. It is generally recommended to
enter a value of 1 for the "Standard" option. If your data was captured with a
professional camera that automatically applied a radial distortion correction (i.e.,
the original distortion value is extremely accurate), enter 0 for "Use input value". -
->
                <RadialAccuracy>1</RadialAccuracy>
                <!-- (Optional) Tangential accuracy. It is generally recommended
to enter a value of 1 for the "Standard" option. If your data was captured with a
professional camera that automatically applied a radial distortion correction (i.e.,
the original distortion value is extremely accurate), enter 0 for "Use input value". -
->
                <TangentialAccuracy>1</TangentialAccuracy>
                <!-- (Optional) Enter a value of 1 to ignore the provided
orientation value. -->
                <IgnoreOrientation>1</IgnoreOrientation>
            </Accuracy>
            <!-- Optional Parameters. Don't include if you don't want the
collectionProperties.xml to override the information in the metadata encoded in the
photo files. -->
            <Bands>Red,Green,Custom_1</Bands>
            <!-- (Optional) Focal length in millimeters. -->
            <FocalLength>80.0</FocalLength>
            <!-- (Optional) Size of a physical pixel on the camera sensor in
millimeters. -->
            <PixelSize>0.0052</PixelSize>
            <!-- (Mandatory) Origin and units of the principal point. The possible

```

```

values are: topleftpixels, topleftmm, centerpixels, centermm. Use "centermm" if you
don't have this information. -->

<PrincipalPointReferenceMethod>centerpixels</PrincipalPointReferenceMethod>
  <!-- (Optional) PrincipalPoint x and y values, in the units set by
PrincipalPointReferenceMethod. -->
  <PrincipalPoint>
    <x>7360</x>
    <y>4912</y>
  </PrincipalPoint>
  <!-- Use yaw, pitch, roll values for orientation -->
  <UseYPR>
    <!-- (Optional) Offset along the Phi axis in degrees.-->
    <PhiOffset>90</PhiOffset>
    <!-- (Optional) Offset along the Kappa axis in degrees.-->
    <KappaOffset>-180</KappaOffset>
    <!-- (Optional) Offset along the Omega axis in degrees. -->
    <OmegaOffset>0.0</OmegaOffset>
    <!-- (Optional) Offset from the current altitude value in the
coordinate system units. If Lat-Long, use meters. -->
    <AltitudeOffset>-10.0</AltitudeOffset>
    <!-- (Optional) Flips the Phi axis.-->
    <FlipPhi>false</FlipPhi>
    <!-- (Optional) Flips the Kappa axis.-->
    <FlipKappa>false</FlipKappa>
    <!-- (Optional) Flips the Omega axis.-->
    <FlipOmega>false</FlipOmega>
    <!-- (Optional) Boolean that determines whether to include this
collection in the point cloud.-->
    <UseInDpc>false</UseInDpc>
    <!-- (Optional) Boolean that determines whether to include this
collection in the textured model.-->
    <UseInModelTexture>true</UseInModelTexture>
    <!-- (Optional) Boolean that determines whether to include this
collection in the orthophoto. -->
    <UseInOrthoTexture>true</UseInOrthoTexture>
    <!-- (Optional) Radial and tangential distortion coefficients
calculated using the Brown-Conrady model. See here for more information. -->
    <Distortion>
      <!-- (Optional) Defines the 1st radial distortion coefficient.
-->
      <K1>0.0</K1>
      <!-- (Optional) Defines the 2nd radial distortion coefficient.
-->
      <K2>0.0</K2>
      <!-- (Optional) Defines the 3rd radial distortion coefficient.
-->
      <K3>0.0</K3>
      <!-- (Optional) Defines the 1st tangential distortion
coefficient. -->
      <P1>0.0</P1>
      <!-- (Optional) Defines the 2nd tangential distortion
coefficient. -->
      <P2>0.0</P2>
    </Distortion>
  </UseYPR>
</Photogroup>
</Photogroups>
</Block>
</BlocksExchange>

```

To define collection properties using a collectionProperties.xml file:

1. Place the collectionProperties.xml file in the corresponding collection folder.
2. Load the collection folder. See "[Loading Photo Files from Disk](#)" in this chapter for information.

Loading a Photo List File

Instead of manually entering all photo and photo collection information (e.g., sensor width, focal length, and aerotriangulation data), you can load an Excel or ContextCapture XML file that references the photos and includes this data. You also have the option to load a local CSV file that includes only XYZ and OPK/YPR data. Excel and ContextCapture XML photo list files can reference photos stored locally or on an HTTP or HTTPS server, or Amazon S3. The path in the photo list file can be relative or absolute, e.g., `s3://skyline-data/Frederick_xlsx/CF012316.tif` OR

`http://www.skylineglobe.com/Photos_LiDARs/Photos/tux-1771.jpg`

Note: In the photo list file, the photos' path should point directly to the photos' path on the server or S3. The photo list file itself must be located on and loaded from a local or shared drive.

A sample Excel (PhotoMeshSample.xlsx) file is included in the application's folder, which can serve as a template for creating such a file as well as provide basic property definitions (in the Excel's notes). You can also see "[Collection Property Sheet](#)" and "[Photo List](#)" in this chapter for a description of the Excel/XML/CSV parameters.

The Excel file should include the following sheets:

- The first sheet named Collections should detail the group properties for each of the "collections", i.e., groups of photos. For optimal performance, each collection should include only photographs taken by the same physical camera with identical focal length and principal point. These group properties are the same as the properties that are listed in the Collection property sheet. See "[Collection Property Sheet](#)" in this chapter. Each row after the heading row (with all the property names) represents a different collection.
- A separate sheet for each of the collections, listing the properties of each photo in the collection. The name of each collection sheet must be identical to the collection name defined in the Name field of the Excel's Collections sheet. The photo properties are the same as the properties in the photo list. See "[Photo List](#)" in this chapter.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Name	Focal Length	Focal Length 35	Focal Length Units	Pixel Size	Sensor Width	Principal Point X	Principal Point Y	Principal Point Units	Principal Point Origin	Width	Height	Altitude Base	Coord
2	A	51.8004		mm		36.064			pixels	Top-Left	7360	4912	WGS84 ELLIPSOID	GS 84
3	B	84.8513		mm		36.064			pixels	Top-Left	7360	4912	GEOID	GS 84

Excel File – Collection Property Sheet

Path	X	Y	Altitude	Omega	Phi	Kappa
C:\skyline\data\Frederick\Images\13_2_239_A.jpg	-77.41471	39.4117622	546.20636	-1.57382	-40.4607	84.137
C:\skyline\data\Frederick\Images\13_2_240_A.jpg	-77.41468	39.4125328	546.6153	2.948473	-44.11862	88.372
C:\skyline\data\Frederick\Images\13_2_243_A.jpg	-77.41468	39.4149437	543.0982	-4.01135	-45.51225	84.531
C:\skyline\data\Frederick\Images\13_2_244_A.jpg	-77.41471	39.4157143	540.58136	-1.84912	-38.56183	87.436
C:\skyline\data\Frederick\Images\13_2_245_A.jpg	-77.4147	39.4165268	539.212646	-3.47045	-47.07546	84.857
C:\skyline\data\Frederick\Images\14_2_259_A.jpg	-77.41569	39.4117279	497.463959	-3.54981	-44.754	87.436
C:\skyline\data\Frederick\Images\14_2_260_A.jpg	-77.41571	39.4125481	496.3526	-4.06477	-44.60705	86.866
C:\skyline\data\Frederick\Images\14_2_261_A.jpg	-77.41579	39.41332	469.36673	-0.52508	-50.41594	87.792
C:\skyline\data\Frederick\Images\14_2_262_A.jpg	-77.41577	39.4141273	493.204834	-3.02948	-44.9635	87.1
C:\skyline\data\Frederick\Images\14_2_263_A.jpg	-77.4157944	39.4149323	493.079224	-2.34766	-45.31397	87.602
C:\skyline\data\Frederick\Images\14_2_264_A.jpg	-77.41583	39.4157372	492.870483	-2.31804	-45.10428	87.605

Excel File – Photo Property Sheet for Specific Collection

Collection,Path,Y,X,Z,yaw,pitch,roll

```

1,lower_left_DSC04405.JPG,-19.2812836,-48.4160783,862.72,187.22,-9.47,-2.57
1,lower_right_DSC04405.JPG,-19.2812836,-48.4160783,862.72,187.22,-9.47,-2.57
1,upper_left_DSC04405.JPG,-19.2812836,-48.4160783,862.72,187.22,-9.47,-2.57
1,upper_right_DSC04405.JPG,-19.2812836,-48.4160783,862.72,187.22,-9.47,-2.57
1,lower_left_DSC04406.JPG,-19.2814442,-48.4161162,863.24,186.84,-8.4,-2.38
1,lower_right_DSC04406.JPG,-19.2814442,-48.4161162,863.24,186.84,-8.4,-2.38
1,upper_left_DSC04406.JPG,-19.2814442,-48.4161162,863.24,186.84,-8.4,-2.38
1,upper_right_DSC04406.JPG,-19.2814442,-48.4161162,863.24,186.84,-8.4,-2.38
1,lower_left_DSC04407.JPG,-19.2815949,-48.4161432,863.61,188.29,-7.58,-1.93
1,lower_right_DSC04407.JPG,-19.2815949,-48.4161432,863.61,188.29,-7.58,-1.93
1,upper_left_DSC04407.JPG,-19.2815949,-48.4161432,863.61,188.29,-7.58,-1.93
1,upper_right_DSC04407.JPG,-19.2815949,-48.4161432,863.61,188.29,-7.58,-1.93
1,lower_left_DSC04408.JPG,-19.2817591,-48.4161703,863.88,189.58,-7.58,-0.51
1,lower_right_DSC04408.JPG,-19.2817591,-48.4161703,863.88,189.58,-7.58,-0.51
1,upper_left_DSC04408.JPG,-19.2817591,-48.4161703,863.88,189.58,-7.58,-0.51
1,upper_right_DSC04408.JPG,-19.2817591,-48.4161703,863.88,189.58,-7.58,-0.51
2,lower_left_DSC04409.JPG,-19.2819152,-48.4161977,864.35,189.86,-6.37,-0.62
2,lower_right_DSC04409.JPG,-19.2819152,-48.4161977,864.35,189.86,-6.37,-0.62
2,upper_left_DSC04409.JPG,-19.2819152,-48.4161977,864.35,189.86,-6.37,-0.62
2,upper_right_DSC04409.JPG,-19.2819152,-48.4161977,864.35,189.86,-6.37,-0.62
2,lower_left_DSC04410.JPG,-19.2820761,-48.4162257,864.59,191.08,-7.19,0.47
2,lower_right_DSC04410.JPG,-19.2820761,-48.4162257,864.59,191.08,-7.19,0.47
2,upper_left_DSC04410.JPG,-19.2820761,-48.4162257,864.59,191.08,-7.19,0.47
2,upper_right_DSC04410.JPG,-19.2820761,-48.4162257,864.59,191.08,-7.19,0.47

```

CSV File – Photo Properties for Multiple Collections

To add photo files using an Excel or XML file:

1. On the **Home** tab, in the **Add** group, click **Photos**, and then from the **Add Photos Wizard**, select **Load Photos from List**. The Load a Photo List File dialog is displayed.

- Browse to the Excel or XML file. The photo files are loaded into the project. After loading the photos, the photo collections' basic properties are displayed in the Add Photos Wizard. Edit the properties as required. See ["Photo Wizard Properties"](#) for information on the wizard properties and ["Modifying Photo and Photo Collection Properties"](#) in this chapter for information on additional photo and collection properties.

To add photo files using a CSV file:

- On the **Home** tab, in the **Add** group, click **Photos**, and then from the **Add Photos Wizard**, select **Load Photos from List**. The Load a Photo List File dialog is displayed.
- Browse to the CSV file. The CSV/TXT Importer dialog is displayed.

Load CSV File

Load CSV File

Sample records

0. lower_left_1_DSC00987.JPG,36.4870466,-120.43400530000001,288.3,2.04,-4.24,4.62
1. lower_right_1_DSC00987.JPG,36.4870466,-120.43400530000001,288.3,2.04,-4.24,4.62
2. upper_left_1_DSC00987.JPG,36.4870466,-120.43400530000001,288.3,2.04,-4.24,4.62
3. upper_right_1_DSC00987.JPG,36.4870466,-120.43400530000001,288.3,2.04,-4.24,4.62
4. lower_left_1_DSC00988.JPG,36.487179600000005,-120.4340058,288.03,3.87,-7.56,2.73

Select the CSV column name to map to each of the photo properties.
Remember to set the correct collection parameters in the collection properties sheet.

Parsed csv file starting at row: with the following delimiter:

CSV Column: Column 1	CSV Column: Column 2	CSV Column: Column 3	CSV Column: Column 4	CSV Column: Column 5
Photo Property: ImagePatl	Photo Property: Y	Photo Property: X	Photo Property: Altitude	Photo Property: Yaw
lower_left_1_DSC00987.JPG	36.4870466	-120.43400530000001	288.3	2.04
lower_right_1_DSC00987.JPG	36.4870466	-120.43400530000001	288.3	2.04
upper_left_1_DSC00987.JPG	36.4870466	-120.43400530000001	288.3	2.04
upper_right_1_DSC00987.JPG	36.4870466	-120.43400530000001	288.3	2.04
lower_left_1_DSC00988.JPG	36.487179600000005	-120.4340058	288.03	3.87
lower_right_1_DSC00988.JPG	36.487179600000005	-120.4340058	288.03	3.87

Load

Cancel

CSV/TXT Importer

- Enter the row at which to start import, and select a **Delimiter**.
- Select the CSV column that corresponds to each of the photo properties. PhotoMesh remembers the column mapping from your most recent CSV import, so only modifications from the last import need to be adjusted.
- Click **Load**.
- After loading the photos, the photo collections' basic properties are displayed in the Add Photos Wizard. Edit the properties as required. See ["Photo Wizard Properties"](#) for information on the wizard

properties and ["Modifying Photo and Photo Collection Properties"](#) in this chapter for information on additional photo and collection properties.

Loading Photos from a Video

PhotoMesh can extract individual frames from video and save them as photo files. Supported video formats include: MPEG, AVI, MOV, and WMV.

To load photos from a video:

- 1. On the **Home** tab, in the **Add** group, click **Photos**, and then from the **Add Photos Wizard**, select **Load Video**. The Load Video dialog is displayed.
- 2. Browse to the video file, and click **Open**. The Load Video Editor is displayed.

Start:00:00:00.000End:00:00:05.000

Frame Every:5Seconds

Focal Length Normalization:Focal length and sensor width

Focal Length:mm

Sensor Width:mm

Pixel Size:mm

Zoom:Fixed

Load Video Editor

- 3. Enter the following information:

Start	Start time from which video frame extraction should begin. Type in the time or drag the Start marker of the Video Cropping bar to the required position.
End	End time for video frame extraction. Type in the time or drag the End marker of the Video Cropping bar to the required position.
Frame Every	Sampling interval, i.e., every how many seconds a frame should be captured.
Focal Length Normalization	Camera properties used to set the focal length. Select one of the following options: - Focal length and sensor width - Focal length and pixel size - Focal length 35
Focal Length	Focal length in millimeters.
Sensor Width	Size of the camera sensor in millimeters.
Pixel Size	Size of a physical pixel on the camera sensor in millimeters.
Zoom	How the video was captured:

-
- **Fixed** - Select if the video was captured with a fixed zoom. In this case, PhotoMesh will ensure that all intrinsic parameters for all the video frames are the same.
 - **Not Fixed** - Select if the zoom level varied over the course of the capture, e.g., by using auto-zoom. In this case, PhotoMesh will calculate intrinsic parameters per frame.
-

4. Click **Load**. The video frames are loaded into the project as photo files into a single photo collection.

Loading a PhotoMesh Project

When a PhotoMesh project (.PhotoMeshXML) is loaded into a second project, only the project's photos and photo and collection parameters are loaded. Other project information (e.g. terrain objects, control points, and other project parameters) is not loaded.

To load a PhotoMesh project:

1. On the **Home** tab, in the **Add** group, click **Photos**, and then from the **Add Photos Wizard**, select **Load Photos from List**. The Load a Photo List File dialog is displayed.
2. Browse to the required PhotoMesh project file. The project's photos and photo and collection parameters are loaded into the project.

Importing Photos with Metadata

Import photos with calculated aerotriangulation data and internal camera parameters from INPHO, BINGO, or Stellacore files. When importing Inpho or Bingo results in a **Cartesian** coordinate system, such as ENU or ECEF, the AT step of the build process can be skipped (by selecting "Do not calculate AT (Fully trusted)," in the [Calculation Mode](#) AT setting). Imported AT data in a **projected** coordinate system such as UTM, however, may be unreliable either due to earth curvature or earth curvature corrections performed in the other system. This is particularly true for datasets that contain oblique photos or large frames, or cover a large geographic area. For data of this type, aerotriangulation should be re-performed in PhotoMesh, i.e., use Refine AT instead of Do not calculate AT (Fully trusted) for the [Calculation Mode](#) AT setting.

In PhotoMesh 8.0, radial/tangential distortion values are not imported with INPHO data, so it is recommended to select the "[Refine AT](#)" Calculation Mode AT setting to rerun the aerotriangulation process (performing only bundle adjustment and GCP).

- [Importing Inpho Data](#)
- [Importing Bingo Data](#)
- [Importing Stellacore Data](#)

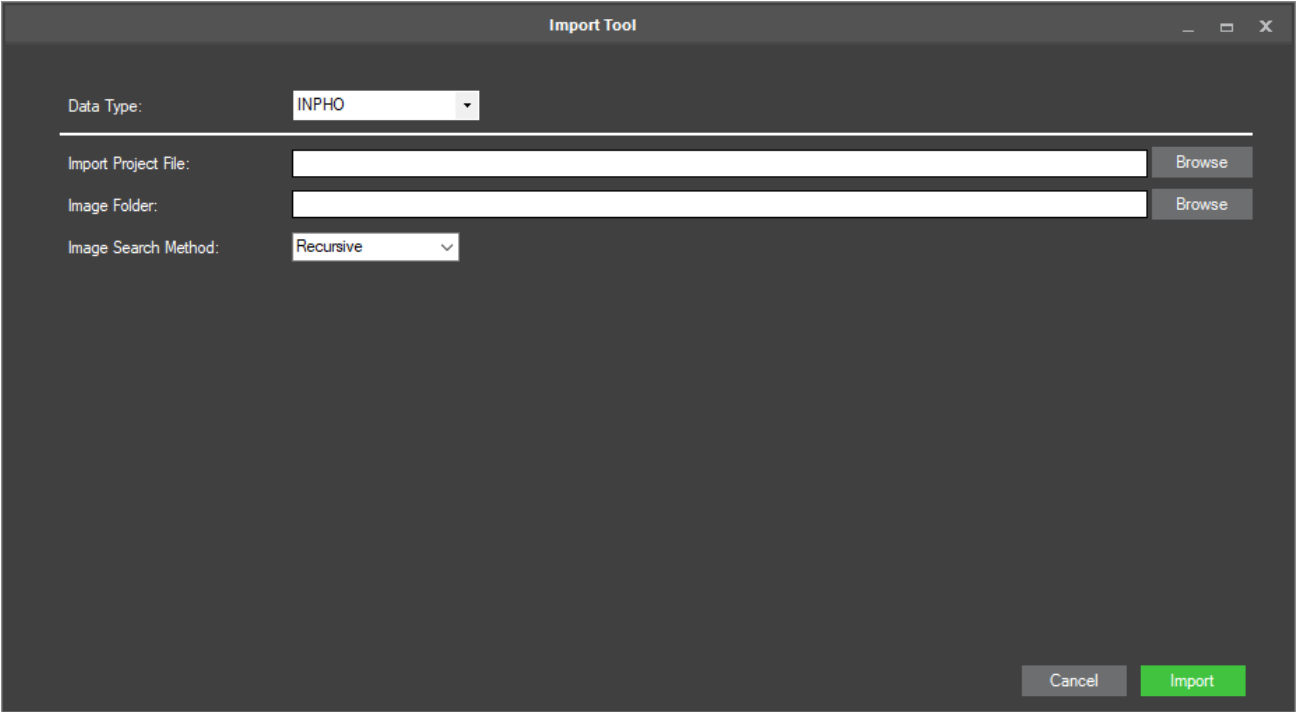
Importing Inpho Data

When importing Inpho results, the AT step of the build process can sometimes be skipped. **See** "[Importing Photos with Metadata](#)" in this chapter for more information.

To import Inpho Data:

1. On the **Home** tab, in the **Add** group, click **Photos**, and then from the **Add Photos Wizard**, select **Import Photos with Metadata**. The Import Tool dialog is displayed.

Note: The fields displayed vary depending on the type of data selected.



Import Tool

2. Enter the following information:

Field	Description
Data Type	Select the data type.
Import Project File	Click Browse . Then in the Import Project File dialog, browse to the required file, and click Open .
Image Folder	Click Browse . Then in the Import Image Folder dialog, browse to the required folder, and click Select Folder .
Image Search Method	Select one of the following options: - Recursive – To include image files in all subdirectories - Top Directory Only – To include image files only from the top level directory

3. Click **Import**.

Importing Bingo Data

When importing Bingo results, the AT step of the build process can sometimes be skipped. See ["Importing Photos with Metadata"](#) in this chapter for more information.

To import Bingo Data:

- On the **Home** tab, in the **Add** group, click **Photos**, and then from the **Add Photos Wizard**, select **Import Photos with Metadata**. The Import Tool dialog is displayed.

Note: The fields displayed vary depending on the type of data selected.

Import Tool

Data Type: BINGO

Import Project File: Browse

Image Folder: Browse

Image Search Method: Recursive

Angle Units: Gons (400°)

Rotation Order: Phi-Omega-Kappa

WKT/ESPG: WGS 84 Set

	Camera ID	Columns [px]	Rows [px]	Pixel Size [mm]	Grid File
•					

Camera Definition:

Cancel Import

Import Tool

2. Enter the following information:

Field	Description
Data Type	Select the data type.
Import Project File	Click Browse . Then in the Import Project File dialog, browse to the required file, and click Open .
Image Folder	Click Browse . Then in the Import Image Folder dialog, browse to the required folder, and click Select Folder .
Image Search Method	Select one of the following options: - Recursive – To include image files in all subdirectories - Top Directory Only – To include image files only from the top level directory
Angle Units	Select the data's angle units (Gons or Degrees).
Rotation Order	Select the rotation order (Phi-Omega-Kappa or Omega-Phi-Kappa).
WKT/ESPG	Click Set to set the data's coordinate system. See "Coordinate System" in the "Basic Concepts" chapter for more information.
Camera Definition	For each camera in the BINGO project file, enter the following information: Camera ID – Camera name as it appears in the BINGO project file. This field is automatically filled in after the Import Project File is selected.

- **Columns** – Number of pixel columns in a photo taken with this camera (image width).
- **Rows** – Number of pixel rows in a photo taken with this camera (image height).
- **PixelSize** – Camera sensor's pixel size.
- **GridFile (Optional)**: File with camera distortion information that may be supplied by your data provider together with the BINGO file. This information can be applied during import to reduce image distortion. Click the button to open the Select Grid Calibration File dialog. Then browse for the grid file.

3. Click **Import**.

Importing Stellacore Data

To import Stellacore Data:

1. On the **Home** tab, in the **Add** group, click **Photos**, and then from the **Add Photos Wizard**, select **Import Photos with Metadata**. The Import Tool dialog is displayed.

Note: The fields displayed vary depending on the type of data selected.

The screenshot shows the 'Import Tool' dialog box. At the top, the title bar says 'Import Tool'. Below it, there's a 'Data Type:' label followed by a dropdown menu currently showing 'Stellacore'. A horizontal line separates this from the main content area. In the main area, there are three rows of controls. The first row is for 'Image Folder:', followed by a text input field, a dropdown menu set to 'Recursive', and a 'Browse' button. The second row is for 'EO Folder:', followed by a text input field, a dropdown menu set to 'Top Directory Only', and a 'Browse' button. The third row is for 'Camera IO Folder:', followed by a text input field, a dropdown menu set to 'Top Directory Only', and a 'Browse' button. At the bottom right of the dialog, there are two buttons: 'Cancel' and 'Import'.

Import Tool

2. Enter the following information:

Field	Description
Data Type	Select the data type.

Image Folder	A folder containing the images. Click Browse. Then in the Select Image Folder dialog, browse to the required folder, and click Select Folder. Select one of the following search options: ▪ Recursive – To include image files in all subdirectories ▪ Top Directory Only – To include image files only from the top level directory
EO Folder	A folder containing the external orientation information. Click Browse. Then in the Import EO Base Folder dialog, browse to the required folder, and click Select Folder. Select one of the following search options: ▪ Recursive – To include files in all subdirectories ▪ Top Directory Only – To include files only from the top level directory
Camera IO Folder	A folder containing the camera's internal orientation information. Click Browse. Then in the Import Camera IO Base Folder dialog, browse to the required folder, and click Select Folder. Select one of the following search options: ▪ Recursive – To include files in all subdirectories ▪ Top Directory Only – To include files only from the top level directory

3. Click **Import**.

Selecting Photos

Photos can be selected either in the 3D Window or from the Photos list to perform various commands on them, e.g., delete or exclude a photo, open in photo viewer, or modify photo properties.

To select photos:

1. In the **Project Panel**, select the **Photos** group.
2. In the **3D Window**, click the photo you want to select, or select the photo(s) in the **Photos** list. You can multi-select photos using the Photos list commands.

Note: If you turned on global selection, you can select a photo in the 3D Window even without first selecting the Photos group in the Project Panel as long as the photos are visible in the 3D Window.

See ["Showing and Hiding Group Items"](#) and ["Selecting Items Using the Item List"](#) in the "Using the Project Panel and Item List" chapter and ["Selecting Objects in the 3D Window"](#) in the "Basic Concepts" chapter for information.

Excluding a Photo

To exclude a photo from the project:

Note: Photos can only be excluded when Review mode is off. See ["About Reviewing Your Build"](#) in the "Reviewing Your Build" chapter for more information about Review mode.

1. Select the required photo. See ["Selecting Photos"](#) in this chapter for information.
2. On the **Photo** tab, in the **Selection** group, click **Exclude**. You can also exclude a photo by selecting its **Exclude** check box in the photo list.

Deleting a Photo

To delete a photo from the project:

Note: Photos can only be deleted when Review mode is off. See ["About Reviewing Your Build"](#) in the "Reviewing Your Build" chapter for more information about Review mode.

Note: When a photo is **deleted** from a project, the photo is no longer available for inclusion within the same project.

1. Select the required photo. See ["Selecting Photos"](#) in this chapter for information.
2. On the **Photo** tab, in the **Selection** group, click **Delete**. The photo is deleted from the photo list.

Modifying Photo and Photo Collection Properties

Photo data can be viewed and modified in the photo collection property sheet and photo list. Information displayed in the property sheet refers to the collection that is currently selected in the collections list, while information in the photo list is specific to each photo.

Information about the photo collections and photos (e.g., coordinate system, sensor width, focal length, photo position information) is generally read from a loaded Excel or XML file that includes this information. These parameters can then be adjusted in the collection property sheet and photo list.

Properties can be edited even after aerotriangulation, up until the reconstruction step starts. When rebuilding, if any change was made to photo/collection properties, the build is restarted from pre-aerotriangulation.

To display and modify photo collection properties:

1. In the **Project Panel**, select the **Photos** group.
2. In the collection property sheet, modify the property values as required. See ["Collection Property Sheet"](#) in this chapter for more information.

To display and modify photo properties:

1. In the **Project Panel**, select the **Photos** group.
2. In the **Photos** list, select the required photos, and modify their properties as required. See ["Photos List"](#) in this chapter for information.

Note: You can also search the list for the required photos. See ["Selecting Items Using the Item List"](#) in the "Basic Concepts" chapter for information.

Collection Property Sheet

Properties	
Quick Search	
Collection	
Name	CAM3
Coordinate System	WGS 84 + EGM96 geoid height
Orientation Method	Omega Phi Kappa
Altitude Type	Absolute
Relative Altitude Reference	0
Camera	
Focal Length Normalization	Focal length and sensor width
Sensor Width	35.9168
Pixel Size	0.0048800000000000007
Focal Length	84.9112465541587
Focal Length 35	
Principal Point Reference	Top left in pixels
Principal Point X	3640.0301108239541
Principal Point Y	2485.9500336083788
Width	7360
Height	4912
Gamma	1
Effective Range	
Calculated Effective Range	869.47053554680951
Crop Area	100 %
Use In Textured Model	☒
Use In Orthophoto	☒
Use In Point Cloud	☒
Bands	
Hyperspectral Bands	Red, Green, Blue
1	Red
2	Green
3	Blue
Camera Distortion	
K1	0.04401660034628193
K2	0.19618285711837311
K3	3.8859124931693629
P1	0
P2	0
Advanced Position	
Altitude Offset	
Omega Offset	
Phi Offset	
Kappa Offset	
Flip Omega	☐
Flip Phi	☐
Flip Kappa	☐
Rotate Yaw	
Rotate Pitch	
Rotate Roll	

Note: If one or more of the camera properties that are used to set the focal length (These properties vary depending on the Focal Length Normalization property setting set in the Collection property sheet) is missing, an orange question mark  is displayed next to the property.

Property	Description
Collection	
Name	Name of the collection in the Photos list.
Coordinate System	Coordinate system of the collection. Click Select  to set a different coordinate system. See "Coordinate System" in the "Basic Concepts" chapter for more information.
Orientation Method	Select which system to use to set orientation: OPK (Omega, Phi, Kappa) or YPR (Yaw, Pitch, and Roll). This selection determines which fields are displayed in the Advanced Position section of the property sheet.
Altitude Type	When both relative and absolute altitude data were provided, this property sets the altitude method to be used by the collection. Select Relative to Terrain to place the collection at a specified altitude above the ground. Select Absolute to place the collection at a specified altitude above the terrain database vertical datum base ellipsoid.
Relative Altitude Reference	If "Relative" was selected as the Altitude Type, this defines the altitude above the ground.
Camera	
Focal Length Normalization	Camera properties used to set the focal length. Select one of the following options: ▪ Focal length and sensor width ▪ Focal length and pixel size ▪ Focal length 35
Sensor Width	Size of the camera sensor in millimeters.
Pixel Size	Size of a physical pixel on the camera sensor in millimeters.
Focal Length	Focal length in millimeters.
Focal Length 35	Equivalent focal length in millimeters for a 35 mm film camera.
Principal Point Reference	Origin and units of the principal point. Select one of the following: ▪ Top left in pixels – Positive x-direction is to the right and positive y-direction is downwards. ▪ Top left in mm – Positive x-direction is to the right and positive y-direction is downwards. ▪ Center in pixels – Positive x-direction is to the right and positive y-direction is upwards. ▪ Center in mm – Positive x-direction is to the right and positive y-direction is upwards.

Principal Point X	Principal point X value in pixels or mm according to principal point reference.
Principal Point Y	Principal point Y value in pixels or mm according to principal point reference.
Width	Width of photo in pixels.
Height	Height of photo in pixels.
Gamma	Brightness of the photo's colors. The default value of 1 leaves the color unchanged. Values >1 brighten the colors, while values <1 darken them.
Effective Range	Maximum distance from the camera position for which photo information is considered valid for the aerotriangulation.
Calculated Effective Range	Whenever the Effective Range parameter value is not provided, PhotoMesh calculates the maximum distance from the camera for which photo information is considered valid for the aerotriangulation, and then displays and uses this value.
Crop Area	Percentage of the photo to use in the project. Blurry or obstructed areas on the edges of a photo will be cropped based on this property value.
Use in Textured Model	Select the check box to include this collection in the textured model.
Use in Orthophoto	Select the check box to include this collection in the orthophoto.
Use as Reference Only	Select the check box to include this collection only in the reference project. See "Adding a Reference Project to Use in Aerotriangulation" in the "Control Points" chapter for information.
Use in Point Cloud	Select the check box to include this collection in the point cloud output.

Bands – Multispectral

Note: The band information is retrieved from the collection information in the photo list Excel or XML, the collectionProperties XML, or from the first photo in the collection. **See** ["Loading a Photo List File"](#) and ["Loading Photo Files from Disk"](#) in the "Photo Management" chapter. You can manually set the value for each band in the property sheet.

Multispectral Bands	Multispectral bands in the collection's photos. These bands can be used in texturing the 3D model and in generating the orthophoto. See "Setting Advanced Reconstruction Parameters" in the "Building" chapter.
1	Select the first band.
2	Select the second band.
3	Select the third band.

Camera Distortion (Optional)

Radial and tangential distortion calculated using the Brown-Conrady model. See [here](#) for more information.

K1	Defines the 1 st radial distortion coefficient.
K2	Defines the 2 nd radial distortion coefficient.
K3	Defines the 3 rd radial distortion coefficient.

P1	Defines the 1 st tangential distortion coefficient.
P2	Defines the 2 nd tangential distortion coefficient.
Advanced Position (Either OPK or YPR offset and flip values will be displayed, depending on your selection above for Orientation Method)	
Altitude Offset	Offset from the current altitude value.
Omega Offset	Offset in degrees along the Omega axis from the photo's provided angle. Applied before any additional rotation and prior to calculating position/orientation, this property helps correct orientation discrepancies between a camera and the plane on which it is mounted.
Phi Offset	Offset in degrees along the Phi axis from the photo's provided angle. Applied before any additional rotation and prior to calculating position/orientation, this property helps correct orientation discrepancies between a camera and the plane on which it is mounted.
Kappa Offset	Offset in degrees along the Kappa axis from the photo's provided angle. Applied before any additional rotation and prior to calculating position/orientation, this property helps correct orientation discrepancies between a camera and the plane on which it is mounted.
Yaw Offset	Offset in degrees along the Yaw axis from the photo's provided angle. Applied before any additional rotation and prior to calculating position/orientation, this property helps correct orientation discrepancies between a camera and the plane on which it is mounted.
Pitch Offset	Offset in degrees along the Pitch axis from the photo's provided angle. Applied before any additional rotation and prior to calculating position/orientation, this property helps correct orientation discrepancies between a camera and the plane on which it is mounted.
Roll Offset	Offset in degrees along the Roll axis from the photo's provided angle. Applied before any additional rotation and prior to calculating position/orientation, this property helps correct orientation discrepancies between a camera and the plane on which it is mounted.
Flip Omega	Flips the Omega axis.
Flip Phi	Flips the Phi axis.
Flip Kappa	Flips the Kappa axis.
Rotate Yaw	Additional rotation along the Yaw axis in degrees, that is performed in YPR order after the photos' OPK or YPR orientation and offsets were applied
Rotate Pitch	Additional rotation along the Pitch axis in degrees, that is performed in YPR order after the photos' OPK or YPR orientation and offsets were applied
Rotate Roll	Additional rotation along the Roll axis in degrees, that is performed in YPR order after the photos' OPK or YPR orientation and offsets were applied

Accuracy

Note: The "Use build settings" option for the properties below sets the accuracy value based on the build settings for the project. For more information on this and other accuracy parameter options, see "[Setting Advanced Aerotriangulation Parameters](#)" in the "Building" chapter for information.

Ignore Orientation	Ignore the orientation value.
Horizontal Accuracy Factor	Weight to assign to the photos' provided XY coordinates in determining the calculated XY values. This ranges from Very High - External AT (Factor = 100) to Unreliable (Factor = 0).
Vertical Accuracy Factor	Weight to assign to the photos' provided altitude in determining the calculated altitude value. This ranges from Very High - External AT (Factor = 100) to Unreliable (Factor = 0).
Focal Length Accuracy	A percentage that indicates the allowable deviation from the provided focal length value when calculating the AT. This represents the accuracy of the provided focal length. This ranges from Use Input Value to Unreliable.
Principal Point Accuracy	A percentage that indicates the allowable deviation from the provided principal point when calculating the AT. This represents the accuracy of the provided principal point. This ranges from Use Input Value to Unreliable.
Radial Accuracy	Radial accuracy. It is generally recommended to select the "Standard" option. If your data was captured with a professional camera that automatically applied a radial distortion correction (i.e., the original distortion value is extremely accurate), select "Use input value".
Tangential Accuracy	Tangential accuracy. It is generally recommended to select the "Standard" option. If your data was captured with a professional camera that automatically applied a tangential distortion correction (i.e., the original distortion value is extremely accurate), select "Use input value".
Calculate Camera Parameters	Method of calculating the camera's intrinsic parameters. - Use Build Settings – Use the method selected in the "Calculate Camera Parameters" AT build parameter. - Entire collection – Calculate the AT so that the camera intrinsic properties are the same for all photos in the collection. - Per photo – Calculate the camera intrinsics individually for each photo in the collection. Note: Parameters should be calculated per photo when not all of a collection's photos were taken by the same physical camera with identical focal length and principal point.

Photo Collections List

In the Collections list, you can view collection properties. These properties can be edited in the [Collection Property Sheet](#). See "[About the Item List](#)" in the "Using the Project Panel and Item List" chapter for information on working with the Collections list.

Collections													
Photos Collections													
0 out of 5 selected													
Enter text to search...													
Selection	Visible	Exclude	Name	Number of Photos	Sensor Width	Pixel Size	Focal Length	Focal Length 35	Width	Height	Principal Point X	Principal Point Y	Altitude Offset
☒	☒	☐	1down	1017	36.064	0.0049	51.77		7360	4912	0	0	
☐	☒	☐	2back	1069	36.064	0.0049	84.985		7360	4912	0	0	
☐	☒	☐	3left	1028	36.064	0.0049	84.911		7360	4912	0	0	
☐	☒	☐	4fore	1016	36.064	0.0049	84.937		7360	4912	0	0	
☐	☒	☐	5right	1081	36.064	0.0049	84.894		7360	4912	0	0	

Collections List

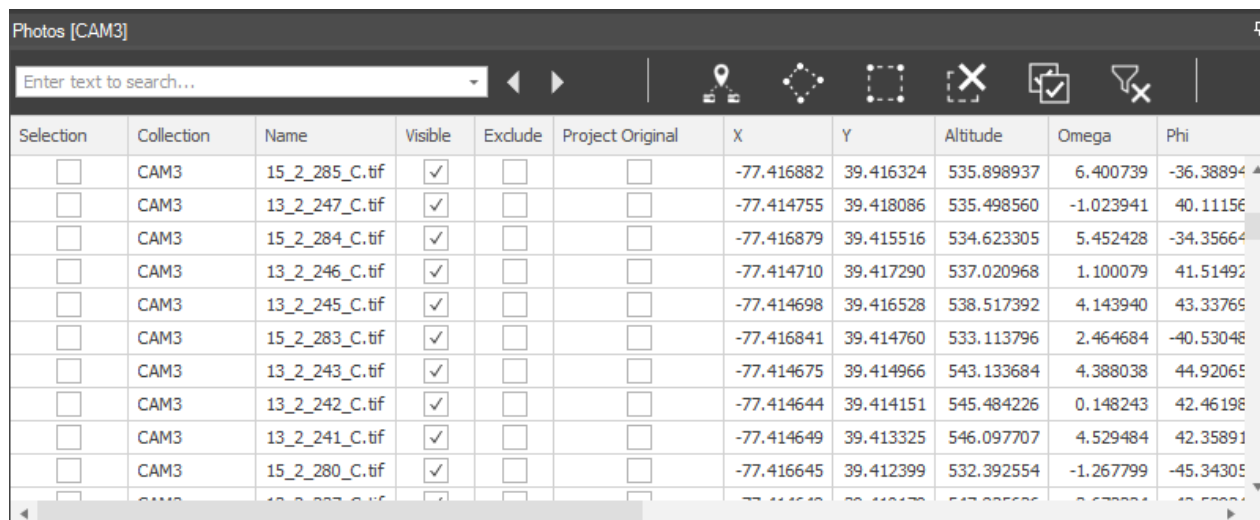
To open the Collections list:

- In the **Project Panel**, select the **Photos** group. The Photos list is displayed showing two tabs. Click the **Collections** tab. The Collections list lists all the collections in the project, with the following information/options:

Property	Description
Selection	Select the collection on which to perform an action.
Visible	Select the check box to show the collection in the 3D Window.
Exclude	Select the check box to exclude the collection from the project.
Name	Name of the photo collection.
Number of Photos	Number of photos in the collection.
Sensor Width	Size of the camera sensor in millimeters.
Pixel Size	Size of a physical pixel on the camera sensor in millimeters.
Focal Length	Focal length in millimeters.
Focal Length 35	Equivalent focal length in millimeters for a 35 mm film camera.
Width	Width of photos in pixels.
Height	Height of photos in pixels.
Principal Point X	Principal point X value in pixels or mm according to principal point reference.
Principal Point Y	Principal point Y value in pixels or mm according to principal point reference.
Altitude Offset	Offset from the current altitude value.
Omega Offset	Offset along the Omega axis in degrees.
Phi Offset	Offset along the Phi axis in degrees.
Kappa Offset	Offset along the Kappa axis in degrees.
Horizontal Accuracy	Accuracy of the photo's provided X and Y coordinates.
Vertical Accuracy	Accuracy of the photo's provided altitude.
Focal Length Accuracy	Accuracy of the camera's inputted focal length.

Photos List

In the Photos list, you can view and set photo properties, as well as set photo visibility and inclusion in the project. When calculated camera positions are displayed, you can also view calculated property information for a selected photo. See ["About the Item List"](#) in the "Using the Project Panel and Item List" chapter for information on working with the Photos list.



Selection	Collection	Name	Visible	Exclude	Project Original	X	Y	Altitude	Omega	Phi
☐	CAM3	15_2_285_C.tif	☒	☐	☐	-77.416882	39.416324	535.898937	6.400739	-36.38894
☐	CAM3	13_2_247_C.tif	☒	☐	☐	-77.414755	39.418086	535.498560	-1.023941	40.11156
☐	CAM3	15_2_284_C.tif	☒	☐	☐	-77.416879	39.415516	534.623305	5.452428	-34.35664
☐	CAM3	13_2_246_C.tif	☒	☐	☐	-77.414710	39.417290	537.020968	1.100079	41.51492
☐	CAM3	13_2_245_C.tif	☒	☐	☐	-77.414698	39.416528	538.517392	4.143940	43.33769
☐	CAM3	15_2_283_C.tif	☒	☐	☐	-77.416841	39.414760	533.113796	2.464684	-40.53048
☐	CAM3	13_2_243_C.tif	☒	☐	☐	-77.414675	39.414966	543.133684	4.388038	44.92069
☐	CAM3	13_2_242_C.tif	☒	☐	☐	-77.414644	39.414151	545.484226	0.148243	42.46198
☐	CAM3	13_2_241_C.tif	☒	☐	☐	-77.414649	39.413325	546.097707	4.529484	42.35891
☐	CAM3	15_2_280_C.tif	☒	☐	☐	-77.416645	39.412399	532.392554	-1.267799	-45.34305
☐	CAM3	13_2_237_C.tif	☒	☐	☐	-77.414640	39.412370	543.025500	0.073304	40.53004

Photo List

To open the Photos list:

- In the **Project Panel**, select the **Photos** group. The Photos list is displayed showing two tabs. Click the **Photos** tab to view a list of all the photo files in the project, with the following information/options:

Property	Description
Selection	Select the photos on which to perform an action, e.g., create tie point, show/hide camera symbol, exclude, delete.
Collection	Name of the photo collection (Read-only).
Name	Name of the photo.
Visible	Select the check box to show the photo in the 3D Window.
Exclude	Select the check box to exclude the photo from the project. This option is not available in Review mode.
Project Original	Select the check box to project the photo on the terrain in its provided camera position. A photo can be projected on the terrain to see if it fits the terrain imagery or to better understand the position of the photo. Only photos that have positioning (XY coordinates and altitude) and orientation (Omega, Phi, and Kappa) information can be projected. Note: Projecting multiple photos simultaneously can take several minutes.
X	East-west coordinate of the camera. (Required)

Y	North-south coordinate of the camera. (Required)
Altitude	Altitude of the camera in meters. (Required)
Omega	Omega Euler angle in degrees.
Phi	Phi Euler angle in degrees.
Kappa	Kappa Euler angle in degrees.
Yaw	Direction angle of the camera along the vertical axis relative to north.
Pitch	Tilt angle of the camera along its lateral axis relative to the horizon.
Roll	Roll angle of the camera along its longitudinal (front-to-back) axis.
Path	Full path of the photo.
K1	Defines the 1 st radial distortion coefficient.
K2	Defines the 2 nd radial distortion coefficient.
K3	Defines the 3 rd radial distortion coefficient.
P1	Defines the 1 st tangential distortion coefficient.
P2	Defines the 2 nd tangential distortion coefficient.
Calculated Cameral Position Information (Additional columns)	
Note: If you want to view calculated information for a selected photo, click the Calculated button on the Tile tab.	
Project Calculated	Select the check box to project the photo on the terrain in its calculated camera position.
Median Error	Median of the triangulation errors of all the features in the photo. The triangulation error is the distance, in pixels, for each of the features in the photo, between the automatically identified feature and its projected triangulation position.
X Calculated	Calculated east-west coordinate of the camera.
Y Calculated	Calculated north-south coordinate of the camera.
Altitude Calculated	Calculated altitude of the camera in meters.
Omega Calculated	Calculated Omega Euler angle in degrees.
Phi Calculated	Calculated Phi Euler angle in degrees.
Kappa Calculated	Calculated Kappa Euler angle in degrees.
K1 Calculated	Calculated 1 st radial distortion coefficient.
K2 Calculated	Calculated 2 nd radial distortion coefficient.
K3 Calculated	Calculated 3 rd radial distortion coefficient.
P1 Calculated	Calculated 1 st tangential distortion coefficient.
P2 Calculated	Calculated 2 nd tangential distortion coefficient.

Adding Photos to a Collection

To add a photo to a collection:

1. In the **Collections** list, select the required collection and click  **Load Photo File**. The Load Photo dialog box is displayed.
2. Browse to the required photo, and click **Open**.

CHAPTER 11 LIDAR MANAGEMENT

About LiDAR Management

LiDAR (light imaging, detection and ranging) is a remote sensing method that uses pulsed laser light to measure terrain and objects. This data together with other information recorded by the sensor system is used to generate a densely spaced point cloud of highly accurate geo-referenced points.

LiDAR data can be loaded into a PhotoMesh project and integrated with project photos to refine and enhance the 3D model, producing cleaner results and preventing artifacts, while increasing the level of accuracy of the complete PhotoMesh process. The benefits of incorporating LiDAR data include:

- Enhanced mesh precision: LiDAR data has a lower noise-to-signal ratio compared to image-based methods, resulting in smoother and more defined geometries such as roads and building facades.
- Improved homogeneous area modeling: In texture-sparse areas, like unmarked roads or uniform vegetation, LiDAR enhances model fidelity.
- Access to challenging angles: LiDAR's capability to measure with a single laser pulse allows for accurate data collection in tight urban spaces, where it is challenging to capture two images with enough angular disparity for image correlation.
- Enhanced street level data: By capturing data beneath vegetation and where trees get close to buildings, LiDAR ensures a more comprehensive street-level view.
- Improved detail of complex structures: LiDAR improves the fine details of complex objects, including scaffolding, tower tops, drawbridges, and cranes.
- Geographic accuracy: LiDAR offers reliable geographic control, to achieve a high level of absolute accuracy in the final 3D mesh and additional geospatial outputs.

The LiDAR data's proper alignment with the project's photos can be verified and corrected using ground control points and through projection of the photos on the LiDARs. See ["Projecting a Photo on the Terrain"](#) in the "Preparing the Project" chapter and ["Creating Ground Control Points from Identifiable Locations on the Terrain"](#) in the "Control Points" chapter for more information.

This chapter deals with basic LiDAR management tasks:

- [Loading of LiDAR data](#)
- [Modifying LiDAR properties](#)
- [Associating a LiDAR point cloud with its trajectory data](#)
- [Showing a LiDAR point cloud on the terrain](#)
- [Navigating to a LiDAR](#)
- [Excluding/deleting LiDAR files from the project](#)

Loading LiDAR Data

LiDAR data can be loaded into a PhotoMesh project and integrated with project photos to supplement project data and increase model accuracy. PhotoMesh supports LiDAR data in .las, .laz or .e57 formats. To improve the reconstruction of the model, LiDAR data can be loaded together with a trajectory file that provides information about the scanner's position while capturing the point cloud. Some .e57 files include internal trajectory information and therefore do not require an external one. LiDAR are converted to CPT, PhotoMesh's internal format, during the build's data preparation step or by Showing Point Cloud. **See** ["About Building"](#) in the "Building" chapter and ["Showing a LiDAR Point Cloud on the Terrain"](#) in this chapter for more information.

If LiDAR files are loaded after aerotriangulation is performed and reconstruction has already started, all reconstruction tiles that intersect with the new LiDAR's bounding box are restored to the point cloud step. This ensures that all necessary processing is performed when the Build Start From step is set to Auto. **See** ["Setting Build Steps"](#) in the "Building" chapter for more information.

To load LiDAR files from your local disk:

1. On the **Home** tab, in the **Add** group, click the arrow next to **LiDAR**, and select one of the following:
 - **Load files** – To add selected files.
 - **Load folder** – To add an entire folder. All supported LiDAR files in the folder are loaded.
2. Browse to the required folder/file(s) and click **Open**.
3. Set the LiDAR's coordinate system including its vertical datum.

Note: If incorrect coordinate system information is entered, the *.CPT created for the LiDAR should be manually deleted from the project's cpts directory: .\[PROJECT_NAME]\cpts\
4. Add a trajectory file and modify LiDAR properties, as required, in the **LiDARs** list. **See** ["Modifying LiDAR Properties"](#) and ["Associating a LiDAR Point Cloud with its Trajectory Data"](#) in this chapter for information.

Modifying LiDAR Properties

LiDAR data can be viewed and modified in the LiDARs list and point cloud properties dialog. This dialog is accessed from a right-click menu as described below.

To display and modify LiDAR point cloud properties:

1. In the **Project Panel** select the **LiDARs** group, and then in the **LiDARs** list, select the required LiDAR.
2. For basic properties such as path and coordinate system, in the **LiDARs** list, select the check boxes of the required LiDARs, and modify their properties as required. **See** ["LiDARs List"](#) in this chapter for more information.

Note: You can also search the list for the required LiDAR. **See** ["Selecting Items Using the Item List"](#) in the "Basic Concepts" chapter for information.
3. For point styling properties and other properties relating to how the point cloud is displayed, right-click the required LiDAR, and select **Style Point Cloud**. The Point Cloud Properties dialog is displayed. **See** ["LiDAR Point Cloud Properties Dialog"](#) in this chapter for more information.

Note: A LiDAR must be converted to CPT, PhotoMesh's internal format before it can be styled. You can convert the LiDAR by performing the build's data preparation step or by **Showing Point Cloud**. **See** ["About Building"](#) in the "Building" chapter and ["Showing a LiDAR Point Cloud on the Terrain"](#) in this chapter for more information.

LiDAR Point Cloud Properties Dialog

Property	Description
Style	
Point Size	Size, in pixels, of each of the LiDAR point cloud's points. If you want to base the point size on attribute fields, click the Field by Attribute button . In the Field by Attribute dialog that opens, select the attribute on which to base point size. Then for each attribute value, type the value and the corresponding point size.
Point Color	Color of the LiDAR point cloud points. This can be set to either of the following: - A set color value for all the points. To select a color, click Edit and select a color. - Based on a specific attribute field - To base each point of the point cloud's color on a specific attribute field's value, click the Field by Attribute button . In the Field by Attribute dialog that opens, select the attribute on which to base point color. Then for each attribute value, type the value and the corresponding point color, e.g., color each point of the point cloud according to the CPT's RGB attribute values or according to the CPT's intensity attribute values.
Point Visibility	Determines if the LiDAR point cloud's points are visible. To set the visibility of points based on their attribute values, click the Field by Attribute button . In the Field by Attribute dialog that opens, select the attribute on which to base visibility. Then for each attribute value, type the value and the corresponding visibility setting.
Intensity Blend	Determines how much the LiDAR point cloud's intensity value is factored in colorizing the point cloud, where 0% colorizes the point cloud model by point color only, and 100% colorizes by intensity only.
Oversampling	The LiDAR point cloud's Level of Detail (LOD). When set to Normal, PhotoMesh determines the optimal quality to performance balance based on the current view. The "High" and "Best" settings will load more points but are also more resource-intensive.
Flatten Underlying Terrain	Determines if the terrain under the model is leveled to the model's elevation value. This option is enabled only if the Altitude Method property in the point cloud property sheet is set to Absolute.
Hide Underlying Terrain	Hides the terrain texture under the LiDAR point cloud model (the terrain appears as a black image).

LiDARs List

In the LiDARs list, you can view and set the LiDAR's coordinate system, path and trajectory path, as well as set LiDAR visibility and inclusion in the project. See ["About the Item List"](#) in the "Using the Project Panel and Item List" chapter for information on working with the LiDARs list.

Selection	Lidar Group ID	#	Name	Excluded	Box	Point Cloud	Coordinate System	Path	Trajectory Path	Tag
☒	0_5619	1	Full_LiDAR_000001.pcd	☐	☐	☐	...	...	...	...
☒	0_5619	2	Full_LiDAR_000002.pcd	☐	☐	☐	...	...	...	...
☐	0_5619	3	Full_LiDAR_000003.pcd	☐	☐	☐	...	...	...	...
☐	0_5619	4	Full_LiDAR_000004.pcd	☐	☐	☐	...	...	...	...
☐	0_5619	5	Full_LiDAR_000005.pcd	☐	☐	☐	...	...	...	...
☐	0_5619	6	Full_LiDAR_000006.pcd	☐	☐	☐	...	...	...	...
☐	0_5619	7	Full_LiDAR_000007.pcd	☐	☐	☐	...	...	...	...

LiDARs List

To open the LiDARs list:

In the **Project Panel**, select the **LiDARs** group. The LiDARs list includes all the LiDAR files in the project, and the following information:

Property	Description
Selection	Select the check box to select the LiDAR in the 3D Window.
LiDAR Group ID	ID of the LiDAR group.
#	Number of the LiDAR.
Name	Name of LiDAR file.
Excluded	Select the check box to exclude the LiDAR point cloud from the project.
Box	Select the check box to show the LiDAR's bounding box.
Point Cloud	Select the check box to show a LiDAR point cloud on the terrain.
Coordinate System	Coordinate system of the LiDAR point cloud. Click Select  to set a different coordinate system. See "Coordinate System" in the "Basic Concepts" chapter for more information.
Path	Full path to the LiDAR point cloud.
Trajectory Path	Full path to file that provides information about the scanner's position while capturing the point cloud. See "Associating a LiDAR Point Cloud with its Trajectory Data" in this chapter for more information on trajectory files.
Tag	Free text about the LiDAR

Associating a LiDAR Point Cloud with its Trajectory Data

Trajectory data provides information about the scanner's position while capturing the LiDAR point cloud. The scanner positions should be ordered chronologically in the file.

Note: Some .e57 files include internal trajectory information and therefore do not require an external one.

Trajectory data can be imported in a tab delimited .trjt (text format) file that includes the following lines:

- First line: Define the LiDAR version (Should always be 1.0)

```
#version=1.0
```

- Second line: Define the LiDAR's coordinate system in WKT format.

```
#wkt=.....
```

Note: The WKT information must be in a single line without any line breaks. You can use the Coordinate System Generator to generate a WKT string defining the required coordinate system. See ["Coordinate System"](#) in the "Basic Concepts" chapter for more information.

- Third line (optional): Indicates a time offset that should be added to the GPS time field in each record (default is 0).

```
#timebase=
```

- For each known scanner position, add a separate line with the scanner's position (X, Y, Altitude) and the time the position was recorded (GPS Time) in tab delimited format. The scanner positions should be ordered chronologically.

```
[x coordinate] <TAB> [y coordinate] <TAB> [altitude] <TAB> [GPS time]
```

Note: The GPS Time is the double floating point time tag value at which the scanner's position was recorded. This data must be in the same units and offset as the time data in the LiDAR file.

Example:

```
#version=1.0
#wkt=GEOGCS["WGS 84",DATUM["WGS_1984",SPHEROID["WGS
84",6378137,298.257223563,AUTHORITY["EPSG","7030"]],AUTHORITY["EPSG","6326
"]],PRIMEM["Greenwich",0,AUTHORITY["EPSG","8901"]],UNIT["degree",0.0174532
925199433,AUTHORITY["EPSG","9122"]],AUTHORITY["EPSG","4326"]]
#timebase=0

9.63390819201651 <TAB> 47.4233597051724 <TAB> 401.463806152344 <TAB>
583.161876711994
9.63390818337074 <TAB> 47.4233597019402 <TAB> 401.463793239556 <TAB>
583.161876711994
9.63390817652904 <TAB> 47.4233597069347 <TAB> 401.463349871528 <TAB>
583.161876711994
9.63390818337074 <TAB> 47.4233597019402 <TAB> 401.463407984571 <TAB>
583.160397257656
```

To associate a LiDAR file with its trajectory file:

- In the **Project Panel** select the **LiDARs** group, and then in the **LiDARs** list, select the required LiDAR and type the trajectory file's path in the **Trajectory Path** field.

Showing a LiDAR Point Cloud on the Terrain

A LiDAR can be shown on the terrain to see if it fits with the project's photos, other LiDAR, ground control points, etc.

Note: The LiDAR data's proper alignment with the project's photos can be verified and corrected using ground control points and through projection of the photos on the LiDARs. See ["Projecting a Photo on the Terrain"](#) in the "Preparing the Project" chapter and ["Creating Ground Control Points from Identifiable Locations on the Terrain"](#) in the "Control Points" chapter for more information.

To show a LiDAR on the terrain:

1. In the **Project Panel** select the **LiDARs** group, and then in the **LiDARs** list, select the **Point Cloud** check box for each of the LiDARs you want to display. The LiDAR (contextual) tab is displayed.

Note: If a particular LiDAR wasn't converted previously to CPT, PhotoMesh's internal format, it will first be converted and then shown on the terrain.

Note: If you want to show more than one LiDAR point cloud on the terrain, it is generally recommended to build the project, performing only the Data Preparation step since the conversion to CPT performed via View > Point Cloud may take a long time.
2. If you want to show a LiDAR's trajectory data, select the LiDAR and then on the **LiDAR** tab, in the **View** group, toggle on **Trajectory**.
3. If you want to hide a LiDAR's bounding box, select the LiDAR and then on the **LiDAR** tab, in the **View** group, toggle off **Bounding Box**.

Navigating to a LiDAR

To fly or jump to a LiDAR:

1. In the **Project Panel** select the **LiDARs** group, and then in the **LiDARs** list, select the required LiDAR.
2. On the **LiDAR** tab, in the **Navigation** group, select **Fly to/ Jump to**.

Excluding a LiDAR File

To exclude a LiDAR file from the project:

1. In the **Project Panel** select the **LiDARs** group, and then in the **LiDARs** list, select the required LiDAR.
2. On the **LiDAR** tab, in the **Selection** group, click **Exclude**. You can also exclude a LiDAR by selecting its **Excluded** check box in the **LiDARs** list.

Deleting a LiDAR File

To delete a LiDAR from the project:

1. In the **Project Panel** select the **LiDARs** group, and then in the **LiDARs** list, select the required LiDAR.
2. On the **LiDAR** tab, in the **Selection** group, click **Delete**. The LiDAR is deleted from the LiDARs list.

CHAPTER 12 PREPARING THE PROJECT

About Preparing the Project

After photos have been loaded, do any of the following to prepare for the build:

Set display of photos

- [Select the photos for which symbols are displayed](#)
- [Select the size and color coding of the camera symbols that display for each photo](#)

Study photos and modify photo properties

- [Navigate to a photo](#)
- [View photos in the photo viewer](#)
- [Project photos on the terrain to see if they fit the terrain imagery or to better understand their position](#)
- [Show a color-coded map analyzing photo directional coverage in every point of the AT area](#)
- [Show a color-coded map displaying the degree of overlap, i.e., for each point within the specified AT area, the number of photographs in which the point is visible.](#)
- [Show a color-coded connection map that indicates the level of photo interconnection](#)
- [Modify photo and photo collection properties](#) (in "Photo Management" chapter)
- [Use ground control points to position the model by associating specific locations \(x, y, altitude\) on the terrain with corresponding locations \(in pixels\) in three or more photos](#) (in "Control Points" chapter)
- [Use tie points to establishing a correspondence between project photos and improve aerotriangulation results through identification of the same physical point in three or more photos](#) (in "Control Points" chapter)

Generate a flat surface with correct texturing over a user-defined water body

- [Create a water body polygon](#)
- [Import a water body polygon](#)

Set the aerotriangulation and reconstruction areas

- [Set the AT area](#)
- [Set the reconstruction area](#)

Displaying and Studying Photos

Do any of the following:

- [Display camera symbols](#)
- [Navigate to a photo](#)
- [View photos in the photo viewer](#)
- [Project photos on the terrain](#)

- [Show the directions map](#)
- [Show the overlap map](#)
- [Show the connection map](#)

Displaying Camera Symbols

Each photo's camera position and orientation (if known) can be displayed on the terrain using camera symbols. The symbol's color also indicates either the photo's collection or its status (in original position, in calculated position, failed to calculate, or excluded by user). You can show all photos or you can filter which photos are displayed:

- Display only the photos individually selected in the Photos list or the collections selected in the Collections list.
- Display only photos that cover a selected point on the terrain.
- Display only photos that were not excluded from the project.

To select specific photos for viewing on the terrain, do either of the following:

- In the **Project Panel**, select the **Photos** group, and then in the **Photos** list, select the **Visible** check box of the required photos.
- In the **Project Panel**, select the **Photos** group, and then in the **Collections** list, select the **Visible** check box of the required collections. All of the selected collections' photos will be displayed.

To select photos for viewing that cover a selected point on the terrain, do the following:

1. Do either of the following:
 - If you want to display all photos that intersect with a selected point based on original photo positions, in the **Project Panel**, select the **photos** group. Then in the **Photos** list, click **Show Original Photos in Selected Point** .
 - If you want to display all photos that intersect with a selected point based on aerotriangulation calculations, in the **Project Panel**, select the group. Then in the **Photos** list, click **Show Calculated Photos in Selected Point** .
2. In the **3D Window**, click the point of interest. All intersecting photos are listed in the Photos list and the first photo in the list automatically opens in the Photo Viewer. See "[Viewing Photos in the Photo Viewer](#)" in the "Preparing the Project" chapter.
3. To display all photos, in the **Photos** list, click **Show All Photos** .

To display only photos that were not excluded from the project:

1. On the **View** tab, in the **Photos** group, select **Do not show excluded**. All excluded photos and photos outside the currently defined AT area are hidden. See "[Excluding a Photo](#)" in the "Photo Management" chapter for information about excluding photos.
2. To (re)display excluded photos, on the **View** tab, in the **Photos** group, select **Show excluded**.

Selecting Camera Symbol Size and Color Coding

To select a camera symbol size:

- On the **View** tab, in the **Photos** group, in the **Symbol size** dropdown, select a size for the camera symbols that display for each photo.

To select a camera color coding option:

- On the **View** tab, in the **Photos** group, in the **Photo color** dropdown, select either of the following:
 - **Color by status** – Camera symbols are color coded based on photo status (in original position, in calculated position, failed to calculate, or excluded by user), according to the colors defined in Options' settings. See "[Setting PhotoMesh Options](#)" in the "Basic Concepts" chapter for more information.
 - **Color by collection** – Camera symbols are color coded according to collection, with a random color assigned to each collection's camera symbol.

Navigating to a Photo

To fly or jump to a photo:

1. Select the required photo. See "[Selecting Photos](#)" in this chapter for information.
2. On the **Photo** tab, in the **Navigation** group, select **Fly to/ Jump to**.

Note: You can also fly to the selected photo by double-clicking it in the photo list.

Viewing Photos in the Photo Viewer

The photo viewer allows you to get a better view of a particular photo as well as highlights and provides a graphic display of the strength of each of the photo features, i.e., how many other photos contain each of the features that were automatically identified in this photo. You can display either the matched features identified in the preliminary match process of the AT step or the matched sparse point cloud features, whose match quality was validated and position calculated during the AT's bundle adjustment process. This information is only available after the AT build step is complete.

In addition to a single docked viewer, multiple instances of a floating viewer can be opened simultaneously.

To view a photo in the Photo Viewer:

1. Select the required photo. See "[Selecting Photos](#)" in this chapter for information. Then on the ribbon's **Photo** tab, in the **View** group, click **Photo Viewer**. The Photo Viewer opens with the selected photo displayed. If a different photo is selected, then it replaces the previously displayed photo in the viewer.

Note: To open a photo in an additional floating photo viewer, without closing or replacing the photos in any currently open viewers, in the **Project Panel**, select the **Photos** group, and then in the **Photos** list, right-click the required photo, and select **Open in a Floating Photo Viewer**. A floating photo viewer opens with the selected photo displayed in a photo viewing window.

2. Use the **Photo Viewer** ribbon commands to zoom in  and out .
3. If you want to highlight matched features in the photo and color-code them according to the number of times they were found in other photos, do the following:
 - a. Select which of the following matched features to show:
 - **Show Features from Match**  - Display all the matched features identified in the preliminary match process of the AT step.
 - **Show Features from Sparse**  - Display all matched sparse point cloud features, whose match quality was validated and position calculated during the AT's bundle adjustment process.

- b. A photos panel is added to the Photo Viewer with all the photos which have features in common with the selected photo based on the AT build step. In the photo viewing window, the photo's features are color coded according to the number of other photos in which they appear:
- Gray – Feature wasn't found in any other photo (no match)
 - Yellow – Feature was found in one other photo
 - Green – Feature was found in two or more photos



Color-Coded Features

- c. Select a viewing format for the **photos** panel:
-  – View as thumbnails
 -  – View in table format with information about name, collection, and number of matches for each photo.
- d. In the photos panel, select which features to display in the photo viewing window:
- All features in the selected photo (including "gray" features that weren't found in other photos)
 - All features in the selected photo that were found in at least one more photo
 - Only the matched features between the photo selected in the photo panel and the main photo. Double-click a photo in the Photos panel to select this photo as the main photo in the viewer.

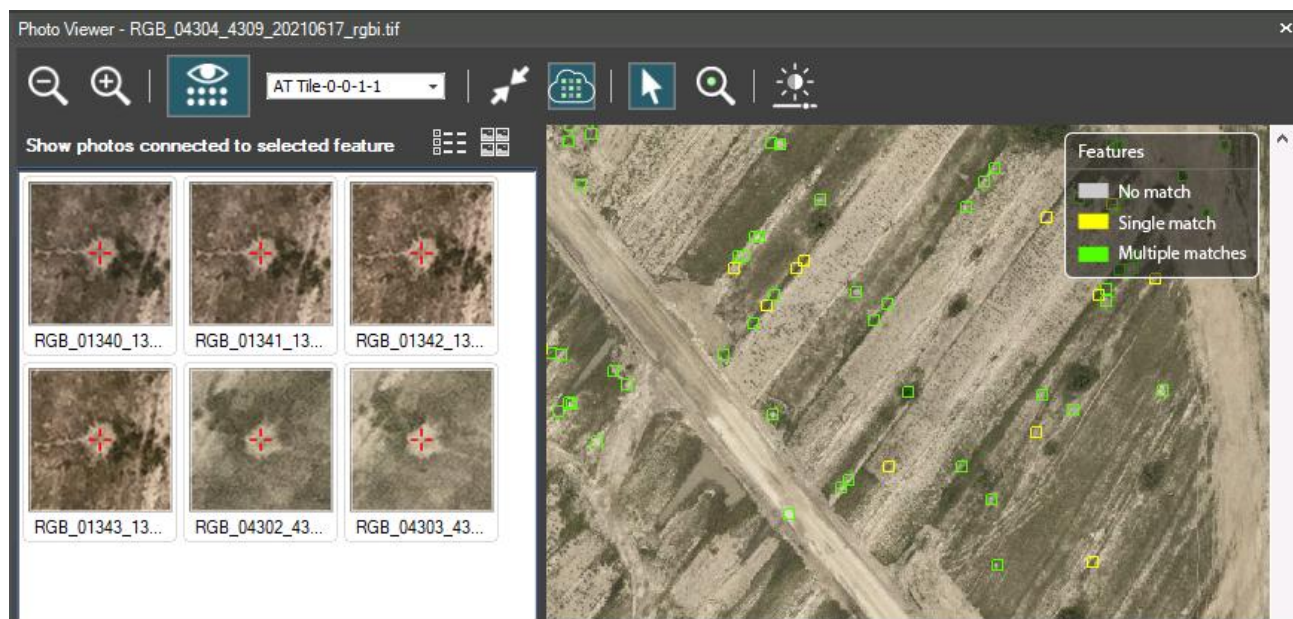
Note: This information is only available in Review mode after aerotriangulation was completed. See "[Setting Build Steps, Parameters, and Outputs](#)" in the "Building" chapter. Build steps after aerotriangulation do not need to be completed to view this information.

Show features connected to

	Name	Collection	Matches
▶	All with matches	-	-
	All	-	-
	20210717_RGBA-1...	Collection8	99
	RGB_01339_1344...	Collection7	512
	RGB_01340_1345...	Collection7	113
	RGB_01341_1346...	Collection7	315
	RGB_01342_1347...	Collection7	803
	RGB_01343_1348...	Collection7	168
	RGB_01344_1349...	Collection7	478
	RGB_04301_4306...	Collection7	793
	RGB_04302_4307...	Collection7	1376
	RGB_04303_4308...	Collection7	2059

Show Features

- e. To darken the image so that the feature markers are more noticeable, click **Dim mode** .
- f. If you want to identify all photos that contain a specific feature, click **Select** , and select a feature in the photo viewing window. The photo panel displays all photos that include this feature. In thumbnail mode, the thumbnails are zoomed in to the point of interest and display a red plus  sign where the feature is located. Click **Select**  to toggle the selection mode off.



Show Photos Connected to Selected Feature

Projecting a Photo on the Terrain

A photo can be projected on the terrain (including LiDAR) to see if it fits the terrain imagery or to better understand the position of the photo. The projection of the photo can be based either on its original position or its calculated position. Only photos that have positioning (XY coordinates and altitude) and orientation (Omega, Phi, and Kappa) information can be projected based on original positions.

To project a photo on the terrain in its original camera position:

1. In the **Project Panel**, select the **Photos** group, and then in the **Photos** list, select the required photo. The Photo (contextual) ribbon tab is displayed.
2. On the **Photo** tab, in the **View** group, select **Project Original**. The photo is shown on the terrain. You can also project the photo by selecting the photo's **Project Original** check box in the **photo** list.

To project photos on the terrain in the camera position calculated through aerotriangulation:

1. To project a specific photo, do the following:
 - a. Select the required tile. See "[Selecting Tiles](#)" in the "Reviewing Your Build" chapter. The Tile tab is displayed.
 - b. On the **Tile** tab, in the **Calculated Photos** group, toggle on **Calculated**.
2. To project all photos in a selected point, do the following:
 - a. In the **Project Panel**, select the **Photos** group.
 - b. In the **Photos** list, click **Show Calculated Photos in Selected Point**. Then in the **3D Window**, click the point of interest. All photos that cover the selected point are displayed.
3. Select the required photo(s). See "[Selecting Photos](#)" in the "Photo Management" chapter for information.
4. On the **Photo** tab, in the **View** group, select **Project Calculated**. The photo is shown on the terrain. You can also project the photo by selecting the photo's **Project Calculated** check box in the **photo** list.

Clearing Review Data from the Terrain

To clear review data:

- On the **Home** tab, in the **Review** group, click **Clean Review Data**. This clears any photo projections from the terrain.

Showing the Directions Map

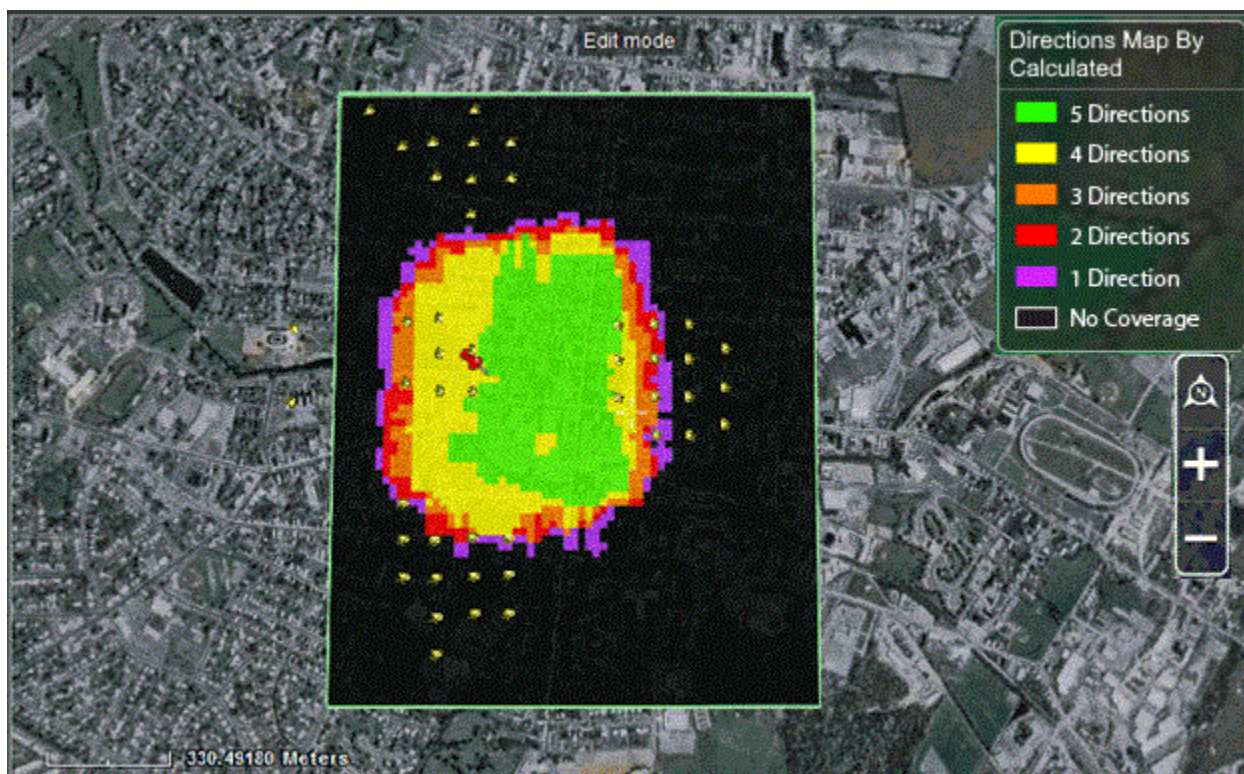
The directions map displays the directional coverage in every point of the defined AT area. Each point in the defined AT area is color coded based on the number of directions from which it was captured in the photos whose frustum intersects with the point. This enables you to identify which areas of the project have coverage from sufficient directions, and which require more. The AT area can be defined to include all the project's photos or restricted to only the photos within a specific area on the terrain. See "[Setting the AT \(Aerotriangulation\) Area](#)" in this chapter for more information.

When in Review mode, the map displays coverage based on the calculated positions of the project's photos. Otherwise, the map displays coverage based on the original positions.

Note: The directions map may take some time to display the first time it is shown or after making changes to the project as PhotoMesh (re)calculates the photo coverage.

To show a directions map:

- On the **Home** tab, in the **Photos** group, click the arrow next to **Coverage Map**. and select **Directions Map**.



Directions Map and Legend

In the directions map, the AT area is color coded based on directional coverage.

- **Green** – Best coverage from all 5 directions: nadir and 4 oblique.
- **Yellow** – Sufficient coverage of 4 directions.
- **Orange** – Three coverage directions, the minimum required.
- **Red** – Two coverage directions.
- **Purple** – One coverage direction.
- **Black** – No coverage.

Showing the Overlap Map

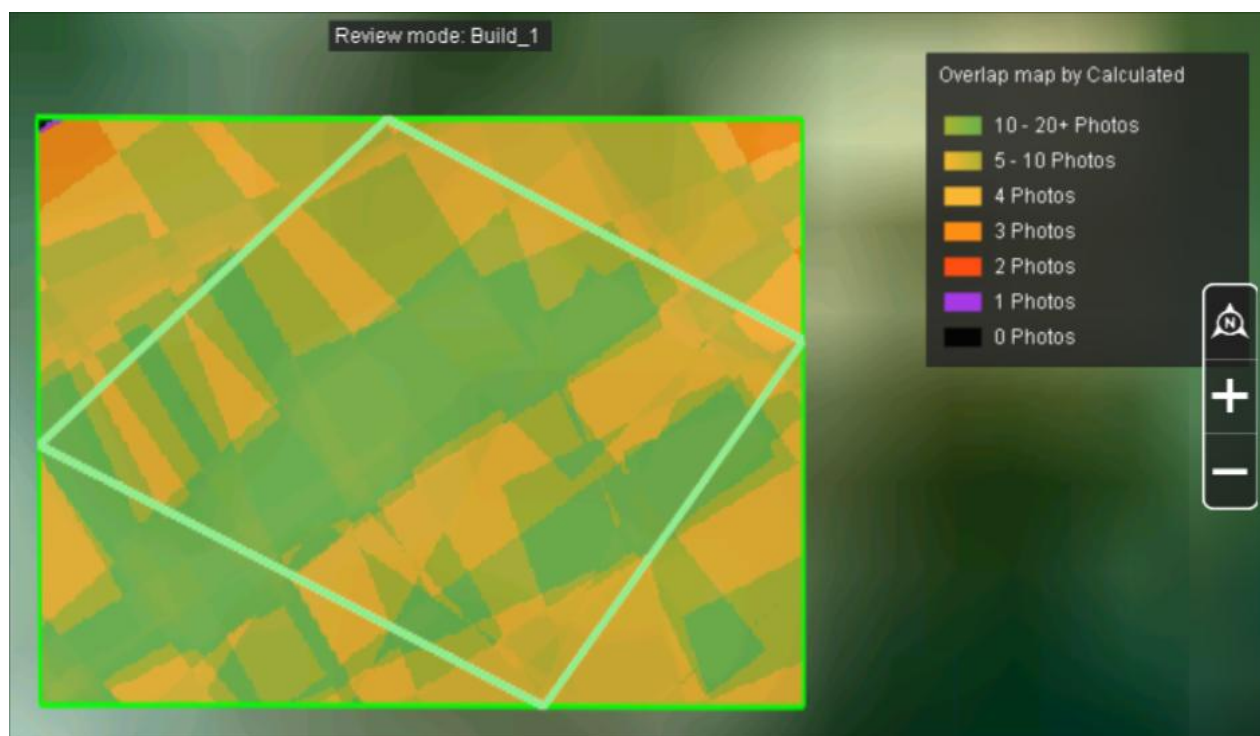
The overlap map represents the degree of overlap by indicating, for each point within the specified AT area, the number of photographs in which the point is visible. Each point in the defined AT area is color coded based on the number of photos whose frustum intersects with the point. This enables you to identify which areas of the project have sufficient overlap, and which require more. Generally, each point in the area of interest should be visible by at least, three cameras. In mobile photography, it is recommended that most points should appear in at least five photos. This provides PhotoMesh with the data to fill in holes in planes and facades that are partially obstructed by objects (such as trees and street furniture) which are closer to the camera. The AT area can be defined to include all the project's photos or restricted to only the photos within a specific area on the terrain. See "[Setting the AT \(Aerotriangulation\) Area](#)" in this chapter for more information.

When in Review mode, the map displays overlap based on the calculated positions of the project's photos. Otherwise, the map displays overlap based on the original positions.

Note: The overlap map may take some time to display the first time it is shown or after making changes to the project as PhotoMesh (re)calculates the photo coverage.

To show an overlap map:

- On the **Home** tab, in the **Photos** group, click the arrow next to **Coverage Map**. and select **Overlap Map**.



Overlap Map and Legend

In the overlap map, the AT area is color coded based on number of overlapping photos.

- **Green** – 10-20+ photos.
- **Yellow** – 5-10 photos.
- **Light Orange** – Four photos.
- **Dark Orange** – Three photos.
- **Red** – Two photos.
- **Purple** – One photo.
- **Black** – Zero photos.

Showing the Connection Map

The connection map is a color-coded map that enables you to evaluate the quality of the calculated AT (this map is only available in Review mode). In the connection map, each point in the defined AT area is colored based on the level of interconnection of the photos that intersect with this point. A group of photos is considered interconnected if each photo in the group was directly or indirectly matched with every other photo. Even photos that do not precisely intersect with the point in the defined AT area can be used to create an indirect connection, as long as they come close to intersecting with the point (i.e., if photos A and D precisely intersect with the point in the AT area, and photos B and C nearly intersect with the point, photo A can be connected to photo D via photos B and C, where A intersects with B which intersects with C, which intersects with D).

- Green – All of the photos that intersect with this point are interconnected.
- Red – 50% of the photos that intersect with this point are interconnected

Each point in the defined AT area is colored on the gradient between green and red, based on the percentage of photos that intersect with the point.

If the connection map indicates that insufficient photo matches were generated for certain points in the AT area, tie points can be used to manually create matches for these points. See ["About Control Points"](#) in the "Control Points" chapter for information.

You can also display the number of matches for each photo by selecting Number of Matches as the color coding option for the photos. See ["Displaying Camera Positions for Tiles"](#) in the "Reviewing Your Build" chapter for more information.

To show a connection map:

- On the **Home** tab, in the **Photos** group, click the arrow next to **Coverage Map**. and select **Connection Map from Calculated**.

Setting the AT (Aerotriangulation) Area

Setting the AT area defines the area of the project on which PhotoMesh should perform aerotriangulation. Aerotriangulation is the process for determining the correct position and orientation of each of the project photos, based on inputted photo information, ground control points and tie points. If a project's photos do not have fully trusted camera positioning and orientation information, this process must be performed before generation of the 3D mesh model. PhotoMesh uses the highest-performance photogrammetry, computer vision, and computational geometry algorithms in performing aerotriangulation.

The defined AT area can include all the project's photos or be restricted to only the photos within a specific area on the terrain. This enables you to perform aerotriangulation on a small subset of the project to validate camera parameters and build settings and check the resulting AT and reconstruction tiles before building the full project. The area for the aerotriangulation can be defined by either drawing or importing an AT polygon.

If project photos have no positioning information, you should manually set an initial AT area based on an estimation of the geographical area of the project, which will then be adjusted by PhotoMesh in the aerotriangulation step.

There are two indicators in the 3D Window that photos are outside the current aerotriangulation area:

- When photos are "colored by status" (See ["Selecting Camera Symbol Size and Color"](#) in this chapter), the camera symbols for these photos will be colored in grey.
- When "Do not show excluded" is selected (See ["Displaying Camera Symbols"](#) in this chapter), the camera symbols for these photos are not displayed.

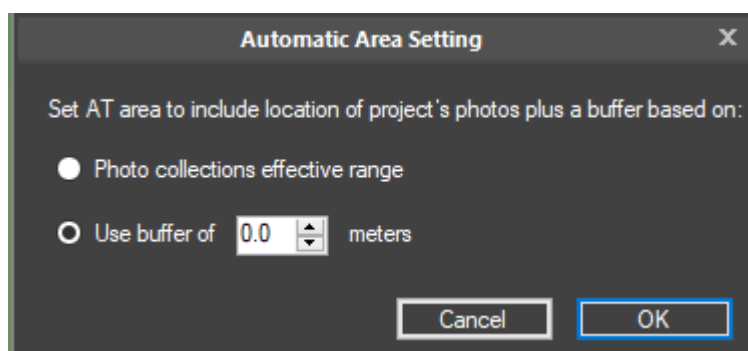
Note: Photos that are outside the currently defined AT area are still considered part of the project and therefore the Exclude check box remains cleared in the photo list.

Setting the AT Area to Include All Photos

The AT area can be calculated automatically based on all the project's photos plus a buffer. This buffer is defined either by the collections' effective range—representing the furthest distance from the camera at which the photo data remains reliable for aerotriangulation—or by a buffer value that you specify.

To set the AT area to include all the project's photos:

1. On the **Home** tab, in the **Area** group, click the arrow under **AT Area** and select **Change Automatic Setting**. The Automatic Area Setting dialog is displayed.



Automatic Area Setting Dialog

2. Select one of the following options:
 - **Photo Collections' Effective Range** - Calculate the AT area based on the positions of all the project's photos plus a buffer that is based on the collections' effective range.
 - **Use Buffer of** - Calculate the AT area based on the positions of all the project's photos plus the buffer that you set.
3. On the **Home** tab, in the **Area** group, click the arrow under **AT Area** and select **Automatic Using Effective Range/Automatic Using Buffer**. The option that is displayed corresponds to the option selected in step 2.

Drawing a New AT Area

To set an AT area within a specific area of interest, do the following:

Note: An AT Area polygon can only be redrawn when Review mode is off. See ["About Reviewing Your Build"](#) in the "Reviewing Your Build" chapter for more information about Review mode.

1. On the **Home** tab, in the **Area** group, click the arrow under **AT Area** and select **Redraw AT Area**.
2. In the **3D Window**, draw an AT Area polygon around the required area by clicking in the required locations. You must place at least three points. Right-click to finish the polygon creation.

Importing an AT Area Polygon

To import an AT area polygon;

1. On the **Home** tab, in the **Area** group, click the arrow under **AT Area**, and select **Import**. The Import Aerotriangulation Area Polygon Layer dialog is displayed.
2. Browse to the shapefile with the required polygon, and click **Open**.

Note: The first polygon found in the file is used as the aerotriangulation area polygon.

Editing an AT Area Polygon

To edit the current AT area:

Note: An AT Area polygon can only be edited when Review mode is off. See ["About Reviewing Your Build"](#) in the "Reviewing Your Build" chapter for more information about Review mode.

1. On the **Home** tab, in the **Area** group, click the arrow under **AT Area** and select **Edit AT Area**.

2. In the **3D Window**, place the cursor over one of the nodes of the AT Area polygon. The node turns blue, and the cursor changes to **Move** mode .
3. Click the node and drag it to the required location.
4. Release the mouse button to set the node at the current location.
5. Red dots on the polygon's line segments indicate where a node can be added. Click to add a node, and then drag it to the required location.
6. Right-click to finish editing the polygon.

Navigating to the AT Area

To fly or jump to the AT area:

- On the **Home** tab, in the **Area** group, click the arrow under **AT Area** and select **Fly to Area/Jump to Area**.

Setting the Reconstruction Area

Setting the reconstruction area defines the area of the project on which PhotoMesh should perform the reconstruction steps of the build process, namely: point cloud creation, model creation, model texturing, and project output. The reconstruction area can be defined before building the project (after identifying which areas of the project have sufficient coverage using the [coverage map](#)) and also after the build is complete if modifications to the build area are necessary. The same area set for the aerotriangulation can be automatically set for the reconstruction area, or a different area can be defined by either drawing or importing a reconstruction polygon.

Setting a Reconstruction Area Equal to the AT Area

To set a reconstruction area that is the same as the AT area:

- On the **Home** tab, in the **Area** group, click the arrow under **Reconstruction Area**, and select **Same as AT Area**.

Drawing a New Reconstruction Area

To draw a reconstruction area polygon:

1. On the **Home** tab, in the **Area** group, click the arrow under **Reconstruction Area**, and select **Redraw Reconstruction Area**.
2. In the **3D Window**, draw a reconstruction area polygon around the required area by clicking in the desired locations. You must place at least three points. Right-click to finish the polygon creation.

Importing a Reconstruction Area Polygon

To import a reconstruction area polygon:

1. On the **Home** tab, in the **Area** group, click the arrow under **Reconstruction Area**, and select **Import**. The Import Reconstruction Area Polygon Layer dialog is displayed.
2. Browse to the shapefile or KML with the required polygon, and click **Open**.

Note: The first polygon found in the file is used as the reconstruction area polygon.

Editing a Reconstruction Area Polygon

To edit a current reconstruction area polygon:

1. On the **Home** tab, in the **Area** group, click the arrow under **Reconstruction Area**, and select **Edit Reconstruction Area**.
2. In the **3D Window**, place the cursor over one of the nodes of the reconstruction area polygon. The node turns blue, and the cursor changes to **Move** mode  .
3. Click the node and drag it to the required location.
4. Release the mouse button to set the node at the current location.
5. Red dots on the polygon's line segments indicate where a node can be added. Click to add a node, and then drag it to the required location.
6. Right-click to finish the polygon editing.

Exporting a Reconstruction Area Polygon

To export a reconstruction area polygon:

1. On the **Home** tab, in the **Area** group, click the arrow under **Reconstruction Area**, and select **Export**. The Export Reconstruction Area Polygon dialog is displayed.
2. Browse to the location you want to save to and click **Save**.

Navigating to the Reconstruction Area

To fly or jump to the reconstruction area:

- On the **Home** tab, in the **Area** group, click the arrow under **Reconstruction Area** and select **Fly to Area/Jump to Area**.

CHAPTER 13 CONTROL POINTS

About Control Points

PhotoMesh provides three types of control points:

- **Ground control points (GCP)** – Improve the accuracy of the model's position by associating real-world X, Y, Z coordinates (generally obtained through surveying or sampling methods) with corresponding locations (in pixels) in photos. In order to enable your control points to be fully utilized in the AT process, it is recommended to add at least 3-5 control points per AT tile. When only two control points are provided, simple vertical adjustment, scaling and rotation are performed. When only one control point is provided, only vertical adjustment is performed.
- **Check points** – Ground control points that are not used to improve the AT calculation, but rather only to sample the AT accuracy in the result. The point is included in the quality report and shows calculated errors for accuracy validation purposes.
- **Tie points** – Establish a correspondence between project photos and improve aerotriangulation results through identification of the same physical point in three or more photos. The XYZ position of the physical point is not necessarily known. Generally, it is recommended to run an initial aerotriangulation to identify photos that either weren't calculated in AT or that have high median errors, as well as areas of insufficient matches between groups of photos.

When a control point is added, a control point symbol is displayed in the 3D Window in the place where it was added. When selected, the control point turns to purple.



Control Point Symbol

Do any of the following:

- Create control points:
 - [Ground control points](#)
 - [Tie points](#)
- [Navigate to a control point](#)
- [Edit a control point](#)
- [Mark a control point in photos](#)
- [Hide a control point symbol](#)
- [Exclude a control point](#)

- [Delete a control point](#)
- [Export control point data to a .gcp \(text format\) file](#)
- [Import control point data in a .gcp \(text format\) file](#)
- [Use control point as a check point to sample the AT accuracy in the result](#)

After creating a control point, the Control Point Editor opens so that you can mark the control point in the relevant photos. [See "Marking and Auto-Marking Control Points in Photos"](#) in this chapter for information. The Control Point Editor opens, by default, docked to the right of the 3D Window, showing the name of the selected control point in its title bar.

Control point property information is viewed and edited from the **Control and Tie Points** list. [See "Editing Control Points"](#) in this chapter for information. When the Control Point Editor is open, selecting a control point in the Control and Tie Points list opens that point in the Editor.

Recommendations for Ground Control Point Placement

The following guidelines are recommended for placement of control points in your AT tiles:

- Distribute control points uniformly across the AT tile.
- Make sure to add control points in the tile areas with the highest and lowest altitudes.
- Begin control point creation by adding control points around the perimeter of the tile.
- Control points should be created on a flat terrain or mesh that has no obstructions close by.

Creating Ground Control Points

Ground control points can be created/added in any of the following ways:

- [Create a control point at a known precise position, generally a surveyed point](#)
- [Create a control point by clicking on an identifiable location on the terrain or LiDAR \(e.g. road intersection or street corner\)](#)
- [Import a .gcp \(text format\) file with control point information](#)

After creating a ground control point, you mark it in the relevant photos using the Control Point Editor. [See "Marking Control Points in Photos"](#) in this chapter for information.

Creating Ground Control Points at Known Coordinates

To create a ground control point at known coordinates:

1. On the **Home** tab, in the **Add** group, click the arrow under **Control Points** and select **Create Ground Control Point**. The Control Point Editor is displayed with the newly-created control point selected in the control point table at the bottom of the Editor.

Note: You can also create a control point at known coordinates from the Control Point Editor: On the **Home** tab of the Control Point Editor, in the **Control Point** group, click **Add New**.
2. Set the control point coordinates and properties. [See "Editing Control Points"](#) in this chapter for information.
3. Mark the control point in the relevant photos, and then save the project. [See "Marking Control Points in Photos"](#) in this chapter for information.

Creating Ground Control Points from Identifiable Locations on the Terrain

To create a ground control point from a specific location on the terrain (including LiDAR):

1. On the **Home** tab, in the **Add** group, click the arrow under **Control Points** and select **Create Ground Control Point from Terrain**.
2. Click in the **3D Window** to define the control point's location. The Control Point Editor is displayed showing a photo viewing window, a photos panel, a control point table, and a sampled photos panel - that gradually becomes populated with photo thumbnails as you mark the control point in photos.
3. Mark the control point in the relevant photos. See "[Marking Control Points in Photos](#)" in this chapter for information.
4. Modify the control point properties as required, and then save the project. See "[Editing Control Points](#)" in this chapter for information.

Creating Tie Points

Tie points can be created in either of the following ways:

- By selecting in the Photos list the photos between which you want to establish a correspondence. The Control Point Editor opens with the Photos panel populated by the selected photos. You can also select photos after opening the Editor. See "[Creating Tie Points in Selected Photos](#)" in this chapter for information.
- By creating a tie point from an easily identifiable location on the terrain. The Control Point Editor opens with the Photos panel populated by the photos whose frustum intersects with the created point. See "[Creating Tie Points from Identifiable Locations on the Terrain](#)" in this chapter for information.

After creating a tie point, you mark it in the relevant photos using the Control Point Editor. See "[Marking Control Points in Photos](#)" in this chapter for information.

Creating Tie Points in Selected Photos

To create a tie point in selected photos:

1. In the **Project Panel**, select the **Photos** group, and then in the **Photos** list, select the required photos.
2. On the **Photo** tab, in the **Tie Point** group, click **Create Tie Point**. The Control Point Editor is displayed with the newly-created tie point selected in the control point table at the bottom of the Editor, and with a Custom selection in the Photos panel that includes all the photos selected in the Photos list.

Note: You can also create tie points and filter the photos for sampling from the Editor itself: On the **Home** tab, in the **Add** group, click the arrow under **Control Points** and select **Open Control Point Editor**. Then on the **Home** tab of the Control Point Editor, in the **Control Point** group, click the arrow under **Add New** and select **Add New Tie Point**. See "[Marking Control Points in Photos](#)" in this chapter for information on filtering the photos for sampling.
3. Set the tie point properties. See "[Editing Control Points](#)" in this chapter for information.
4. Mark the tie point in the relevant photos, and then save the project. See "[Marking Control Points in Photos](#)" in this chapter for information. The Sampled Photos panel gradually becomes populated with photo thumbnails as you mark the tie point in photos.

Creating Tie Points from Identifiable Locations on the Terrain

To create a tie point from a specific location on the terrain:

1. On the **Home** tab, in the **Add** group, click the arrow under **Control Points** and select **Create Tie Point from Terrain**.
2. Click in the **3D Window** to define the tie point's position. The Control Point Editor is displayed with the Photos panel populated by photos whose frustum intersects with the tie point created.
3. Mark the tie point in the relevant photos. See "[Marking Control Points in Photos](#)" in this chapter for information. The Sampled Photos panel gradually becomes populated with photo thumbnails as you mark the tie point in photos.

Creating Check Points from Identifiable Locations on the Terrain

To create a check point from a specific location on the terrain (including LiDAR):

1. On the **Home** tab, in the **Add** group, click the arrow under **Control Points** and select **Create Check Point from Terrain**.
2. Click in the **3D Window** to define the check point's location. The Control Point Editor is displayed showing a photo viewing window, a photos panel, a control point table, and a sampled photos panel - that gradually becomes populated with photo thumbnails as you mark the control point in photos.
3. Mark the check point in the relevant photos. See "[Marking Control Points in Photos](#)" in this chapter for information.
4. Modify the check point properties as required, and then save the project. See "[Editing Control Points](#)" in this chapter for information.

Selecting Control Points

Control points can be selected either in the 3D Window or from the Control and Tie Points list to perform various commands on them, e.g., hide, exclude, delete, export, or modify their properties.

To select a control point:

1. In the **Project Panel**, select the **Control and Tie Points** group.
2. In the **3D Window**, click the control point you want to select, or select the control point(s) in the **Control and Tie Points** list. You can multi-select control points using the Control and Tie Points list commands.

Note: If you turned on global selection, you can select a control point in the 3D Window even without first selecting the Control and Tie Points group in the Project Panel as long as the control points are visible in the 3D Window.

See "[Showing and Hiding Group Items](#)" and "[Selecting Items Using the Item List](#)" in the "Using the Project Panel and Item List" chapter and "[Selecting Objects in the 3D Window](#)" in the "Basic Concepts" chapter for information.

Navigating to a Control Point

To fly or jump to a control point:

1. Select the required control point. See "[Selecting Control Points](#)" in this chapter.
2. On the **Control Point** tab, in the **Navigation** group, click **Fly To/ Jump To**.

Editing Control Points

To edit a control point:

1. In the **Project Panel**, select the **Control and Tie Points** group, and then in the **Control and Tie Points** list, select the control point and edit its properties as required.

Parameter	Definition
General	
Name	Name of the control point (as it appears in Quality Report, Control Point Editor, 3D Window, etc.).
Point Type	Select the point type: control point, tie point, or check point.
Auto-GCP	Automatically identify this control point's marker in your photos and mark the photos in PhotoMesh accordingly. This will not override any manual marking of control points. See "Marking and Auto-Marking Control Points in Photos" step 9 in this chapter for more information.
Exclude	Select the check box to exclude the control point from the project. See "Excluding Control Points" in this chapter for more information.
Visible	Select the check box to make control point visible in project. See "Hiding Control Point Symbols" in this chapter for information.
AT Tile	AT Tile in which this control point is included.
Sampled Photos	Number of photos in which this control point was marked. If fewer than three photos were sampled, this field is marked in red (Read-only).
Calculated Photos	Number of photos in which the position of this control point was calculated (Read-only).
Horizontal Accuracy (meters)	Accuracy of the control point's X and Y coordinates. The AT is adjusted to the control point within the specified accuracy margin of error.
Vertical Accuracy (meters)	Accuracy of the control point's altitude. The AT is adjusted to the control point within the specified accuracy margin of error.
Horizontal Error	If the horizontal error is greater than the margin of error defined by the Horizontal Accuracy value, this field is marked in red (Read-only).
Vertical Error	If the vertical error is greater than the margin of error defined by the Vertical Accuracy value, this field is marked in red (Read-only).
Sampling Error	If the sampling error is greater than 1 pixel, this field is marked in red (Read-only).
X	East-west coordinate of the control point.
Y	North-south coordinate of the control point.
Altitude	Altitude of the control point in meters.
Coordinate System	Coordinate system of the control point. Click Select  to set a different coordinate system. See "Coordinate System" in the "Basic Concepts" chapter for more information.

2. Save the project.

Marking and Auto-Marking Control Points in Photos

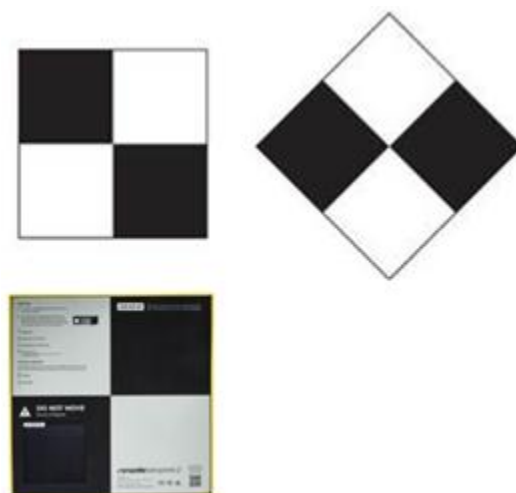
Control points are marked in photos using the Control Point Editor. Each control point should be marked in a few photos from each collection or photo direction. It is recommended to have at least 10 sampled photos for each control point.

In addition to the ribbon, the Control Point Editor includes a photo viewing window, a photos panel, and a sampled photos panel - that gradually becomes populated with photo thumbnails as you mark the control point in photos. The photos panel displays the photos from which to select the photos for marking control points, either as thumbnails or in table format with relevant photo information. You can also manually add photos to this list as described in the steps below.

Note: In order to enable your ground control points to be fully utilized in the AT process, it is recommended to add at least 3-5 ground control points per AT tile. When only two ground control points are provided, simple vertical adjustment, scaling and rotation are performed. When only one ground control point is provided, only vertical adjustment is performed. Any control point marked in fewer than three photos will be marked in red in the Control Point list.

Auto-Marking Photos: If your photos include control point markers, PhotoMesh can automatically identify them and mark control points in photos based on them. You can select the template that corresponds to your marker type, e.g., checkboard or cross. After selecting a control point for auto-marking, this setting can be overridden by manually marking this control point in one or more photos. The automatic markers are added as regular markers to the photos only after aerotriangulation is performed.

Note: It is recommended to review the photos in Control Point Editor after performing aerotriangulation to see which were auto-marked with control points, so that you can add manual markers, if necessary.



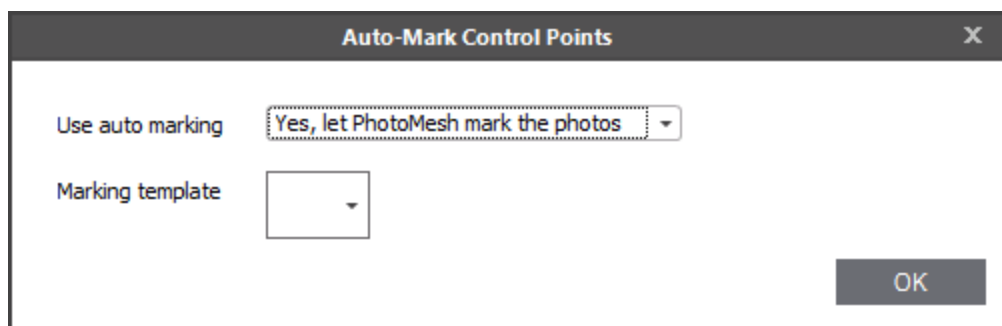
Control Point Markers

To mark a control point in photos:

1. If the control point was not already created, create the control point. See "[Creating Ground Control Points](#)" and "[Creating Tie Points](#)" in this chapter for information. The Control Point Editor is displayed.
2. If the control point was previously created, in the **Project Panel**, select the **Control and Tie Points** group, and then in the **Control and Tie Points** list, right-click the required control point, and select **Edit**. The Control Point Editor is displayed.

Note: If the Control Point Editor is already open, selecting a control point in the Control and Tie Points list opens that point in the Editor.

3. The Control Point Editor by default is docked in the right-hand side of the PhotoMesh window. If you want to undock it, click **Toggle Editor Docking** . Click again to redock the Editor.
4. From the **Photos** filter, select which photos should be listed or displayed in the Photos panel:
 - **All in frustum (Calculated)** – All the photos whose calculated frustum intersects the control point.
 - **All in range (Calculated)** – All photos with the selected control point in their calculated range (according to their [Effective Range](#)).
 - **All in frustum (Original)** – All the photos whose frustum intersects the control point.
Note: For photos with positioning and orientation information, this is determined based on frustum, and for photos with only positioning, according to the effective range set (See ["Collection Property Sheet"](#) in the "Photo Management" chapter for information).
 - **All in range (Original)** – All photos with the selected control point in their range (according to their [Effective Range](#)).
Note: This option is only available if photos have positioning information.
 - **All in project (Original)** – All photos in project (up to 2000 photos).
 - **Custom selection (Best)** – All photos selected in the Photos list before creating the tie point from the Photo tab. See ["Creating Tie Points in Selected Photos"](#) in this chapter for more information.
5. Select a viewing format for the **Photos** panel:
 -  – View as thumbnails
 -  – View in table format with information about distance, sampling error, etc. for each photo
6. If viewing the photos in table format, select the column by which you want to sort the table (in ascending order). Select the column a second time to switch to descending order. For filtering and advanced sorting options, see ["Filtering Data in the Control Point Editor"](#) in this chapter for information.
7. To view a photo using the Photo Viewer, select the photo from either the **Photos** panel or **Sampled Photos** panel, and click **View in Photo Viewer** . See ["Viewing Photos in the Photo Viewer"](#) in the "Preparing the Project" chapter. The Photo Viewer opens in the Control Point window. Click **Back to GCP Editor** to close the Photo Viewer and redisplay the editor.
8. In the **Photos** panel, select the photo on which you want to mark the position of the selected control point. The photo is displayed in the photo viewing window with a white cross hairs indicating the projected location of the control point that PhotoMesh automatically calculated. Type 'n' to select the next photo, or 'p' to select the previous one.
Note: Any photo thumbnails that are outside of the current build area are displayed in strikethrough font.
9. To have PhotoMesh automatically identify control point markers in your photos and mark the photos in PhotoMesh accordingly, do the following:
 - a. Click  **Auto-Mark**. The Auto-Mark Control Points dialog is displayed.



Auto-Mark Control Points Dialog

- b. In the **Use auto marking** field, select **Yes, let PhotoMesh mark the photos**.
- c. Select a **Marking template**. The template determines what marker type will be automatically identified by PhotoMesh as a control point, e.g., checkboard or cross. A dialog box is displayed asking whether you want to perform auto-marking of photos using the selected template for all the control points.
- d. Click **Yes** if you want to auto-mark all the control points in all the photos. After selecting a control point for auto-marking, this setting can be overridden by manually marking this control point in one or more photos. If any of the photos a control point is found in were set for auto-marking, the Auto-GCP check box is selected for this control point in the Control and Tie Points list. The automatic markers are added as regular markers to the photos only after aerotriangulation is performed.

Note: It is recommended to review the photos in the Control Point Editor after performing aerotriangulation to see which were auto-marked with control points, so that you can add manual markers, if necessary.

10. If manually marking control points, find the correct position in the photo in which to mark the control point. Use the mouse to zoom in and out and pan in the photo or use the ribbon commands:

	Zooms in.
	Zooms out
	Zoom in to the marked control point. This command is only available after a control point was marked in the photo.

11. Click **Sample in Photo**, and then click on the photo where you want to set the position of the control point. You can also type 's' on the keyboard to toggle between navigation and sampling modes. A red crosshairs is displayed where you clicked in the 3D Window and in the corresponding position on the photo thumbnail which is added to the "Sampled Photos" panel. The photo's XY pixel coordinates are displayed in the Mark X and Mark Y fields on the Control Point Editor ribbon in the Control Point group. If the Photos panel is displayed in table format, the check box in the Marked column is selected (If displayed as thumbnails, a check is displayed on top of the reviewed photo). The following information is displayed in the photo viewing window:

	User sample for position of control point.
	Projected location of control point.

	Projected triangulation – Projection of the triangulation center (adaptive weighted median of all ray intersections) back to the photos.
	Sampling error - RMS
Sampling Error - Photo	Distance, in pixels, between the projected triangulation and the user sample in this photo.
Sampling Errors - RMS	Root mean square (RMS) of all distances, in pixels, between the projected triangulation and the user sample in all photos.
Horizontal Error	Horizontal distance, in meters, between the projected triangulation and the control point position.
Vertical Error	Vertical distance, in meters, between the projected triangulation and the control point position.

- If you want to modify the photo's XY pixel coordinates, in the **Control Point Editor**, in the **Mark X** and **Mark Y** fields, type the required values.
- If you want to clear the previously marked position of a control point, in the **Control Point Editor**, click  **Clear from Photo**. The photo is removed from the "Sampled Photos" panel and the check is cleared from that photo thumbnail in the Photos panel.
Note: Only one position can be marked in each photo. If an additional position is marked, the previously marked position is automatically deleted and replaced by the new position.
- Repeat steps 7-11 for each of the photos in which you want to mark the selected control point.

Filtering Data in the Control Point Editor

To filter control point lists in the Control Point Editor:

- Point to the column by which you want to filter, and click the **Filter** icon . See "Filtering Data" and "Filter Editor" in the "Using the Project Panel and Item List" chapter for information.
- Click **Close**. The applied filter(s) are listed at the bottom of the Photos panel, e.g.,

 ☒ [Sampling Error (px)] >= '10.864' And [Sampling Error (px)] <= '12.493' And [Marked] = 'Checked' 

Click X to remove a filter.

- To access additional sorting commands, right-click any column header. Select any of the following commands:

Command	Description
Sort Ascending	Sort all data in the table by this column in ascending order.
Sort Descending	Sort all data in the table by this column in ascending order.
Clear All Sorting	If multiple columns were sorted, this option is displayed to clear all.
Group by This Column	Group the data by this column. Note: If you want to clear the grouping, right click above the column headers, and select Clear Grouping .
Show Group by Box	Display boxes, above the photo table, indicating how the data is grouped in the table. Note: If you want to clear the Group by boxes, right click above the column headers, and select Hide Group by Box .

Hide This Column	Hide the selected column
Column Chooser	Customize which columns show in the table. Drag and drop columns to and from the Customization dialog.
Best Fit	Adjust the selected column's width for the best fit for its contents.
Best Fit (all columns)	Adjust the all the table's column widths for the overall best fit.
Filter Editor	Open the Filter Editor to edit the table filter. You can use the Filter Editor to add and remove conditions. A search expression can consist of more the one condition. In this case, you need to insert a connector between the conditions to specify their relations. In addition to AND, you can use the OR, NOT and [] connectors to define the relations between conditions.
Show Find Panel	Show a Find panel to search for a specific text. Note: To hide the Find panel, click the X next to it.
Show Auto Filter Row	Display a row with a filter option for each of the columns.

Control Point Editor Keyboard Shortcuts

The following keyboard shortcuts are available:

Shortcut	Description
s	Toggle between navigation and sampling modes.
n	Selects the next photo in the Photos panel and displays it in the photo viewing window.
p	Selects the previous photo in the Photos panel and displays it in the photo viewing window.
N	Selects the next control point in the Control Points table.
P	Selects the previous control point in the Control Points table.

Hiding Control Point Symbols

To hide a control point symbol:

- In the **Project Panel**, select the **Control and Tie Points** group, and then in the **Control and Tie Points** list, clear the control point's **Visible** check box.

Excluding Control Points

To exclude a control point from the project:

- In the **Project Panel**, select the **Control and Tie Points** group, and then in the **Control and Tie Points** list, select the control point's **Exclude** check box.

Deleting Control Points

To delete a control point:

- In the **Project Panel**, select the **Control and Tie Points** group, and then in the **Control and Tie**

Points list, right-click the required control point, and select **Delete**.

Exporting Control Points

Control point data can be exported to a .gcp (text format) file. Auto-marked control point data is only exported in the .gcp file after aerotriangulation was performed. See ["Marking Control Points in Photos"](#) step 9 in this chapter for more information.

Note: All control points are exported in EGM96 geoid vertical coordinate system even if they were defined in "WGS84ELLIPSOID".

To export a project's control point data:

1. On the **Home** tab, in the **Add** group, click the arrow under **Control Points** and select **Export**. The Export Control Points dialog is displayed.
2. Type a name for the .gcp (text format) file and browse to the required location in which to save the file.
3. Click **OK**.

Importing Control Points

Control point data can be imported in a tab delimited .gcp (text format) file:

- First line (optional): Define the control points' coordinate system in WKT format. (Default: geographic Lat-Long WGS84.)

```
wkt=.....
```

Note: The WKT information must be in a single line without any line breaks. You can use the Coordinate System Generator to generate a WKT string defining the required coordinate system. See ["Coordinate System"](#) in the "Basic Concepts" chapter for more information.

- Second line: sep=\t:
This sets the file as tab delimited.
- Third line (optional): Select the template file for auto-marking of control points. The template file determines what marker type will be automatically identified in photos as a control point, e.g., checkboard or cross. See ["Marking and Auto-Marking Control Points in Photos"](#) in this chapter for more information. The template is saved in the AGCPTemplates folder under the application's Fuser folder. Currently, the only template available is *aeropoint.agcp*.

```
template=aeropoint.agcp
```

- Fourth line (optional): Indicate the number of control points. If this information is not provided, each subsequent line in the file is interpreted as providing positioning information for a distinct control point. If the file includes this information, all lines after the number of lines in cp=[number] are interpreted as mapping between control points and photos.

```
cp=[number]
```

- For each control point, set the control point's position on the terrain (X, Y, Altitude (Optional), Name (Optional), Type (Optional; Possible values: 1-control; 2-check; 3-tie) in tab delimited format.

```
[x coordinate] <TAB> [y coordinate] <TAB> [altitude] <TAB> [name] <TAB> [type]
```

Note: The Type parameter can only be passed together with the Name parameter.

- Specify the position (in pixel coordinates) in a specified photo to associate with a particular control point (referenced by index) (optional).

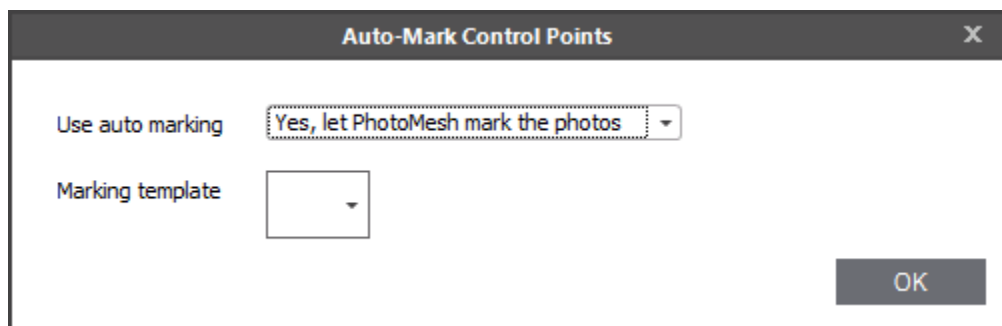
```
[control point index] <TAB> [photo file path] <TAB> [x pixel coordinate]
<TAB> [y pixel coordinate]
```

Example:

```
wkt=GEOGCS["WGS 84",DATUM["WGS_1984",SPHEROID["WGS
84",6378137,298.257223563,AUTHORITY["EPSG","7030"]],AUTHORITY["EPSG","6326
"]],PRIMEM["Greenwich",0,AUTHORITY["EPSG","8901"]],UNIT["degree",0.0174532
925199433,AUTHORITY["EPSG","9122"]],AUTHORITY["EPSG","4326"]]
sep=\t
cp=2
9.63390819201651 <TAB> 47.4233597051724 <TAB> 401.463806152344 <TAB>
Control point1
9.63390818337074 <TAB> 47.4233597019402 <TAB> 401.463793239556 <TAB>
Control point2
0 <TAB> C:\Photos\1.tif <TAB> 384 <TAB> 5456
0 <TAB> C:\Photos\2.tif <TAB> 7689 <TAB> 4761
0 <TAB> C:\Photos\3.tif <TAB> 8738 <TAB> 4340
1 <TAB> C:\Photos\4.tif <TAB> 388 <TAB> 5450
1 <TAB> C:\Photos\5.tif <TAB> 7690 <TAB> 4761
1 <TAB> C:\Photos\6.tif <TAB> 8741<TAB> 4326
```

To import control points:

- On the **Home** tab, in the **Add** group, click the arrow under **Control Points** and select **Import**. The Import Control Point File dialog is displayed.
- Browse to the required .gcp file and click **Open**. If the control point file only contains information about the control point's position on the terrain but not about either the positions (in pixel coordinates) in photos to associate with each control point or the auto-marking template, then upon import of the file, the Auto-Mark Control Points dialog box is displayed asking if you want to auto-mark the photos. Select whether to use auto-marking and if Yes, what **Marking template** to use. See "[Marking Control Points in Photos](#)" step 9 in this chapter for more information.



Auto-Mark Control Points Dialog

- Modify the control point properties as required, and then save the project.

Adding a Reference Project to Use in Aerotriangulation

Adding a reference project is the process of adjusting photos in your project to the geographic location of photos of the same geographic area in a different project. When building a project that overlaps partially or entirely with a different project that will be overlaid on the same map in a client viewer, it is important to ensure that matching areas align properly. This can be achieved by adding the older "base" project as a reference project for the new "target" project. After adding the reference project, the Reference Project group is listed in the Project Panel and each project's photos are accessed from the Reference Photos list, but its photos are not displayed in the 3D Window. When building the new project, PhotoMesh establishes a correspondence between points in the reference project and target project, so that the positions of identified features in the target project match the geographic positions of the same features in the reference project.

This tool can be particularly useful for multiple flights over the same area. Ground control points or high accuracy GPS locations (RTK, PPK) can be used for the first flight and then the reference project mechanism can be used to accurately register the other flights, even without GCP's or accurate location information (e.g., in GPS-denied areas).

To add a reference project:

1. On the **Home** tab, in the **Add** group, click the arrow under **Control Points** and select **Add Reference Project**. The Add Reference Project dialog is displayed.
2. Browse to the required reference project, i.e., the calculated AT of that build (ATResult*.sqlite), and click **Open**. The project's photos and photo and collection parameters are loaded into the project. When the aerotriangulation step of the build process is performed, the project will be automatically aligned to the reference project.

Reference Photos List

In the Reference Photos list, you can view but not edit the properties of the photos in your reference projects. See ["Modifying Photo and Photo Collection Properties"](#) in the "Photo Management" chapter for information about the photo properties.

Reference Photos									
Photos		Collections		1 out of 1603 selected					
Selection	Collection	Exclude	Project Original	X	Y	Altitude	Altitude Relative	Omega	Phi
☒	Ref: Image Series - 01 - ...	☐	☐	-5.748730	-4.035088	392.445033	0.000000	23.802043	-25.885836
☐	Ref: Image Series - 01 - ...	☐	☐	4.373993	7.899670	352.839037	0.000000	-39.776785	10.708313
☐	Ref: Image Series - 01 - ...	☐	☐	-14.979374	13.390186	412.695221	0.000000	-15.564220	-26.929706
☐	Ref: Image Series - 01 - ...	☐	☐	-5.288355	-2.885113	366.717920	0.000000	50.648300	-37.381717
☐	Ref: Image Series - 01 - ...	☐	☐	6.304705	0.362859	398.565240	0.000000	-5.424908	34.243921
☐	Ref: Image Series - 01 - ...	☐	☐	3.907432	5.243674	392.336240	0.000000	-73.519758	41.077054
☐	Ref: Image Series - 01 - ...	☐	☐	5.412754	-3.637769	374.665679	0.000000	42.332064	46.288495
☐	Ref: Image Series - 01 - ...	☐	☐	6.075575	0.920781	359.190733	0.000000	-17.223771	58.181044

Reference Photos List

CHAPTER 14 FUSERS

About Fusers

PhotoMesh can take advantage of computer clusters to accelerate database creation. Fusers allow you to share the demanding processing requirements of the different build steps (data preparation, AT, point cloud, mesh model, texture, 3DML/other outputs) between several computers on the same network. The master computer utilizes the computing power of the network's client computers by connecting to PhotoMesh fusers on the client machines.

The fuser remains idle until used by the master computer on the network. Once started, the fuser runs on the client machines, with only an icon in the system tray that opens a status log and a dialog to stop or suspend fusers. You can install the PhotoMesh Fuser on the same computer with the PhotoMesh Build Manager and/or on other computers.

In addition to standard fuser use, PhotoMesh also offers two options for auto-scaling of fusers to enable dynamic allocation of fusers based on project requirements:

- **Amazon Web Services (AWS) Fusers** – AWS fuser instances can be dynamically launched whenever there are unassigned pending fuser tasks, to further accelerate model creation.
- **Fusers Pool** – Fusers can be shared between several PhotoMesh Build Managers in your organization, enabling each machine running PhotoMesh to use the combined resources of all the machines in the pool. A distributed management approach is used to manage the pool fusers, whereby each PhotoMesh Build Manager manages its own fuser needs and resources by requesting fusers from the pool when generating a build and returning the fusers when they are no longer needed.

When beginning the build process the PhotoMesh Build Manager application determines what build steps are required and assigns them to the different fusers. PhotoMesh Build Manager continuously monitors the fusers' progress and assigns new processing tasks based on fuser availability.

Both PhotoMesh Build Manager and the PhotoMesh fusers require read-write access to the working and project folders.

Fuser Management (on the Master computer)

All available fusers are listed in the Fusers section of PhotoMesh Build Manager. As the build process is progressing, PhotoMesh Build Manager displays fuser information (which fusers are in use, which aren't, etc.). See ["Monitoring a Build"](#) in the "Building" chapter for more information.

PhotoMesh Build Manager - \\10.0.1.9 \c\Automation\Projects\ 21_60m_67\project.xml

Data Preparation Aerotriangulation **Reconstruction Tiles (Point Clouds, Model, Texture)** Project Output

Working Folder: \\10.0.1.9\\c\\PM-Work\\21_60m_67_ppk [Browse...](#)

Automatically use new available fusers: ☒

Use AWS Instances ([Change](#)): **ON** Running AWS instances: 10 (Required: 30 max: 10).

Use Fusers Pool ([Change](#)): **OFF** [Pool Manager](#)

Fusers Online: 10 (local 0, AWS 10) Selected: 10 (local 0, AWS 10) Working: 10 Filter: **All**

Use	Fuser	Version	Running Step	Step Elapsed Time	Progress	Status
☒	IP-10-0-0-250(10.0.0.250)	7.7.0.7089	Create point cloud [Tile-1-2-1-1]	00:01:00	0%	Online - Working
☒	IP-10-0-0-41(10.0.0.41)	7.7.0.7089	Create point cloud [Tile-2-2-1-1]	00:01:00	0%	Online - Working
☒	IP-10-0-0-58(10.0.0.58)	7.7.0.7089	Create point cloud [Tile-6-1-1-1]	00:01:00	0%	Online - Working
☒	IP-10-0-1-165(10.0.1.165)	7.7.0.7089	Create point cloud [Tile-5-1-1-1]	00:01:00	0%	Online - Working
☒	IP-10-0-1-236(10.0.1.236)	7.7.0.7089	Create point cloud [Tile-5-2-1-1]	00:00:55	0%	Online - Working
☒	IP-10-0-2-216(10.0.2.216)	7.7.0.7089	Create point cloud [Tile-0-2-1-1]	00:01:00	0%	Online - Working
☒	IP-10-0-2-25(10.0.2.25)	7.7.0.7089	Create point cloud [Tile-4-2-1-1]	00:01:00	0%	Online - Working
☒	IP-10-0-2-89(10.0.2.89)	7.7.0.7089	Create point cloud [Tile-4-1-1-1]	00:01:00	0%	Online - Working
☒	IP-10-0-2-95(10.0.2.95)	7.7.0.7089	Create point cloud [Tile-3-2-1-1]	00:01:00	0%	Online - Working
☒	IP-10-0-3-75(10.0.3.75)	7.7.0.7089	Create point cloud [Tile-3-1-1-1]	00:01:00	0%	Online - Working

AT Tiles: 1 Reconstruction Tiles: 40 Filter: **Errors**

Task	Last Step	Running Step	Status Description	Progress	Fusers
------	-----------	--------------	--------------------	----------	--------

[Production Report](#) [Show Settings](#) [Cancel](#) [Build](#)

Setting the Working Folder on the Master Computer

The working folder is the folder that PhotoMesh Build Manager (on the Master computer) and PhotoMesh fusers use to communicate: to deposit and collect pending tasks, and share information about fuser availability, status and progress.

The working folder should be set to a network folder with read-write access permissions, for both the master and client computers. It is recommended to use UNC format (`\\[UNC Master] \folder\WorkingFolder`, e.g.: `\\Machine1\c\PhotoMesh7.0\WorkingFolder`), though a mapped network drive can also be used as long as the same letter is mapped to the same share on all computers.

Note: When using fusers, the project must also be created in a network folder with read-write access permissions, for both the master and client computers. It is recommended to use UNC format (`\\[UNC Master] \folder\ProjectFolder`, e.g.: `\\Machine1\c\PhotoMesh\ProjectFolder`), though a mapped network drive can also be used as long as the same letter is mapped to the same share on all computers. See ["Creating and Opening a PhotoMesh Project"](#) in the "Project Management" chapter for more information.

The working folder must be on a device that supports the number of concurrent network connections you require, e.g., Windows 10 limits the number of supported concurrent connections to 20.

It is recommended to create the working folder on a fast drive that is different from the one used for the project data, most preferably on a local solid-state drive (SSD) on the master machine that is shared with the other fuser machines.

To set the working folder on the master computer, do either of the following:

- In the **Options** dialog, set the working folder. See ["Setting PhotoMesh Options"](#) in the "Basic Concepts" chapter for more information.

OR

- In **PhotoMesh Build Manager**, next to the **Working Folder** field, click **Browse**. Then browse to the required location.

Using Fusers

To use fusers on the master computer:

1. In **PhotoMesh Build Manager**, select the required fusers.
2. If you want PhotoMesh Build Manager to automatically use any new fusers that connect to the project's working folder, select **Automatically use new available fusers** from the dropdown list.
3. If there are no available fusers, a dialog is displayed asking if you want to start a local fuser. Click **Yes** to start a local fuser. Click **No** to start the build without adding a fuser. Processing will begin automatically when fusers are added.
4. If you want to change the working folder that PhotoMesh Build Manager and PhotoMesh fusers use to communicate (to deposit and collect pending tasks, and share information about fuser availability, status and progress), click **Browse**. Then browse to the required location.

Note: The working folder in the PhotoMesh Fuser dialog on the client computer should be set to the same location as the location set here. See ["Fuser Management \(on the client Computer\)"](#) in this chapter for information.

Fuser Management (on the client computer)

The owner of the fuser computer has full control over fuser usage. For information on installing and uninstalling fusers, see ["Installing PhotoMesh Fusers"](#) and ["Uninstalling PhotoMesh Fusers"](#) in the "Getting Started" chapter. Fusers can be installed locally, or run from a shared network path.

Setting the Working Folder on a Client Computer

The working folder is the folder that PhotoMesh Build Manager (on the master computer) and PhotoMesh fusers use to communicate: to deposit and collect pending tasks, and share information about fuser availability, status and progress.

The working folder on each of the client computers must be set to the same network folder as was set for the master computer. See "[Setting the Working Folder \(on the Master Computer\)](#)" in this chapter for more information.

To set the working folder on the client computer:

1. In the system tray, click the **PhotoMesh Fuser icon**, and then click **Open PhotoMesh Fuser**. The PhotoMesh Fuser dialog is displayed.
2. To the right of the **Working Folder** field, click **Change**. The Select Folder dialog is displayed.
3. Browse to the required folder and click **Select Folder**.

Note: If the working folder is changed during processing, processing is stopped.

Starting Fusers

Once a fuser is started on a client computer, it is added to the list of available fusers on the master computer. Fusers can be installed locally, or run from a shared network path.

To start a fuser:

- Click **Start > Skyline PhotoMesh > PhotoMesh Fuser**. A PhotoMesh fuser icon is placed in the Windows system tray.

Note: In the full PhotoMesh installation, the name of the folder containing the fuser application is **Skyline PhotoMesh**, while in the standalone fuser installation, the folder name is **Skyline PhotoMesh Fuser**.

Fuser Log

To open the fuser log:

1. In the system tray, click the **PhotoMesh Fuser icon**, and then click **Open PhotoMesh Fuser**. The PhotoMesh Fuser dialog is displayed, showing the fuser log in the lower half of the dialog.
2. If you want to clear the log, click **Clear Log**.
3. If you want to stop the log from automatically scrolling to the last line of the log, click **Scroll Lock**.

Disabling a Fuser

Fusers can be temporarily disabled.

To disable a fuser:

- In the system tray, click the **PhotoMesh Fuser icon**, and then click **Disable**. When you want to restart the fuser, click the **PhotoMesh Fuser icon** and then click **Enable**.

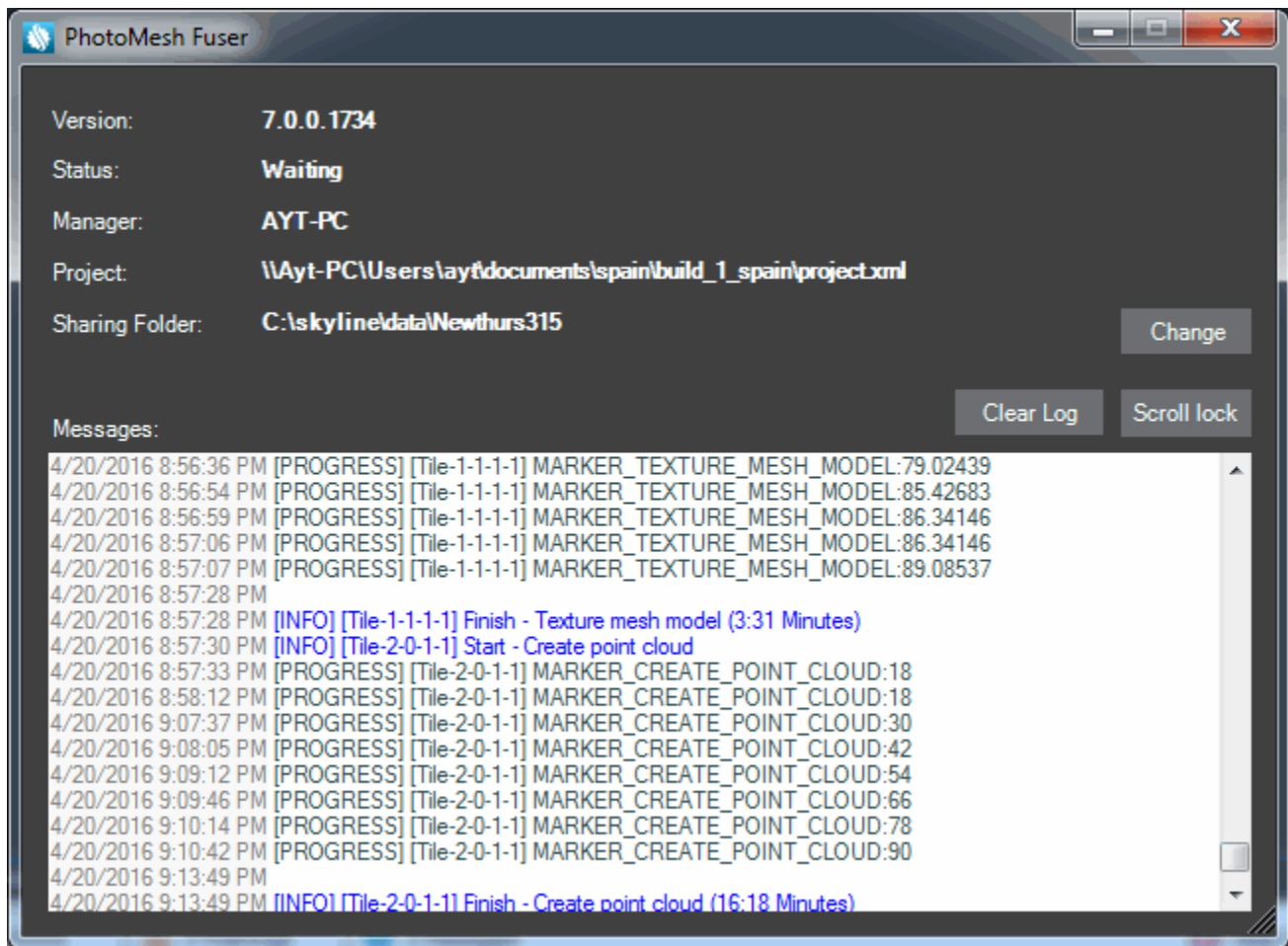
Note: The Master computer reattempts to use the fuser every five minutes. The fuser is used only if its owner has enabled it.

Exiting a Fuser

You can permanently exit a fuser.

To exit a fuser:

- In the system tray, click the **PhotoMesh Fuser icon**, and then click **Exit**.



PhotoMesh Fuser Log

Running Fusers Upon Login

You can schedule fusers to be run automatically upon login, either individual fusers or the fuser service (PMFuserService.exe), if using a fusers pool. See ["Fusers Pools"](#) in this chapter for information.

To run fusers upon login using the Task Scheduler

1. Right-click Task Scheduler Library and select Create Task.
2. **General** > **Name**: Start PM Fuser OR Start PM Fuser Service.
Note: If using a fusers pool, then start the fuser service (PMFuserService.exe) rather than a fuser. See ["Fusers Pools"](#) in this chapter for information.
3. **Triggers**:
 - a. Click **New**.
 - b. **Begin the task**: Select **At log on**.
 - c. **Settings**: Specific user.

- d. Click **OK**.
4. Actions:
 - a. Click **New**.
 - b. **Program/script:** `\\10.0.0.10\\d\\PhotoMesh\\fuser\\PhotoMeshFuser.exe` OR `\\10.0.0.10\\d\\PhotoMesh\\fuser\\PMFuserService.exe`.

Note: If using a fusers pool, then run the fuser service (PMFuserService.exe) rather than a fuser. See "[Fusers Pools](#)" in this chapter for information.
 - c. Click **OK**.
5. Settings:
 - a. Select **if the task fails, restart every**, and keep the default 1 minute and 3 times.
6. Click **OK**.

If you want to log in automatically upon start up, do the following:

1. Click the **Start**  button, type "cmd", and then select **Command Prompt** from the list. In the command prompt window type the following:
`control userpasswords2`
2. Select the user that you want automatically logged in, e.g., Administrator, and clear the **Users must enter a user name and password to use this computer checkbox**, and click **OK**.
3. Enter and confirm the password for the fuser computer.

Fuser Auto-Scaling

PhotoMesh offers two options to scale project production up and down based on resource needs:

- [Amazon Web Services \(AWS\) Fusers](#) – AWS fuser instances can be dynamically launched whenever there are unassigned pending fuser tasks, to further accelerate model creation.
- [Fusers Pool](#) – Fusers can be shared between several PhotoMesh Build Managers in your organization, to dynamically allocate and free fusers based on each project's resource needs.

These options can be used together or separately. When either option is used, PhotoMesh Build Manager continuously monitors the fusers' progress and status and the processing requirements and allocates or frees fusers as required.

Using Amazon Web Services (AWS) Fusers

PhotoMesh can work with Amazon Web Services to scale project production up and down based on resource needs. AWS fuser instances can be dynamically launched whenever there are unassigned pending fuser tasks, to further accelerate model creation. PhotoMesh Build Manager continuously monitors the fusers' progress and status and the processing requirements of the current step, adding and terminating AWS fusers as required. See "[Getting Started with AWS](#)" for more information.

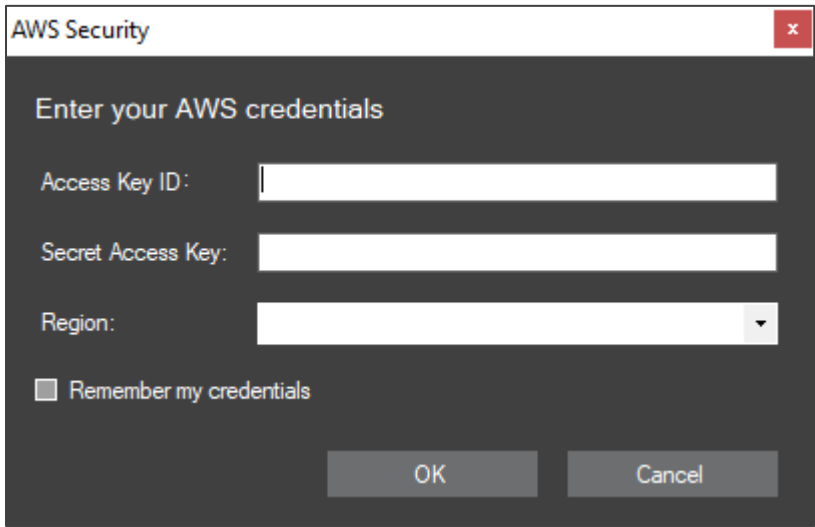
To use AWS fusers:

1. In the **Build Manager**, toggle on **Use AWS Instances**. You can also access these settings from the Tools tab (AWS Instance Settings command) and from the Start page's [Automation Dashboard](#).

Note: The **Automatically use new available fusers** check box must also be selected, so that PhotoMesh will use the AWS fuser instances after they are launched.

2. If PhotoMesh Build Manager is running on a local computer, and this is the first time connecting to the AWS fusers, the AWS Security dialog is displayed. Enter your AWS security information, and click **OK**.

Note: If PhotoMesh Build Manager is running on an AWS instance with an IAM (Identity and Access Management) role, security credentials are not required. See ["Creating an IAM User for Running PhotoMesh Project Using an On-Premises Master"](#) in the "Getting started with AWS" chapter for information.



AWS Security Dialog

Access Key ID	Token that allows PhotoMesh to communicate with your AWS account. Note: You can find your Secret Access Key on the My Security Credentials link in the AWS Management Console.
Secret Access Key	Token that allows PhotoMesh to communicate with your AWS account. Note: You can find your Secret Access Key on the My Security Credentials link in the AWS Management Console.
Region	Select an AWS region.
Remember my credentials	Select this option to save the password. The password will be saved and will be displayed using asterisks the next time the dialog is opened.

3. In the **Auto-Scaling Fusers Settings** dialog, select the **Use AWS Instances** check box, and then enter the following information:

Maximum Price per Spot Instance	Set the maximum price you want to pay for an EC2 fuser spot instance.
Maximum Instances	Set the maximum number of fuser instances to launch.
Launch Rate	Set the maximum number of fuser instances to launch per minute, e.g. if Maximum Instances was set to 200, and the Launch Rate to 50, the Build Manager will spread the launch of the fusers over 4 minutes. This prevents a potential bottleneck that could be caused

	by a large number of fusers attempting to read the build data at the same time.
Terminate AWS instances when build finished	Select the check box if you want to terminate AWS fuser instances when the build is complete.
Select fuser instance launch parameters	
Region	The region the master is running in, or the region set in the AWS Security dialog (Read-only).
Configuration Name	Name of the JSON configuration file PhotoMesh should use. The JSON file contains an ordered array of fuser instance types based on preferability. During production, PhotoMesh iterates over the elements of the array, beginning with the first instance type, until it finds an available instance. Priority is given to the instances within the subnets listed in the Subnet ID (below), before moving to the next machine type. To modify the configuration, click Edit or open the file in a text editor See "Recommendations Based on PhotoMesh Step" in the "Getting Started with AWS" chapter for information. Additional JSON configuration files can be created and placed under <code>../photomesh/AWSConfigurations</code> where the two example JSON files are also saved. Each instance object in the JSON file's array has four properties: ▪ <code>ami</code> - ID of the Amazon Machine Image (AMI). ▪ <code>instanceType</code> - Defines the hardware of the host computer used for your instance. Each instance type offers different compute, memory, and storage capabilities. ▪ <code>demandType</code> - Determines the pricing arrangement: "onDemand" or "spot." ▪ <code>fusersCount</code> - Number of PhotoMesh fusers to run on each instance. More powerful instances have the capability to run multiple fusers simultaneously, thereby enhancing cost efficiency. Example: <pre>{ "configurations": [{ "ami": "ami-12345", "instanceType": "g4ad.4xlarge", "demandType": "spot", "fusersCount": 4 }, { "ami": "ami-12345", "instanceType": "g4ad.2xlarge", "demandType": "spot", "fusersCount": 2 }], }</pre>

	<pre>{ "ami": "ami-12345", "instanceType": "g4dn.4xlarge", "demandType": "spot", "fusersCount": 4 }, { "ami": "ami-12345", "instanceType": "g4dn.2xlarge", "demandType": "spot", "fusersCount": 2 }], "Name": "GPU" }</pre>
Build Priority	Select either Choose all listed options , to use all machine types listed in the JSON file, with priority based on the order they are listed in the file, or select a specific machine.
Security Group ID	Select the instance's security group ID. Security groups control inbound and outbound traffic using rules.
Subnet ID	Select the subnet of the virtual private cloud (VPC) dedicated to your AWS account, into which to launch your fuser instances. If you want to use additional subnets, click Additional Subnets , and type a list of all the subnet ID's you want to use. See "Building a PhotoMesh Project Using the Master Instance" in the "Getting Started with AWS" chapter for more information.
Key Pair Name	The name of the Key Pair used to configure the PhotoMesh AWS fuser Amazon Machine Images (AMI's). Key Pairs are used to encrypt and decrypt login information.
Additional Per Project Fuser Instance Tags	
Add tags to help you easily identify a fuser instance in the Amazon EC2 console after it is launched. These tags are saved in the JSON configuration file.	
Customer	Name of customer who commissioned the project.
Description	Free text describing the project.
Shutdown inactive fuser instance...	Select one of the options for shutting down inactive fuser instances: ▪ Shut down inactive fuser instances before the next billing hour ▪ Shut down inactive fuser instances after XX minutes

4. If you want to use another account, click **Use another AWS account**, and enter the AWS security information (see step 2 for more information).
5. If you also want to use pool fusers, toggle on the **Use Fusers Pool** link, and enter the required information. See ["Using a Fusers Pool"](#) in this chapter for information.
6. Click **OK**.

Fusers Pools

The fusers pool is used to share fusers between several PhotoMesh master computers in your organization. When working with a pool, the PhotoMesh master computer's Build Manager requests fusers from the pool, and returns them when they are no longer needed. A distributed management approach is used to manage the pool fusers, whereby each PhotoMesh master computer manages its own fuser needs and resources. Fusers are allocated on a first come first served basis, i.e., to the first PhotoMesh Build Managers that request fusers from the pool. If a manager submits a request for a user, but no fusers are currently available in the pool, the manager continually checks the fusers pool for a fuser that was returned to the pool, at which point it assigns the fuser to its project's working folder.

Fuser allocation can be manually set from the Pool Manager page. From this page, you can monitor and manage all the fusers in the pool, e.g., assign fusers to specific projects' working folders and prevent their being returned to the pool.

Running the Fuser Service

The PMFuserService.exe should be run on all fuser computers in the pool since it is responsible for running the required fuser (PhotoMeshFuser.exe) version for the requesting manager and assigning it to the manager's working folder. This enables different PhotoMesh versions to use the same fusers pool. The PMFuserService.exe can be run from the fuser folder under the installation folder (.\Application Files\Fuser\PhotoMeshFuser.exe) or from a file share or network drive that is accessible to all fuser computers. It is recommended to run the fuser service on startup using auto-login. See ["Running Fusers Upon Login"](#) in this chapter for information.

Creating a Fusers Pool

To create a fusers pool:

1. Designate a pool directory which will hold the pool.json file created in the next step as well as the working folders that will be created automatically for each of the projects using the fusers pool.

Note: If you are also using local fusers for a project, the working folder must be the one automatically created in the pool directory. It is recommended to create the working folder on a fast drive that is different from the one used for the project data, most preferably on a solid-state drive (SSD) that is shared with the master machine and other fuser machines.

2. Create a pool.json file with the name and IP of all fusers in the pool. Use the following structure and syntax in creating the file:

- Create an array called "Pool" (enclosed in square brackets).
- Enclose within curly braces each of the fuser objects. Each fuser object is composed of two name/value pairs separated by a comma: Name: FuserName, and IP: FuserIP.

```
{
  "Pool": [
    {
      "Name": "pm-020",
      "IP": "http://192.168.204.20:8085"
    },
    {
      "Name": "pm-021",
      "IP": "http://192.168.204.21:8085"
    }
  ]
}
```

```

    }
  ],
  "Name": "ProductionPool"
}

```

- When using more than a single fuser on a machine (See ["Launching Multiple Fusers"](#) in this chapter), a unique /X must be added after the IP address for each of the fusers, e.g., "IP": "http://192.168.200.001:8085/**1**", "IP": "http://192.168.200.001:8085/**2**", "IP": "http://192.168.200.001:8085/**3**" will launch three fusers from the same machine.

Note: The number of fusers set to run on a machine should be limited by the amount of memory on the machine. See ["Launching Multiple Fusers"](#) in this chapter for more information.

Note: The launching of multiple fusers on a single machine from a *pool.json* is only supported in PhotoMesh 7.8.3 or higher.

The fusers are launched in the order that they are listed in the json file. So if you want to launch a single fuser on each machine before launching additional fusers, each of the machine IPs should be listed first in the json file with /1 after each IP. To launch additional fusers, enter the machine's IP followed by a unique /X. The example below will launch fusers in the following order: a fuser from computer 1, a fuser from computer 2, a fuser from computer 3, a second fuser from computer 1, and finally a second fuser from computer 2.

```

{
  "Pool": [
    {
      "Name": "fuserX1",
      "IP": "X.X.X.X/1"
    },
    {
      "Name": "fuserY1",
      "IP": "Y.Y.Y.Y/1"
    },
    {
      "Name": "fuserZ3",
      "IP": "Z.Z.Z.Z/1"
    },
    {
      "Name": "fuserX2",
      "IP": "X.X.X.X/2"
    },
    {
      "Name": "fuserY2",
      "IP": "Y.Y.Y.Y/2"
    }
  ],
}

```

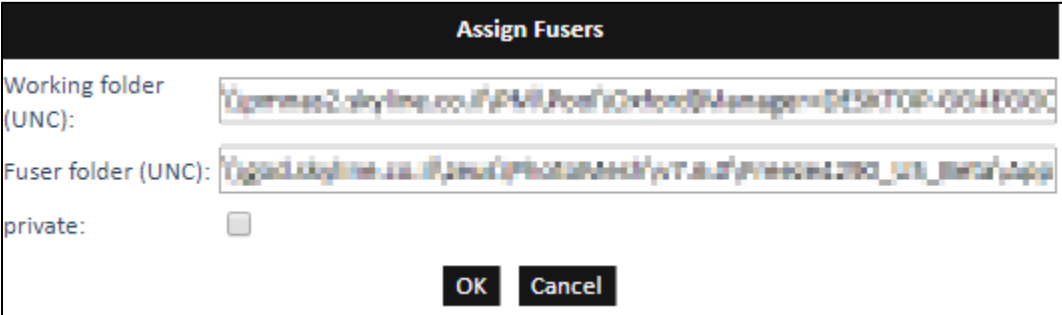
```
"Name": "ProductionPool-Multiple fusers on machines"
}
```

Managing a Fusers Pool

Fusers are allocated on a first come first served basis, i.e., to the first PhotoMesh Build Manager that requests fusers from the pool. If a subsequent manager submits a request for a user, but no fusers are currently available in the pool, the manager continually checks the fusers pool for a fuser that was returned to the pool, at which point it assigns the fuser to its project's working folder. If you want to manually allocate fusers, you can do so from the Pool Manager page. From this page, you can monitor and manage all the fusers in the pool, e.g., assign fusers to specific projects' working folders and prevent their being returned to the pool.

To manage a fusers pool:

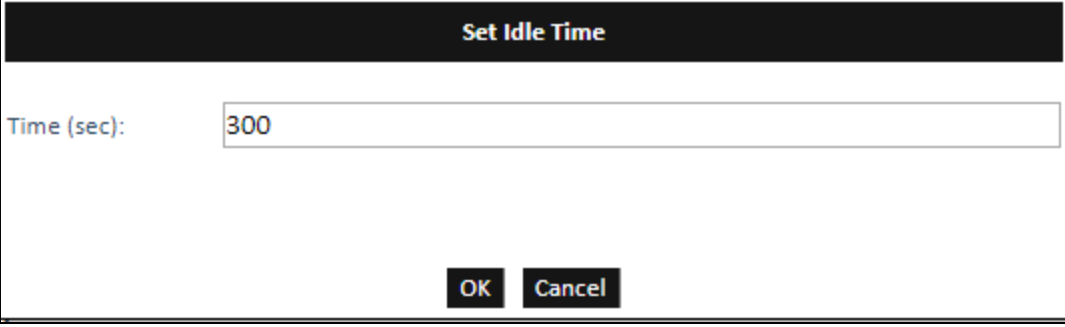
1. In the **Build Manager** dialog, in the Auto Scaling section, click the **Pool Manager** link. The PhotoMesh – Pool Manager page is displayed, with information on all the fusers in the pool. You can also access the Pool Manager from the Tools tab and from the Start page's [Automation Dashboard](#). You can set any of the following for either selected or all fusers:
 - Assign to a project
 - Modify their idle timeout
 - Set private or public status
 - Add a comment
2. If you want to manually assign fusers to a project's working folder, do the following:
 - a. Select the fusers' check boxes, and click **Assign**. The Assign Fusers dialog is displayed.



The image shows a dialog box titled "Assign Fusers". It contains two text input fields. The first field is labeled "Working folder (UNC):" and contains the path "C:\photos2.dryline.co.uk\PM\Pool\Working\Manager-0458TOP-0046000". The second field is labeled "Fuser folder (UNC):" and contains the path "C:\photos2.dryline.co.uk\PM\Pool\Working\Manager-0458TOP-0046000". Below these fields is a checkbox labeled "private:" which is currently unchecked. At the bottom right of the dialog are two buttons: "OK" and "Cancel".

Assign Fusers

- b. Enter the working folder and fuser folders.
 - c. If you want to prevent these fusers from being restored to the pool after completing their current tasks, select the **private** check box.
 - d. Click **OK**.
3. If you want to modify the idle timeout set for certain fusers, select the fusers' check boxes, and click **Set Idle Time**. Then enter the time in seconds, and click **OK**.

A dialog box titled "Set Idle Time" with a black header. Below the header, the text "Time (sec):" is followed by a text input field containing the number "300". At the bottom right, there are two buttons: "OK" and "Cancel".

Set Idle Time

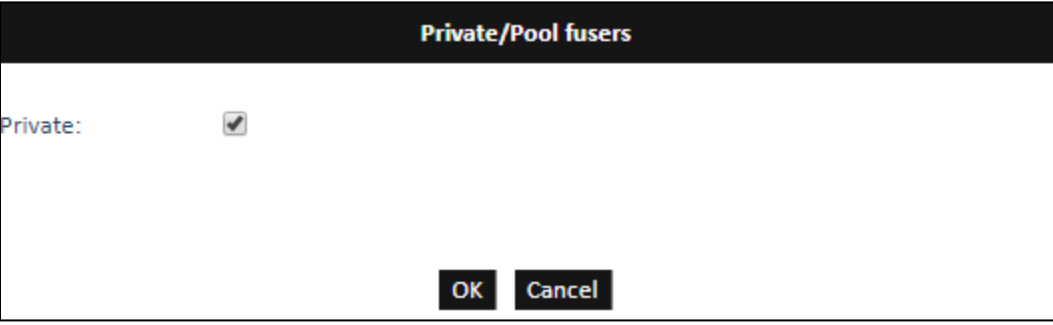
Time (sec): 300

OK Cancel

Set Idle Time

- Fusers can be made available to all master computers in a pool or restricted to a specific project's working folder. If you want to change fusers' availability in the pool, i.e., to either make them available in the pool (public) or restrict their use to a specific project (private), select the fusers' check boxes, and click **Private/Pool**. Then select the private check box to make them private (or clear to make them public), and click **OK**.

Note: Private fusers are not restored to the pool even after completing their current tasks. If you want to assign fusers to a specific project's working folder, but have them returned to the pool after they complete their tasks, clear this check box, and see step 3.

A dialog box titled "Private/Pool fusers" with a black header. Below the header, the text "Private:" is followed by a checked checkbox. At the bottom right, there are two buttons: "OK" and "Cancel".

Private/Pool fusers

Private: ☒

OK Cancel

- If you want to write a comment for certain fusers, select the fusers' check boxes, and click **Comment**. Then type the comment, and click **OK**.

A dialog box titled "Comment" with a black header. Below the header, the text "Comment:" is followed by a text input field. At the bottom right, there are two buttons: "OK" and "Cancel".

Comment

Comment:

OK Cancel

Using a Fusers Pool

The PhotoMesh master should be run from a shared network drive that is accessible to all fuser computers, to enable the PMFuserService.exe to launch a fuser that is the same version as the master.

To use a fusers pool:

1. Copy the PhotoMesh application files from the PhotoMesh master computer to a shared network drive that is accessible to all fuser computers, to enable the PMFuserService.exe to launch a fuser that is the same version as the master. Then run PhotoMesh from that path on the master computer.
2. In the **Build Manager**, toggle on **Use Fusers Pool**. The **Auto Scaling Fusers Settings** dialog is displayed. You can also access these settings from the Tools tab (Fuser Pool Settings command).

Note: The **Automatically use new available fusers** check box must also be selected, so that PhotoMesh will use the pool fusers when they become available.

3. Select the **Use Fusers Pool** check box, and enter the following information:

Fusers Pool Folder	Click Change to browse to the location of the pool folder. The pool folder must contain the pool.json file. See " Creating a Fusers Pool " in this chapter for more information. Note: If you are also using local fusers for a project, the working folder for the entire project, i.e. for both the local and pool fusers must be located in the pool directory.
Fuser Folder	Click Change to browse to the location of the fuser folder. This folder is set, by default, to the UNC path of the folder from where PhotoMesh is running so that the Fuser service launches the fuser version required by the master. Make sure that this path is accessible to all the fusers, e.g., by running PhotoMesh from a file share or network drive that is accessible to all fuser computers.
Return inactive fuser to pool after	Set the number of seconds after which inactive fusers should be returned to the pool.
Maximum Fusers	Type the maximum number of fusers to run. If fusers are available in the pool, PhotoMesh Build Manager adds pool fusers up to the maximum number set. Note: Make sure your pool limit does not exceed your license fuser limit, to avoid rejection of your fuser request by the pool.

4. If you also want to use AWS fusers, select the **Use AWS Instances** check box, and then enter the required information. See "[Using Amazon Web Services \(AWS\) Fusers](#)" in this chapter for information.
5. Click **OK**.
6. If you want to monitor and manage all pool fusers, e.g., assign fusers to specific projects' working folders and prevent their being restored to the pool, in the **Build Manager** dialog, click the **Pool Manager** link. See "[Managing a Fusers Pool](#)" in this chapter for information. You can also access the Pool Manager from the Tools tab.

Launching Multiple Fusers

PhotoMesh supports the running of multiple fusers on powerful machines for potentially enhanced performance. To run multiple fusers, a computer requires a high number of CPU cores (16 or more) and a large amount of RAM (32 GB or more). It is recommended that a minimum of 16 GB RAM and 8 cores be available per fuser. Multiple fusers on the same machine cannot be currently used when using Auto-Scaling AWS instances or a Fusers Pool.

As a general guideline, the optimal number of PhotoMesh fusers per machine can be calculated by taking the lower of – the total number of CPU cores in the machine divided by 8 and the total RAM (in GB)

divided by 16 (rounded down). It is highly recommended, however, to perform tests with different numbers of fusers since various factors such as I/O speed, network, memory cache can affect performance.

It is also recommended to use a dedicated graphics processing unit (GPU) when running multiple fusers on a single machine.

To launch multiple fusers from the command line:

- Add a unique ID parameter to the PhotoMeshFuser.exe command line.

Example:

To launch three PhotoMesh fusers in parallel on one machine, run these command lines:

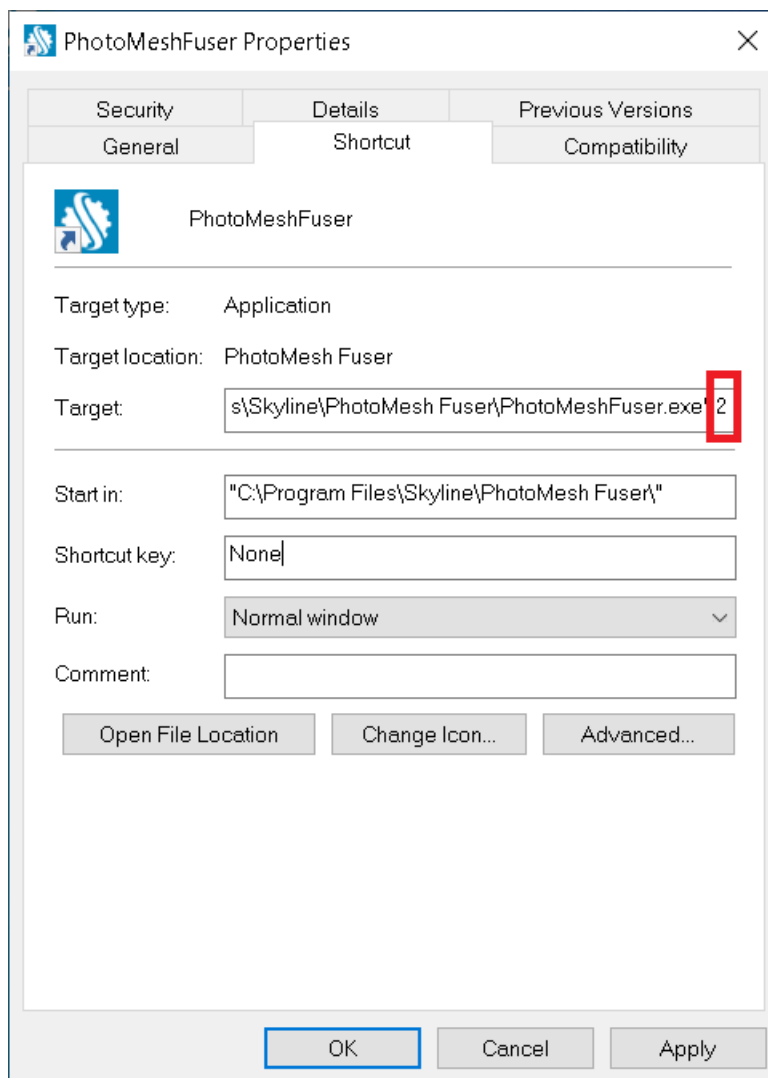
```
>"C:\Program Files\Skyline\PhotoMesh\Fuser\PhotoMeshFuser.exe" 1  
>"C:\Program Files\Skyline\PhotoMesh\Fuser\PhotoMeshFuser.exe" 2  
>"C:\Program Files\Skyline\PhotoMesh\Fuser\PhotoMeshFuser.exe" 3
```

A PhotoMesh fuser icon is placed in the Windows system tray for each of the launched fusers (in our example, the first fuser plus two more).



To launch additional fusers from the user interface, perform the following steps for each fuser that you want to add:

1. Make a copy of the fuser shortcut on your computer Desktop.
2. Right-click your new fuser shortcut and select **Properties**. The Properties dialog is displayed.



3. On the **Shortcut** tab, in the **Target** field, add the following at the end of the URL:

`"[unique numerical identifier]" "[path to working folder]"`

e.g.,

`"C:\Program Files\Skyline\PhotoMesh Fuser\PhotoMeshFuser.exe" "2" "[path to working folder]"`

4. Double-click the fuser shortcut to start the fuser. A PhotoMesh fuser icon is placed in the Windows system tray.

CHAPTER 15 GETTING STARTED WITH AWS

Overview

PhotoMesh is designed and built to fully exploit computer clusters and cloud computing. Cloud computing provides the flexibility to quickly scale up and down based on resource needs - even within a single project. Projects (or even steps within a single project) with demanding processing requirements can be run simultaneously on hundreds of virtual fuser machines (instances), vastly accelerating mesh model creation, while for less demanding projects (or project steps), users can quickly scale down. Users only pay-per-use, thus avoiding any significant upfront investment or wasted resources.

[Amazon Web Services](#) (AWS) is a secure cloud services platform that provides a perfect fit for PhotoMesh users interested in leveraging cloud products for scalable processing power with only minimal investment in infrastructure.

When setting up an AWS environment for a Linux fuser, you have two principal options: using AWS CloudFormation for an automated deployment, or manually configuring each component for more detailed control over the setup process. See "[Automated Deployment Using AWS CloudFormation](#)" for more information on automated deployment.

The general workflow involves the following steps:

- **Step 1:** Setting up an [Amazon Virtual Private Cloud \(VPC\)](#).
- **Step 2:** Creating the Master PhotoMesh machine, by launching and customizing [a Windows instance](#) from an existing [Amazon Machine Image \(AMI\)](#) and setting up storage on [Amazon EBS Volumes](#).
- **Step 3:** Creating a Linux AMI for a fuser computer by launching and customizing an instance from an existing AMI, and then creating a new AMI from the instance.
- **Step 4:** Building a PhotoMesh project by using the configured environment to create and then build a project.

The workflow outlined below is intended only as a recommendation for a typical workflow, and it can be modified based on other AWS and IT knowledge and preferences.

Note: The steps below based on [Amazon's AWS documentation](#), adapted for PhotoMesh's specific requirements.

Automated Deployment Using AWS CloudFormation

The AWS CloudFormation service can automate the process of creating a virtual private cloud and of creating/connecting to a master instance (steps 1 and 2 below).

Creating a Cloud Stack (Including configuring a VPC and launching an initial instance)

To create a cloud stack:

1. From the **AWS Management Console**, open the **CloudFormation** service.
2. Click **Create Stack** and select **Choose an existing template** (default).
3. In the **Amazon S3 URL** field, paste the following URL: https://skylinesoft-resources.s3.amazonaws.com/photomesh/cloudformation/skyline-pm-template_server2022.yaml
4. Click **Next** to continue.
5. Under **Stack** name, enter **PM-Master**.

6. Set the following values:
 - **masterInstanceStorageVolume**: Set the volume based on your expected need (E.g. 100 GB).
Note: This is the EBS storage that will hold your data, projects, PM files, etc. The EBS volumes will incur additional charges. For more information, see [Amazon EBS Volumes](#) in Amazon's AWS documentation.
 - **masterInstanceType**: g4dn.xlarge
 - **remoteAccess**: Enter the approved IP range. If you don't know this address range, you can use 0.0.0.0/0.
Note: For the rules above, when you use 0.0.0.0/0 for remote access, you enable all IP addresses to access your instance using RDP, SSH, or VNC. This is suitable for a short exercise, but it is unsafe for production environments. In production, you will want to authorize only a specific IP address or range of addresses to access your instance.
 - **vpcCIDR**: 10.0.0.0/16
 - **vpcName**: PM VPC.
7. Click **Next** to open the stack options configuration page.
8. **Optional:** Under **Tags**, add a key-value pair:
 - **Key**: Customer
 - **Value**: [Customer Name]
9. Click **Next** to review your settings.
10. Before submitting, confirm by checking the box stating: "I acknowledge that AWS CloudFormation might create IAM resources with custom names."
11. Click **Submit** to initiate stack creation. Stack creation will usually take between 10-15 minutes. You can monitor progress in the Events tab, which will open automatically.
12. Once the stack status is CREATE_COMPLETE, your CloudFormation stack is ready to use, and you can skip to [Step 3](#).
Note: If at any point, you need to delete a stack: navigate to the **CloudFormation** service, select the stack you want to delete in the **CloudFormation** dashboard, and click **Delete**.

Connecting to an Instance

To connect to a PM_Master instance:

1. **Access the Parameter Store:** From the **AWS Management Console**, navigate to the **AWS Systems Manager** service, and click **Parameter Store**.
2. **Retrieve the Key:** From the **Parameter Store**, click the specific key name related to the PM Master. Select **Show** to reveal the key, and copy the entire key text displayed.
3. Navigate to the **AWS EC2** service, and from the menu select **Instances**.
4. Right-click the EC2 instance you want to access and choose **Connect**. The **Connect To Your Instance** dialog box is displayed.
5. Select the **RDP client** tab, and click the **Get password** tab at the bottom of the screen.
6. Paste the copied key into the **Private key contents** field.
7. Click **Decrypt password** to unlock the instance password, and copy the decrypted password.
8. Click **Download remote desktop** to download the RDP file, and then open the downloaded file.
9. When prompted, log in to the instance, using the provided username ("Administrator") along with the decrypted password you copied previously.

10. Open **PhotoMesh** using the desktop shortcut (or from the shared path: \\10.0.0.10\d\PhotoMesh\PhotoMesh.exe)
11. In the dialog that is displayed, click **Run license Manager**. If a warning message is displayed, click **OK**.
12. Enter your license token in **File -> Set floating license server**

Step 1: Manual Deployment - Create a Virtual Private Cloud (VPC)

To create a Virtual Private Cloud for PhotoMesh production, configure a VPC and then create a security group.

See: [What is Amazon VPC](#) in Amazon's AWS documentation for more information.

Configuring a VPC

1. Open the Amazon VPC console at <https://console.aws.amazon.com/vpc/>.
2. On the dashboard choose **Create VPC**.
3. On the configuration page, enter the following information:
 - a. **VPC settings:** *VPC and more*
 - b. **Name tag auto-generation:** *PM VPC*
 - c. **IPv4 CIDR block:** X.X.X.X/16
This leaves 65,531 available addresses.
 - d. **Number of Availability Zones (AZs):** 1
 - e. **Number of public subnets:** 1
 - f. **Number of private subnets:** 0
 - g. **Customize subnets CIDR blocks:** Public subnet CIDR block: Change X.X.X.X/26 to X.X.X.X/22 (E.g., The default 10.0.0.0/22). This allows up to 1019 instances on your network.
 - h. **NAT gateways (\$):** none
 - i. **VPC endpoints:** none
4. Click **Create VPC**.
5. In the navigation pane, choose **Subnets**.
6. Select **PM VPC subnet** (E.g., PM VPC-subnet-public1-us-east-1a), choose **Actions**, and then **Edit Subnet Settings**.
7. Select the **Enable Auto-assign Public IPv4 address** check box.

Note: Enabling the Auto-assign Public IPv4 address option will result in additional charges. For more information, see the AWS documentation on [public IPv4 address charges](#).
8. Click **Save**.

Creating a Security Group

See: [Create a Security Group](#) in Amazon's AWS documentation for more information.

1. Open the Amazon VPC console at <https://console.aws.amazon.com/vpc/>.
2. In the navigation pane, choose **Security Groups**.
3. Choose **Create Security Group**.

4. In the **Security group name** field, enter *PM_SG* as the name of the security group, and type a description.
5. Select the ID of your PM VPC from the **VPC** menu, and then choose **Create**.
6. Select the **PM_SG** security group that you just created (you can view its name in the Group Name column).
7. Click **Actions** and then select **Edit inbound rules**.
8. On the **Inbound Rules** tab, click **Add rule** and input the parameters as follows below. Click **Save Rules** when you're done:
 - a. Select **RDP** (Remote Desktop Protocol) from the **Type** list, and enter your network's public IP address range in the **Source** field. If you don't know this address range, you can use 0.0.0.0/0.
Note: You can find your network's public IP address [here](#)
 - b. Choose **Add Rule**, and select **SSH** from the **Type** list. Then enter your network's public IP address range in the **Source** field. If you don't know this address range, you can use 0.0.0.0/0.
 - c. Choose **Add Rule**, select **Custom TCP Rule** from the **Type** list, and enter 5900-5901 (default VNC ports) in the **Port Range** field. Then enter your network's public IP address range in the **Source** field. If you don't know this address range, you can use 0.0.0.0/0.
Note: For the rules above, when you use 0.0.0.0/0 for the Source field, you enable all IP addresses to access your instance using RDP, SSH, or VNC. This is suitable for a short exercise, but it is unsafe for production environments. In production, you will want to authorize only a specific IP address or range of addresses to access your instance.
 - d. Choose **Add Rule**, select **Custom TCP Rule**, from the **Type** list, and enter 445 in the **Port Range** field. In the **Source** field, start typing *sg*, and select the group ID of your security group.
Note: You can also add another identical rule for your network's IP address range, to allow file sharing directly from your computer. If you don't know this address range, you can use 0.0.0.0/0, but keep in mind that this will enable all IP addresses to access your instance using file sharing. This is suitable for a short exercise, but it is unsafe for production environments. In production, you will want to authorize only a specific IP address or range of addresses to access your instance.

Step 2: Manual Deployment - Create a Master Instance

Creating an Amazon Machine Image (AMI) for the master computer is a multi-part step that entails: launching an instance from an existing AMI and customization of the instance.

To create a master Instance, follow the steps below. For more information, see: [Getting Started with Amazon EC2 Windows Instances](#) in Amazon's AWS documentation. If you are using a hybrid cloud architecture and running your PhotoMesh Manager locally, there is no need to create a master instance.

Creating an IAM Policy and Role

1. Open the IAM console at <https://console.aws.amazon.com/iam/>.
2. In the navigation pane on the left, choose **Policies**.
If this is your first time choosing **Policies**, the **Welcome to Managed Policies** page appears. Choose **Get Started**.
3. Choose **Create policy**.
4. Choose the **JSON** tab.
Note: You can switch between the **JSON** and **Visual editor** tabs any time and add each of the permissions below individually.

- Paste the following JSON policy document:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "VisualEditor0",
      "Effect": "Allow",
      "Action": [
        "ec2:DescribeInstances",
        "ec2:TerminateInstances",
        "ec2:RequestSpotInstances",
        "ec2:DescribeTags",
        "ec2:CreateTags",
        "ec2:DescribeRegions",
        "ec2:RunInstances",
        "s3:ListBucket",
        "ec2:DescribeSpotInstanceRequests",
        "ec2:DescribeSecurityGroups",
        "ec2:DescribeSpotPriceHistory",
        "ec2:DescribeImages",
        "ec2:CancelSpotInstanceRequests",
        "s3:GetObject",
        "iam:CreateServiceLinkedRole",
        "ec2:DescribeSubnets",
        "ec2:DescribeKeyPairs",
        "ec2:DescribeInstanceStatus"
      ],
      "Resource": "*"
    }
  ]
}
```

- When you are finished, choose **Next**.
- On the **Review and create** page, Policy name: enter *PM_RUNFUSER_POLICY*. Review the Permissions defined in this policy section to see the permissions that are granted by your policy. Then choose **Create policy** to save your work.
- In the navigation pane, choose **Roles**, and then choose **Create role**.
- For Trusted entity type, choose AWS service.
- For **Services or use case**, choose **EC2**. Then choose **Next**.
- In the list of policies, select the **PM_RUNFUSER_POLICY** policy. Then click **Next**. You can use the Filter menu to filter the list of policies.
- In the **Role name**, enter *PM_RUNFUSER_ROLE*.
- Review the role and then choose **Create role**.

Launching an Initial Instance and Creating Storage

To launch an initial instance and create storage:

- Open the Amazon EC2 console at <https://console.aws.amazon.com/ec2/>.

2. From the Amazon EC2 console dashboard, choose **Launch Instance**.
3. In the **Name and tags** section, tag your instance to help you identify it in the Amazon EC2 console after you launch it. Enter the following information: Under **Name** add: **PM Master**.
4. In the Application and OS Images (Amazon Machine Image) section, search for the following AMI from the AWS Marketplace list: Microsoft Windows Server 2022 Base, and then choose Select.
5. In the **Instance Type** section, select the **g4dn.xlarge** type.

Note: Using a GPU instance is recommended for normal operation of PhotoMesh Master over RDP connection. For productions with hundreds of fusers, other types can be used such as g4dn.8xlarge, which has faster network performance. For more information, see [Instance Types](#) in Amazon's AWS documentation.
6. In the **Key pair (login)** section,
 - a. Choose **Create a new key pair**.
 - b. Key pair name: *PM_KeyPair*.
 - c. Choose **Create key pair**
A key pair (*PM_KeyPair.pem*) will be downloaded.

Note: Store the file in a secure and accessible location since you will need the contents of the private key to connect to your instance after it is launched.
7. In the **Network settings** section, click **Edit** and then do the following:
 - a. **VPC – required:** Choose the **PM VPC** VPC created in step 4 of "[Configuring a VPC](#)".
 - b. Choose **Select an existing security group**.
 - c. Select the **PM_SG** group.
 - d. Open the **Advanced network configuration** tab, and enter the following information:
Network interface 1: Set the **Primary IP** to 10.0.0.10.
8. In the **Configure storage** section, enter the following information:
 - a. Change the **Volume Type** of the **Root** volume to **General Purpose SSD (gp3)**
 - b. Click **Add New Volume**. Then set **Size** based on your expected need (E.g. 100 GiB). This is the EBS storage that will hold your data, projects, PM files, etc.

Note: The EBS volumes will incur additional charges. For more information, see [Amazon EBS Volumes](#) in Amazon's AWS documentation.

 - c. **Volume Type:** Choose a volume type based on your expected need (e.g., General Purpose SSD (gp2)), and use the default IOS and throughput for this volume type. It is recommended to select **General Purpose SSD (gp3)**. For more information, see [Amazon EBS Volume Types](#).
9. In the **Advanced details** section, enter the following information:
 - a. **IAM instance profile:** Choose the **PM_RUNFUSER_ROLE** created in step 14 of "[Creating an IAM Policy and Role](#)".
 - b. **Termination protection:** Select **Enable**.
10. On the **Summary** side panel, check the details of your instance, and make any changes if necessary. When you are ready, choose **Launch instance**.

Connecting to the Instance

See: [Connecting to Your Windows Instance Using RDP](#) in Amazon's AWS documentation for more information.

1. Open the Amazon EC2 console at <https://console.aws.amazon.com/ec2/>

2. In the navigation pane, choose **Instances**.
3. Select the instance, and then choose **Connect**.
4. In the **Connect To Your Instance** dialog, choose **RDP client tab** (it will take a few minutes after the instance is launched before the password is available).
5. Choose **Get password -> Upload private key file** and navigate to the private key file you created when you launched the instance. Select the file and choose **Open** to copy the entire contents of the file into the contents box.
6. Choose **Decrypt Password**. The console displays the default administrator password for the instance in the **Connect To Your Instance** dialog, replacing the link to **Get Password** shown previously with the actual password.
7. Record the default administrator password, or copy it to the clipboard. You need this password to connect to the instance.
8. Choose **Download Remote Desktop File**. Your browser prompts you to either open or save the .rdp file. Either option is fine. When you have finished, you can choose **Close** to dismiss the **Connect To Your Instance** dialog.
9. You may get a warning that the publisher of the remote connection is unknown. Choose **Connect** to connect to your instance.
10. When prompted, log in to the instance, using the administrator account for the operating system and the password that you recorded or copied previously. If your **Remote Desktop Connection** already has an administrator account set up, you might have to choose the **Use another account** option and enter the user name and password manually.
11. Due to the nature of self-signed certificates, you may get a warning that the security certificate could not be authenticated. Choose **Yes** or **Continue** to continue.
12. A Windows **Networks** message is displayed asking if "you want to allow your PC to be discoverable by other PCs and devices on this network?" Click **Yes**.
13. After you connect, we recommend that you change the administrator password from the default value. You change the password while logged on to the instance itself, just as you would on any other Windows Server.

Note: Due to the Remote Desktop Protocol (RDP), the recommended method for changing the password is to press CTRL+ALT+END, and then select **Change a password**.

Customizing the Instance

See: [Making an Amazon EBS Volume Available for Use on Windows](#) and [Windows GPU Instances](#) in Amazon's AWS documentation for more information.

1. While connected to the Master instance using RDP, start **Windows File Explorer**.
2. Start the Disk Management utility. On the taskbar, open the context (right-click) menu for the Windows logo and choose **Disk Management**.
3. If Disk 1 requires initialization, right-click on it and choose **Initialize Disk**.
4. Open the context (right-click) menu for the right panel for Disk 1 and choose **New Simple Volume**. Complete the wizard with the default settings.
5. Give share permissions with **Full Control** to **Everyone** on the D drive: In **Windows File Explorer**, right-click on the D drive and select **Properties**. Click the **Sharing** tab, and then click the **Advanced Sharing** button. Select the **Share this folder** check box. Click **Permissions**, and then click **Add**, and type **Everyone**. Select **Full Control**. Then click **Apply**.

6. Install PhotoMesh using the standard installation to *D:\PhotoMesh*.
7. Create the folder *D:\PMWorkingFolder*, and then then the subfolder *A* directly below it, so that you have the directory structure: *D:\PMWorkingFolder\A*.
8. Create the folder *D:\PMProjects*.
9. Start PhotoMesh from "*\\10.0.0.10\D\PhotoMesh\PhotoMesh.exe*".
10. From the **File** menu, select **Options (F9)**.
11. Change the **Working Folder** to *\\10.0.0.10\D\PMWorkingFolder\A*.
12. Click **OK** and then **close** PhotoMesh.
13. Install NDVIDIA drivers on Windows for your instance. See "[NVIDIA gaming drivers \(G5 and G4dn instances\)](#)" in Amazon's AWS documentation for more information.

Step 3: Create a Fuser AMI (Linux)

Creating a Linux AMI for a fuser computer is a multi-part step that entails: launching an instance from an existing AMI, customizing the instance, and finally creating a new AMI from the instance.

To create a fuser AMI, follow the steps below. For more information, see: [Getting Started with Amazon EC2 Linux Instances](#) in Amazon's AWS documentation.

Linux fuser AMI's are generally recommended over Windows AMI's since Linux instances are more cost effective to run.

Launching an Initial Instance

1. Open the Amazon EC2 console at <https://console.aws.amazon.com/ec2/>.
2. From the Amazon EC2 console dashboard, choose **Launch Instance**.
3. In the **Name and Tags** section, enter **Name: PM Linux Fuser Initial**.
4. In the **Application and OS Images (Amazon Machine Image)** section, choose **Ubuntu** from the **Quick Start** list (Amazon Machine Image (AMI): **Ubuntu Server 24.04 LTS (HVM), SSD Volume Type**).
5. In the **Instance Type** section, select the **g4dn.xlarge** type.

Note: The instance type selected here depends on the fusers you intend to use to build the project. The **g4dn.xlarge** instance type should be used to create an AMI for G4, G5 and G6 instances.
6. In the **Select Key pair (login)** section, choose the following dropdown: **Key pair name:** *PM_KeyPair*.
7. In the **Network Settings** section click **EDIT**:
 - a. **VPC dropdown:** Choose the **PM VPC** VPC created in step 4 of "[Configuring a VPC](#)".
 - b. **Firewall (security groups):** Select "**Select existing security group**" and choose the **PM_SG** group.
8. In the **Configure storage** section, do the following:
 - a. Change the default **Root** volume **Size** to at least 70 GiB.

Note: The EBS volumes will incur additional charges. For more information see [Amazon EBS Volumes](#) in Amazon's AWS documentation.
 - b. Change the **Volume Type** to **General Purpose SSD (gp3)**.

9. In the **Advanced details** section, do the following: In the **IAM instance profile** field, choose the IAM role "PM_RUNFUSER_POLICY" created in "[Creating an IAM Policy and Role](#)".
10. To launch your instance, click **Launch Instance**.

Convert Your Private Key Using PuTTYgen

See: [Connecting to Your Linux Instance from Windows Using PuTTY](#) in Amazon's AWS documentation for more information.

Note: For information on additional ways to connect to your Linux instance, see:

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/AccessingInstances.html>

1. Download and install PuTTY from the [PuTTY download page](#). If you already have an older version of PuTTY installed, we recommend that you download the latest version. Be sure to install the entire suite.
2. From the **Start** menu -> **All Programs** -> **PuTTY** -> **PuTTYgen**.
3. Under **Type of key to generate**, choose **RSA**. If you're using an older version of PuTTYgen, choose **SSH-2 RSA**.
4. Choose **Load**. By default, PuTTYgen displays only files with the extension .ppk. To locate your .pem file, choose the option to display files of all types.
5. Select your .pem file for the key pair that you specified when you launched your instance and choose **Open**. PuTTYgen displays a notice that the .pem file was successfully imported. Choose **OK**.
6. To save the key in the format that PuTTY can use, choose **Save private key**. PuTTYgen displays a warning about saving the key without a passphrase. Choose **Yes**.

Note: Using a passphrase on a private key adds an additional layer of security. Even if someone gains access to your private key, they cannot use it without the passphrase. However, the downside of using a passphrase is that it makes automation harder because manual input is required to log on to an instance, or to copy files to an instance.

7. Specify the same name for the key that you used for the key pair (for example, *PM_KeyPair*) and choose **Save**. PuTTY automatically adds the .ppk file extension.
Your private key is now in the correct format for use with PuTTY. You can now connect to your instance using PuTTY's SSH client.

Connecting to the Instance

To connect:

1. Start PuTTY (from the **Start** menu -> **All Programs** -> **PuTTY** -> **PuTTY**).
2. In the **Category** pane, choose **Session** and complete the following fields:
 - a. To connect using your instance's public DNS, in the **Host Name** box, enter *ubuntu@public_dns_name*.
You can get the public DNS for your instance using the Amazon EC2 console. Check the **Public DNS (IPv4)** column. If this column is hidden, choose the **Show/Hide** icon and select **Public DNS (IPv4)**.
 - b. Ensure that the **Port** value is 22.
 - c. Under **Connection type**, select **SSH**.
3. (Optional) You can configure PuTTY to automatically send 'keepalive' data at regular intervals to keep the session active. This is useful to avoid disconnecting from your instance due to session inactivity. In the **Category** pane, choose **Connection**, and then enter the required interval in the **Seconds**

between keepalives field. For example, if your session disconnects after 10 minutes of inactivity, enter **180** to configure PuTTY to send keepalive data every 3 minutes.

4. In the **Category** pane, expand **Connection > SSH > Auth >** choose **Credentials**. Enter the following:
 - a. In the **Private key file for authentication** field, choose **Browse**.
 - b. Select the .ppk file that you generated for your key pair and choose **Open**.
 - c. (Optional) If you plan to start this session again later, you can save the session information for future use. Under **Category**, choose **Session**, enter a name for the session in **Saved Sessions**, and then choose **Save**.
 - d. Choose **Open**.
5. If this is the first time you have connected to this instance, PuTTY displays a security alert dialog asking whether you trust the host to which you are connecting. Choose **Accept**. A window opens and you are connected to your instance.
6. Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, **ubuntu**.

Customizing the Instance

See: [Linux Accelerated Computing Instances](#) in Amazon's AWS documentation for more information.

1. Add 32-bit support and update the instance: In the PuTTY terminal window, type:

```
sudo dpkg --add-architecture i386
sudo apt-get -y update
sudo apt-get -y upgrade
```

2. Download all necessary packages, drivers and software (This step can take a while): In the PuTTY terminal window, type:

```
sudo apt-get -y install pkg-config unzip x11vnc xterm fvwm dkms xserver-xorg xvfb ubuntu-drivers-common cifs-utils nvidia-driver-550 wine winetricks winbind mesa-utils
```

3. Restart the instance:

```
sudo reboot
```

The following message is displayed: "Remote side unexpectedly closed network connection". Click **OK**.

4. Right-click the title bar of the PuTTY terminal window, and select **Restart Session**.
5. (Optional) To test the successful Nvidia driver installation, in the PuTTY terminal window, type:

```
nvidia-smi
```

6. To configure the GPU and start the X window system, in the PuTTY terminal window, type:

```
sudo nvidia-xconfig -a
sudo /usr/bin/X :0 &
export DISPLAY=:0
```

7. (Optional) To configure VNC in order to verify X runs properly, in the PuTTY terminal window type:

- a. **x11vnc -storepasswd** Then type and verify the password.
- b. **y**
This confirms writing the password to `/home/Ubuntu/.vnc/passwd`.
- c. **x11vnc -rfbauth ~/.vnc/passwd &**
- d. **fvwm &**

8. (Optional) Connect with VNC client using your instance's public DNS and the password set in step 7 above.
You can get the public DNS for your instance using the Amazon EC2 console. Check the **Public DNS (IPv4)** column. If this column is hidden, choose the **Show/Hide** icon and select **Public DNS (IPv4)**.
9. To configure [Wine](#), do the following:
 - a. Click the **Start**  button, type "cmd", and then select **Command Prompt** from the list.
 - b. In the command prompt window type the following and click **Enter**:

```
pscp -i <.ppk file> <wineinst.tar full path> ubuntu@<fuser IP>:/home/ubuntu/wineinst.tar
```

 - Replace <.ppk file> with the full path to the ppk file you created in "[Convert Your Private Key Using PuTTYgen](#)."
 - Replace < wineinst.tar full path> with the full path to the wineinst.tar file included in the zip folder that can be [downloaded from here](#).
 - Replace <fuser IP> with the Public DNS (IPv4) of the instance.
 - c. In the PuTTY terminal window, type:

```
tar -xvf wineinst.tar
sh fuser.sh
```

Note: This process can take a few minutes to complete.
10. Once finished, shutdown the instance:

```
sudo shutdown -h now
```
11. In the Amazon EC2 console, after the instance has stopped, right-click the instance, point to **Instance Settings**, and select **Change Instance Type**. The Change Instance Type dialog is displayed.
12. From the **Instance Type** dropdown list, select **g4ad.4xlarge**, and click **Apply**.
13. Restart the instance by right-clicking the instance in the Amazon EC2 console and selecting **Start Instance**.
14. Reconnect to the instance via PuTTY by performing the steps in "[Connecting to the Instance](#)".
15. To download and install AMD drivers, follow these steps in the PuTTY terminal window:
 - a. Install AWS CLI by typing the following command:

```
sudo apt install awscli
```

Note: When prompted, type 'Y' to confirm and continue the installation.
 - b. Download and extract the AMD drivers and install additional kernel modules by typing the following commands:

```
aws s3 cp --recursive s3://ec2-amd-linux-drivers/latest/ .
tar -xvf amdgpu-pro*ubuntu*.xz
sudo apt install linux-modules-extra-$(uname -r) -y
```

Note: After extracting the driver, change to the folder containing the extracted files:

 - i. Type the following (without pressing Enter):

```
cd amd
```
 - ii. Press tab once to auto-complete the folder name.
 - iii. Press Enter.
 - c. Add the AMD CPG key by typing:

```
cat RPM-GPG-KEY-amdgpu | sudo apt-key add -
```

- d. Install the AMD drivers by typing:


```
./amdgpu-pro-install -y --opencl=pal,legacy
```
- e. Reboot the system to apply the changes:


```
sudo reboot
```
16. The following message is displayed: "Remote side unexpectedly closed network connection". Click **OK**.
17. Right-click the title bar of the PuTTY window, and select **Restart Session**.
18. Test that the installation was successful by typing the following in the PuTTY terminal window:


```
dmesg | grep amdgpu
```
19. Check the response and look for "Initialized amdgpu". This indicates that the installation was successful.

Saving a Fuser AMI

See: [Creating an Amazon EBS-Backed Linux AMI](#) in Amazon's AWS documentation for more information.

1. In the navigation pane, choose **Instances** and select the **PM Linux Fuser Initial instance**.
2. Click the instance ID to display the Instance summary screen.
3. Click the **Storage** tab and click the volume ID.
4. Select the volume. Then from the **Actions** dropdown list at the top left, select **Create snapshot**.
5. In the description, enter **PM Linux Fuser Initial snapshot** (this process takes several minutes).
6. Notice the green ribbon at the top of the screen, and click the ID of the snapshot which is being created.
7. Copy the snapshot ID.
8. Use the following command in [CloudShell](#) to create a new AMI. Replace <Snapshot_ID> with the one copied in the previous step:

```
aws ec2 register-image \
  --name "PM Linux Fuser Image" \
  --root-device-name "/dev/sda1" \
  --block-device-mappings
  "DeviceName=/dev/sda1,Ebs={SnapshotId=<Snapshot_ID>,VolumeType=gp3}" \
  --architecture x86_64
```

9. Close **CloudShell**.
10. Choose **AMIs** in the navigation pane and record the AMI ID of the created AMI, or copy it to the clipboard. You need this AMI ID for PhotoMesh to launch fuser instances.
11. In the navigation pane, choose **Instances** and select the **PM Linux Fuser Initial** instance.
12. Choose **Actions**, select **Instance State**, and then choose **Terminate**.

Step 4: Build a PhotoMesh Project

This step outlines the workflow for building a PhotoMesh project. The option to run the entire PhotoMesh production with only cloud resources is the main workflow discussed above, and then in "[Building a PhotoMesh Project Using the Master Instance](#)" below. It is also possible to apply a hybrid cloud architecture that mixes cloud and on-premises computing power and storage solutions. This option is covered in "[Creating an IAM User for Running PhotoMesh Project Using an On-Premises Master](#)" and "[Building a PhotoMesh Project Using an On-Premises Master and Storage](#)".

There are a number of variables involved in the selection of fuser instances to process a build:

- **CPU/GPU** – The hardware of the host computer used for your instance determines its compute, memory, storage, and graphic capabilities. GPU-based instances provide access to NVIDIA GPUs to accelerate processing. The need for GPUs is generally determined based on build step. See the table below.
- **Price/Performance** – Optimized balance between machine price and its computing power. See here for more information: <https://aws.amazon.com/ec2/pricing/>. See the table below for Skyline recommendations (based on our experience in the US East – N. Virginia region).
- **Availability Zones-Subnets** – The availability of the required number of instances and their price should be considered when selecting an availability zone. Since you are also charged for data transfer from one availability zone to another, if you require many fusers, it is recommended to launch the master in an availability zone with good spot availability to avoid significant cross-availability zone charges. A subnet must be created for each availability zone. For more information, consult the [Amazon EC2 console as described below](#).
- **onDemand/ Spot** – onDemand instances provide guaranteed compute resources, while spot instances are dependent on the availability of unused instances: spot instances will only be launched if an instance is available and instance use may be interrupted unexpectedly if capacity is no longer available. onDemand instances also have a static price, while spot instance price varies based on demand. For more information see [Amazon EC2 Using Spot Instances](#) in Amazon's AWS documentation.
- **Number of Fusers** – The recommendations below are for instances with 1 fuser. Stronger instances can run more than 1 fuser, which allows for more efficient use of the instance's resources. As a general guideline, instances with 16GB RAM can run 1 fuser, 32GB RAM can run 2 fusers and so on. Running multiple fusers on stronger instances can potentially result in better performance per price, depending on the price ratio between different sizes of the same instance type. For example, if the price of m7a.4xl (2 fusers) is less than twice the price of m7a.2xl (1 fuser), it might be more cost-efficient. For more information, see [here](#).

Analyzing the price history of spot instances in [Amazon EC2 console's Pricing History](#) can help you determine the availability of different spot instance types in different availability zones. Price is generally determined by the law of supply and demand, so if the price for a particular instance type is low in a specific zone, this usually indicates higher availability of spot instances of that type in the zone. No price is listed for a particular zone if there are no spot instances of that specific machine type in this zone.

See the chart below for our recommendations for each PhotoMesh step.

Recommendations Based on PhotoMesh Step

PhotoMesh Step	I/O (Disk IOPS and Network bandwidth)	CPU	GPU	Recommended # of fusers / 1Gbps	Recommended Machine Types	Preferred Method for Hybrid
Data Preparation	Intensive	Intensive	Low	5	M7a/i.xlarge C7a/i.2xlarge M6a/i.xlarge C6a/i.2xlarge	Local
AT*	Intensive	Intensive	No	20	M7a/i.xlarge C7a/i.2xlarge M6a/i.xlarge C6a/i.2xlarge	Local
Point Cloud	Low	High	No	200	M7a/i.xlarge C7a/i.2xlarge M6a/i.xlarge C6a/i.2xlarge	AWS
Model	Low	High	No	200	M7a/i.xlarge C7a/i.2xlarge M6a/i.xlarge C6a/i.2xlarge	AWS
Texture Mesh	High	Low	High	50	G6.xlarge G4ad.xlarge G4dn.xlarge G5.xlarge	AWS
Output	High	Low	High	20	G6.xlarge G4ad.xlarge	Local

G4dn.xlarge
e
G5.xlarge

Building a PhotoMesh Project Using the Master Instance

To use the configured environment to produce PhotoMesh projects, follow the steps below:

1. Connect to the **PM Master** instance.

Note: If the PM Master instance is not already running, choose **Actions**, select **Instance State**, and then choose **Start**.

2. Create the directory: "D:\PhotoMesh\AWSConfigurations", and then copy the JSON files ("DataPrep_AT.json", "Reconstruction_CPU.json", and "Reconstruction_GPU.json") from the zip file downloaded in step 6b of "[Customizing the \(Fuser\) Instance](#)" to this directory.

3. Start **PhotoMesh** and **create** a project.

Note: The project must be created in a location that is accessible to both master and fuser instances. It is recommended to organize all projects and sources in the \\10.0.0.10\d\PMProjects\ directory.

4. On the **Tools** tab, in the **Fuser** Automation group, click **AWS Instance Settings**.

5. In the **Auto-Scaling Fusers Settings** dialog, do the following:

- a. Check the **Use AWS Instances** check box.
- b. **Maximum Price Per Spot Instance:** Set the maximum price you want to pay per EC2 spot instance. For more information, see [Amazon EC2 Spot Instance Pricing](#) in Amazon's AWS documentation.
- c. **Maximum Instances:** Set the number based on the number of EC2 instances you want to run.
- d. **Launch Rate:** Set the maximum number of fuser instances to launch per minute, e.g. if Maximum Instances was set to 200, and the Launch Rate to 50, the Build Manager will spread the launch of the fusers over 4 minutes. This prevents a potential bottleneck that could be caused by a large number of fusers attempting to read the build data at the same time.
- e. **Terminate AWS instances when build finished:** Select the check box if you want to terminate AWS fuser instances when the build is complete.
- f. **Configuration Name:** Select based on the current processing step. Select the GPU_Reconstruction option if you want to run an entire build without needing to reselect the configuration before each build step, since this option can be used for all build steps.
 - GPU_Reconstruction – For texture and output steps.
 - AT_MultiCoreCPU – Multi-core CPU for data preparation and aerotriangulation steps.
 - Reconstruction_CPU – For point cloud and model steps.

Note: These configuration options are read from the JSON files copied in step 2 above. Each of these JSON files contains an array of instance types that PhotoMesh should use, ordered based on preferability. During production, PhotoMesh iterates over the elements of the array, beginning with the first instance type, until it finds an available instance. PhotoMesh will always attempt first to launch instances from the same availability zone as the master, and only if no instances are available, will it launch an instance from a different zone.

Each instance object in the array has four properties: ami, instanceType, demandType, and fusersCount. The instanceType defines the hardware of the host computer used for your instance. Each instance type offers different compute, memory, and storage

capabilities. The demandType determines the pricing arrangement: "onDemand" or "spot." The fusersCount determines the number of PhotoMesh processes to run on each instance. See the [overview](#) to step 4 for some of the issues to consider when selecting instance types.

- g. **Build Priority:** Select either **Choose all listed options**, to use all machine types listed in the JSON file, with priority based on the order they are listed in the file, or select a specific machine.
 - h. **Security Group ID:** Select **PM_SG**.
 - i. **Subnet ID:** Click **Additional Subnets...** and type a list of all the subnet ID's you want to use. These were created in "[Configuring a VPC](#)". A subnet must be created for each availability zone. Since you are charged for data transfer from one availability zone to another, it is recommended to launch the master in an availability zone with good spot availability. PhotoMesh will always attempt first to launch instances from the same availability zone as the master, and only if no instances are available, will it launch an instance from a different zone. See the [overview](#) to step 4 for some of the issues to consider when selecting a subnet/availability zone.
 - j. **Key Pair Name:** Select **PM_KeyPair**.
 - k. Click **Edit** (next to **Configuration Name**), insert in all fields the AMI ID of the PM Linux Fuser Image (created in "[Saving a Fuser AMI](#)"), and then click **OK**.
 - l. Select **Shutdown inactive fuser instances after** and set its value to "1".
 - m. Click **Edit Startup Script** and paste the template script from the zip file downloaded in step 6b of "[Customizing the \(Fuser\) Instance](#)" (*Startup_Script_awsQG_Template*). Edit the script so it includes the username and the password of your PM Master machine set in step 13 of "[Connecting to the \(Master\) Instance](#)" (In the script, replace all <pass> and <user> placeholders with the required value).
 - n. Click **OK**.
6. Click **Build**.
- Note:** If a message is displayed that "PhotoMesh's fuser request was rejected by AWS, since it exceeded the instance limit", you need to request a quota increase for "Spot instance requests". **See:** [Amazon EC2 Service Limits](#) in Amazon's AWS documentation for more information.
7. In the **PhotoMesh Build Manager** dialog, you can monitor the creation and automatic use of the automatically launched fuser instances. You can also monitor the creation of your instances by choosing **Instances** in the navigation pane of the Amazon EC2 console, and reviewing the custom tags added to PhotoMesh launched instances: Build start time, Owner, Project, Type, and Working folder.
8. When processing is complete, stop the PM Master instance.
- Note:** Do not terminate the master instance; only stop and start as needed.
- Note:** It is recommended to verify that the instances are terminated according to the settings defined in the **AWS Cloud Settings** dialog, particularly the first time they are terminated.

Creating an IAM User for Running PhotoMesh Project Using an On-Premises Master

To allow PhotoMesh Master running on an on-premises computer to launch EC2 instances, create an IAM user and assign permissions.

See: [Creating IAM Users \(Console\)](#) in Amazon's AWS documentation for more information.

1. Open the Amazon IAM console at <https://console.aws.amazon.com/iam/>.
2. In the navigation pane, choose **Users** and then choose **Add user**.

3. Type the user name for the new user.
4. Select the **Programmatic access** check box. Then choose **Next: Permissions**.
5. On the Set Permissions page, choose Attach policies directly.
6. In the list of policies, select the **PM_RUNFUSER_POLICY** policy. You can use the Filter menu to filter the list of policies.
7. Choose Next: Review and Create.
8. Review the role and then choose **Create user**.
9. To view the users' access keys (access key IDs and secret access keys), choose **Show** next to each password and access key that you want to see. To save the access keys, choose **Download .csv** and then save the file to a safe location.

Note: This is your only opportunity to view or download the secret access keys. Save the user's new access key ID and secret access key in a safe and secure place. You will not have access to the secret keys again after this step.

Building a PhotoMesh Project Using an On-Premises Master and Storage

Before beginning a build using an on-premises master and storage, ensure that your fuser machines have access to your local storage, e.g., via [AWS Direct Connect](#), [Site-to-Site VPN](#), or Security Group and firewall rules. Sufficient bandwidth for the data transfer between your on-premises storage and the fuser machines significantly affects production performance. See the [overview](#) to step 4 for some of the issues to consider regarding the hybrid architecture.

To use the configured environment to produce PhotoMesh projects using on-premises storage and a PhotoMesh that is installed on an on-premises computer while utilizing AWS fuser instances, follow the steps below:

1. Make sure your on-premises computer has PhotoMesh installed.

Note: File access can be achieved in different ways, e.g. via [AWS Direct Connect](#), [Site-to-Site VPN](#), or Security Group and firewall rules. Consult AWS documentation, support, your IT person, or Skyline support for more information.
2. Create the directory: ".\PhotoMesh\AWSConfigurations", and copy the JSON files ("DataPrep_AT.json", "Reconstruction_CPU.json", and "Reconstruction_GPU.json") from the zip file downloaded in step 6b of "[Customizing the \(Fuser\) Instance](#)" to this directory.
3. Start **PhotoMesh** and **create** a project.

Note: The project must be created in a location that is accessible to all fuser instances.
4. On the **Home** tab, in the **Process** group, click **Build**, then enter the Build Parameters and click **Build**.
5. In the PhotoMesh Build Manger dialog, toggle on Use AWS Instances (Change).
6. In the **AWS Security** dialog, enter your AWS credentials created in "[Creating an IAM User for Running PhotoMesh Project Using an On-Premises Master](#)".
7. In the **Auto-Scaling Fusers Settings** dialog, do the following:
 - a. Check the **Use AWS Instances** check box.
 - b. **Maximum Price Per Spot Instance:** Set the maximum price you want to pay per EC2 spot instance. For more information see [Amazon EC2 Spot Instance Pricing](#) in Amazon's AWS documentation.
 - c. **Maximum Instances:** Set the number based on the number of EC2 instances you want to run.
 - d. **Launch Rate:** Set the maximum number of fuser instances to launch per minute, e.g. if Maximum Instances was set to 200, and the Launch Rate to 50, the Build Manager will spread the launch of

the fusers over 4 minutes. This prevents a potential bottleneck that could be caused by a large number of fusers attempting to read the build data at the same time.

- e. **Terminate AWS instances when build finished:** Select the check box if you want to terminate AWS fuser instances when the build is complete.
- f. **Build Priority:** Select either **Choose all listed options**, to use all machine types listed in the JSON file, with priority based on the order they are listed in the file, or select a specific machine.
- g. **Security Group ID:** Select **PM_SG**.
- h. **Subnet ID:** Click **Additional Subnets...** and type a list of all the subnet ID's you want to use. These were created in "[Configuring a VPC](#)". A subnet must be created for each availability zone. See the [overview](#) to step 4 for a comparison of onDemand/spot instances and information on using pricing history to determine instance availability.
- i. **Key Pair Name:** Select **PM_KeyPair**.
- j. **Configuration Name:** Select based on the current processing step. Select the **Reconstruction_GPU** option if you want to run an entire build without needing to reselect the configuration before each build step, since this option can be used for all build steps.
 - **Reconstruction_GPU** – For texture and output steps.
 - **AT_MultiCoreCPU** – Multi-core CPU for data preparation and aerotriangulation steps.
 - **Reconstruction_CPU** – For point cloud and model steps.

Note: These configuration options are derived from the JSON files copied in step 2 of "[Building a PhotoMesh Project Using the Master Instance](#)". Each of these JSON files contains an array of instance types that PhotoMesh should use, ordered based on preferability. During production, PhotoMesh iterates over the elements of the array, beginning with the first instance type, until it finds an available instance.

Each instance object in the array has four properties: `ami`, `instanceType`, `demandType`, and `fusersCount`. The `instanceType` defines the hardware of the host computer used for your instance. Each instance type offers different compute, memory, and storage capabilities. The `demandType` determines the pricing arrangement: `onDemand` or `spot`. The `fusersCount` determines the number of PhotoMesh processes to run on each instance. See the [overview](#) to step 4 for some of the issues to consider when selecting instance types.
- k. Click **Edit** (next to **Configuration Name**), insert in all fields the AMI ID of the PM Linux Fuser Image (created in "[Saving a Fuser AMI](#)"), and then click **OK**.
- l. Select **Shutdown inactive fuser instances after** and set its value to "1".
- m. Click **Edit Startup Script** and paste the template script from the zip file downloaded in step 6b of "[Customizing the \(Fuser\) Instance](#)" (*Startup_Script_Template.txt*). Edit the script so it includes the username and the password of your PM Master machine set in step 13 of "[Connecting to the \(Master\) Instance](#)" (In the script, replace all `<serverip >`, `<share>`, `<pass>` and `<user>` placeholders with the required value).
- n. Click **OK**.
8. Click **Build**.

Note: A message is displayed informing you that no available fusers are currently running and asking if you want to start a local fuser. If you want to start a fuser on the master machine to take advantage of its computing resources, click **Yes**. If you are running many spot fuser instances, however, it is recommended to click **No** to free up computing resources for management of the build and file server tasks.

Note: If a message is displayed that "PhotoMesh's fuser request was rejected by AWS, since it exceeded the instance limit", you need to request a limit increase for "Spot instance requests". **See:** [Amazon EC2 Service Limits](#) in Amazon's AWS documentation for more information.

9. In the **PhotoMesh Build Manger** dialog, you can monitor the creation and automatic use of the automatically launched fuser instances. You can also monitor the creation of your instances by choosing **Instances** in the navigation pane of the Amazon EC2 console, and reviewing the custom Tags added to PhotoMesh launched instances: Build start time, Owner, Project, Type, and Working folder.
10. It is recommended to verify that the instances are terminated according to the settings defined in the **AWS Cloud Settings** dialog, particularly the first time they are terminated.

CHAPTER 16 BUILDING

About Building

After all photos are added to the project and all properties are set as required, you can begin the build process. The build process provides flexibility with regard to the AT area, reconstruction area, build steps that are to be completed, and output formats.

The build process entails a series of sequential steps beginning with data preparation and aerotriangulation and followed by three reconstruction steps: point cloud creation, model creation, and model texturing, and then project output. The required build steps are selected in the Build Parameters dialog. The final output can be created in any of the following output formats: 3DML (for loading directly into TerraExplorer and streaming with SkylineGlobe Server), DAE, OBJ, 3D Tiles, SLPK (I3S), OSGB, Orthophoto, DSM, DTM/DEM, and LAS (Point cloud public file format). **See "Setting Build Steps, Parameters, and Outputs"** in this chapter for information on the different file formats.

As the build process is progressing, the 3D Window displays the interim build results and PhotoMesh Build Manager displays information about the current step and active fusers.

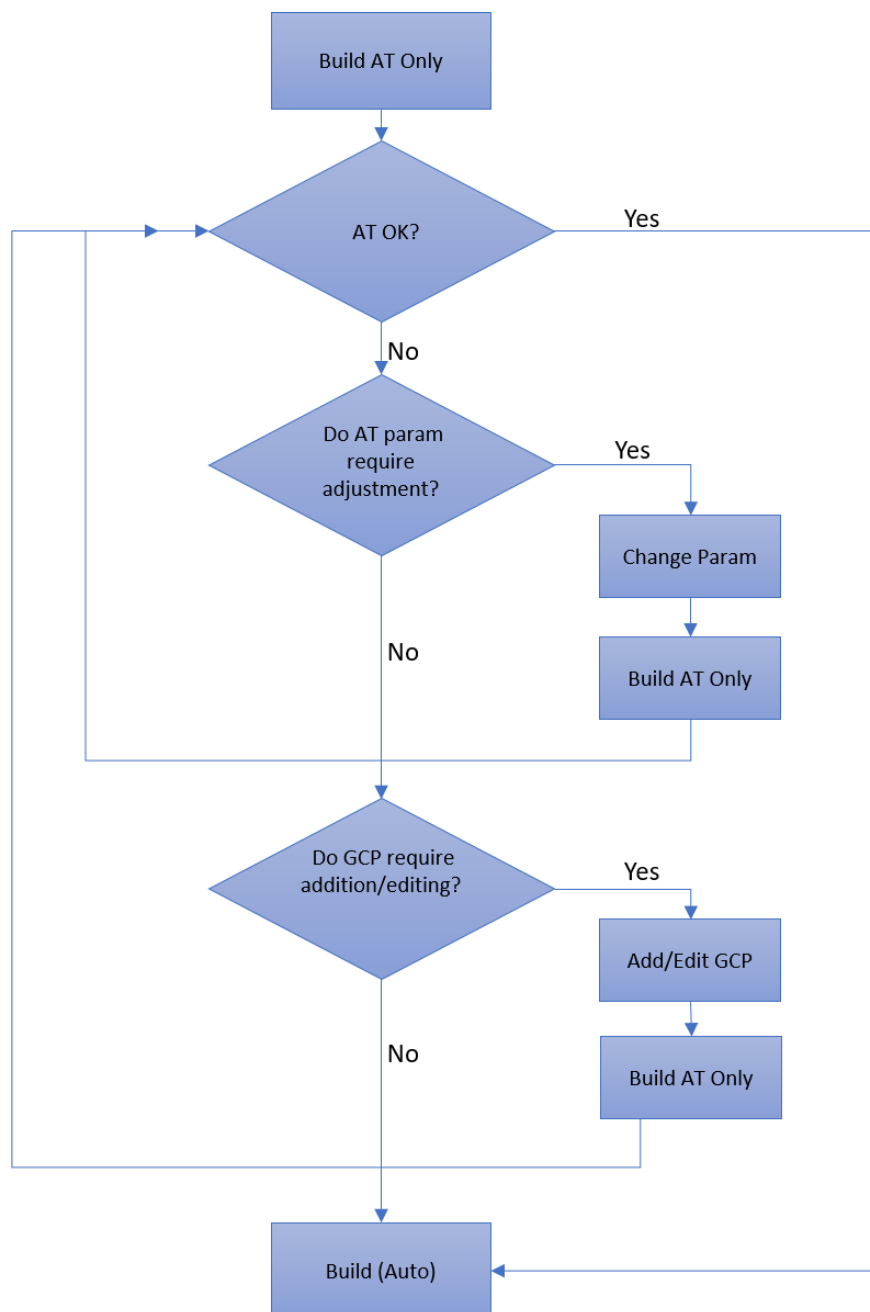
If a previously built project requires modification, instead of building the entire project again from the first step of the build process, you have the option to rebuild, incorporating some of a previous build's results. Rebuilding provides flexibility both with regard to the project section (entire project, only error tiles or selected tiles) and project steps that are to be rebuilt. **See "Rebuilding"** in this chapter for more information.

Recommended Aerotriangulation Workflow

For optimal AT results, it is recommended to initially perform only the aerotriangulation step ("AT Only"), and then review the results, to ensure a precise AT which includes all project's photos that intersect with the AT area, before proceeding with the rest of the build. If the results of the AT Only were in fact satisfactory, you can continue with the rest of the build process. PhotoMesh will automatically determine what processing is required to apply the changes to the project, by rebuilding the project from "Auto". **See "Rebuilding"** in this chapter for more information.

If the results of the initial AT Only build were unsatisfactory, modify the AT parameters or collection properties and/or edit/add control points based on the aerotriangulation result. Then repeat the aerotriangulation by selecting AT Only, and review your results. When repeating the aerotriangulation, PhotoMesh determines precisely which tiles and which specific AT sub-processes were affected by the modification of properties or control points, enabling a much lighter and faster AT process. This iterative process should be repeated until satisfactory AT results are achieved.

The diagram below outlines the recommended aerotriangulation workflow.



Defining the Aerotriangulation (AT) Area

Before starting a build, you should define the aerotriangulation (AT) area.

To define the aerotriangulation area:

1. If you want to perform aerotriangulation on the entire project, on the **Home** tab, in the **Area** group, click the arrow under **AT Area** and select **Automatic**.
2. If you want to perform aerotriangulation on a subset of the project define the AT area (See "[Setting the AT Area](#)" in the "Preparing the Project" chapter for more information).

Note: It is generally recommended to perform aerotriangulation on a preview model of a small subset of the project to validate camera parameters and build settings and check the resulting AT tiles before building the full project.

Defining the Reconstruction Area

By default, the reconstruction area is set to the same as the AT area.

To define the reconstruction area:

1. If you want to perform reconstruction on the same area as aerotriangulation, on the **Home** tab, in the **Area** group, click the arrow under **Reconstruction Area** and select **Same as AT Area**.
2. If you want to perform reconstruction on a different area than the area set for aerotriangulation, set the reconstruction area (See "[Setting the Reconstruction Area](#)" in the "Preparing the Project" chapter for more information).

Setting Build Steps, Parameters and Outputs

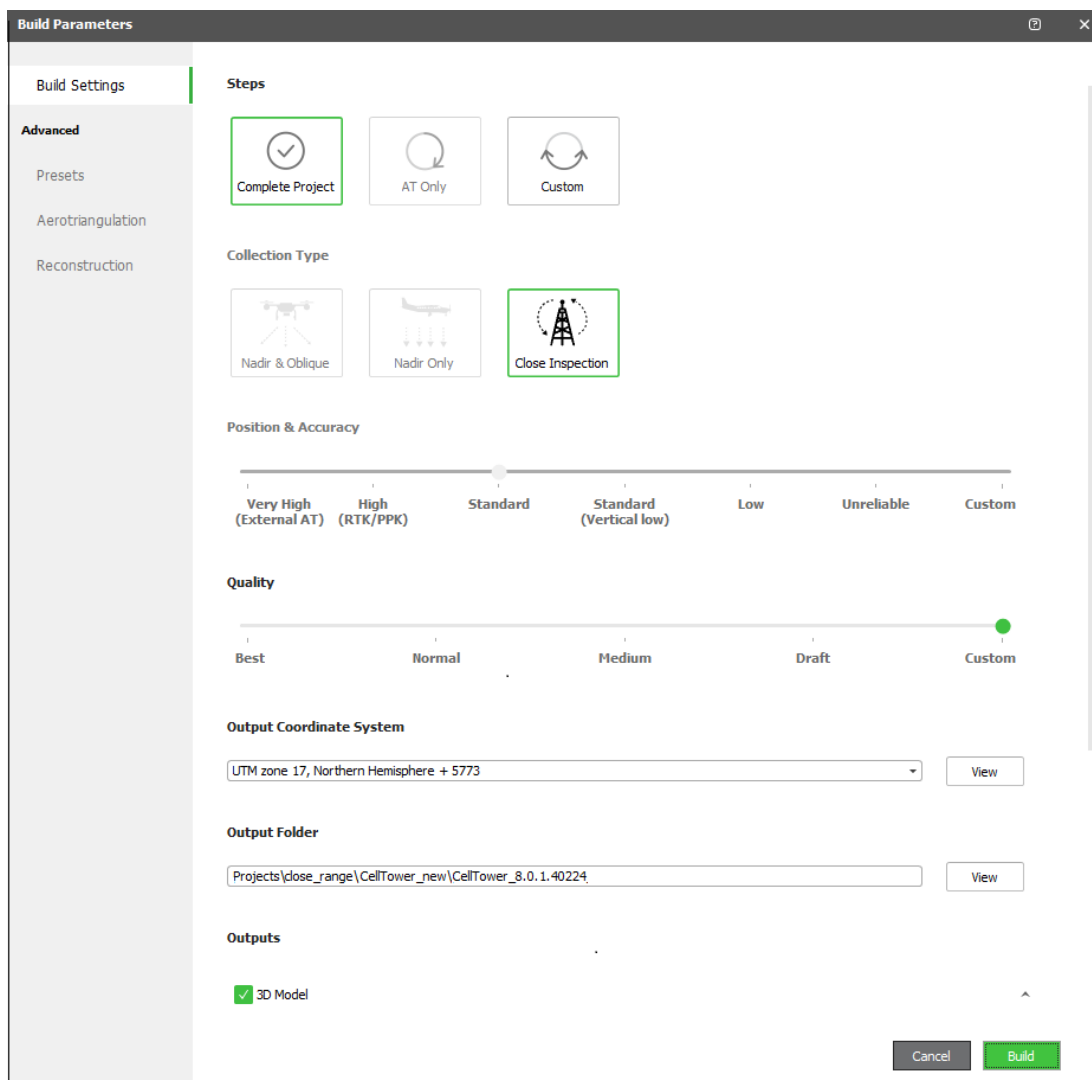
Before PhotoMesh can build the project, build steps, project parameters and project outputs are set in the Build Parameters dialog, using either a simplified view, or an advanced view that enables manual setting of relevant parameters. The simplified view consolidates all the main settings under a single tab ("Build Settings").

When using the advanced settings, you can select from a list of Build Presets that make it easy to automatically set the required parameters to generate the optimum output for the project's data as well as to share user-defined build settings between projects. Presets only include changes from the default settings. See "[Working with Presets](#)" in this chapter for information.

If you want to manually set the AT and reconstruction settings, click the Aerotriangulation and Reconstruction tabs. See "[Setting Advanced Aerotriangulation Parameters](#)" and "[Setting Advanced Reconstruction Parameters](#)" in this chapter.

To set build steps, parameters, and outputs:

1. On the **Home** tab, in the **Process** group, click **Build**. The Build Parameters dialog is displayed.



Build Parameters Dialog

2. Select the Build Steps to perform:

- **Complete Project** – Perform all build steps. PhotoMesh automatically determines what processing must be performed to build the outputs and complete the project, avoiding unnecessary rebuilding. E.g., when rebuilding a project after modifying a water polygon, only affected tiles are rebuilt from the point cloud step. Similarly, when rebuilding after reimporting exported models, only affected tiles are rebuilt, either from the texturing step, if just models were imported, or from the output step, if both models and textures were imported. If any change is made to photo/collection properties, the build is restarted from pre-aerotriangulation.
- **AT Only** – Perform only AT so you can review results before proceeding with the rest of the build to ensure a precise AT which includes all project's photos that intersect with the AT area. When an AT Only build is performed and then repeated following project adjustments, PhotoMesh determines precisely which tiles and which specific AT sub-processes were affected by the modification of properties or control points, enabling a much lighter and faster AT process.
- **Custom** – Perform selected build steps. To select the steps:

- i. In the **Start From** field, select the step from which you want to start the build. Select "Auto" to have PhotoMesh automatically determine what processing must be performed to complete the project, avoiding unnecessary rebuilding. E.g., when rebuilding a project after modifying a water polygon, only affected tiles are rebuilt from the point cloud step. Similarly, when rebuilding after reimporting exported models, only affected tiles are rebuilt, either from the texturing step, if just models were imported, or from the output step, if both models and textures were imported.
 - ii. In the **Build Until** field, select the step until which you want to build.
3. Select the **Collection Type**. This determines what project presets are set to achieve optimum output for the project's data.
 - **Nadir & Oblique** – For photo collections with vertical and diagonal perspectives. PhotoMesh performs a full 3D reconstruction of terrain and surface areas.
 - **Nadir Only** – For photo collections with vertical perspective only. PhotoMesh reconstructs the terrain and textures the horizontal surfaces. PhotoMesh increases verticality of above ground features for an improved 3D model and True Ortho.
 - **Close Inspection** – For projects with a single object captured from all perspectives. PhotoMesh focuses on the foreground object to achieve best image correlation relative to the close-range object and optimizes 3D reconstruction for modeling of thin features and complex geometry.
4. Set the Position & Accuracy:

Note: The accuracy of each individual collection's position and orientation data is set in the collection's property sheet in the [Accuracy](#) section. There, you have the option to either input custom values or to select the "Use build settings" alternative to apply the general accuracy settings set in this dialog. If some collections have custom settings, a notification will display in this dialog informing you of this.

 - **Very High (External AT)** - PhotoMesh assigns a very strong weight to provided camera positioning (AT Result) when performing aerotriangulation. This option is recommended when reliable AT data in a **Cartesian** coordinate system, such as ENU or ECEF, is added to PM from other AT systems, e.g., [Inpho](#) and [Bingo](#).

Note: AT data in a **projected** coordinate system such as UTM may be unreliable either due to earth curvature or earth curvature corrections performed in the other system. This is particularly true for datasets that contain oblique photos or large frames, or cover a large geographic area. For data of this type, aerotriangulation should be re-performed in PhotoMesh, i.e., use the **High Accuracy** option.
 - **High (RTK/PPK)** - PhotoMesh assigns a strong weight to provided camera positioning when performing aerotriangulation. If the inputted camera positions are inaccurate or inconsistent with any provided control points, it may be advisable to use the Standard GPS option. This option is recommended for collections captured by professional systems with accurate positioning and orientation information (e.g., using DGPS and INS).
 - **Standard** - PhotoMesh assigns a moderate weight to provided camera positioning. This option is recommended for collections using standard GPS with or without orientation information (e.g., standard drones such as those produced by DJI).
 - **Standard (Vertical Low)** - PhotoMesh assigns a moderate weight to provided camera horizontal positioning and a low weight to the camera's vertical positioning. This option is recommended for use with photo collections captured in flights that involve reinitialization of the drone's GPS system due to battery changes.

- **Low** – PhotoMesh assigns a low weight to provided camera positioning data. This option is recommended for collections whose GPS accuracy is compromised due to factors like weather conditions or varying elevation.
 - **Unreliable** – PhotoMesh performs a robust calculation of positioning and orientation information and camera parameters. This option is recommended when the difference between the input position and the actual position may be greater than 50m. In most cases this is caused by a lag between the acquisition of the positioning information and the photo capture time, or when multiple collections are captured at different times with inaccurate positioning hardware.
 - **Custom** –The level will be automatically set to Custom if custom accuracy settings were set on the Advanced - Aerotriangulation tab.
5. Set the **Output Quality** level of the point cloud, mesh model, and texture. Click **Draft** mode for faster processing and low quality outputs. The level will be automatically set to Custom if custom Quality and Performance settings were set on the Advanced - Reconstruction tab.
 6. Set the project's **Output Coordinate System** to which the project data should be reprojected. Select the **Custom** option to create a new custom coordinate system. See "[Coordinate System](#)" in the "Basic Concepts" chapter for more information. Click **View** to view the full Well-Known Text (WKT) for the coordinate system.

Note: 3D Tiles output is always produced in ECEF coordinate system and SLPK (I3S) output in WGS84 coordinate system with EGM96 vertical datum, regardless of the coordinate system selected here.

Note: All outputs are saved to the output folder, with a WKT to identify the output coordinate system that was selected in this parameter and special WKT's for the coordinate systems of the 3D Tiles and SLPK (I3S) output (Output-WKT_Cesium.txt and WKT_I3S.txt).
 7. The Output Folder is automatically populated with the path set for the project folder when creating a project. See "[Creating and Opening a PhotoMesh Project](#)" in the "Project Management" chapter for information. Click **View** to open the output folder in your File Explorer.
 8. Select the required **Output Formats**:

Note: The available output formats vary depending on your license level. Contact Skyline Support for more information.

3D Model

To generate 3D model output:

1. Select **3D Model**.
2. Select all required 3D Mesh output formats:
 - **3DML** – Stream-optimized, high-resolution 3D Mesh Layer database for loading directly into TerraExplorer and streaming with SkylineGlobe Server.
 - **3D Tiles** – 3D Tiles streamline the delivery of large 3D datasets by organizing them into a multi-resolution format, using GLB to store the actual model data for streaming to Cesium clients. PhotoMesh supports 3D Tiles version 1.0 and GLB 2.0. If you want to reproject the elevation values to ellipsoid (as required by the Cesium 3D Tiles standard) for compatibility with terrain that uses ellipsoid elevation, select **Reproject to ellipsoid**. This option should only be selected for terrain providers that use TRUE ellipsoid elevation.

Note: Some applications use a [TerrainProvider](#) and other data with geoid elevation values even though they are declared as ellipsoid when delivered to Cesium. Thus, they are actually compatible without reprojection.

Note: PhotoMesh outputs can easily be published to Cesium viewers by publishing the resulting 3DML to SGS. SGS will then make them available to TerraExplorer clients as 3DML as well as to Cesium by converting them on-the-fly to 3D Tiles. Use the 3D tiles output if you need to produce a full 3D Tiles output for a non-SGS web-server.

- **o3DML** – 3D Tiles-based, hierarchically structured format designed for streaming and rendering massive 3D geospatial content. The Open3DML format packages all the mesh data files, with their full paths and names as unique IDs, within a single SQLite database table, that can be easily copied and managed.
- **OSGB** – Multi resolution textured model for use in [OpenSceneGraph](#). The file pyramid \p-00000.osgb is the root of the hierarchy that can be used to load all data.
- **DAE** – Textured Collada model per reconstruction tile.
- **OBJ** – Textured OBJ model per reconstruction tile.

Note: If you want to reference the DAE/OBJ model's coordinates to the project's center, select **Center pivot to project**. This setting improves the stability of 3D model display in 3rd party viewers that are limited regarding display of large coordinates. This option does not need to be selected for OSGB since OSGB models' coordinates are by default referenced to the project center. When selected, only DAE, OBJ, and OSGB can be produced. If you want to set

	a custom pivot, click More Options . See "Setting a DAE/OBJ/OSGB Model Pivot" in this chapter for information. If you need the project center's coordinate information to reference the model in external software, this information can be extracted from the Output-CenterPivotOrigin JSON file created in the build's output folder when the output is generated. The Origin and WKT name/value pairs in this file contain the coordinates of the project's center to which the model is referenced and their coordinate system WKT. ▪ SLPK (I3S) – Compressed package of ESRI's Indexed 3D Scene (I3S) Layers delivery format used to stream very large 3D geospatial datasets to mobile, web and desktop clients. SLPK v1.7 is generated.
Point Cloud	LiDAR LAS file with X,Y,Z position and color information for each point of the point cloud To generate point cloud output: 1. Select Point Cloud. 2. Select one of the following options: ▪ Sampled refined 3D model – To create a point cloud by sampling the final textured model. Then in the Resolution field, enter the density for sampling the model. ▪ Points from photo correlation - To generate a point cloud based on the correlation of points in the photos. 3. Select a format: LAS or LAZ. 4. Type a resolution in meters per pixel.
Orthophoto	True-orthophoto created from the resulting textured mesh in GeoTiff format. To generate: 1. Select Orthophoto. 2. Set the following properties: ▪ Resolution – The raster resolution in meters per pixel. Select Auto to have PhotoMesh automatically select the optimal resolution based on the inputted data. ▪ Image Quality – Select the check box to compress the orthophoto output, and then enter a JPEG compression level (0-100%). Select 99% to apply an LZW lossless compression. ▪ Tile Size – Set the maximum tile size in pixels. Only values between 1000-40000 are accepted. The required RAM in bytes for processing can be roughly calculated as follows: Tile size (in pixels) * Number of bands. E.g., generation of an RGB orthophoto with tile size of 50,000 pixels will use about 7.5 GB (50,000*50,000*3= 7,500,000,000 = about 7.2 GB). Generating this orthophoto on fuser computers

with about 16 GB RAM leaves sufficient RAM for other usages.

- **Overlap Between Tiles** – Overlap in pixels between tiles.
- **Texture Blending** – Determines whether to use data blended from multiple photos in texturing the orthophoto, or only data from the optimal photo. Select **Auto** to base this setting on Photo Priority, where "Nadir Direction" will set blending to No, and "Used in Reconstruction" will set blending to Yes. Select **No** if blending results in ghosting artifacts.
- **Photo Priority** – Determines what type of photo is preferred when selecting the optimal photo for texturing the orthophoto. Select "Nadir Direction" to give priority to photos photographed from above. This option is recommended to avoid objects such as trees being textured from multiple photos photographed from different directions. Select "Used in Reconstruction" to give priority to the photo used to reconstruct the mesh for this area. This option is recommended when there are moving objects in the photos, to avoid having parts of objects appear in different areas of the orthophoto.
- **No-Data Color** – Determines the color to assign to areas of the orthophoto for which there is no data or which are outside the reconstruction area.
- **Bands and Collections** – Click the **More Options**  button to select the bands to include in the orthophoto, and the collections to include in generating the orthophoto. See ["Setting the Collections and Bands to Include in the Build Outputs"](#) in this chapter for information.
- **Tile Grid Origin** – Set the X and Y coordinates of the origin of the tile grid in the output coordinate system. This allows you to set a common grid for multiple projects. Select **Auto** to have PhotoMesh automatically select the optimal grid origin based on the inputted data.

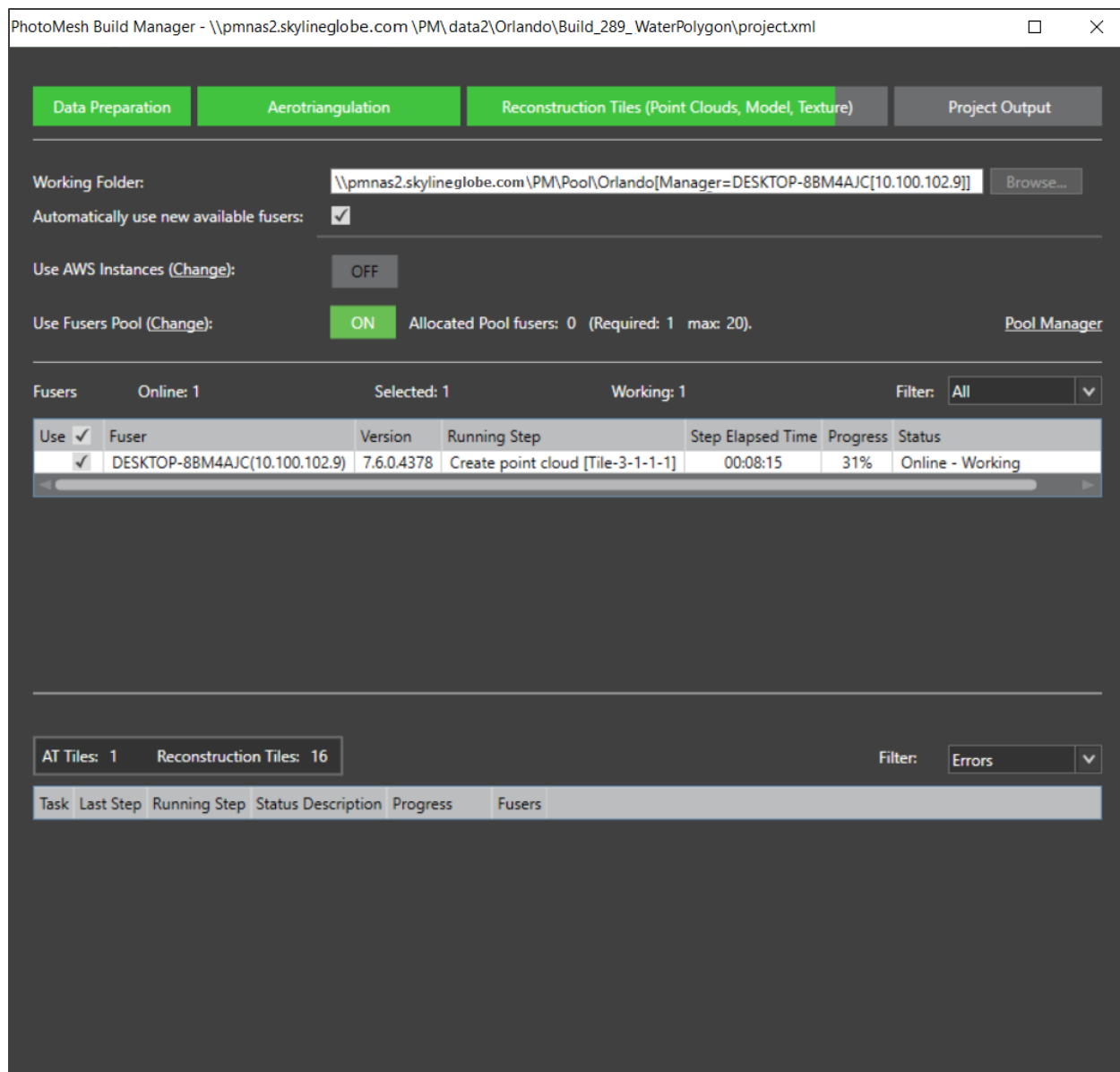
DSM (Digital Surface Model)

Elevation model in GeoTiff format that includes all elevation data, including buildings and reconstructed objects in the mesh.

To generate:

1. Select **DSM**.
2. Enter the following values:
 - **Resolution** – The raster resolution in meters per pixel. Select **Auto** to have PhotoMesh automatically select the optimal resolution based on the inputted data.
 - **Tile Size** – Set the maximum tile size in pixels. Only values between 1000-40000 are accepted. The required RAM in bytes for processing can be roughly calculated as follows: Tile size (in pixels) *

- | | |
|--|--|
| | <p>Number of bands. E.g., generation of a DSM with tile size of 45,000 pixels will use about 7.7 GB (45,000*45,000*4= 8,100,000,000 = about 7.7 GB). Generating this DSM on fuser computers with about 16 GB RAM leaves sufficient RAM for other usages.</p> <ul style="list-style-type: none"> ▪ Overlap Between Tiles – Overlap in pixels between tiles. ▪ Tile Grid Origin – Set the X and Y coordinates of the origin of the tile grid in the output coordinate system. This allows you to set a common grid for several projects. Select Auto to have PhotoMesh automatically select the optimal grid origin based on the inputted data. |
|--|--|
- DTM (Digital Terrain Model) An elevation model that represents the bare ground surface without any objects such as plants and buildings.
- To generate:
1. Select **DTM**.
 2. Enter the **Resolution** – The raster resolution in meters per pixel. Select **Auto** to have PhotoMesh automatically select the optimal resolution based on the inputted data.
-
9. Click **Build**. The New build description dialog is displayed. Enter a build name and description and click **Ok**. The PhotoMesh Build Manager is displayed.



PhotoMesh Build Manager – Starting a Build

10. If you want to change the working folder, in the **Working Folder** field, click **Browse** and in the dialog that opens browse to the required folder. See "[Setting the Working Folder \(on the Master Computer\)](#)" in the "Fusers" chapter for information.
11. If you want PhotoMesh to automatically use any new fusers that connect to the project's working folder, select **Automatically use new available fusers**. See "[Using Fusers](#)" in the "Fusers" chapter for information.
Note: This setting should generally be selected when using AWS and pool fusers.
12. If you want to use AWS fusers, toggle on **Use AWS Instances**. See "[Using Amazon Web Services \(AWS\) Fusers](#)" in the "Fusers" chapter for information.
13. If you want to use pool fusers, toggle on **Use Fusers Pool**, and enter the required information. See "[Using a Fusers Pool](#)" in this chapter for information

14. Select the required fusers. **See** ["Fuser Management \(on the Master computer\)"](#) in the "Fusers" chapter and ["Monitoring a Build"](#) in this chapter for more information on fusers.

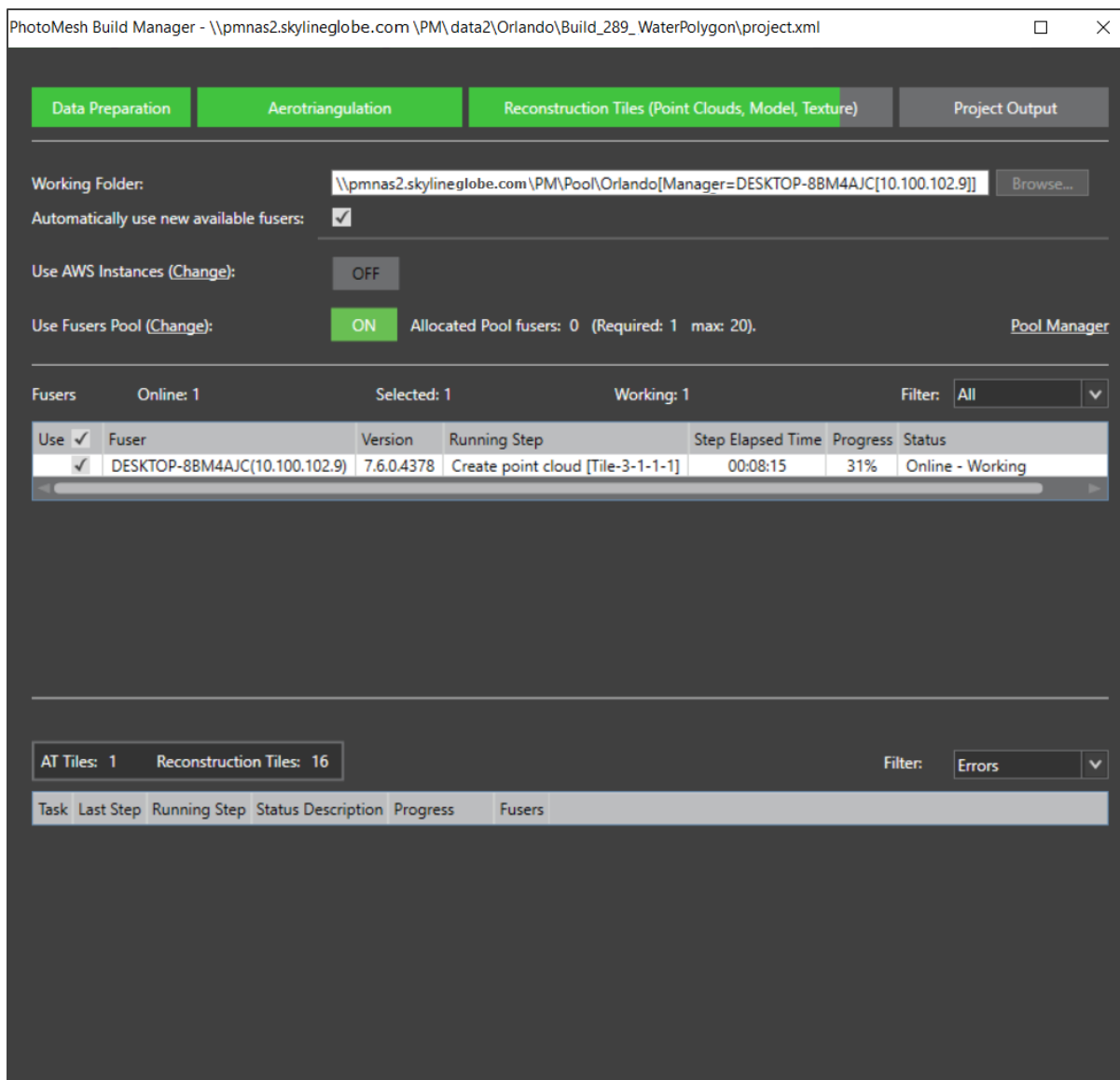
Note: If there are currently no available fusers, a dialog is displayed asking if you want to start a local fuser. Click **Yes** to start a local fuser. Click **No** to start the build without adding a fuser; processing will begin automatically when fusers are added. If you want to cancel the build, click **Cancel**.

15. Click **Build**.

Setting Build Parameters – Advanced Settings

To manually set build parameters using advanced settings:

1. On the **Home** tab, in the **Process** group, click **Build**. The Build Parameters dialog is displayed.
2. To set build parameters by using one of the available presets, click the **Presets** tab. Click **Select and Manage** to open the Preset Manager, and then select the required presets, and click **OK**. **See** ["Selecting Presets"](#) in this chapter for information.
3. To manually configure aerotriangulation and reconstruction parameters, click the corresponding tab and set the required parameters:
 - [Aerotriangulation](#)
 - [Reconstruction](#)
4. To select the collections to include when generating the mesh model texture, orthophoto, or point cloud, as well as the bands to include in the orthophoto and to assign to each RGB of the mesh model texture, click the **Reconstruction** tab, and define the required settings in the **Bands and Collections** section. **See** ["Selecting the Collections and Bands to Include in the Build Outputs"](#).
5. Click **Build**. The New build description dialog is displayed. Enter a build name and click **Ok**. The PhotoMesh Build Manager is displayed.



PhotoMesh Build Manager – Starting a Build

- If you want to change the working folder, in the **Working Folder** field, click **Browse** and in the dialog that opens browse to the required folder. See "[Setting the Working Folder \(on the Master Computer\)](#)" in the "Fusers" chapter for information.
- If you want PhotoMesh to automatically use any new fusers that connect to the project's working folder, select **Automatically use new available fusers**. See "[Using Fusers](#)" in the "Fusers" chapter for information.
Note: This setting should generally be selected when using AWS and pool fusers.
- If you want to use AWS fusers, toggle on **Use AWS Instances**. See "[Using Amazon Web Services \(AWS\) Fusers](#)" in the "Fusers" chapter for information.
- If you want to use pool fusers, toggle on **Use Fusers Pool**, and enter the required information. See "[Using a Fusers Pool](#)" in this chapter for information

10. Select the required fusers. **See** "[Fuser Management \(on the Master computer\)](#)" in the "Fusers" chapter and "[Monitoring a Build](#)" in this chapter for more information on fusers.

Note: If there are currently no available fusers, a dialog is displayed asking if you want to start a local fuser. Click **Yes** to start a local fuser. Click **No** to start the build without adding a fuser; processing will begin automatically when fusers are added. If you want to cancel the build, click **Cancel**.

11. Click **Build**.

Setting Advanced Aerotriangulation Parameters

Advanced aerotriangulation settings can be set when building a project or creating a new build version. When rebuilding, aerotriangulation parameters can only be set if the reconstruction step was not started in the previous build. **See** "[Rebuilding](#)" and "[Creating a New Build Version](#)" in this chapter for more information.

To set advanced aerotriangulation parameters:

1. Click the **Aerotriangulation** tab.

Coordinate System

Coordinate System

ENU (0.999060471553424,45.008464915865) +5773

View

Aerotriangulation

Calculation mode

Auto

Target photos per AT tile

5000

Accuracy

Horizontal accuracy factor

Standard (0.1)

Vertical accuracy factor

Standard (0.1)

Focal length accuracy

Custom (40%)

Principal point accuracy

Standard (5%)

Radial accuracy

Standard

Tangential accuracy

Standard

Calculate camera parameters

Entire Collection

Ignore Orientation

☐

Advanced

Ground control point factor

☒ Auto

10

Multiple AT groups in tile

Use only largest group

Align photos to camera position

Per AT tile

Skip data preparation

☐

AT Flags

Advanced AT Parameters

2. Set the parameters on the AT tab:

Parameter	Description
Coordinate System	
Coordinate System	Project's internal coordinate system, in which the aerotriangulation and reconstruction are performed. By default, this is set to the coordinate system of the input data. If the inputted data uses multiple coordinate systems, or you want to use a different coordinate system from your input's, select the required coordinate system. Select the Custom option to create a new custom coordinate system. See "Coordinate System" in the "Basic Concepts" chapter for more information. Note: Click View to view the full Well-Known Text (WKT) for the coordinate system.
Aerotriangulation	

Calculation Mode	The AT calculation mode that PhotoMesh should use: ▪ Auto – PhotoMesh automatically determines the required calculation mode. If no AT was previously performed, then PM performs a full aerotriangulation ("Calculate AT"). If an AT was previously performed, then it refines the existing AT ("Refine AT"). ▪ Calculate AT – PhotoMesh performs a full aerotriangulation (including matching, bundle adjustment, and GCP) taking into account provided orientation and position information. All previous aerotriangulation calculations performed within PhotoMesh are ignored. ▪ Refine AT – PhotoMesh reruns the aerotriangulation process (performing only bundle adjustment and GCP). This option is recommended for refining the previously calculated AT or after making changes to the project that can affect the calculated AT, i.e., adding/modifying control points. Refine AT can only be performed if an AT was previously performed or imported from an external source. ▪ Do Not Calculate AT (Fully Trusted) – PhotoMesh uses inputted camera positioning and orientation information and camera parameters (including focal length, principal point and radial/tangential distortion) without performing aerotriangulation. This option is recommended when reliable AT data in a Cartesian coordinate system, such as ENU or ECEF, is added to PM from other AT systems, e.g, Inpho and Bingo. Note: AT data in a projected coordinate system such as UTM may be unreliable either due to earth curvature or earth curvature corrections performed in the other system. This is particularly true for datasets that contain oblique photos or large frames, or cover a large geographic area. For data of this type, aerotriangulation should be re-performed in PhotoMesh, i.e., use the Refine AT option.
Target Photos per AT Tile	Number of photos to be included in each AT tile. The default setting of 5,000 photos is recommended for most projects and can be efficiently processed on 16 GB computers. For projects requiring more than 5,000 photos per AT tile, the following RAM guidelines are recommended. Note that the actual RAM needed may be up to twice the suggested amount, varying based on the type of collection, inputted photo information, and specific settings used. Maximum Photos/AT Tile – Recommended Fuser RAM ▪ 7.5K – 16 GB ▪ 15K – 32 GB ▪ 30K – 64 GB

- 50K – 128 GB

Certain settings or AT-related presets may require more RAM than recommended above. E.g., it is recommended to set the Target Photos per AT Tile to 30% less than the value recommended above in both of the following scenarios:

- When using the aggressive match preset
- When photo overlap is higher than 75%

When running a large AT, it is highly recommended to leave the CPU free of other processes, since the AT process utilizes a large percentage of the available processing power.

PhotoMesh includes photos in an AT tile based on the photos' frustum, if orientation information is available, or based on its position and [effective range](#) in cases where no orientation information is provided. If the effective range is large and there is a very high overlap of photos, it may be advisable to set the orientation filter to Auto. This filter will try to accelerate the AT process by filtering out photos after a rough calculation of their orientation before proceeding with the remaining aerotriangulation steps. [See "Collection Property Sheet"](#) in the "Photo Management" chapter for information about the effective range property. [See "Setting Advanced Aerotriangulation Parameters"](#) in the "Building" chapter for more information about the orientation filter.

Larger AT tiles with more photos may be necessary if the default AT tiles have fewer than five ground control points that fall in or close to each tile, as is recommended for good accuracy. [See the "Control Points"](#) chapter for more information.

Note: An AT with a large number of photos may take many hours to complete, and if interrupted will need to be rebuilt.

Accuracy

The accuracy of each collection's position and orientation data is set in the collection's property sheet in the [Accuracy](#) section. There, you have the option to either input custom values or to select the "Use build settings" alternative to apply the general accuracy settings set in this dialog. If some collections have custom settings, a red notification will display in this dialog informing you of this.

Horizontal Accuracy Factor	Weight to assign to the photos' provided XY coordinates in determining the calculated XY values. This ranges from Very High - External AT (Factor = 10) to Unreliable (Factor = 0).
Vertical Accuracy Factor	Weight to assign to the photos' provided altitude in determining the calculated altitude value. This ranges from Very High - External AT (Factor = 10) to Unreliable (Factor = 0).
Focal Length Accuracy	A percentage that indicates the allowable deviation from the provided focal length value when calculating the AT. This represents the accuracy of the provided focal length. This ranges from Use Input Value to Unreliable.

Principal Point Accuracy	A percentage that indicates the allowable deviation from the provided principal point when calculating the AT. This represents the accuracy of the provided principal point. This ranges from Use Input Value to Unreliable.
Radial Accuracy	Camera's radial accuracy. It is generally recommended to select the "Standard" option. If your data was captured with a professional camera that automatically applied a radial distortion correction (i.e., the original distortion value is extremely accurate), select "Use input value".
Tangential Accuracy	Camera's tangential accuracy. It is generally recommended to select the "Standard" option. If your data was captured with a professional camera that automatically applied a tangential distortion correction (i.e., the original distortion value is extremely accurate), select "Use input value".
Calculate Camera Parameters	Method of calculating the camera's intrinsic parameters. - Entire collection – Calculate the AT so that the camera intrinsic properties are the same for all photos in the collection. - Per photo – Calculate the camera intrinsics individually for each photo in the collection. Note: Parameters should be calculated per photo when not all of a collection's photos were taken by the same physical camera with identical focal length and principal point.
Ignore Orientation	Select the check box to ignore the provided orientation value.
Advanced	
Ground Control Point Factor	Weight factor to assign to the ground control points (0-10000) when performing the aerotriangulation. Select Auto to use the default value of 1000.
Multiple AT Groups in Tile	Select whether to include only a tile's largest AT group (photo block) or all AT groups in AT Calculated Photos.
Align Photos to Camera Position	Select one of the following options: - Per AT Tile – Each AT tile is individually aligned to its respective camera positions and ground control points. This option is recommended for projects where most of the AT tiles have enough control points (generally 3 or more) to provide a reliable basis for adjusting the tile's position. Those AT tiles with fewer than three points are adjusted in relation to stable tiles which have an adequate number of control points. This option is recommended for projects with accurate camera positions as well as for nadir and corridor (linear) projects (e.g., roads or rivers) where the AT tile may come out arched. - Entire Project – The entire project is aligned as a single block to the camera positions and ground control points. This option is recommended for projects where very few of the AT tiles have sufficient control points (3 or more) to provide a reliable basis for adjusting the tile's position.

Skip Data Preparation	When this option is selected, PhotoMesh skips the data preparation step entirely, saving the time that would otherwise be necessary to verify that it had been performed. Only select this option if this step was previously performed, and there has been no change to the AT area or to the project.
AT Flags	Flags that specify alternate behaviors and settings that PhotoMesh should apply. These flags can be obtained from Skyline support.

3. Click the **Reconstruction** tab to set advanced reconstruction parameters (see "[Setting Advanced Reconstruction Parameters](#)" in this chapter), or click **Build** to open PhotoMesh Build Manager to start processing (see "[Monitoring a Build](#)" in this chapter).

Setting Advanced Reconstruction Parameters

To set advanced reconstruction parameters

1. Click the **Reconstruction** tab.

Tiles

Tile split method:

☐ Fixed 2D grid
 ☒ Adaptive 3D boxes

Reconstruction tile size:

☒ Auto
 ☐ Size

100

Meters

Quality & Performance

Photo sampling interval:

Best

Mesh model resolution:

Best

Texture resolution:

Best

Bands & Collections

Use for point cloud and model:

Red,Green,Blue [4/4 Collections]

...

Use for texture:

Red,Green,Blue [4/4 Collections]

...

Point Cloud

Min. number of matches:

Auto

Correlation window size:

Auto

Pixels

LiDAR

Merge LiDAR points:

Auto

LiDAR collection mode:

Auto

Use LiDAR colors:

Only Photos

Model

Smooth surfaces:

Yes

Enhance edges:

Yes

Fill in ground:

Yes

Fill in walls:

No

Smooth walls:

No

Remove floating components:

None

Tile vertical clipping buffer:

0

Meters

Texture

Color saturation:

Vivid

Texture blending:

Yes

Sharpening:

Normal

Advanced Reconstruction Parameters

213

Parameter	Description
Tiles	
Tile Split Method	Select either of the following options: ▪ Fixed 2D grid – The reconstruction area is divided into uniformly-sized tiles. ▪ Adaptive 3D boxes – The reconstruction area is divided into 3D tiles based on the memory requirements of each section of the reconstruction area. For example, sections of high resolution or close range, or areas with greater photo coverage, will be split into smaller tiles. The reconstruction area is adaptively divided along the horizontal, vertical and depth axes, allowing for complex three-dimensional data such as tall buildings or utility poles to be split into multiple tiles. When reconstruction tiles are created as adaptive 3D boxes, the display style option is automatically set to "3D Dynamic". In this display mode, the tiles and their boxes are displayed in absolute heights. Tiles that were created one on top of each other will appear as if they are floating in this mode. See "Setting a Reconstruction Tile's Display Style" in the "Reviewing Your Build" chapter for more information.
Reconstruction Tile Size	Size of the reconstruction tiles. This can be set to Auto or to a specified size in meters. It is generally recommended to use the Auto option. When tile division results were non-optimal, setting a size in meters is recommended. Note: The size of the reconstruction tiles can only be set when using the 2D grid tile split method. If you want to control tile size based on size in meters, three factors should be taken into account: RAM, resolution and overlap. For projects with standard resolution (5-10 cm photos) 100-150 meters is recommended on computers with 16 GB of RAM. Larger reconstruction tiles require more memory. On computers with over 64 GB of RAM, a tile size up to 400 meters can be used. As a general rule, the reconstruction tile size should be the ground resolution of the data multiplied by 2000. For high overlap datasets, the ratio between the reconstruction tile size and the ground resolution should be lower. Note: If point cloud build steps are repeatedly failing, and you observe a very high RAM usage towards the end of the point cloud step, it is recommended to reduce memory requirements or reconstruction tile size.
Quality & Performance	
Photo sampling interval	The resolution of the photos at which 3D points are extracted for the point cloud: ▪ Best: Extracts points at up to the full resolution of the photo, ensuring the most detailed and high-quality point cloud. This option is recommended in scenarios where the highest quality output is required, even at the cost of

	extended processing times. ▪ Optimize: Balances quality and efficiency by extracting points at a medium resolution (between 1/4 to 1/2 of the photo's native pixel count). This option reduces processing time to approximately one-third of what is required for "Best" quality output. Note that this setting's impact on quality can vary in comparison to "Best" processing: although medium resolution correlation sometimes leads to slightly reduced quality, in instances where full resolution correlation isn't suited to the image's geometry, it can actually improve output quality. ▪ Low: Offers significantly faster processing compared to the 'Optimize' option by employing low-resolution correlation (at 1/8 of the native pixel resolution), capturing broad geometric details. ▪ Draft: Generates a lower quality point cloud in the fastest processing time. This option provides a broad overview of coverage for preliminary project evaluations.
Mesh model resolution	Level of mesh complexity. The Low option will aggressively simplify the mesh.
Texture resolution	The resolution from the photos to use for texturing the model: Selecting Best applies textures at the same quality as the original photos, while. Medium and Low down-sample mesh textures.
Bands & Collections	
Use for point cloud and model	Click the More Options  button to select the collections and bands to include in the point cloud and 3D model. See "Setting the Collections and Bands to Include in the Build Outputs" in this chapter for information.
Use for texture	Click the More Options  button to select the bands to assign to each RGB of the mesh model texture, and the collections to include in generating the texture. See "Setting the Collections and Bands to Include in the Build Outputs" in this chapter for information.
Point Cloud	
Min. number of matches	The minimum number of matches required to generate each 3D point. When the point cloud generated is sparse with little overlap of photos, consider reducing the minimum number of matches. When there is noise in the point cloud output, it is recommended to increase the minimum number of matches, providing there is high overlap of the photos.
Correlation window size	Size in pixels of the window used to perform the correlation between photos and extraction of 3D points from them.
LiDAR	
Merge LiDAR Points	The data to use in creating the point cloud: only LiDAR points, only points generated from photos, or Auto – supplement

	LiDAR with points generated from photos where LiDAR information is insufficient.
LiDAR collection mode	The direction of LiDAR collection assists in calculating surfaces from the point cloud. When there is no trajectory file and the LiDAR collection is known to be airborne, select Aerial . Otherwise, select Auto .
Use LiDAR colors	The data to use in texturing the model: ▪ Auto – Supplement photo texture with color from LiDAR points where photo texturing information is insufficient. ▪ Only LiDAR colors – Only color from LiDAR points. ▪ Only photos – Only photo texture.
Model	
Smooth surfaces	Determines whether surfaces are flattened.
Enhance edges	Determines whether 2D edges in the project's photos are correlated and 3D edges extracted from them, to produce the 3D model. This setting produces better-defined 3D models with sharper edges, which is important for urban area reconstructions and thin structures such as poles and wires.
Fill in ground	Fills in ground gaps in areas with low coverage by incorporating calculated ground surface. Recommended to improve build results in low overlap, nadir projects.
Fill in walls	When PhotoMesh detects a vertical gap, it automatically fills in the missing sections to form a complete and continuous vertical surface. This option is only available when "Fill in Ground" is set to Yes.
Smooth walls	Smooths out vertical surfaces.
Remove floating components	Level of removal of small floating components in the mesh model (e.g. noise, electric lines, lamp poles).
Tile vertical clipping buffer	Vertical buffer to apply to the tile. Adding a buffer with a positive value is useful when there are objects that were cut off in a specific tile. A buffer with a negative value can be used to crop areas with small floating components.
Texture	
Color saturation	Vividness of the colors of the textured model.
Texture blending	Determines whether to use data blended from multiple photos in texturing the model, or only data from the optimal photo. Select No if blending results in ghosting artifacts. Select Extra if Yes produces color patches in areas with strong color variance.
Sharpening	Level of image sharpening applied to the final result.
Advanced	
Reconstruction flags	Flags that specify alternate behaviors and settings that PhotoMesh should apply. These flags can be obtained from Skyline support.

2. Click **Next** to set output formats (see "Setting Build Steps, Parameters, and Outputs" in this chapter), or click **Build** to open PhotoMesh Build Manager to start processing (see "Monitoring a Build" in this chapter).
3. To save your building parameters as a Build Preset, click **Save as Preset** (see "Creating a New Preset" in this chapter) or click **Build** to begin the build (see "Setting Build Steps, Parameters, and Outputs" in this chapter).

Selecting the Collections and Bands to Include in the Build Outputs

You can select the collections and bands to include when generating the model, point cloud, and mesh model texture in the Build Parameters – Reconstruction tab. The collections and bands to include in an orthophoto are selected from the Build Parameters – Build Settings tab in the orthophoto output settings. See "Setting Build Steps, Parameters and Outputs" in this chapter for more information.

To select the collections and bands to include when generating the model or point cloud:

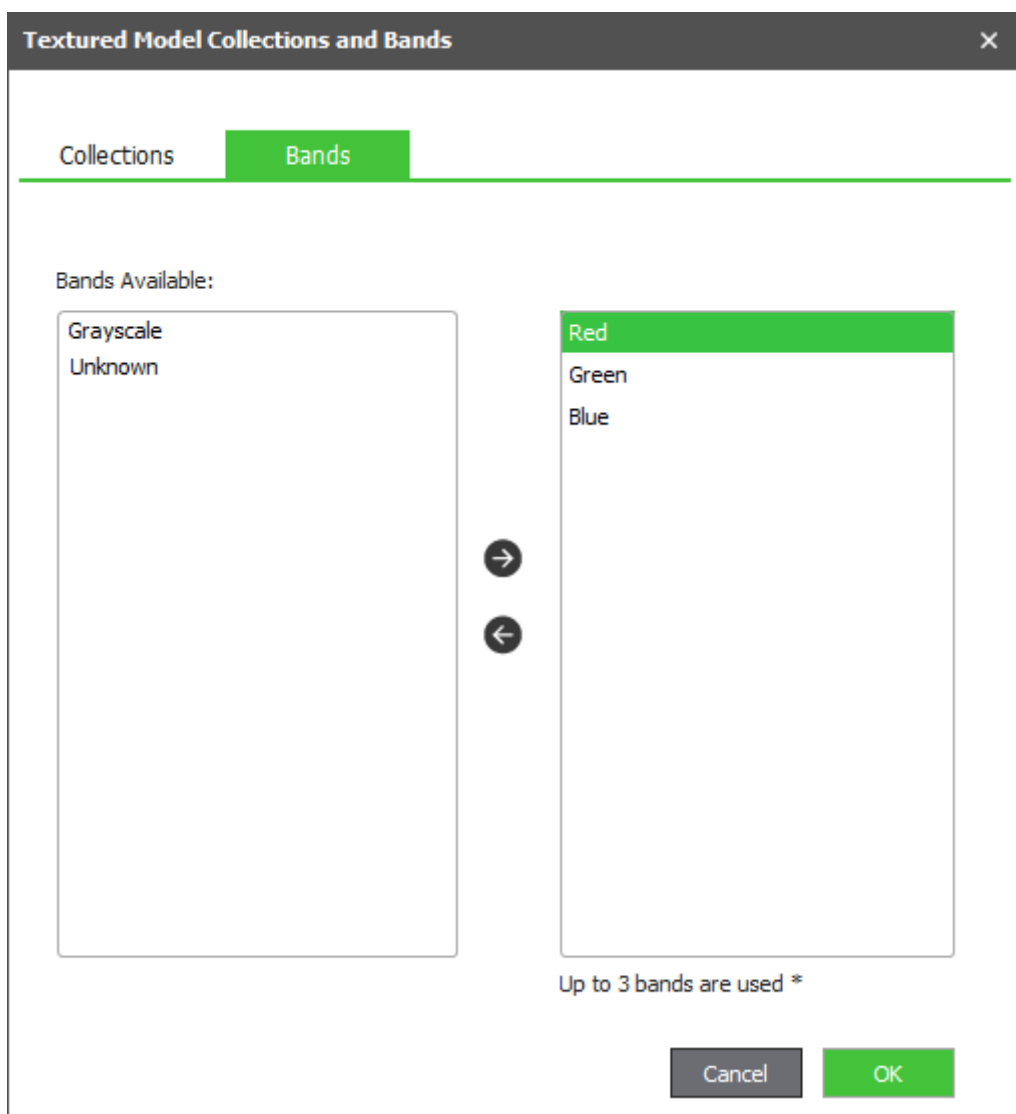
1. On the **Reconstruction** tab, in the **Bands and Collections** section, in the **Use for point cloud and model** field, click **More Options** .
2. On the **Collections** tab, select the collections that you want to include and click **OK**.
3. On the **Bands** tab, from the **Bands Available** section, select a required band, and click **Insert**  to insert it in the **Bands Used** section. Insert the bands in the required order. If you want to remove a band from the Bands Used section, select the band, and click **Remove** .
4. Click **OK**.

To select the collections and bands to include when generating the mesh model texture:

1. On the **Reconstruction** tab, in the **Bands and Collections** section, in the **Use for texture** field, click **More Options** .
2. On the **Collections** tab, select the collections that you want to include and click **OK**.
3. On the **Bands** tab, from the **Bands Available** section, select a required band, and click **Insert**  to insert it in the **Bands Used** section. Insert the bands in the required order. Three bands are required. If you want to remove a band from the Bands Used section, select the band, and click **Remove** .
4. Click **OK**.

To select the collections and bands to include when generating the orthophoto:

1. On the **Build Settings** tab, under the **Outputs-Orthophoto** format, in the **Bands and Collections** field, click **More Options** .
2. On the **Collections** tab, select the collections that you want to include and click **OK**.
3. On the **Bands** tab, from the **Bands Available** section, select a required band, and click **Insert**  to insert it in the **Bands Used** section. Insert the bands in the required order. If you want to remove a band from the Bands Used section, select the band, and click **Remove** .
4. Click **OK**.



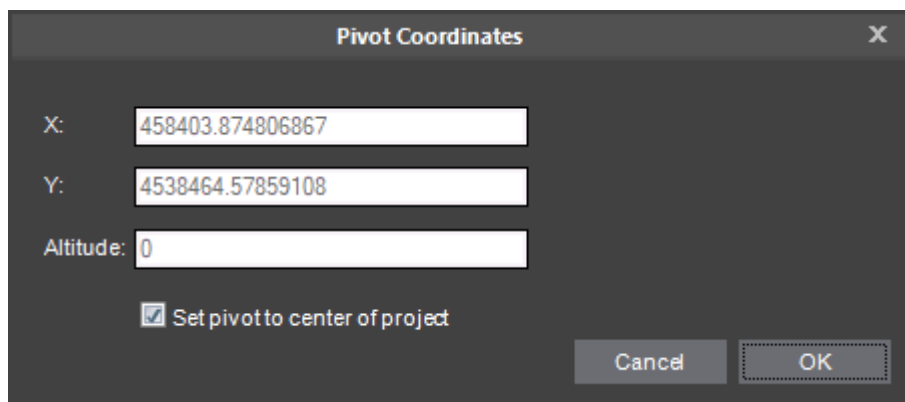
Selecting the Textured Model Collections and Bands

Setting a DAE/OBJ/OSGB Model Pivot

Setting the pivot for a DAE/OBJ/OSGB model improves the stability of its display in 3rd party viewers that are limited regarding display of large coordinates. A DAE/OBJ/OSGB model's pivot can be set to the project's center or set to custom coordinates.

To set a DAE/OBJ/OSGB model's pivot:

1. In the **Build Parameters** dialog, on the **Build Settings** tab, in the **Outputs** section, select **Center Pivot to Project**.
2. If you want to set a custom pivot, click **More Options** . The Pivot Coordinates dialog is displayed.



Pivot Coordinates Dialog

3. If you want to set the pivot to the center of the project, select the check box.
4. If you want to set a custom pivot, enter its coordinates.
5. Click **OK**.

Applying Gigapixel Credits

PhotoMesh Gigapixel (GX) is a licensing mechanism that allows you to apply gigapixel credits towards the building of your PhotoMesh projects based on the number of gigapixels in each project. Within a single project, multiple build versions in all output types can be created using the same gigapixel credits. This mechanism can be implemented by pre-purchasing a bank of gigapixel credits, loading them to your local Skyline Floating License Server (SFLS), and then applying them to project photos when building a project. You can also send a request to Skyline, using an online form or request file, for gigapixel credits to apply to photos in a project.

If you have a PhotoMesh GX license (see "[PhotoMesh License Options](#)" in the Getting Started chapter for information), PhotoMesh automatically launches the Apply Gigapixel Credit (GPC) dialog when starting a build process. From this dialog, you can apply the required gigapixel credits to the project's photos, as well as access the License Server (SFLS) to load additional gigapixel credits, if required. PhotoMesh calculates the number of gigapixels in the project based on the number of gigapixels in the selected AT area. When rebuilding a project that gigapixel credits were already applied to, additional credits are only required for photos that were added or if the AT area was expanded to include additional photos.

Apply Gigapixel Credit (GPC)

Project Name:

Orlando-GX-4443

Number of Photos in AT:

46

Total Gigapixels in AT:

2.0149504

GP Credits Already Applied to Project:

0

GP Credits Required:

2.0149504 (46 Photos)

Request Type:

☐ From Server (SFLS)

☐ Send Request

Company Name:

Contact Person:

Email Address:

Phone Number:

Country:

GP Credits Request Key:

[[[000000F4Sq2FWcYrWDBiGUM2Ph2JWZMDbUGoWr3xbUsX.UTT.TA2Pr2H

The build key is a unique key generated automatically based on the current build, and does not contain private data, project coordinates, or client or computer information of any sort.

Save Request File

Fill Online Form

GP Credits License Key:

Apply GP Credits

Apply Gigapixel Credit Dialog

To apply gigapixel credits:

1. Review the following settings in the Apply Gigapixel Credit (GPC) dialog:

Setting	Description
Project Name	Name of project.
Number of Photos in AT	Number of photos in the AT area.
Total Gigapixels in AT	Total number of gigapixels in the photos that intersect with the AT area.
GP Credits Already Applied to Project	Number of gigapixel credits already applied to photos in this project. Additional credits are only required if photos were

	added to the project or if the AT area was expanded to include additional photos.
GP Credits Required	Number of gigapixel credits required to build the project. If gigapixel credits were already applied to photos in this project, additional credits are only required for photos that were added or if the AT area was expanded to include additional photos.

- If you want to apply gigapixel credits from the available credits on your Skyline Floating License Server (SFLS), to photos in this project, select **From Server (SFLS)**. Then enter the following:

License Server (SFLS)	URL to your Skyline Floating License Server.
Available GP Credits on Server	Number of gigapixel credits in your server (SFLS) balance, that could be applied towards building this project.

- If you want to send a request to Skyline, using an online form or request file, for gigapixel credits to apply to photos in this project, select **Send Request**. Your request includes a key that includes information about the total number of gigapixels in your AT. The following additional fields are displayed:

Company Name	Name of company.
Contact Person	Name of contact person in company.
Email Address	Email address.
Phone Number	Phone number.
Country	Location of company.
GP Credits Request Key	A key that includes information about the total number of gigapixels in your AT, which is used to request gigapixel credits from Skyline. This key is automatically passed to Skyline when you use the Save Request File or Fill Online Form. The build key is a unique key generated automatically based on the current build, and does not contain private data, project coordinates, or client or computer information of any sort.

- If you want to submit your request offline, click **Save Request File**, enter the contact information in the fields that are displayed, and send it to Licensing@SkylineGlobe.com. After you receive your build license code from Skyline, enter the code in the **GP Credits License Key** field.
- If you want to submit your request online, click **Fill Online Form**, and enter the required contact information in the online form that is displayed. After you receive your build license code from Skyline, enter the code in the **GP Credits License Key** field.
- Click **Apply GP Credits**.

Working with Presets

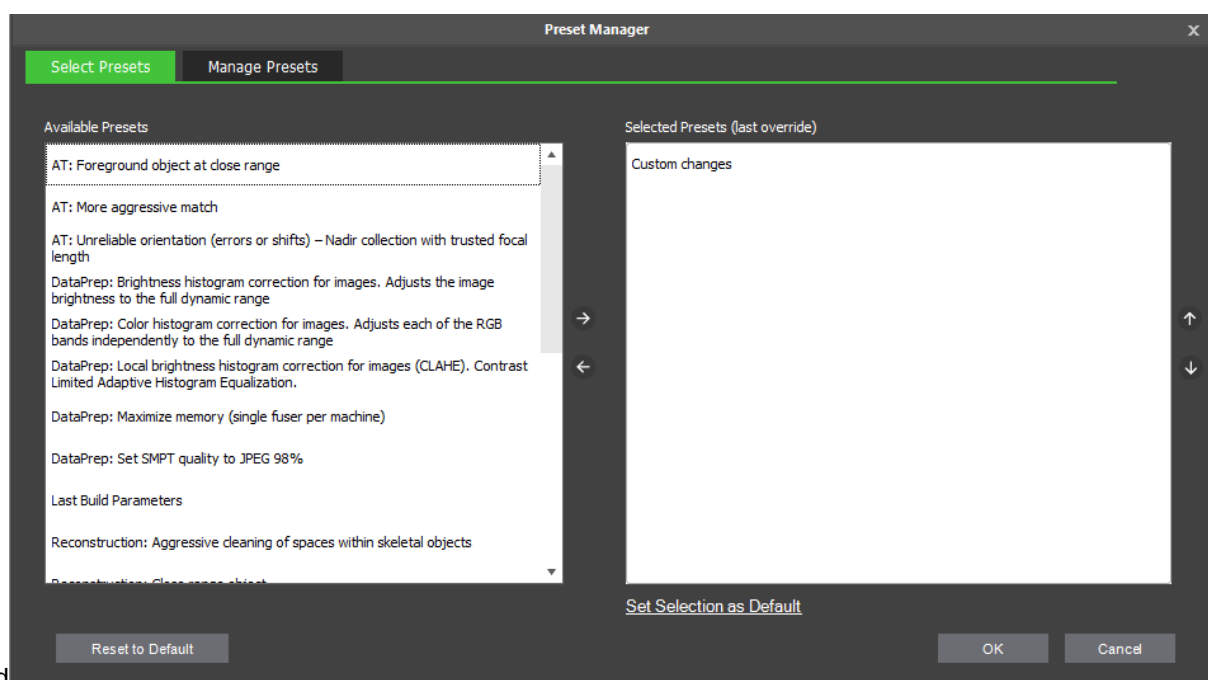
Build parameter presets make it easy to automatically set the required parameters to generate the optimum output for the project's data as well as to share user-defined build settings between projects. New presets can easily be created by saving your build settings to a preset file. You have the flexibility to save all settings to the preset, or only selected ones. Presets only include changes from the default settings. This allows multiple presets to be selected for a build, each of which defines settings for a different subset of the build steps. If selected preset files have conflicting properties or flags, the value

from the last file on the list is applied. Presets are managed (e.g., loaded, deleted, edited, and set as the default) from the Preset Manager.

Selecting Presets

To select presets for your build:

1. On the **Home** tab, in the **Process** group, click **Build**. The Build Parameters dialog is displayed.
2. Click the **Presets** tab, and then click **Select and Manage**. The Preset Manager is displayed.



Preset Manager

3. On the **Select Presets** tab, on the left side of the Preset Manager is a list of available presets. Select one or more presets from this list, and click **Add** . If you want to change the selected preset(s), use the **Remove**  icon to remove a preset from the **Selected Presets** side. If selected preset files have conflicting properties or flags, the value from the last file on the list is applied. The selected Presets side also includes "Custom changes" with all custom changes that were made to AT, reconstruction, and output settings. The selected presets and custom changes are listed in the Preset section of the Build Parameters dialog.

Note: The Last Build Parameters preset includes all the Build settings from the last build.

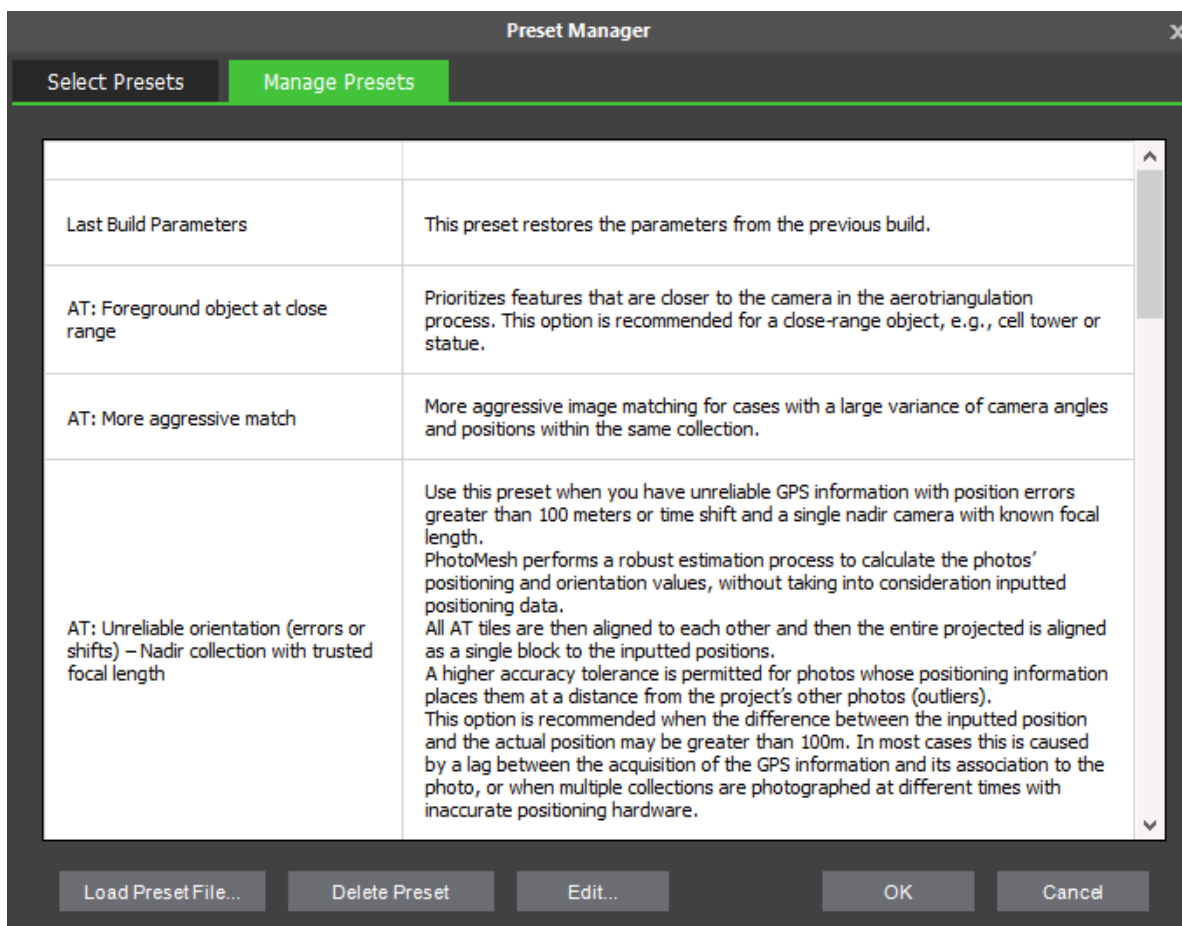
Note: The PhotoMesh Default preset is designed to provide optimal results for the majority of projects.

4. If you want to set your selections as the default presets for new projects, click **Set Selection as Default**.
5. If you want to clear your selections and restore the last saved default, click **Reset to Default**.
6. Click **OK**.

Loading a Preset from File

To load a preset from file:

1. On the **Home** tab, in the **Process** group, click **Build**. The Build Parameters dialog is displayed.
2. Click the **Presets** tab, and then click **Select and Manage**. The Preset Manager is displayed.



Preset Manager

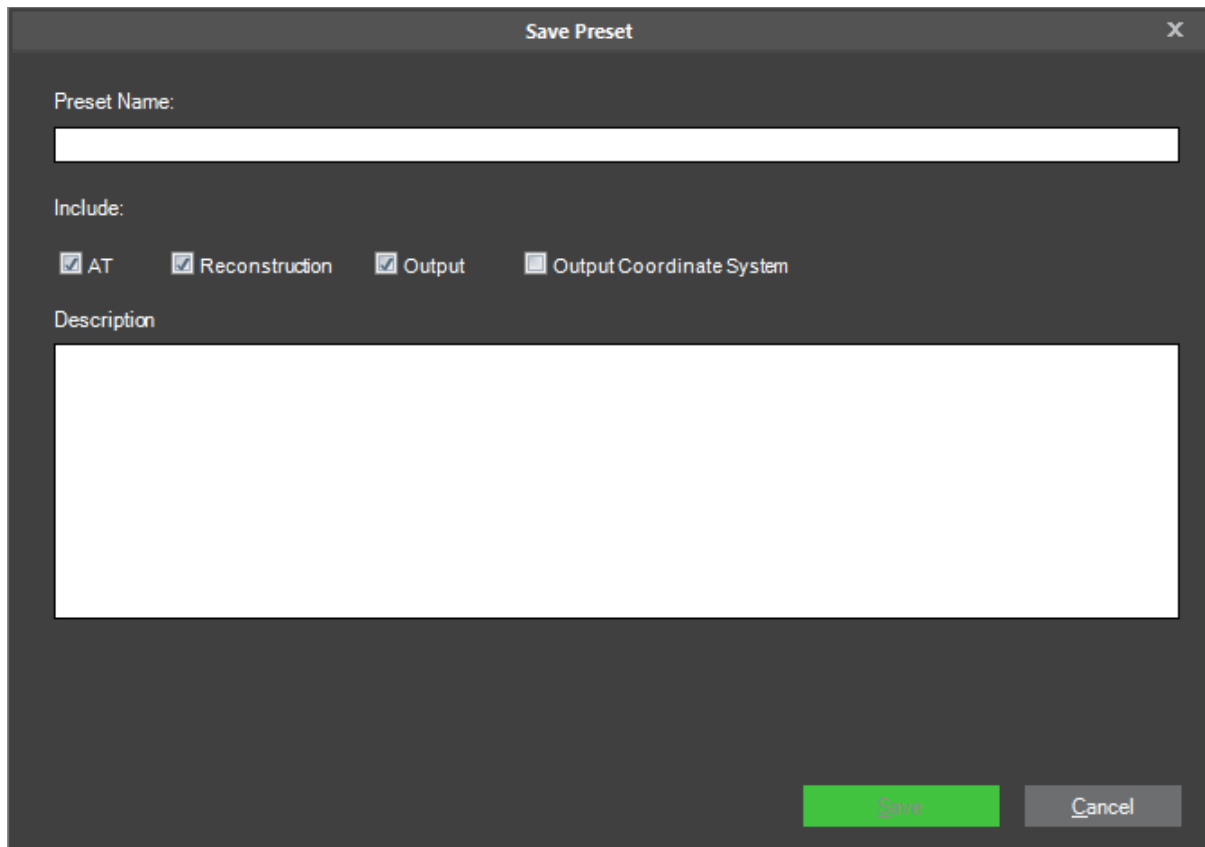
3. On the **Manage Presets** tab, click **Load Preset File**. The Load Preset from File dialog is displayed.
4. Browse to the required file, and click **Open**.

Creating a New Preset

A new preset can be easily created by saving your build settings to a preset file. You can save all your build settings, or only the settings from specific build steps. Only changes to the default settings are saved.

To create a new preset:

1. Set the build parameters as required. See "[Setting Build Steps, Parameters, and Outputs](#)" in this chapter for information.
2. In the **Build Parameters** dialog, click the **Presets** tab, and then **Save as Preset**. The Save Preset dialog is displayed.

The image shows a 'Save Preset' dialog box with a dark gray background. At the top, the title bar says 'Save Preset' with a close button (X) on the right. Below the title bar, there is a label 'Preset Name:' followed by a white text input field. Underneath that is a label 'Include:' followed by four checkboxes: 'AT' (checked), 'Reconstruction' (checked), 'Output' (checked), and 'Output Coordinate System' (unchecked). Below the checkboxes is a label 'Description' followed by a large white text area. At the bottom right of the dialog, there are two buttons: a green 'Save' button and a gray 'Cancel' button.

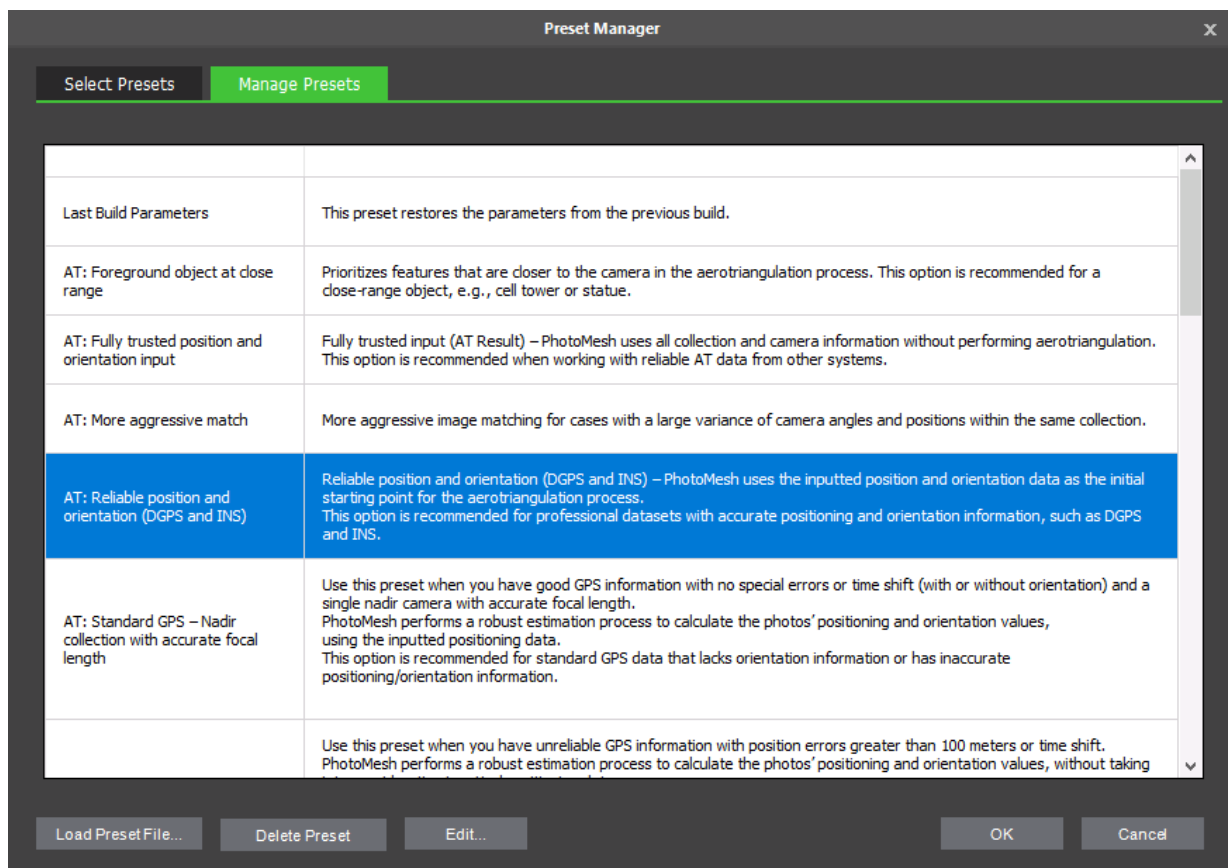
Save Preset Dialog

3. Type a **Preset Name** for the preset and a free text **Description**.
4. Select the check boxes of the build steps whose settings you want to save in your preset file.
5. Click **Save**.

Deleting Presets

To delete a preset file:

1. On the **Home** tab, in the **Process** group, click **Build**. The Build Parameters dialog is displayed.
2. Click the **Presets** tab, and then click **Select and Manage**. The Preset Manager is displayed.



Preset Manager

- On the **Manage Presets** tab, select the required preset from the list, and click **Delete Preset**.

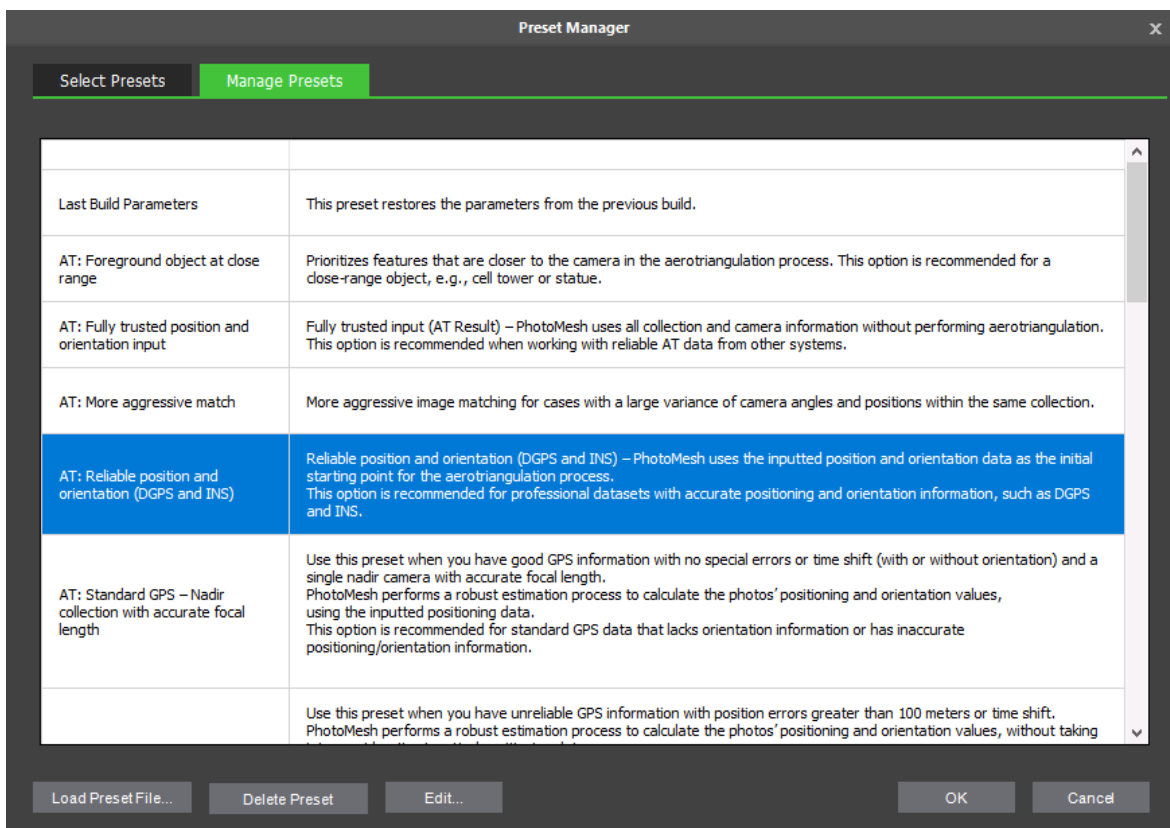
Note: Built-in presets cannot be deleted.

- Click **OK**.

Editing Presets

To edit a preset file:

- On the **Home** tab, in the **Process** group, click **Build**. The Build Parameters dialog is displayed.
- Click the **Presets** tab, and then click **Select and Manage**. The Preset Manager is displayed.



Preset Manager

- On the **Manage Presets** tab, select the file you want to edit from the list, and click **Edit**. The preset XML file is opened in a text editor

Note: Built-in presets cannot be edited.

- Edit the file in the text editor, and save your changes.
- In the **Preset Manager** dialog, click **OK**.

Monitoring a Build

PhotoMesh Build Manager has two sections for monitoring the current build:

- The upper monitoring section of PhotoMesh Build Manager displays information for each fuser, including status, its currently running step, and its progress for that step.
- The lower monitoring section of PhotoMesh Build Manager displays information for each aerotriangulation and reconstruction tile, including currently running step, last step, progress, and currently running fusers.

To monitor the build:

- In the upper monitoring section of PhotoMesh Build Manager, from the **Filter** dropdown list, select one of the following:
 - All** – To view all fuser information
 - Selected** – To view only information related to selected fusers.
- The following data is available in the upper monitoring section for each of the available fusers:

Field	Description
Fuser	Fuser name.
Running Step	Step that the fuser is currently processing.
Step Elapsed Time	Time that has elapsed since the currently processing step was started.
Progress	Percentage of currently running step that has been completed.
Status	Fuser status: ▪ Online – Working – Online and processing ▪ Online – Waiting – Online but not currently processing ▪ Offline – Will be used when online – Fuser was either disabled or closed.

- In the lower monitoring section of PhotoMesh Build Manager, from the **Filter** dropdown list, select one of the following:
 - **All** – to view all task information.
 - **Errors** – to view only information related to task errors.
- The following data is available in the lower monitoring section:

Field	Description
Task	All project tasks including a list of AT and reconstruction tiles.
Last Step	The previous step that was performed
Running Step	Step that the fuser is currently processing.
Status Description	Information about the build status.
Progress	Percentage of currently running step that has been completed.
Fusers	Fuser that is currently processing this task.

PhotoMesh Build Manager - \\pmnas2.skylineglobe.com \PM\sfm\data2\ZedekPort\Build_2_lbe0_3183\project.xml

[Data Preparation](#)
[Aerotriangulation](#)
[Reconstruction Tiles \(Point Clouds, Model, Texture\)](#)
[Project Output](#)

Working Folder: \\pmnas2.skylineglobe.com \Pool\ZedekPort[Manager=DESKTOP-8BM4AJC[10.100.102.9]] [Browse...](#)

Automatically use new available fusers: ☒

Use AWS Instances (Change): [OFF](#)

Use Fusers Pool (Change): [ON](#) Allocated Pool fusers: 5 (Required: 6 max: 20). [Pool Manager](#)

Fusers Online: 6 Selected: 7 Working: 1 Filter: All

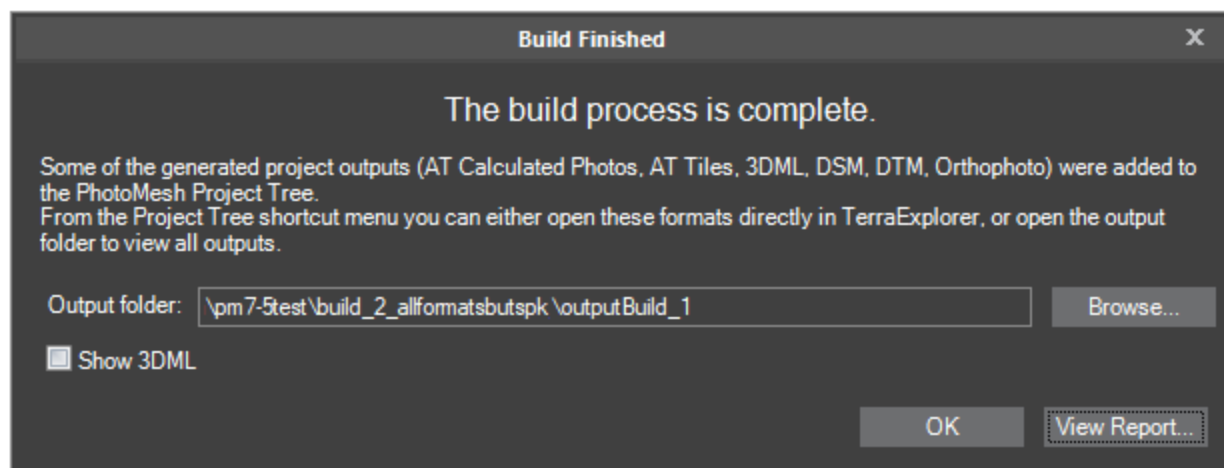
Use	Fuser	Version	Running Step	Step Elapsed Time	Progress	Status
☒	DESKTOP-8BM4AJC(10.100.102.9)	7.6.0.4367	Create mesh model [Tile-1-1-1-1]	00:02:10	0%	Online - Working
☒	PM022(192.168.204.22)	7.6.0.4367				Online - Waiting
☒	PM023(192.168.204.23)	7.6.0.4367			0%	Offline - Will be used when
☒	PM024(192.168.204.24)	7.6.0.4367				Online - Waiting
☒	PM056(192.168.204.56)	7.6.0.4367				Online - Waiting
☒	PM057(192.168.204.57)	7.6.0.4367				Online - Waiting
☒	PM060(192.168.204.60)	7.6.0.4367				Online - Waiting

AT Tiles: 1 Reconstruction Tiles: 6 Filter: All

Task	Last Step	Running Step	Status Description	Progress	Fusers
AT Tile-0-0-1-1	Calculate camera positions				
Build_2_lbe0_3183	Merge				
Tile-0-0-1-1	Convert Tile				
Tile-0-1-1-1	Convert Tile				
Tile-0-2-1-1	Convert Tile				
Tile-1-0-1-1	Convert Tile				
Tile-1-1-1-1	Create point cloud	Create mesh model			DESKTOP-8BM4AJC(10.100.102.9)
Tile-1-2-1-1	Convert Tile				

PhotoMesh Build Manager – Monitoring a Build

- If you want to view a production report, click **Production Report**. See "[Viewing a Production Report](#)" in the "Reviewing Your Build" chapter.
- If you want to view build settings (read-only), click **Show Settings**.
- When the build process is complete, the Build Finished dialog is displayed.



Build Finished Dialog

8. If you want to open the output folder, select the link in the **Output Folder** field, and open it in your browser.
Note: The output folder can also be opened at any time from the Build Outputs list. See ["Opening Build Output in TerraExplorer"](#) in this chapter for more information.
9. If a 3DML was created in this build, and you want to show it in the 3D Window, click **Show 3DML**.
Note: The 3DML's display can also be turned on or off by toggling the Visible check box in the Build Outputs list.
10. If you want to view a production report, click **View Report**.

Canceling a Build

To cancel a build:

- On the bottom right of **PhotoMesh Build Manager**, click **Cancel**.

Resuming a Build

A PhotoMesh build process can be resumed in any of the following cases:

- The initial build process was canceled by the operator in the middle.
- Unexpected hardware or software failure occurred during the build causing the build to stop

To resume a build:

Note: You can only resume a build when you are in Review mode for that build.

1. If you are not in Review mode, on the **Home** tab, in the **Review** group, click **Review Mode**.
2. On the **Home** tab, in the **Process** group, click the arrow under **Build** and select **Resume Build**.

Note: If you want to create an entirely new build, see ["Creating a New Build"](#) in this chapter for information.

Accessing the Build Outputs

Many outputs can be opened in TerraExplorer after the build is complete. The supported outputs include: 3DML, AT Calculated Photos, AT Tiles, DSM, DTM, and Orthophoto. These outputs are added to the

Build Outputs list when a build is finished. The Build Outputs list is organized by build. Each build can be expanded in the list to show all the outputs generated in that build.

Build Outputs			
Visible	Type	Name	Output Folder
☐		AT Tiles	outputBuild_1_WGS 84 UTM zone 17N + EGM96 geoid height
☒		Orthophoto	outputBuild_1_WGS 84 UTM zone 17N + EGM96 geoid height
▼ Build: Build_605_RapidOrtho_fr40809			
☐		AT Calculated Photos	outputBuild_1_WGS 84 UTM zone 17N + EGM96 geoid height
☐		AT Tiles	outputBuild_1_WGS 84 UTM zone 17N + EGM96 geoid height
☒		Orthophoto rapid	outputBuild_1_WGS 84 UTM zone 17N + EGM96 geoid height
☒		Orthophoto	outputBuild_1_WGS 84 UTM zone 17N + EGM96 geoid height
> Build: Build_607_Rapid_Nadir cameras_40811			
▼ Build: Build_608_4AT's			

To view a build output in PhotoMesh:

- In the **Build Outputs** list, select the **Visible** check box of the output you want to display in the 3D Window.

To open a build output in TerraExplorer:

- In the **Build Outputs** list, select one of the outputs, and click **Open with TerraExplorer** . A FLY file opens in TerraExplorer with the 3DML, AT Calculated Photos, AT Tiles, DSM, DTM, and Orthophoto outputs, if they were generated in the PhotoMesh project.

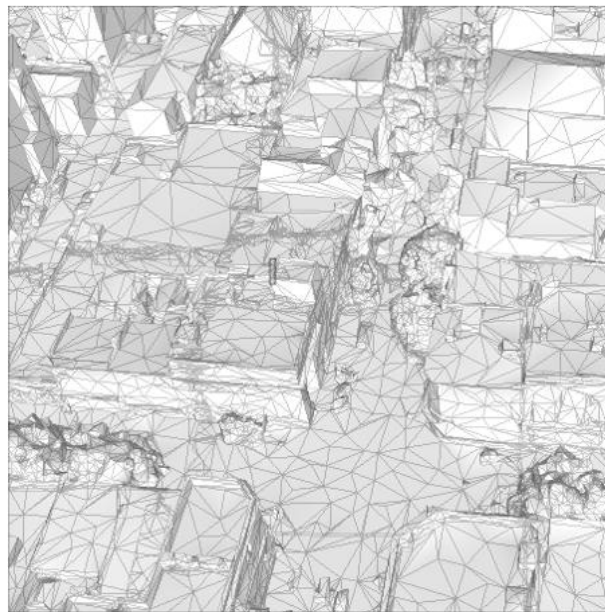
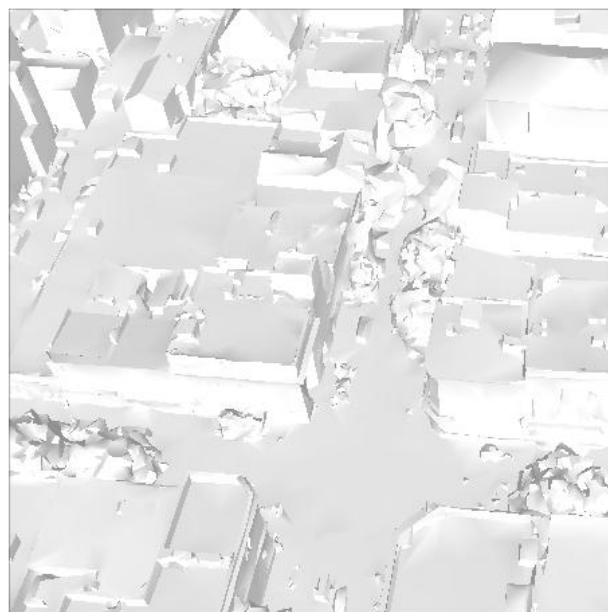
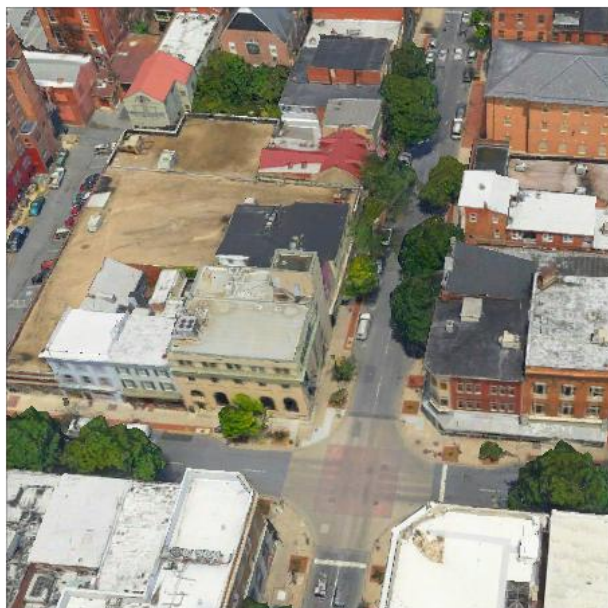
Note: The Build Outputs list also provides an Open Project's Output Folder  option to access any of the project's outputs.

3DML Display Style Options

Several style options are available for displaying the mesh model output. The mesh model can include a wireframe and be textured or solid color.

To select a mesh layer display style:

- On the **View** tab, in the **Mesh Layer** group, select the arrow under **Display Style** and select a style:
 - Texture
 - Texture + wireframe
 - Solid color
 - Solid color + wireframe



Exporting DSM Output to LAS/LAZ

DSM output can be exported to LAS/LAZ format.

To export DSM output to LAS/LAZ:

1. Select the **Build Outputs** group in the Project Panel, and then in the **Build Outputs** list, right-click the required DSM build output, and select **Export to LAS/Export to LAZ**. The Select LAS/LAZ dialog is displayed.
2. Browse to the output folder, and click **Select Folder**.

Rebuilding

If a previously built project requires modification, instead of building the entire project again from the first step of the build process, you have the option to run the build again, incorporating some of a previous build's results. Rebuilding provides flexibility both with regard to the project section (entire project, only error tiles or selected tiles) and project steps that are to be performed. Alternative build parameters can also be set. PhotoMesh will automatically determine what processing is required to apply the changes to the project, by rebuilding the project from "Auto". E.g., when rebuilding a project after modifying a water polygon, only affected tiles are rebuilt from the point cloud step. Similarly, when rebuilding after reimporting exported models, only affected tiles are rebuilt, either from the texturing step, if just models were imported, or from the output step, if both models and textures were imported.

Note: If you want to create an entirely new build, see ["Creating a New Build Version"](#) in this chapter for information.

To rebuild:

1. On the **Home** tab, in the **Process** group, click the arrow next to **Build**, and select the required option:
 - Build
 - Build and Rerun Failed Tasks – Continue the build process including rerunning of any build failed task. Tiles with insufficient photo coverage for reconstruction are not rebuilt. To rebuild these tiles, select the Rebuild Selected Tiles option, and perform all build steps (i.e., select [Complete Project](#)).
 - Rebuild Selected Tiles – Rebuild only the tiles that are currently selected.

The Build Parameters dialog is displayed.

2. Select the project steps and enter the required parameters. See ["Setting Build Steps, Parameters, and Outputs"](#) in this chapter for information.

Note: It is generally recommended to select Auto for the Start From step. In this case, PhotoMesh will automatically determine what processing must be performed to complete the project.

Note: Some of the advanced parameters will only apply if tiles need to be recreated.

Note: Output fields are not available when rebuilding error tiles.

3. PhotoMesh Build Manager then opens. Select the required fusers, and then click **Build**. See ["Using Fusers"](#) in the "Fusers" chapter for more information.

Note: The expected production times for a 10.63 gigapixel project (0.13 km² and 3 cm GSD) on an i7 14th generation, 20 cores and 28 logical cores CPU, with 32 GB RAM, and a local SSD drive are as follows:

- Approximately 6 minutes for photo preparation (238 photos)
- 2 minutes for a single AT tile
- 13 hours and 13 minutes for the reconstruction tiles (20 tiles)

The total machine time adds up to 13 hours and 21 minutes (approximately 47 minutes per gigapixel).

Creating a New Build Version

You can create a new build version with different project steps and different build parameters.

Note: If you want to create a completely new build version with a copy of the AT information from the current build, [see "Creating a New Build with a Copy of the Current Aerotriangulation Results"](#) in this chapter.

To create a new build version:

1. On the **Home** tab, in the **Build Versions** group, click the arrow next to **New Version**, and select **New Build Version**. The Build Parameters dialog is displayed.
2. Select the project steps and then enter the required parameters. [See "Setting Build Steps, Parameters, and Outputs"](#) in this chapter for information. The New build description dialog is displayed with a consecutive build number, e.g. "Build_2", "Build_3" etc.
3. Type a new build description, and click **OK**. PhotoMesh Build Manager then opens. Select the required fusers, and then click **Build**. [See "Using Fusers"](#) in the "Fusers" chapter for more information.

Creating a New Build Version with a Copy of the Current Aerotriangulation Results

When no AT parameters were changed from a previous build, you can create a new build version with different build parameters (e.g. reconstruction tile size) but with a copy of the AT information from the current build, thereby significantly accelerating the build process.

Note: If you want to create a completely new build version without copying the AT information from the current build, [see "Creating a New Build Version"](#) in this chapter.

To create a new build version with a copy of the AT:

1. On the **Home** tab, in the **Build Versions** group, click the arrow next to **New Version**, and select **New Build Version – Copy AT**. The Build Parameters dialog is displayed.
2. Select the Build Steps to perform. [See "Setting Build Steps, Parameters, and Outputs"](#) in this chapter for information. Select **Complete Project** to have PhotoMesh automatically determine what processing must be performed to build the outputs and complete the project, avoiding unnecessary rebuilding.

Note: This build version is using a copy of the AT from the current build, so no aerotriangulation parameters can be modified. This includes the Position & Accuracy and Coordinate System settings on the Build Settings tab as well as all settings on the Aerotriangulation tab.

3. Select the **Collection Type** and other parameters, and click **Build**. [See "Setting Build Steps, Parameters, and Outputs"](#) in this chapter for information.

Note: Some of the parameters will only apply if tiles need to be recreated.

Note: Output fields are not available when rebuilding error tiles.

The New build description dialog is displayed.

4. Type a new build description, and click **OK**. PhotoMesh Build Manager then opens. Select the required fusers, and then click **Build**. [See "Using Fusers"](#) in the "Fusers" chapter for more information.

CHAPTER 17 REVIEWING YOUR BUILD

About Reviewing Your Build

Upon initiating a build, PhotoMesh automatically switches to Review mode, enabling an array of capabilities relating to the processed data. These include:

- Viewing all processed build data in the 3D Window, e.g. AT and reconstruction tiles, calculated camera positions, and point cloud, 3D model, and texture data.
- Reading build and AT reports containing statistical and graphical information about the build process.
- Exporting/importing tiles for manual retouching.
- Rerunning the build (for the entire project or for selected tiles) with different parameters.

As soon as a build step is completed for a tile, its build data becomes immediately available and viewable, e.g., if aerotriangulation was performed, PhotoMesh can display the calculated camera positions in the 3D Window, and if reconstruction tiles were calculated, they can be displayed in the 3D Window, with each tile colorized according to the build step completed.

Upon build completion, a [Quality Report](#) is displayed with information about errors and missing information, e.g., high median errors (larger than 0.5 px), high standard deviation errors (larger than 0.8 px) or inaccurate positioning data.

In addition to these capabilities which are specific to Review mode, you can also perform in Review mode nearly all operations that were available in preparing for the build, e.g. add or edit control points, water polygons, and reconstruction polygons, add terrain objects and layers and perform measurements. The only operations that cannot be performed in Review mode involve changes to the project's photos (addition, exclusion, or deletion) or modifications to the AT area.

Turning on Review Mode

To turn on review mode:

- On the **Home** tab, in the **Review** group, click **Review Mode**.

Note: Review mode is activated automatically after a build was performed.

Clearing Review Data from Terrain

To clear review data:

- On the **Home** tab, in the **Review** group, click **Clean Review Data**. This clears any point clouds, models, or textured models displayed on terrain or photo projection on the terrain.

Reviewing Aerotriangulation (AT) and Reconstruction Tiles

Upon completion of aerotriangulation or any of the build steps, the aerotriangulation/reconstruction tiles can be previewed in the 3D Window, with the tiles colorized according to their processing status.

Additional tile information (tile camera positions and build data) can be viewed from the Tile tab.

The sections that follow describe the activities that can be performed to evaluate the aerotriangulation/reconstruction tile results:

- [View aerotriangulation tiles](#)

- [View reconstruction tiles](#)
- [Navigate to a tile](#)
- [Display original camera positions for tiles](#)
- [Display calculated camera positions for tiles](#)
- [View an aerotriangulation tile's sparse point cloud](#)
- [Display reconstruction tile results](#)
- [View and set tile status](#)
- [Display user status flags](#)
- [Set a reconstruction tile's display style](#)
- [View a tile's production report](#)
- [View an aerotriangulation tile's quality report](#)
- [Export tiles](#)
- [Import tiles](#)

Viewing Aerotriangulation (AT) Tiles

To view aerotriangulation tiles:

Note: This information is also available in the AT Tiles list. **See** ["Selecting Tiles"](#) and ["Viewing Tile Information in the Tile List"](#) in this chapter for information.

1. In the **Project Panel**, select the **AT Tiles** check box. The aerotriangulation tiles are displayed in the 3D Window, colorized according to their aerotriangulation status:
 - **White** tile – Unprocessed
 - **Purple** tile – Aerotriangulation completed
 - **Yellow** tile – Aerotriangulation merged
 - **Orange** border – Aerotriangulation failed (retrying)
 - **Red** border – Aerotriangulation failed
 - **Blue** border - <80% Success

Note: In the AT Tiles legend, the colored squares (e.g. ) indicate colored tiles, while the colored lines (e.g. ) indicate a colored border.

2. If you hover anywhere on the tile, the tile's name and number of photos in the tile are displayed at the top left corner of the 3D Window.
3. In the **3D Window**, click the tile you want to review, or select the tile in the **AT Tiles List**. The Tile tab is displayed from which you can navigate to the tile, display its original and calculated camera positions and sparse point cloud, and view quality and production reports. **See** ["Navigating to a Tile"](#), ["Displaying Camera Positions for Tiles"](#), ["Viewing an Aerotriangulation Tile's Sparse Point Cloud"](#), ["Viewing a Quality Report"](#) and ["Viewing a Production Report"](#) in this chapter for information.

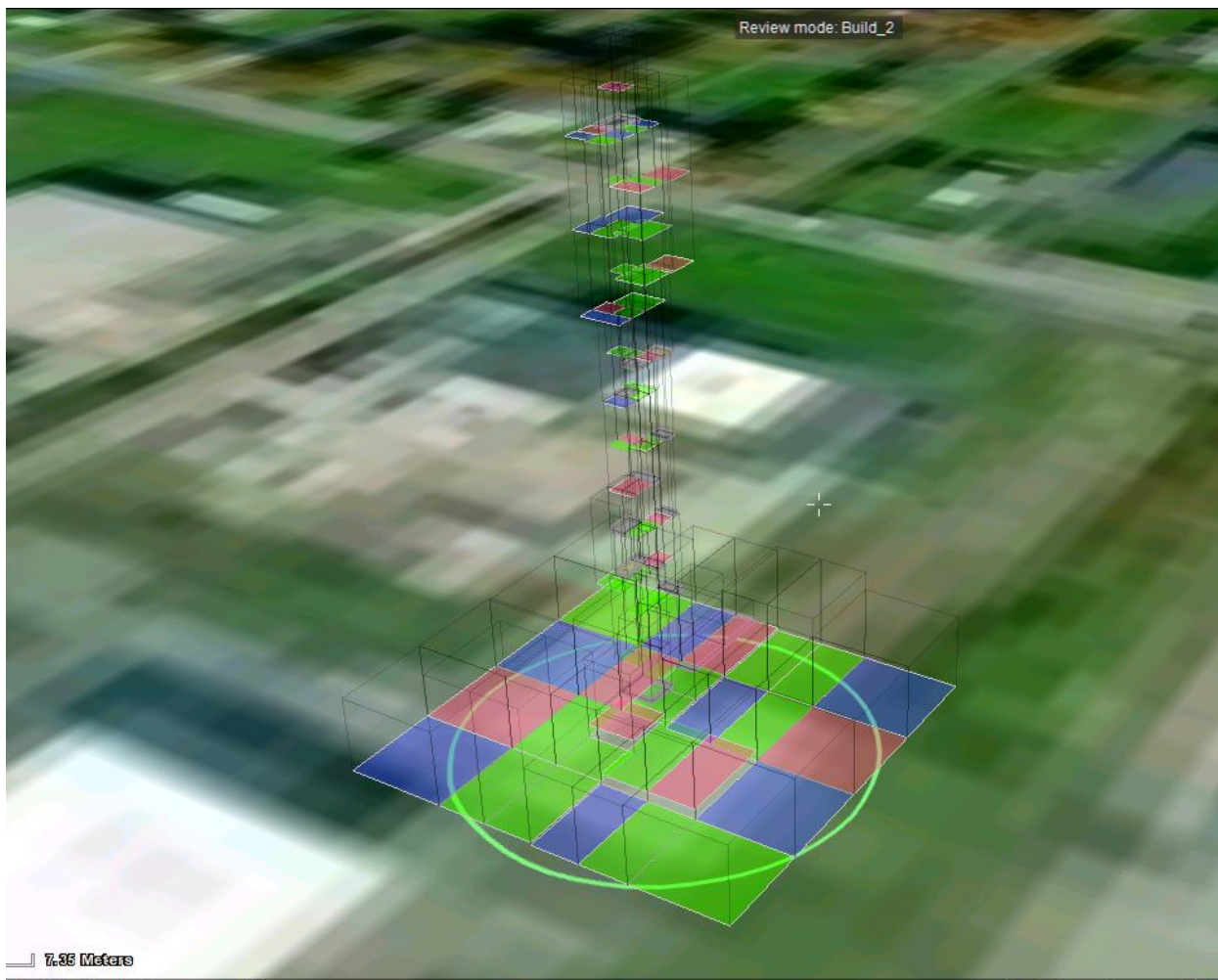
Viewing Reconstruction Tiles

To view the reconstruction tiles:

Note: This information is also available in the Reconstruction Tiles list. See ["Selecting Tiles"](#) and ["Viewing Tile Information in the Tile List"](#) in this chapter for information.

1. In the **Project Panel**, select the **Reconstruction Tiles** check box. The reconstruction tiles are displayed in the 3D Window, colorized according to the last build processing step completed for the tile.
 - **Yellow** tile – Aerotriangulation is complete.
 - **Pink** tile – All steps through point cloud creation are complete.
 - **Dark Blue** tile – All steps through model creation are complete,
 - **Light Blue** tile – All steps through model texturing are complete.
 - **Green** tile – All steps through project output are complete.
 - **Orange** border – An error occurred in the processing step that follows the last successfully completed step and a retry mechanism will be applied soon (e.g., if a dark blue colored tile has an orange border, this indicates that the 3D model creation step completed successfully but that model texturing failed and will be retried).
 - **Red** border – Tile failed.
 - **Dark Red** border – Insufficient photo coverage for reconstruction.

Note: In the Reconstruction Tiles legend, the colored squares (e.g. ) indicate colored tiles, while the colored lines (e.g. ) indicate a colored border.



Reconstruction Tiles

2. If you hover anywhere on the tile, the tile's name and number of photos in the tile are displayed at the top left corner of the 3D Window.
3. In the **3D Window**, click the tile you want to preview, or select the tile in the **Reconstruction Tiles** list. The **Tile** tab is displayed from which you can navigate to the tile, display its original and calculated camera positions and reconstruction tile results, view production reports, and export and import the tile's model. See ["Navigating to a Tile"](#), ["Displaying Camera Positions for Tiles"](#), ["Displaying Reconstruction Tile Results"](#), and ["Viewing a Production Report"](#) in this chapter and the ["Importing and Exporting Models"](#) chapter for information.

Selecting Tiles

Tiles can be selected either in the 3D Window or from the AT or Reconstruction Tiles Lists to perform various commands on them, e.g., jump to tile, show calculated information, or show AT or reconstruction tile results.

Reconstruction Tiles						
 1 out of 86 selected						
Selection	Name	Photo Count	Build Status	Error Status	User Status	Comment
☐	Tile-0-0-1-1	613	■ Output		Unset	
☐	Tile-0-1-1-1	669	■ Output		Unset	
☐	Tile-1-0-1-1	667	■ Output		Unset	
☐	Tile-2-0-1-1	701	■ Output		Unset	
☐	Tile-1-1-1-1	912	■ Output		Unset	
☐	Tile-2-1-4-1	836	■ Output		Unset	
☐	Tile-2-2-2-1	542	■ Output		Unset	
☐	Tile-11-3-6-1	568	■ Output		Unset	
☐	Tile-11-8-6-1	444	■ Output		Unset	

Reconstruction Tiles List

To select tiles:

1. In the **Project Panel**, select the **Reconstruction Tiles** or **AT Tiles** group.
2. In the **3D Window**, click the tile you want to select, or select the tile(s) in the **Reconstruction Tiles** or **AT Tiles** list. You can multi-select tiles using the Tile List commands.

Note: If you turned on global selection, you can select a tile in the 3D Window even without first selecting the Tiles group in the Project Panel as long as the tiles are visible in the 3D Window.

See ["Showing and Hiding Group Items"](#) and ["Selecting Items Using the Item List"](#) in the "Using the Project Panel and Item List" chapter and ["Selecting Objects in the 3D Window"](#) in the "Basic Concepts" chapter for information.

3. If you want to save your selection, right-click in the **Reconstruction Tiles** or **AT Tiles** list, and select **Save Selection**.
4. To clear your selection, right-click, and select **Clear Selection**.

5. To load a tile's *.select* file with selection information, right-click, and select **Load Selection**. See "[Tile Menu](#)" in the "Using the Project Panel and Item List" chapter for more information.

Viewing Tile Information in the Tile List

The AT and reconstruction tiles lists provide information about each tile, including: photo count, build processing status, and error status. See "[Selecting Items Using the Item List](#)" in the "Using the Project Panel and Item List" chapter for more information.

To view tile information:

- In the **Project Panel**, select the **AT Tiles** or **Reconstruction Tiles** group. The tile list displays all the tiles in the selected group, and the following information:

Property	Description
Selection	Select the check box to select the tile in the 3D Window. You can select multiple tiles to perform actions on multiple tiles.
Name	Name of the tile (Read-only).
Photo Count	Number of photos in the tile (Read -only).
Photos Calculated	Number of photos in AT tile for which PhotoMesh successfully calculated camera position (only AT) (Read -only).
Calculated %	Percentage of photos in AT tile for which position information was calculated: Photos Calculated/Photo Count (only AT) (Read -only).
Build Status	For AT tiles: ▪ Unprocessed ▪ AT completed ▪ AT merged For reconstruction tiles: ▪ Initiated ▪ Point Cloud ▪ Model ▪ Textured Model ▪ Output (Read -only) Note: The tiles in the 3D Window are also colorized based on build status. See "Viewing Aerotriangulation (AT) Tiles" and "Viewing Reconstruction Tiles" in this chapter for information.
Error Status	Indicates if there was an error in creating the tile: Failed, Failed (retrying), No coverage (only reconstruction) or <80% success (only AT) (Read -only).

User Status	Click the dropdown arrow, and select the applicable status, indicating what actions still need to be performed on the tile, e.g., rebuild, editing or external retouch. You can also set the user status from the Tile tab > User Status. User status flags can be displayed in the 3D Window on each of the tiles to indicate at a glance the current status of each tile. See "Setting and Displaying User Status Flags and Legend" in this chapter for information. You can customize the name and color associated with each user status in the Options menu. See "Setting PhotoMesh Options" in the "Basic Concepts" chapter for more information. ▪ For AT Tiles: OK, Requires Rebuild, Requires GCP. ▪ For Reconstruction Tiles: OK, Requires Rebuild, Requires Editing, Requires External Retouch. Note: User status can also be set from the Tiles tab (Tiles tab > User Status).
Comments	Free text field where you can add any applicable information about the tile. You can add comments from the Tile tab > User Status.

Navigating to a Tile

To fly or jump to a tile:

1. Select the required tile. See "[Selecting Tiles](#)" in the "Reviewing Your Build" chapter.

Note: If multiple tiles are selected, navigation is to the last selected tile.

2. On the **Tile** tab, in the **View** group, click **Fly to Tile/ Jump To Tile**.

Displaying Camera Positions for Tiles

After the aerotriangulation step has begun, you can display photos' original camera positions. Calculated camera positions for photos can only be displayed after the aerotriangulation step is complete. When calculated camera positions are displayed, you can also view calculated property information in the photo list. See "[Calculated Camera Position Properties](#)" in the "Photo Management" chapter for information.

To display photos in the original camera positions defined in the project:

1. Select the required tile. See "[Selecting Tiles](#)" in this chapter. The Tile tab is displayed.
2. On the **Tile** tab, in the **Original Photos** group, toggle on **Original**.
3. On the **Tile** tab, in the **Original Photos** group, in the **Color Code** dropdown, select one of the following options for color coding of the photos' camera symbols:
 - **Status** – Camera symbols are color coded based on photo status (calculated/ not calculated).
 - **Collection** – Camera symbols are color coded according to collection, with a random color assigned to each collection's camera symbol.

Note: These settings can also be accessed from the View tab.

To display photos in the camera positions calculated through aerotriangulation:

1. Select the required tile. See "[Selecting Tiles](#)" in this chapter. The Tile tab is displayed.
2. On the **Tile** tab, in the **Calculated Photos** group, toggle on **Calculated**. In the 3D Window, the photos are displayed in the camera positions calculated through aerotriangulation, and calculated

positioning information is included in the photo list. See ["Photo List"](#) in the "Photo Management" chapter for information.

3. On the **Tile** tab, in the **Calculated Photos** group, in the **Color Code** dropdown, select one of the following options for color coding of the photos' camera symbols in the 3D Window:
 - **Status** – Camera symbols are color coded based on status (calculated/ not calculated).
 - **Collection** – Camera symbols are color coded according to collection, with a random color assigned to each collection's camera symbol.
 - **Median Error** – Camera symbols are color coded according to median error (px):
 - Green: $px < 0.5$
 - Orange: $0.5 < px < 1$
 - Purple: $px > 1$
 - **Reconstruction Group** – Camera symbols are color coded according to reconstruction group (i.e., photo blocks created in the AT process), with a random color assigned to each reconstruction group's camera symbol. This information is significant in identifying the largest group (i.e., the group PhotoMesh will perform reconstruction on) and determining which additional photos can be connected to this group using tie points and other methods, for optimal output.
 - **Number of Matches** - Camera symbols are color coded according to the number of photo matches:
 - Red: matches ≤ 2
 - Orange: $2 < \text{matches} \leq 4$
 - Green: matches > 4

Viewing an Aerotriangulation Tile's Sparse Point Cloud

A sparse point cloud is a representation of all photo features that were automatically found in two or more project photos, with their computed positions. This provides a quick overview of the aerotriangulation results.

To view an AT tile's sparse point cloud:

1. Select the required tile. See ["Selecting Tiles"](#) in this chapter. The Tile tab is displayed.
2. On the **Tile** tab, in the **Sparse Point Cloud** group, click **Show Point Cloud**.

Note: The sparse point cloud may take some time to display the first time it is shown or after making changes to the project.
3. Select the **Point Size** in pixels.
4. Select the color coding option for classifying the point cloud's points:
 - **Single color** – A single color is used for all points.
 - **Num. tied photos** – Points are color coded according to the number of photos in which this point was found.
 - Orange – Two matches
 - Green – Three or more matches

Displaying Reconstruction Tile Results

To display point cloud, 3D model, or texture results for a tile:

1. Select the required tile. See ["Selecting Tiles"](#) in this chapter. The Tile tab is displayed.

2. On the **Tile** tab, in the **Show** group, toggle on any of the following options:
 - Point Cloud – Click the arrow to select the data to display:
 - Merged – Show point cloud created by supplementing LiDAR with points generated from photos where LiDAR information is insufficient.
 - From Photos – Show point cloud created only from points generated from photos.
 - From Lidars – Show point cloud created only from LiDAR points.
 - Model – Shows only the mesh model
 - Textured Model – Shows the model with texturing. Several style options are available for displaying the textured mesh model. [See "3DML Display Style Options"](#) in the "Building" chapter for information.
 - Orthophoto – If RGB color bands were selected for the orthophoto, they are displayed. Otherwise, only the first band is displayed in grayscale mode. [See "Selecting the Collections and Bands to Include in the Build Outputs"](#) in this chapter for information.

Note: Each of these options is enabled only if the build step for this data was completed.

Setting and Displaying User Status Flags and Legend

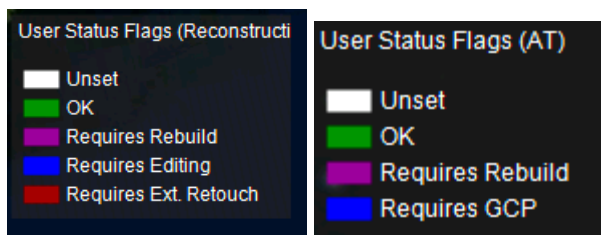
You can set and display the user status of AT and reconstruction tiles. By default the user status options are used to indicate what actions still need to be performed on the tile, e.g., rebuild, editing or external retouch, but they can be customized as required from the Options menu. After setting user status, flags can be displayed in the 3D Window on each of the tiles to indicate at a glance the current status of each tile.

To set and display user status flags:

1. In the **Project Panel**, select the **AT Tiles** or **Reconstruction Tiles** group. The tile list displays all the tiles in the selected group.
2. Select the row of the required tile, and in the **User Status** column, click the dropdown arrow, and select the applicable status, indicating what actions still need to be performed on the tile, e.g., rebuild, editing or external retouch. You can also set the user status by selecting the tile and then selecting the status from the **Tile** tab > **User Status**. [See "Viewing Tile Information in the Tile List"](#) in this chapter for more information.

Note: You can customize the color and name associated with each user status in the Options menu. [See "Setting PhotoMesh Options"](#) in the "Basic Concepts" chapter for more information.

3. On the **View** tab, in the **Reconstruction Tiles** group, click the arrow under **User Status Flags**, and select **Show User Status Flags**.
4. If you want to show a user status legend that indicates what AT and reconstruction tile user status is associated with each color flag, on the same tab, click the arrow under **User Status Flags**, and select **Show User Status Legend**.



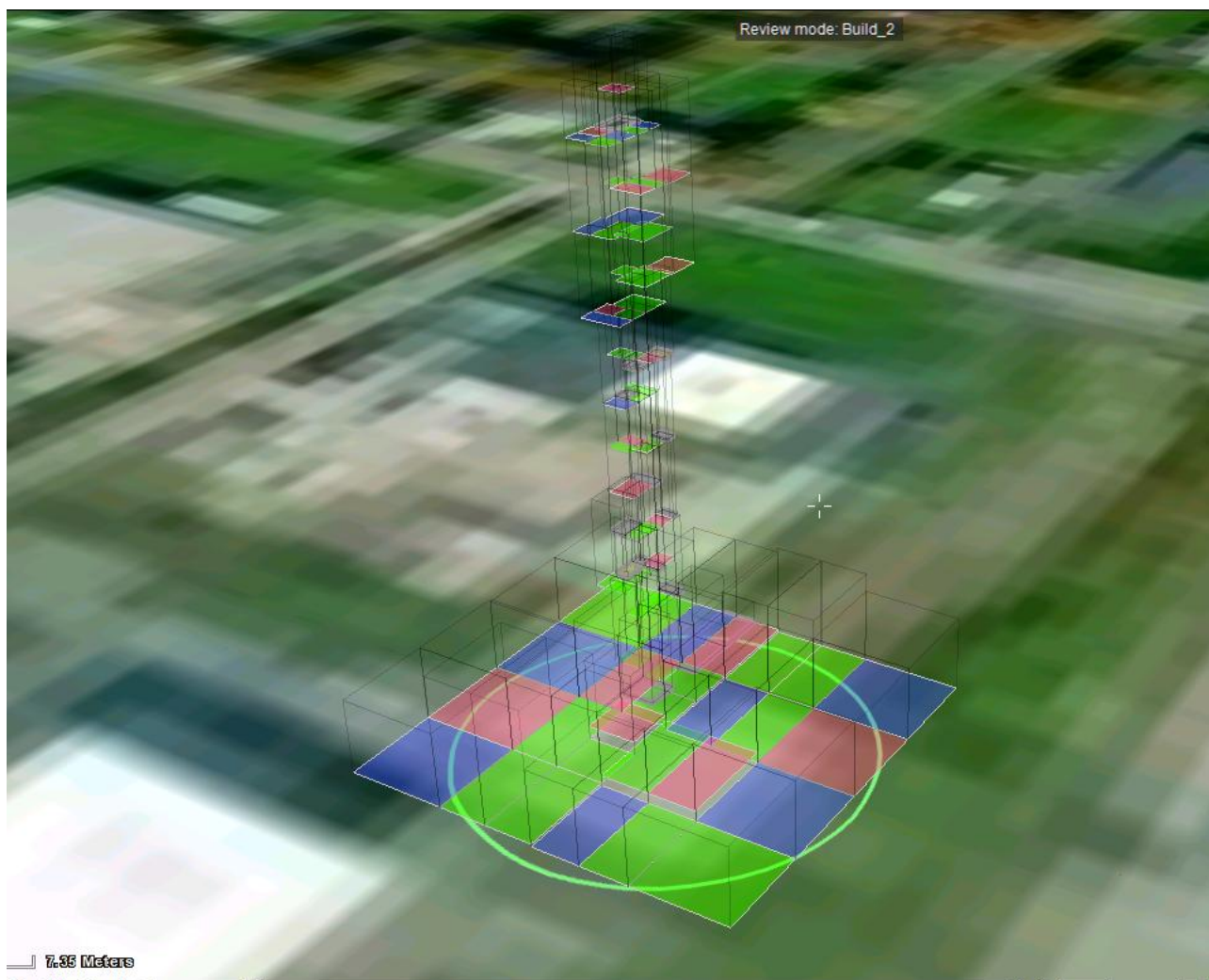
User Status Legend

Setting a Reconstruction Tile's Display Style

Several options are available for the display style of selected reconstruction tiles:

- 2D Rectangle
- 3D Box
- 3D Box Fill
- 3D Dynamic

Only the first three options are available when creating reconstruction tiles using a fixed 2D grid. When reconstruction tiles are created as adaptive 3D boxes, the display style option is automatically set to "3D Dynamic". See "[Setting Advanced Reconstruction Parameters](#)" in the "Building" chapter for more information. In "3D Dynamic" mode, the tiles and their boxes are displayed in their absolute heights. Tiles that were created one on top of each other will appear as if they are floating in this mode.



To set the display style for selected reconstruction tiles:

- On the **View** tab, in the **Reconstruction Tiles** group, click the arrow under **Selection Style** and select the style you want. If you want to display a frame around reconstruction tiles only when selecting a tile, and not when hovering over it, click the arrow under **Selection Style** and select **Hide Tiles Hover**.

To toggle the display of the tile bases of a reconstruction area that is divided into 3D tiles in an adaptive manner:

1. Select the required tile. See ["Selecting Tiles"](#) in the "Reviewing Your Build" chapter.
2. In the **AT Tiles** list or **Reconstruction Tiles** list, select the required tile(s), and toggle the **Tile Base** button  on/off. The color of the base corresponds to the last build processing step completed for that tile. See ["Viewing Reconstruction Tiles"](#) in this chapter for more information. See ["Setting Advanced Reconstruction Parameters"](#) in the "Building" chapter for more information on adaptive 3D boxes.

Viewing a Production Report

The Production report contains statistical and graphical information about the build process. The first section of the report displays summary information: If the report is generated for a specific tile, it will only display information for that tile.

- Basic project information including: project name, number of photos, and total input gigapixels.
- Basic aerotriangulation information including: AT area, number of aerotriangulation tiles, calculated photos, and mean median error. The mean median error is the average of the median reprojection errors for all photos in the collection.
- Basic reconstruction information including: output area, number of reconstruction tiles, and output coordinate system.
- Performance information including: build and fuser time.

This summary section is followed on each report tab by detailed information and graphs relating to the respective tab:

- **(Build Process) Steps** – Details about each build step: steps involved, tiles produced, processing time, and active fusers. When a reconstruction tile is selected, details display about the tile building steps (point cloud, model, texturing). When an aerotriangulation tile is selected, details display about the aerotriangulation.
- **Fusers** – Statistics for each fuser regarding processing time for each build step and each tile in each step.

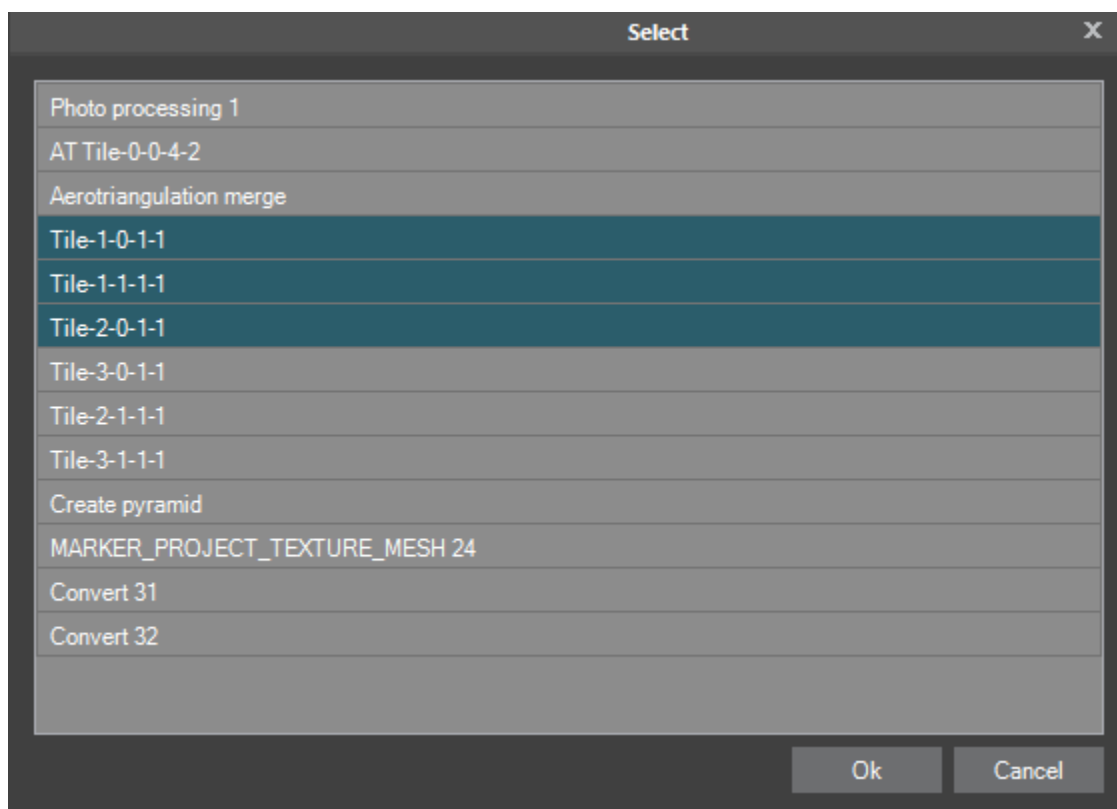
To view a production report:

1. Do any of the following:

Note: The same production report opens in each of these options but with different filter values selected.

- On the **Home** tab, in the **Process** group, click the arrow under **Reports**, and select **Production Report**.
 - Select a tile in the **3D Window** or in the tile list, and then on the **Tile** tab, in the **View** group, click the arrow under **Reports**, and select **Tile Production Report**. The report displayed will display information only for the selected tile.
 - In **PhotoMesh Build Manager**, click the **Production Report** button.
 - In **PhotoMesh Build Manager**, right-click a fuser, and select **Fuser build report**. The report displayed will display information only for the selected fuser.
 - In **PhotoMesh Build Manager**, right-click a tile, and select **Tile build report**. The report displayed will display information only for the selected tile.
2. At the top of the **Production Report**, select a report type. This determines what detailed information and graphs display in the section following the summary section.
 - Steps (Build)
 - Fusers
 3. The detailed information and graph data can be filtered to display information only about tiles with warnings or errors or about specific build process steps, fusers, or aerotriangulation tiles. Click **Filter** and, select one of the following:
 - All

- With warnings
 - With errors
 - User selection
4. If you choose **User selection**, the Select dialog is displayed:
 - a. Select the required step or fuser (depending on the report type being viewed). Use CTRL and SHIFT to multi-select. Then click **OK**.
 - b. If you want to change your selection, click **Select** to reopen the Select dialog.



Select Dialog

5. If you want to expand or collapse parts of the report to make it easier to find and view only the information you want, click **Expand/Collapse All** at the top of the report.
6. If you want to export the Production Report as an HTML file, at the top of the report, click **Export**. Then in the **Export Production Report** dialog, browse to the required file location, and click **Save**.
7. If you want to print the report, at the top of the report, click **Print**.

Viewing a Quality Report

The Quality Report contains statistical and graphical information about the quality of the build and its aerotriangulation and reconstruction results. It highlights key information and alerts you to potential problems such as missing information, triangulation errors, or failed tiles, which are likely to affect the quality of your build output. This enables you to address and resolve the necessary issues. If the report is generated for a specific tile, it will only display information for that tile. The report contains the following sections:

- **Report Summary** – Quick overview of the status of your build and warnings about errors or missing information that may need adjustment to produce a quality build.
- **Project Information** – Basic project information including: project name, number of photos, and total input gigapixels.
- **Photos and Collections** – Information about the photos in each collection including: number of photos, focal length, distortion and coordinate system
- **Control Points** – Basic ground control point, check point, and tie point information including: triangulation error and sampled photos.
- **Reconstruction** – Basic reconstruction information including: reconstruction area, number of reconstruction tiles, number of fusers, and total processing time.
- **Aerotriangulation Results** – Basic aerotriangulation information including: number of aerotriangulation tiles, calculated photos, and mean median error. Aerotriangulation information for each tile is also provided: original and calculated distortion values, error, and principal point, control point information, and maps that graphically depict information about photo matches, level of detail, position accuracy and more. Each map has three views: from the top, front, and left.

To view a Quality Report, do either of the following:

- On the **Home** tab, in the **Process** group, click the arrow under **Reports**, and select **Quality Report**. A PDF is created in the project build's **Report** folder.
- Select the required tile. See "[Selecting Tiles](#)" in this chapter. Then on the **Tile** tab, in the **View** group, click the arrow under **Reports**, and select **Quality Report**. The report displayed will display information only for the selected tile.

Quality Report Error Messages

The Quality Report includes error messages to help you easily identify what are the causes of inadequate output results, so that you can make the necessary corrections before rebuilding.

Below are listed the most common error messages and some recommendations on how to handle them:

- [Input: Photos and Collections](#)
- [Input: Lidar](#)
- [Aerotriangulation Result](#)
- [Control Points](#)

Input: Photos and Collections

Field	Message	Recommendation
Number of Photos	Fewer than 20 photos in collection. / Fewer than 6 photos in collection.	Check whether a collection was mistakenly split into two collections, and if so, merge the split groups into a single one. Homogeneous groups of photographs, all taken by the same physical camera with identical focal length and dimensions can be placed in the same collection.
Image Width	Missing property value. / Invalid property value. Value must be greater than 0.	This information can usually be obtained from your data provider or camera manufacturer, and then entered in the collection property sheet or Excel/XML photo list file.
Image Height	Missing property value. / Invalid property value. Value must be greater than 0.	This information can usually be obtained from your data provider or camera manufacturer, and then entered in the collection property sheet or Excel/XML photo list file.
Sensor Width	Missing property value. This property is necessary in calculating focal length.	This information can usually be obtained from your data provider or camera manufacturer, and then entered in the collection property sheet or Excel/XML photo list file. If you don't have access to this property value, focal length can instead be set based on the pixel size and focal length values or the focal length 35 value.
Pixel Size	Missing property value. This property is necessary in calculating focal length.	This information can usually be obtained from your data provider or camera manufacturer, and then entered in the collection property sheet or Excel/XML photo list file. If you don't have access to this property value, focal length can instead be set based on the sensor width and focal length values or the focal length 35 value.
Focal Length	Missing property value. / Invalid property value. Value must be greater than 0.	This information can usually be obtained from your data provider or camera manufacturer, and then entered in the collection property sheet or Excel/XML photo list file. If you don't have access to this property value, focal length can instead be set based on the focal length 35 value.

Focal Length 35	Missing property value. This property is necessary in calculating focal length. / Invalid property value. Value must be greater than 0.	This information can usually be obtained from your data provider or camera manufacturer, and then entered in the collection property sheet or Excel/XML photo list file. If you don't have access to this value, focal length can instead be set based on focal length together with either pixel size or sensor width. If none of the required values are available, set focal length 35 to the default value of 50.
Principal Point (X,Y)	Principal point is too far from photo center.	Verify that the principal point reference's correct origin and units were selected in the collection property sheet or Excel/XML photo list file.
Input Coordinate System	Missing collection's coordinate system information. / Invalid coordinate system information.	If the XY values seem to be in degrees, i.e., they are in the range from -180 to 180 with 5-6 decimal places, the coordinate system is most likely WGS84 with EGM96 vertical datum. Enter this information in the collection property sheet or Excel/XML photo list file.
Vertical Datum	The WKT of one of your project coordinate systems references a vertical datum (GTX) file that was not found on your computer.	Download the required GTX file. See here for more information.
Name of Photo Collection - Photo Position Data	Missing XYZ position data.	Check if this information is available in an external positions file. If not, you can estimate the missing values by using the values of the adjacent photos, and enter this information in the photo property sheet.
	All of the project's photos have the same XYZ values.	Delete the position data from the property sheet. Verify if this information is available in an external positions file.
	Many of the project's photos are positioned underground.	Verify that the correct geoid/ellipsoid are set in the photo's coordinates system. Also check that the correct altitude mode (relative/absolute) and offset are set in the collection property sheet.
	There are more than 10,000 photos in the project.	Verify that PhotoMesh is running on a sufficiently powerful machine. PhotoMesh can process projects with more than 10,000 photos as long as the machine it is running on has sufficient compute, memory, and storage capabilities.

Name of Photo Collection - Photo Orientation	Invalid value. Orientation values must be in degrees.	If values are in radians and photo information was imported in an Excel file, verify that the Omega/Phi/Kappa Units property is set to "Radians" in the Excel. Otherwise, If orientation values are in gon or radians, convert to degrees, and reimport the photos.
Name of Photo Collection - Band Definition	Mismatch between the number of bands according to the property sheet and the photo files' metadata.	Fix value in property sheet.

Input: Lidar

Field	Message	Recommendation
Coordinate System	Missing Lidar's coordinate system information. / Invalid coordinate system information.	Verify the validity of the WKT in the coordinate system dialog. Fix as required.
Location (bbox)	Lidar point cloud points are all outside the AT or reconstruction area	Verify in the Lidar List that the correct coordinate system and vertical datum are set. If a local coordinate system was set, it must be referenced to a known coordinate system.
Content	Lidar point cloud contains no points	Verify the validity of the WKT in the coordinate system dialog and check that the projected point cloud falls in the expected area. You can also increase the Point Size for better visibility of the points and change Oversampling from Low to Best (right-click menu).

Aerotriangulation Result

Field	Message	Recommendation
Triangulation Error	Triangulation error between 0.5px-1px. / Triangulation error > 1px	Check the sensor parameters. Verify that radial/tangential distortion values are correct in property sheet or Excel/XML photo list file.
Calculated Position	Difference between photos' calculated and inputted position values exceeds the error tolerance defined by the aerotriangulation Calculation Mode parameter. All of the project's photos have the same calculated position.	Verify that the control points were marked accurately and that photos' inputted position and orientation information is correct.

Focal Calculated	Difference between photos' calculated and inputted focal values exceeds the error tolerance defined by the Focal Length Accuracy parameter. / Difference between photos' calculated and inputted focal values is greater than 50%.	Check the input value. Fix if required and then rerun the AT.
Principal point (X,Y) calculated	Difference between photos' calculated and inputted principal point values exceeds the error tolerance defined by the Principal Point Accuracy value. / Difference between photos' calculated and inputted focal values exceeds permitted deviation.	Check the input value. Fix if required and then rerun the AT.

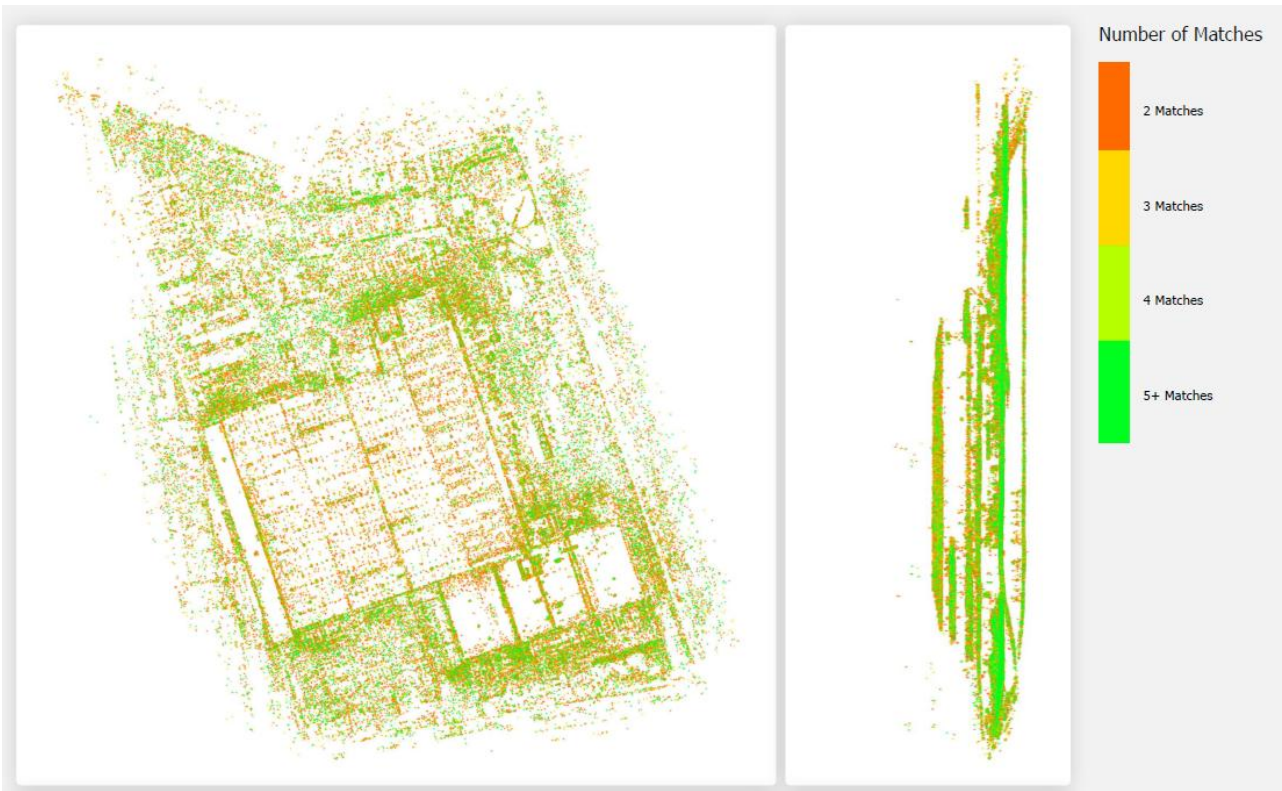
Control Points

Field	Message	Recommendation
Number of GCPs	Only 1 ground control point was added to collection.	It is recommended to add at least 3-5 control points per AT tile, to enable them to be fully utilized in the AT process. When only two control points are provided, simple vertical adjustment, scaling and rotation are performed. When only one control point is provided, only vertical adjustment is performed.
Number of Photos Marked	Control points marked in 2 or fewer photos. / Control points marked in fewer than 2 photos.	Mark each control point in at least three photos.
Number of Photos Used	Number of photos used is <= 2.	Verify the accuracy of control point marking in the photos.
Geographic Error Horizontal (X,Y)	Per GCP: If greater than defined accuracy. / Per project: if >10% or GCPs have errors.	Check that the coordinate system and vertical datum of both the control points and photos are accurate.
Geographic Error Vertical	Per GCP: If greater than defined accuracy. / Per project: if >10% or GCPs have errors.	Check that the coordinate system and vertical datum of both the control points and photos are accurate.

Geographic Error Circular	Per GCP: If greater than defined accuracy. / Per project: if >10% or GCPs have errors.	Check that the coordinate system and vertical datum of both the control points and photos are accurate.
Triangulation Error RMS	Root mean square (RMS) exceeds 0.5 pixels.	Verify the accuracy of control point placement and their marking in photos.
Triangulation Error Maximum	Triangulation error exceeds 1 pixel.	Verify the accuracy of control point placement and their marking in photos.

Maps

Photo Matches Map

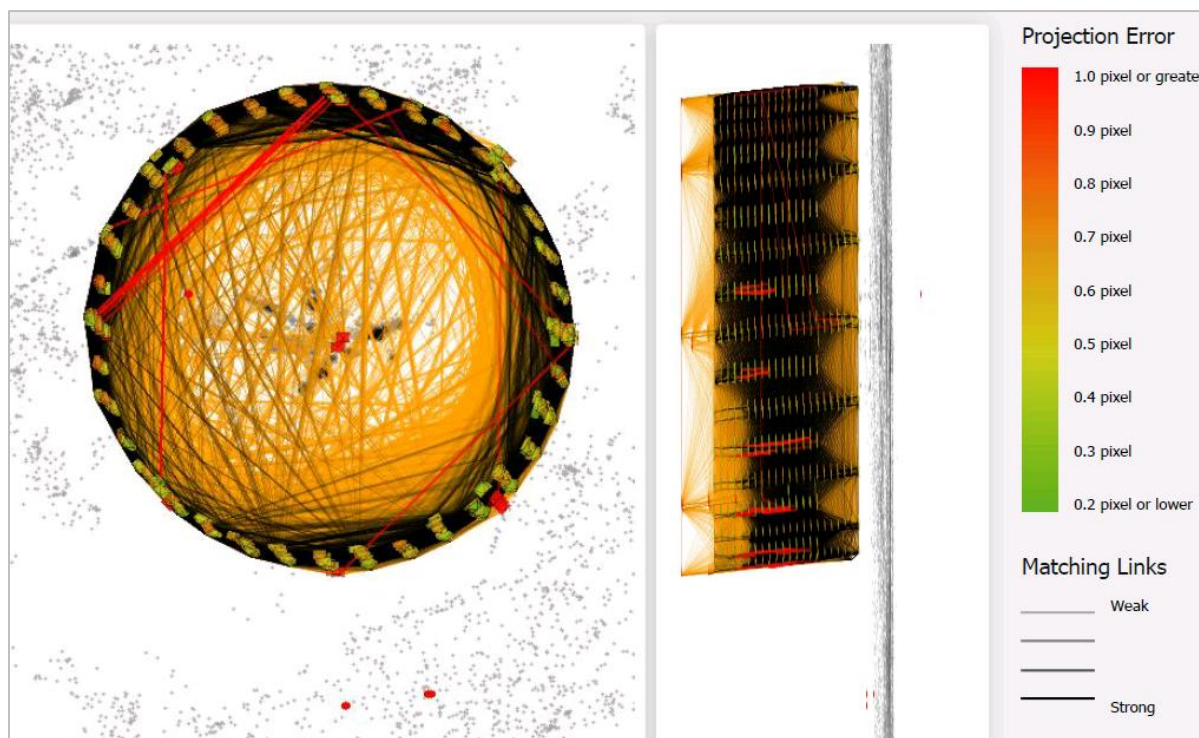


Color-coded map that graphically depicts the degree of photo overlap. Each match (feature that was found in at least two photos) in the defined AT area is colored according to the number of photos in which it was found.

- Red – Minimal coverage (found in 2 photos)
- Yellow – Moderate coverage (found in 3 photos)
- Green – Strong coverage (found in 5 or more photos)

In oblique collections, moderate-strong coverage is generally necessary, but only in the area of interest, while on the project edges, minimal coverage will usually suffice. In nadir collections, minimal coverage even in the area of interest is generally sufficient. To improve matching in cases of weak coverage, it is recommended to use the **AT: More Aggressive Match** preset.

Connection and Projection Error Maps



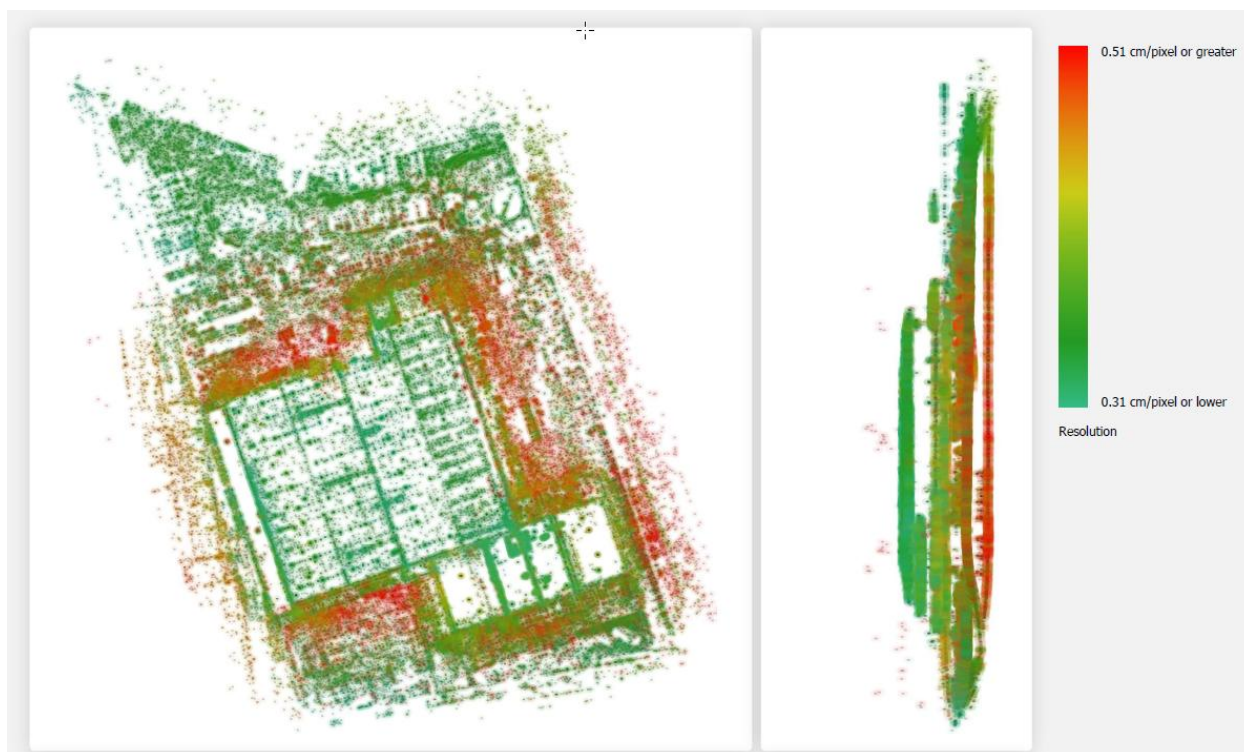
Projection error - Median of the triangulation errors of all the features in each photo. The triangulation error is the distance in pixels for every automatically identified feature, between its original position and the position computed through aerotriangulation. Each camera symbol is color coded, based on the level of projection error. If the projection error is greater than 0.6 pixels in a group of images localized in the project, it could be an indication of a bad control point, constraints in the camera model geometry, or an area with challenging features (forested areas with large time and/or shadow differences, or extremely urban areas with inconsistent coverage). It is recommended to review that area in the project AT in order to verify that there are no errors on the input data (control point or images), and that the images have proper overlap and coverage.

Connection - Lines connect between photos with common automatic tie points, with the weight of the connecting line corresponding to the number of common points. 3D automatic tie points are obtained by computing the positions of features (i.e., characteristic points on photos) that were automatically detected in two or more overlapping photos. If the links between photos are weak, **manual** tie points can be used to establish a stronger correspondence between project photos and improve aerotriangulation results. If the number of links remains low, the **AT: More Aggressive Match** preset can be applied when building the project to increase the likelihood of matches being made. The color of the connection lines provides information about the strength of the connection:

- Black – Strong connections

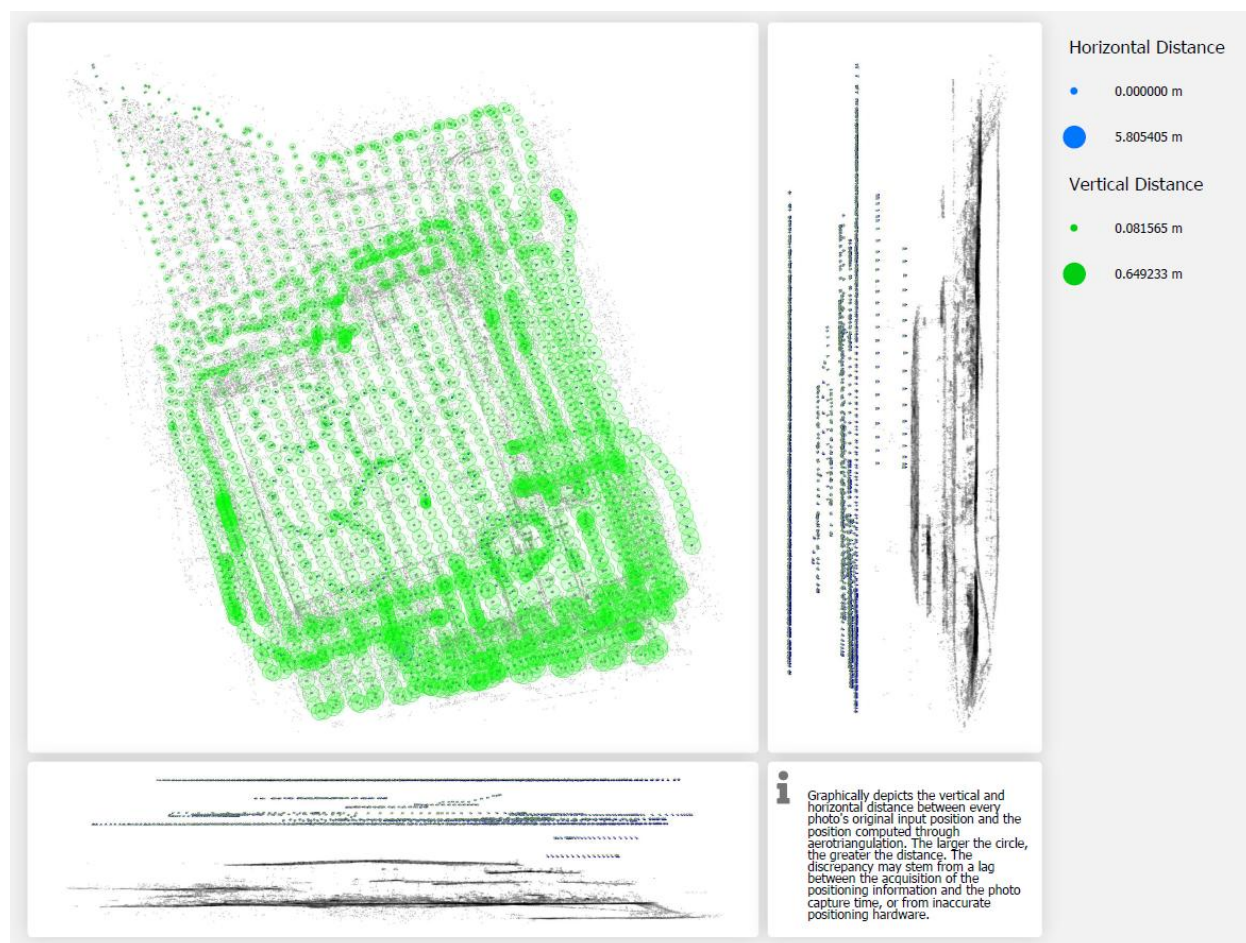
- Brown – Weak connections
- Red – Initial connections that were later disconnected in the AT bundle adjustment process. This indicates that the overall connection in the area was not strong enough to maintain the link between the photos.

Ground Resolution Map



Color-coded map that provides information on the highest resolution that was achieved in cm per pixel for each 3D automatic tie point within the defined AT area. 3D automatic tie points are obtained by computing the positions of features (i.e., characteristic points on photos) that were automatically detected in two or more overlapping photos. Make sure that the expected resolution level is achieved in the area of interest. On the project's edges, lower resolution will usually suffice.

Distance Map



This map graphically depicts the vertical and horizontal distance between every photo's original input position and the position computed through aerotriangulation. Each circle on the map represents a photo, with the size of the circle corresponding to the difference between the input values of the photo and its calculated values, i.e., the greater the difference, the larger the circle. The following are some of the distance map results you may see:

- A consistent shift in length and direction from the input to computed values (depicted by lines), generally in large drone projects - This often indicates a lag between the acquisition of the positioning information and the photo capture time
- A significant discrepancy between input and computed values in a specific stretch of the map, with minimal differences on either side. This may be caused by a consistent GPS offset between two different flights or by an incorrectly placed control point that has caused a significant shift in a specific section of the AT area.
- Arc between original and calculated values may be because of tangential effect. This most commonly occurs with rolling shutter cameras, and low-quality lenses.

To improve positioning in cases of significant discrepancies between inputted and computed positions, it is recommended to use the AT: More Aggressive Match [preset](#).

Exporting Photo List Information

Photo position and orientation properties can be exported to a CSV file. All columns currently displayed in the Photo list will be exported. See ["Exporting Aerotriangulation Results"](#) in this chapter for information on exporting your calculated AT results.

To export photo information:

1. In the **Project Panel**, select the **Photos** group. Then in the **Photos** list, on the **Photos** tab, right-click one of the photos and select **Export to CSV**. The Export Photos to CSV dialog is displayed.
2. Browse to the required folder, enter a file name, and click **Save**.

Exporting Aerotriangulation Results

You can export your aerotriangulation results either as multiple tiles or as a single unified SQLite, XML, or CSV file.

To export your aerotriangulation results:

1. On the **Home** tab, in the **Process** group, click the arrow under **Reports** and select **Export AT**. The Export Aerotriangulation Results dialog is displayed.

Export Aerotriangulation Dialog

2. In the **Method** field, select how to export the AT results:
 - Unified AT – Create a single, unified file from all AT tiles for export of AT information into a single set of AT results for each photo and camera. The unified AT includes the average AT results for photos/cameras that participate in more than one AT tile .
 - Multiple AT Tiles – Export each AT tile to a separate file.
 - Create a unified AT (see description above) + generate undistorted photos based on the distortion parameters calculated in the AT process. When photos are captured, most suffer (radial/tangential) distortion, as a result of the shape of the lens and other lens properties, that causes the photos' straight lines to appear deformed or curved unnaturally, particularly at the photos' edges.
3. Select a **Format**:

- Blocks Exchange XML – An open exchange format for import/export of AT data.
 - SQLite - Skyline internal format for import/export of AT results.
 - CSV – Comma Separated Value format that can be easily created, read or manipulated as well as read in other systems and custom readers.
4. In the **Output File** field, click **Browse**, and then in the **Save Aerotriangulation Results** dialog, do the following:
 - a. Browse to the folder in which you want to save the AT results.
 - b. Type the **File Name** for your results, and click **Save**.

Note: If the results are being exported as multiple tiles, the naming system for each of the tiles is as follows: File Name_AT_Tile-]Tile #[, e.g., FileName_ AT_Tile-0-0-1-1.xml, FileName_ AT_Tile-0-1-1-1.xml, etc.
 5. If you want to **exclude uncalculated photos** from the exported AT results, select the check box. If uncalculated photos are not excluded, they are exported with original position and orientation values.
 6. If you selected CSV format, select the **coordinate system** for the export. The default coordinate system for CSV is ECEF.
 7. Click **Export**.

Exporting Tiles

Selected AT or reconstruction tiles with all their information (e.g., user status, comment, photo count) can be exported to an SQLite file, and then imported into TerraExplorer or standard geospatial applications. This allows you to easily share information about the project's data coverage. The exported tiles can also be reviewed in TerraExplorer or other geospatial applications, and user status and comments added, and then reimported to PhotoMesh. [See "Importing Tiles"](#) in this chapter for information.

To export a tile:

1. Select the required tile. [See "Selecting Tiles"](#) in this chapter.
2. In the **AT Tiles** list or **Reconstruction Tiles** list, select the required tile(s), right-click and select **Export Selected Tiles**. The Export Tiles dialog is displayed. [See "Selecting Tiles"](#) in this chapter for more information.
3. Browse to the required folder, and click **Save**.

Importing Tiles

After exported tiles were reviewed in TerraExplorer or other standard geospatial applications, and user status and comments were added, the tiles can be reimported to PhotoMesh. [See "Exporting Tiles"](#) in this chapter for information.

To import a tile:

1. In the **Project Panel**, select the **AT Tiles** or **Reconstruction Tiles** group.
2. In the **AT Tiles** list or **Reconstruction Tiles** list, right-click and select **Import Tile Status**. The Import Tile Status dialog is displayed.
3. Browse to the required folder, and click **Open**.

Note: After adding water polygons, all affected tiles (e.g., tiles that now are intersected by a water polygon) require rebuilding from the point cloud step. PhotoMesh will automatically determine what processing is required following the change, by rebuilding the project from "Auto".

Working with Water Body Polygons

Water bodies are difficult to model accurately because it is hard to identify and match features due to the uniformity and non-static nature of water. In order to enable PhotoMesh to model and texture a water body successfully, you can define the required water body by drawing or importing a corresponding polygon. PhotoMesh uses this polygon to generate a flat surface over the user-defined water body, and automatically reconstruct any above-water elements – boats, piers, small islands without requiring special user markings. Water bodies can be either of the following:

- **Leveled** – A constant water level (i.e., altitude) is set for all points in the water body polygon, to create level water bodies such as seas, lakes, ponds, and pools. This is especially useful for creating a level water body on top of a non-flat terrain or mesh, or for creating a flat water body based on an imported polygon with variable altitude values.
- **Non-Leveled (Sloped)** – PhotoMesh segments the water body polygon into multiple flat reconstruction tiles, each of which has a different elevation value based on the average elevation within the specific tile. This option should be selected to create an elevation gradient for water bodies such as rivers, with non-uniform water levels.

Polygon data for both imported polygon layers and manually created ones can be viewed and modified in the Water Polygon list. This list has two tabs: "Polygons" which lists all the individual polygons in the project, and "Layers" which lists the polygon layers.

Creating Water Body Polygons

To create a water body polygon:

1. In the **Project Panel**, select the **Water Polygons** group.
2. In the **Water Polygon** list, on the **Polygons** tab, select **Draw Water Body Polygon**  or **Draw Leveled Polygon** .
3. With the cursor in the **3D Window**, click to add the first point of the polygon. Click again to add additional points of the polygon until you have marked the entire shape. You must place at least three points to define the polygon.

Note: It is sometimes easier to draw an accurate water polygon over a textured mesh model. After adding the model, rebuild, setting the **Build Start From** field to **Auto**, so that the required steps are automatically selected.

Note: When creating a water body polygon, the lines of the polygon should not overlap each other.

Note: It is recommended to keep the number of polygon points to the minimum required to define the shape. Adding unnecessary points may adversely affect performance.
4. Right-click to finish the polygon. The polygon name and water average elevation can be modified in the Water Polygon list. See "[Water Polygon List](#)" in this chapter for information. If you want to edit the polygon, see "[Editing a Water Body Polygon's Shape](#)" and "[Setting Water Body Polygon Properties](#)" in this chapter for information.

Importing Water Body Polygons

To import a water polygon shapefile or KML layer:

1. In the **Project Panel**, select the **Water Polygons** group.

2. In the **Water Polygon** list, on the **Layers** tab, select **Import** . The Import Water Body Polygon Layer dialog is displayed.
3. Browse to the water polygon layer you want to import, and click **Open**. If you want to edit the polygons, see ["Editing a Water Body Polygon's Shape"](#) and ["Setting Water Body Polygon Properties"](#) in this chapter for information. The water polygons are reprojected to the coordinate system of the current project.

Exporting Water Body Polygons

To export a water polygon layer to a shapefile:

1. In the **Project Panel**, select the **Water Polygons** group.
2. In the **Water Polygon** list, on the **Layers** tab, select **Export** . The Export Water Body Polygon Layer dialog is displayed.
3. Browse to the folder in which to save the layer, type a **File name**, and click **Save**. The polygons are exported in Lat-Long coordinate system with the vertical coordinate system of the current project's terrain.

Editing a Water Body Polygon

After modifying a water body polygon, all affected tiles (e.g., tiles that now are intersected by the polygon) require rebuilding from the point cloud step. PhotoMesh will automatically determine what processing is required following the water polygon change, by rebuilding the project from "Auto".

To move a water polygon or modify its shape:

1. In the **Project Panel**, select the **Water Polygons** group.
2. In the **Water Polygon** list, on the **Polygons** tab, select the required water polygon, and click **Edit** . Then use the polygon mini toolbar to edit the polygon. See ["Editing Polylines and Polygons"](#) in this chapter for information on editing polygons.

Selecting Water Body Polygons

Polygons can be selected to perform various commands on them, e.g., jump to a polygon, level/unlevel, or set average elevation.

To select water body polygons:

1. In the **Project Panel**, select the **Water Polygons** group.
2. In the **3D Window**, click the polygon you want to select, or select the polygon in the **Water Polygon** list, on the **Polygons** tab. You can multi-select polygons, either in the Water Polygon list or by using CTRL-click or SHIFT-click in the 3D Window.

Note: If you turned on global selection, you can select a polygon in the 3D Window even without first selecting the Water Polygons group in the Project Panel as long as the polygons are visible in the 3D Window.

See ["Showing and Hiding Group Items"](#) and ["Using the Item List"](#) in the "Using the Project Panel and Item List" chapter and ["Selecting Objects in the 3D Window"](#) in the "Basic Concepts" chapter for information.

Navigating to a Water Body Polygon

To fly or jump to a water body polygon:

1. In the **Project Panel**, select the **Water Polygons** group.
2. In the **Water Polygon** list, on the **Polygons** tab, right-click the required polygon, and select **Fly To/Jump To**.

Setting Water Body Polygon Properties

Water body polygon data (e.g., average elevation and elevation offset) can be viewed or modified in the Water Polygon list. See ["About the Item List"](#) in the "Using the Project Panel and Item List" chapter for information on working with the Water Polygon list.

To display and modify water body polygon properties:

1. In the **Project Panel**, select the **Water Polygons** group.
2. For group/layer properties: In the **Water Polygon** list, click the **Layers** tab, and modify the property value as required. See ["Water Polygon List – Layers Tab"](#) in this chapter for more information.
3. For individual polygon properties: In the **Water Polygon** list, click the **Polygons** tab, select the row of the required polygon, and then click in the required column, and modify its properties as required. See ["Water Polygon List"](#) in this chapter for more information.

Note: You can also search the list for the required polygons. See ["Selecting Items Using the Item List"](#) in the "Basic Concepts" chapter for information.

Water Polygon List – Layers Tab

The properties in the Water Polygon list – Layers tab apply to all polygons in the layer.

Property	Description
Selection	Select the check box to select the layer for editing.
Name	Name of the water body layer in the Water Polygon list. All manually created polygons are saved in the same layer.
Fill Color	Fill color for all the water body polygons in the layer.
Visible	Select the check box to show the layer's polygons in the 3D Window.
Excluded	Select the check box to exclude the water body layer from the project.
Leveled	Select the check box to set a constant water level (i.e., altitude) for all points in all the water body polygons in the layer, to create level water bodies such as seas, lakes, ponds, and pools. This is especially useful for creating level water bodies on top of a non-flat terrain or mesh, or for creating flat water bodies based on imported polygons with variable altitude values. If the check box is cleared, PhotoMesh segments the water body polygons into multiple flat reconstruction tiles, each of which has a different elevation value based on the average elevation within the specific tile. This option should be selected to create an elevation gradient for water bodies such as rivers, with non-uniform water levels.
Elevation Offset	Elevation offset from the current elevation value applied to entire water body layer.

Coordinate System	Coordinate system of all the water polygons in the layer (Read-only for manually created polygons). When water polygon layers are imported, they are reprojected to the terrain database coordinate system. Note: When the coordinate system is changed, the layer is reloaded causing all manual modifications, e.g., editing of the polygon or changes to elevation offset, to be lost.
Original Source Path	Original source path from where the water body layer was imported (Read-only).

Water Polygon List – Polygons Tab

In the Water Polygon list – Polygons tab, you can view and set certain polygon properties, as well as set polygon visibility and inclusion in the project. See ["About the Item List"](#) in the "Using the Project Panel and Item List" chapter for information on working with the polygon list.

Property	Description
Selection	Select the check box to select the polygon for editing. You can select multiple polygons to perform actions on multiple polygons.
Name	Name of the water polygon.
Layer	Name of the water polygon layer. All manually created polygons are saved in the same layer. Click the three dots  to open the Layers tab to edit the layer name.
Visible	Select the check box to show the polygon in the 3D Window.
Excluded	Select the check box to exclude the polygon from the project. The polygon continues to display in the 3D Window.
Leveled	Select the check box to set a constant water level (i.e., altitude) for all points in the water body polygon, to create level water bodies such as seas, lakes, ponds, and pools. This is especially useful for creating a level water body on top of a non-flat terrain or mesh, or for creating a flat water body based on an imported polygon with variable altitude values. If the check box is cleared, PhotoMesh segments the water body polygon into multiple flat reconstruction tiles, each of which has a different elevation value based on the average elevation within the specific tile. This option should be selected to create an elevation gradient for water bodies such as rivers, with non-uniform water levels. If a water polygon is leveled, all its nodes are set to the elevation of the first node drawn. Note: If you want to level or unlevel multiple polygons, multi-select the required polygons. Then right-click in the Water Polygon list, and select Level/Unlevel. Unleveling allows you to set different elevations for each polygon node. See "Selecting Water Body Polygons" in this chapter for more information.
Average Elevation	The average elevation of the water polygon. When this value is edited, PhotoMesh proportionally adjusts the elevation value of

each of the individual nodes, so that the ratio between node elevations is still maintained.

Note: If you want to set the average elevation of multiple polygons, multi-select the required polygons. Then right-click in the **Water Polygon** list, and select **Set Average Elevation**. You can also modify the average elevation of multiple polygons by a specified delta. To do so, multi-select the required polygons. Then right-click in the **Water Polygon** list, and select **Set Elevation Delta**. See "[Selecting Water Body Polygons](#)" in this chapter for more information.

Deleting Water Body Polygon Layers

To delete a polygon layer:

1. In the **Project Panel**, select the **Water Polygons** group.
2. In the **Water Polygon** list, on the **Layers** tab, right-click the required layer, and select **Delete**.

Deleting Water Body Polygons

To delete a polygon from a layer:

1. In the **Project Panel**, select the **Water Polygons** group.
2. In the **Water Polygon** list, on the **Polygons** tab, right-click the required polygon, and select **Delete**.

Editing Polylines and Polygons

Water body polygons and retouch polygons and polylines are edited using the commands in the mini toolbar. This toolbar is displayed whenever the polygon/polyline is selected.



To edit a polygon/polyline:

1. Select the polygon/polyline, and then do either of the following:
 - [Move the polygon/polyline](#)
 - [Modify the polygon/polyline shape](#)
2. If you want to select a different object from the one currently being edited, click **Select another object** .
3. If you want to align your polygon/polyline to a surface, click **Align to Surface** . The polyline/polygon's nodes are aligned to the surface.

To move a polygon/polyline:

1. Click the Edit Object icon .
2. Do any of the following:
 - If you want to set position based on mouse **and** snap the polygon/polyline to another object, click **Snap Based on Snapping Options** icon  (Shortcut key: 4). See "[Setting Snapping Options](#)" in this chapter for information.

- If you want to set position based on mouse without snapping the polyline/polygon to another object, click **Set Position Based on Mouse**  (Shortcut key: 3).
 - If you want to move polygon/polyline in the XY plane, click **XY Plane** .
 - If you want to move the polygon/polyline in the Z plane, click **Z Axis** .
3. Click the polygon/polyline, and drag it to the required location/altitude. Release the left mouse button to set the polygon/polyline at the current location/altitude. If snapping to an object, move and snap the polygon/polyline to an edge or vertex of any existing object.

To modify a polygon/polyline's shape by moving its nodes:

1. Click Edit Nodes .
2. Do any of the following:
 - If you want to set position based on mouse **and** snap the polygon/polyline to another object, click **Snap Based on Snapping Options** icon  (Shortcut key: 4). See "[Setting Snapping Options](#)" in this chapter for information.
 - If you want to set position based on mouse without snapping the polygon/polyline to another object, click **Set Position Based on Mouse**  (Shortcut key: 3).
 - If you want to move a node in the XY plane, click **XY Plane** .
 - If you want to move a node in the Z plane, click **Z Axis** .
3. In the **3D Window**, place the cursor over one of the polyline/polygon's nodes. The node turns blue.
4. Click the node and drag it to the required location in the XY/Z Plane. Release the mouse button to set the node at the current location.

Setting Snapping Options

You can set how a node or object snaps to other objects.

To set snapping options:

1. Select a single object for editing.
2. On the top bar of the object's property sheet or mini toolbar, click **Set Snapping Options** .
3. Toggle on any of the following options:
 - **Node Snapping** – Snap object or node to nodes of other objects or features.
 - **Line Segment Snapping** – Snap object or node to line segments of other objects or features.
 - **3D Snapping** – Smart snap object or node to dynamically-identified edges and corners of any element in the 3D World.
 - **Directional Snapping** – Constrain the object line segments currently being drawn to pre-defined angle increments. The angle constraint can either be absolute, i.e., relative to true north or relative to the previous line segment.
 - **Segment Distance on Edit** – Show segment lengths while editing a polygon/polyline.

Retouching a Mesh Layer

Mesh imperfections, such as bumps, irregular surfaces, or floating artifacts, can be flattened, filled, and cleaned up in PhotoMesh using the Manual Retouch tool. This tool allows you to mark the surfaces for

flattening and floating artifacts for removal. The output of the Manual Retouch tool is a feature "retouch layer" with attribute information that defines what retouch action should be performed in each area: clean, flatten, and retexture. When rebuilding the project, PhotoMesh uses the manual retouch layer to add a set of constraints to the reconstruction algorithms. Most of the constraints defined apply to the mesh model, and the modified model is then retextured using the project's photos. There is also an option in the Manual Retouch tool to set a specific file texture, e.g., water or grass, for a marked area, instead of basing texturing on the project's photos.

You can run the manual retouch tool either on a 3D model (the reconstruction tile's 3D textured model or the final 3DML) or on orthophoto output. All retouch operations are listed in the Manual Retouch list.

Note: After performing manual retouch operations, all affected tiles (e.g., tiles that now are intersected by the manual retouch polygon) require rebuilding from the model step. PhotoMesh will automatically determine what processing is required following the change, by rebuilding the project from "Auto".

To use the Manual Retouch tool:

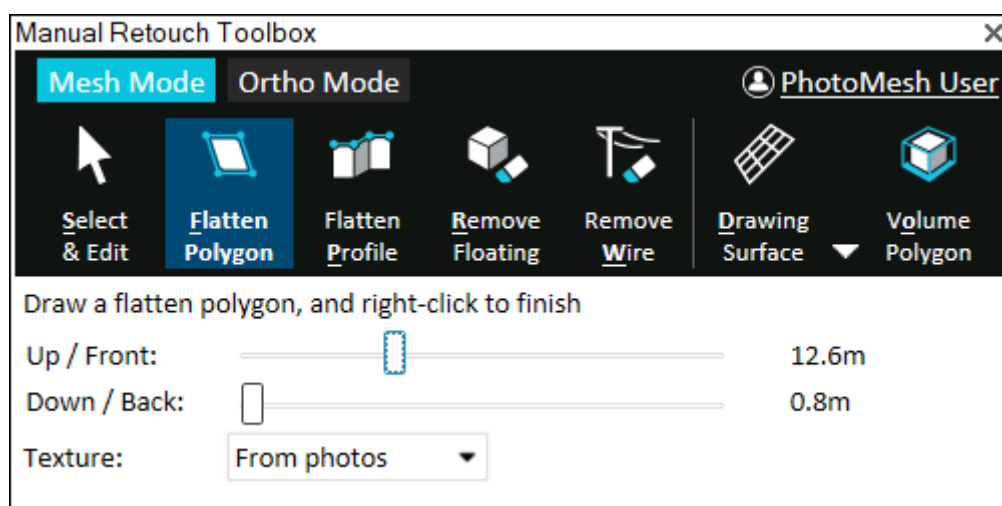
Note: The shortcut keys only work when keyboard focus is on the retouch toolbox or the 3D Window.

1. Do any of the following:

- In **Mesh Mode**:
 - In the **3D Window**, click the required reconstruction tile. Then on the **Tile** tab, in the **Show group**, toggle on **Textured Model**.
 - In the **Build Outputs** list, select the **3DML** output.
- In **Orthophoto Mode**:
 - In the **Build Outputs** list, select the **Orthophoto** output.

2. On the **Home** tab, in the **Add** group, click the arrow under **Retouch**, and select **Open Manual Retouch Toolbox**. The Manual Retouch Toolbox is displayed.

Note: You can also open the Retouch tool by clicking **Manual Retouch Tool** from the **Manual Retouch** list. Alternatively, you can turn on global selection, and then select a manual retouch polygon/polyline in the 3D Window. See ["Selecting Objects in the 3D Window"](#) in the "Basic Concepts" chapter for information.



Manual Retouch

3. If you want to tag your edits with your name, click the user name link, and enter your user name. This information is saved in the layer's username attribute. When multiple users work on the same retouch layer, this attribute enables you to identify the particular user that last worked on each retouch polygon.
4. Select a mode and perform your retouch operations:
 - **Mesh Mode** – Mark your edits on a 3D model output (the reconstruction tile's 3D textured model or the final 3DML mesh model). See "[Working in Mesh Mode](#)" for more information.
 - **Ortho Mode** – Mark your edits on the orthophoto output. See "[Working in Orthophoto Mode](#)" for more information.

Working in Mesh Mode

- Creating a Surface on Which to Draw Your Retouch Polygon/Polyline
- Performing a Retouch Operation

Creating a Surface on Which to Draw Your Retouch Polygon/Polyline

You can create a surface on which to draw your retouch polygon or polyline, e.g., for a model that is not flat or missing corners.

Do the following:

- In the **Manual Retouch Toolbox**, click the arrow next to **Drawing Surface** (Shortcut key: d) and select one of the following options:
 - **Show Last** – Display the last surface that was created.
 - **Edit** – Open the surface property sheet to edit altitude, x, y, yaw, pitch, and roll properties, and use the toolbar to edit the surface in the 3D Window. See "[Editing Polylines and Polygons](#)" in this chapter for information.
 - Click one of the other manual retouch commands and draw a polygon or polyline whose nodes are all lying in the same plane. Selecting Set Position Based on Mouse  or Snap Based on Snapping Options  in the polygon property sheet before drawing makes it easier to draw the polygons/polylines on a vertical plane.
 - **Horizontal** – The surface is aligned horizontally to the area under the cursor.
 - **Vertical** – The surface is aligned vertically to the area under the cursor.
 - **3 Points** – The surface is aligned to the polygon created by the three points you draw.

Performing a Retouch Operation

A range of retouch operations are available so that you easily flatten, fill, or clean up any mesh imperfection:

- **Flatten polygon** – Remove all elements inside a defined volume, and add a flat surface in the area of the 2D polygon, for e.g., filling holes and flattening walls. The area of the 2D polygon can be textured with any of the following:
 - **From photos** – Fill the polygon area using photos from the area.
 - **Remove moving cars** – Fill the polygon area using only photos from the area that don't include moving cars. This option offers a solution to undesirable reconstruction effects that can occur when capturing an area with moving cars (e.g., a car that is partially sunk in the road), since only

some of the captured photos included the cars, while others did not. It is recommended to individually retouch each car you want to remove by applying a unique polygon to it.

- **Single color** – Fill the polygon area using a single color.
- **Texture pattern** – Fill the polygon area with a water or grass texture.
- **Flatten profile** – Create multiple flatten polygons by defining a profile line (usually on a roof top) and the base elevation. A flatten polygon is created for each segment of the profile line (from the line to the base elevation). This method is useful for creating retouch polygons in hard to access places, e.g., between buildings.
- **Remove floating** – Remove all elements inside a defined volume. This option is generally used to remove floating elements that aren't connected to the ground.
- **Remove wire** – Remove thin elements such as power lines by drawing a polyline that marks the wire to be removed. A separate floating polygon is created for each segment of the polyline.

To perform a retouch operation:

1. Select one of the retouch operations in the **Manual Retouch Toolbox**, and then do the following:
 - **Flatten polygon** (Shortcut key: f)
 - i. Set a buffer for the retouch polygon upwards or frontwards and downwards or backwards in the direction of the polygon's normal by sliding the **Up/Front buffer** and **Down/Back buffer** sliders. If a polygon's elevation angle to the ground is less than 20 degrees, the polygon is extruded vertically. If the polygon points were created in a clockwise orientation, then the front is the side facing the user.
 - ii. Select one of the fill options: **From photos**, **Remove moving cars**, **Single color**, or **Texture pattern**.
 - iii. For **Single color** fill, select the color to use. For **Texture pattern** fill, select a **Texture pattern**, and enter a value for the texture tile size.
 - iv. Draw a polygon around the area for flattening. Right-click to complete.
 - **Flatten profile** (Shortcut key: p)
 - i. Set a buffer for the retouch polygon upwards or frontwards and downwards or backwards in the direction of the polygon's normal by sliding the **Up/Front buffer** and **Down/Back buffer** sliders. If a polygon's elevation angle to the ground is less than 20 degrees, the polygon is extruded vertically. If the polygon points were created in a clockwise orientation, then the front is the side facing the user.
 - ii. Draw the profile line, and then right-click to complete. Then click on the model to define the model's base elevation. A flatten polygon is created for each segment of the profile line (from the line to the base elevation).
 - **Remove floating** (Shortcut key: r)
 - i. Click any point on the floating artifact. PhotoMesh automatically creates a polygon that includes the selected point.
 - ii. Set a buffer for the retouch polygon upwards or frontwards and downwards or backwards in the direction of the polygon's normal by sliding the **Up/Front buffer** and **Down/Back buffer** sliders. If a polygon's elevation angle to the ground is less than 20 degrees, the polygon is extruded vertically. If the polygon points were created in a clockwise orientation, then the front is the side facing the user.
 - **Remove wire** (Shortcut key: w)

- i. Set a buffer for the retouch polyline upwards or frontwards and downwards or backwards in the direction of the polyline's normal by sliding the **Front buffer** and **Back buffer** sliders. If a polyline's elevation angle to the ground is less than 20 degrees, the polyline is extruded vertically. If the polyline points were created in a clockwise orientation, then the front is the side facing the user.
 - ii. Draw a polyline along the wire to remove, and then right-click to complete.
2. To select the polygon/polyline for editing after it is drawn, click **Select and Edit** (Shortcut key: s), and then select the retouch object for editing or right-click it in the **Manual Retouch** list, and select **Edit**. When editing a drawn polyline and changing the extrusion length, only the selected line segment is modified. See "[Editing Polylines and Polygons](#)" in this chapter for information.
3. To select how to display all manual retouch polygons, click the **Volume** icon (Shortcut key: o):
 - **Volume Colorize** – Mark the area for retouching by colorizing all parts of the mesh contained within the extruded polygon.
 - **Volume Polygon** – Mark the area for retouching by displaying 3D polygons based on the extruded polygon.
 - **Volume No** – Mark the area for retouching only with a 2D polygon.

Working in Orthophoto Mode

The following retouch operations are available in Ortho mode:

- **Clean Cars** – Remove all elements inside a defined area, and texture the drawn polygon area using only photos from the area that don't include moving cars. This option offers a solution to undesirable effects in the reconstruction (e.g., a car that is partially sunk in the road) that can occur when capturing an area with moving cars, since only some of the captured photos included the cars, while others did not. It is recommended to individually retouch each car you want to remove by applying a unique polygon to it.
- **Clean Edges** – Remove artifacts along the edges of a model.
- **Import Layer** – Import a polyline or polygon feature layer with retouch polylines/polygons for cleaning edges or cars. The following layer formats are supported: .shp, sqlite, and gpkg.

To perform a retouch operation in Ortho mode:

1. Select one of the retouch operations in the **Manual Retouch Toolbox**, and then do the following:
 - **Clean Cars** (Shortcut key: c)
 - i. Draw a polygon around the area that you want to clean.
 - ii. Right-click to complete. The buffers are automatically set to 10 meters above/below the surface.
 - **Clean Edges** (Shortcut key: e)
 - i. Set a buffer for the retouch polygon upwards or frontwards and downwards or backwards in the direction of the polygon's normal by sliding the **Up/Front buffer** and **Down/Back buffer** sliders. If a polygon's elevation angle to the ground is less than 20 degrees, the polygon is extruded vertically. If the polygon points were created in a clockwise orientation, then the front is the side facing the user.

- ii. Draw a polyline along the edge you want to clean. Right-click to complete. A polygon is generated with the base line on the ortho surface and the top line, one meter above the surface. All artifacts with normal vertical to the polygon are cleaned.
- **Import Layer** (Shortcut key: i)
 - i. Click **Import Layer**, and then **Browse**, and browse to the required layer. The following layer formats are supported: .shp, sqlite, and gpkg. The retouch polygons included in the layer are added to the Manual Retouch list.
- 2. To select how to display the drawn polygon, click the **Volume** icon (Shortcut key: o):
 - **Volume Colorize** – Mark the area for retouching by colorizing all parts of the mesh contained within the extruded polygon.
 - **Volume Polygon** – Mark the area for retouching by displaying 3D polygons based on the extruded polygon.
 - **Volume No** – Mark the area for retouching only with a 2D polygon.
- 3. To adjust the length to extrude the retouch polygon upwards or frontwards and downwards or backwards in the direction of the polygon's normal, select the polygon and in the **Manual Retouch Toolbox**, slide the **Front buffer** and **Back buffer** sliders. If a polygon's elevation angle to the ground is less than 20 degrees, the polygon is extruded vertically. If the polygon points were created in a clockwise orientation, then the front is the side facing the user.
- 4. To select the polygon for editing after it is drawn, click **Select and Edit** (Shortcut key: s), and then select the retouch object for editing. Use the mini toolbar to edit the polygon. See ["Editing Polylines and Polygons"](#) in this chapter for information. You can also view and edit the manual retouch object's properties in the Manual Retouch list. See ["Modifying Retouch Objects"](#) in this chapter for information.

Modifying Retouch Objects

All retouch operations can be viewed and edited from the Manual Retouch list. See ["Selecting Items Using the Item List"](#) in the "Using the Project Panel and Item List" chapter for more information.

Manual Retouch					
       0 out of 7 selected					
Sele...	User Name	Operation	Front Buffer	Back Buffer	Texture
☐	PhotoMesh User	Clean, Fill & Single Color	7.6	16.4	#FFFFFF
☐	PhotoMesh User	Clean	30.8	16.4	
☐	PhotoMesh User	Clean	30.8	16.4	
☐	PhotoMesh User	Clean, Fill – 2D mode	30.8	9.2	
☐	PhotoMesh User	Clean, Fill – 2D mode	30.8	9.2	
☐	PhotoMesh User	Clean, Fill – 2D mode	30.8	9.2	
☐	PhotoMesh User	Clean, Fill – 2D mode	30.8	9.2	

Manual Retouch List

To modify a retouch operation:

1. In the **Project Panel**, select the **Manual Retouch** group, and then in the **Manual Retouch** list, select the row with the required retouch object.
2. Perform any of the following actions:
 - **Fly To/Jump To** – Right-click the required row, and select **Fly To/Jump To**.
 - **Delete Polygon** – Right-click the required row, and select **Delete**.
 - **Edit Polygon** – Right-click the required row, and select **Edit**. The retouch polygon is selected in the 3D Window and a mini toolbar is displayed. See ["Editing Polylines and Polygons"](#) in this chapter for information on modifying the polygon shape/position and properties. You can also modify the buffer in the Manual Retouch list.

Exporting and Importing a Retouch Layer

You can export retouch layers with selected retouch polygons to an SQLite file. The layer can be modified in TerraExplorer and reimported to PhotoMesh.

In Ortho mode, you can also import a polyline or polygon feature layer with retouch polylines/polygons for cleaning edges or cars, in shp, sqlite, and gpkg formats. See ["Working in Ortho Mode"](#) in this chapter for more information.

To export a retouch layer:

1. In the **Project Panel**, select the **Manual Retouch** group, and then in the **Manual Retouch** list, select the retouch polygons that you want to export, and click **Export** . The Export Manual Retouch Objects dialog is displayed.
2. Browse to the location where you want to save the layer, and click **Save**. The layer is exported with the selected retouch polygons as an SQLite file.
3. Browse to the required folder, and click **Save**.

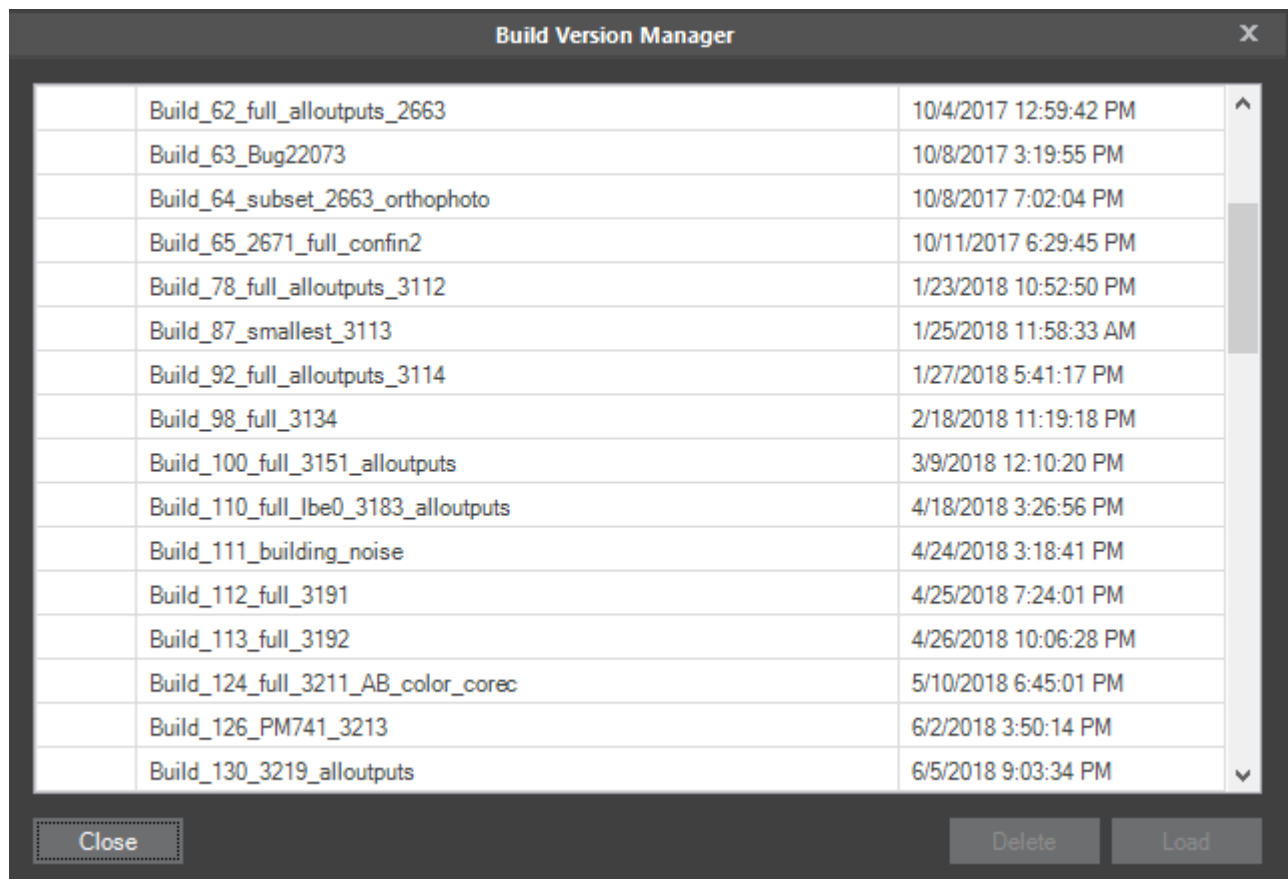
To import a retouch layer:

1. On the **Home** tab, in the **Add** group, open the **Retouch** dropdown, and select **Import Manual Retouch**. The Import Manual Retouch Objects dialog is displayed.
2. Browse to the required SQLite file, and click **Open**. The layer is imported to the project.

Managing Build Versions

To manage your build versions:

1. On the **Home** tab, in the **Build Versions** group, click **Versions Manager**. The Build Versions Manager is displayed.



Build Versions Dialog

2. If you want to load a different version than the current one, select the version and click **Load**. If the current version was not saved, a dialog is displayed asking if you want to save it before the selected version is loaded. Click **Yes** to save.
3. If you want to delete a version, select the version and click **Delete**.

CHAPTER 18 IMPORTING AND EXPORTING MODELS

About Importing and Exporting Models

After a 3D mesh model is created in Skyline PhotoMesh, certain tiles may require editing, e.g. to remove a particular building that is being torn down, flatten surfaces and correct certain imperfections such as bumps or irregular surfaces, or edit textures. These edits, whether they involve mesh or texture retouching, can be performed in external software, such as Autodesk® MeshMixer, by exporting the required OBJ models and then reimporting them to PhotoMesh in order to rebuild the project. When edits are complete, you can reimport precisely the data you want, all in one step:

- Only textures, using the models generated by PhotoMesh before the edits
- Only models (in which case retexturing is required in PhotoMesh)
- Both models and textures

You can import a single model or a folder with multiple model files, in which case each model file is read into its respective tile as long as its original file name is maintained.

Exporting a Point Cloud

To export a point cloud:

1. Select the tile whose point cloud you want to export. See ["Selecting Tiles"](#) in the "Reviewing Your Build" chapter. The Tile tab is displayed.
2. On the **Tile** tab, in the **Export** group, click **Point Cloud**. The Export Point Cloud dialog is displayed.
3. Browse to the required location, and click **Save**. The point cloud is exported in Polygon File Format (PLY).

Exporting a Model

To export a model:

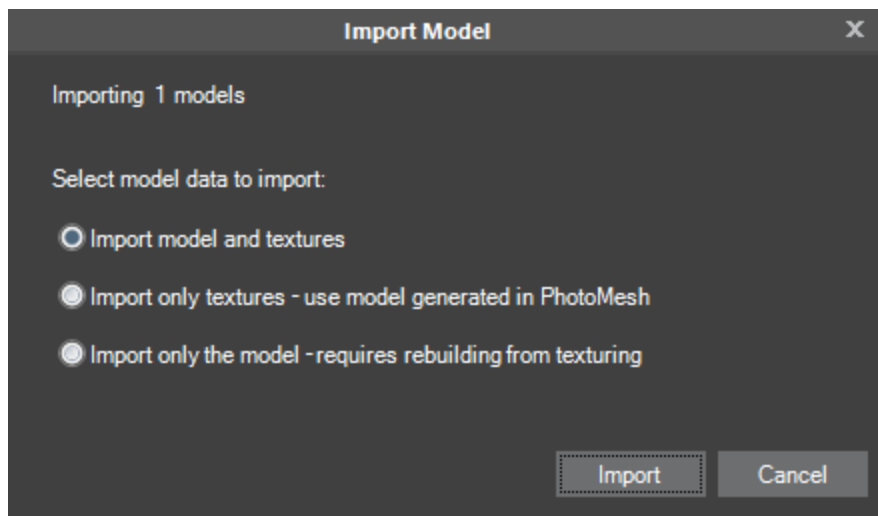
1. Select the tile whose model you want to export. See ["Selecting Tiles"](#) in the "Reviewing Your Build" chapter. The Tile tab is displayed.
2. On the **Tile** tab, in the **Export** group, click **Model**. The Export Model Folder dialog is displayed.
3. Browse to the required location, and click **Save**. The model is exported in OBJ format.

Note: When reimporting an exported model file after editing in external software, the model file must have the original file name with which it was exported to enable PhotoMesh to update the respective tile based on the edited model.

Importing a Model

To import a single model:

1. Select the tile for which you want to import a model. See ["Selecting Tiles"](#) in the "Reviewing Your Build" chapter. The Tile tab is displayed.
2. On the **Tile** tab, in the **Import** group, click **Model**. The Import Model dialog is displayed.
3. Browse to the required model and click **Open**. The Import Model dialog is displayed.



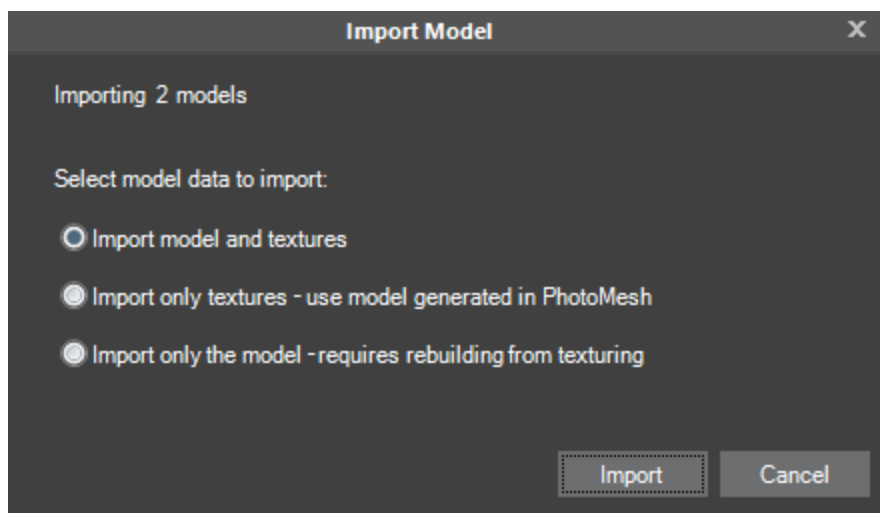
Import Model Dialog

4. Select the model data to import and then click **Import**:
 - Import model and textures
 - Import only textures – use model generated in PhotoMesh
 - Import only the model – requires rebuilding from texturing
5. After reimporting, rebuild using the "Build" command starting from "Auto", to have PhotoMesh automatically determine what processing must be performed to complete the project. **See** ["Rebuilding"](#) in the "Building" chapter for information.

To import a folder with multiple models:

1. On the **Home** tab, in the **Add** group, click the arrow under **Retouch** and select **Import Models**. The Import Model Folder dialog is displayed.
2. Browse to the folder that contains the edited models and click **Select Folder**. The Import Model dialog is displayed.

Note: The model files must have the original file names with which they were exported to enable PhotoMesh to update the respective tiles based on the edited models.



Import Model Dialog

3. Select the model data to import and then click **Import**:
 - Import model and textures
 - Import only textures – use model generated in PhotoMesh
 - Import only the model – requires rebuilding from texturing
4. After reimporting, rebuild using the "Build" command starting from "Auto" and building until "Output", to have PhotoMesh automatically determine what processing must be performed to complete the project. **See** ["Rebuilding"](#) in the "Building" chapter for information.

CHAPTER 19 ADDING TERRAIN OBJECTS AND LAYERS

About Terrain Objects and Layers

Terrain objects and layers can be added to a PhotoMesh project to enable you to visualize how the model will appear in TerraExplorer's 3D World.

You can add any of the following objects and layers:

- [Locations](#)
- [3D Mesh Layers](#)
- [Imagery and Elevation Layers](#)
- [Terrain Objects](#) (Modify Terrain and Hole on Terrain)

Locations

A location defines the geographical coordinates of a point on the terrain as well as the position from which the point is to be viewed.

To create a new location in the current camera position:

- On the **Tools** tab, in the **Objects and Layers** group, click **Create Location**. The location of interest is set to the terrain coordinate at the center of the 3D Window. If there is no terrain at the center of the view (the camera is facing the sky), the location of interest is set to a distance of two times the current altitude of the camera above the terrain. The location is added to the Objects and Layers list.

Note: When you create a location, the point of interest is set only to the terrain coordinate of the center of the 3D Window and is not affected by any 2D or 3D models in front of the terrain.

3D Mesh Layers

3D Mesh Layers (3DML) can be loaded from 3DML files on your own computer or local network or from a database that is served over a network by SkylineGlobe 3D Mesh service.

To load a mesh layer:

1. On the **Tools** tab in the **Objects and Layers** group, click **3DML**. The Browse for 3DML Layer dialog is displayed.
2. Browse to the required file, and click **Open** or select the required server type, and click **Select**. Then enter the information required for the selected server. **See** the "[Working with SkylineGlobe Server](#)" chapter for information.
3. After the layer has been placed on the terrain, you can change its properties using the property sheet. **See** "[3D Mesh Layer Parameters](#)" in the TerraExplorer User Guide for further information.

Imagery and Elevation Layers

Raster layers can be added to your project from either of the following:

- Files on your own computer or on your local network.
- A database that is served over a network by one of these GIS servers:
 - Web Map Service
 - Web Map Tile Service

- Oracle Spatial Database
- ECW Image Web Server
- TerraGate Server
- SkylineGlobe Server (**See** the "[Working with SkylineGlobe Server](#)" chapter for information)
- ArcSDE Raster Server

See the [TerraExplorer User Guide](#) for more information on loading raster layers.

To add an imagery/elevation source to a project:

1. On the **Tools** tab, in the **Objects and Layers** group, click **Imagery Layer/Elevation Layer**. The Load Imagery/Elevation Layer dialog is displayed.
2. Browse to the required file and select it, or select the server type and enter the required information.
3. Click **Open**.

Note: If the source file is geo-referenced, it is located accordingly on the terrain. Otherwise, it is placed at the center of the 3D Window. You can then manually edit its position, or reproject it.

4. After the layer has been placed on the terrain, you can change its properties using the property sheet.

Terrain Objects

The following terrain objects can be added to a PhotoMesh project to enable you to visualize how the model will appear in TerraExplorer's 3D World:

- [Modify terrain](#)
- [Hole on terrain](#)

Modifying the Terrain

Using the Modify Terrain tool, you can modify the elevation values in an area of the terrain by drawing a polygon in the 3D Window and then setting the Modify Terrain object's properties in its property sheet. The surface of the terrain is raised or lowered according to the elevation values of the polygon's points. Different elevation behaviors allow replacing, cropping above or cropping below the original values of the terrain. The interior of the polygon can be flat or irregular in shape.

To modify the terrain:

1. On the **Tools** tab, in the **Objects and Layers** group, click **Modify Terrain**. The Modify Terrain property sheet is displayed.
2. With the cursor in the **3D Window**, click to add the first point of the polygon.
3. Click again to add additional points of the polygon until you have marked the entire shape. You must place at least three points to define the polygon.

Note: When creating a Modify Terrain object, the lines of the polygon should not overlap each other.

Note: It is recommended to keep the number of polygon points to the minimum required to define the shape. Adding unnecessary points may adversely affect performance.

4. Right-click to finish the object.
5. Continue to edit the object's parameters and shape, or close the property sheet to finish the operation. **See** "Modify Terrain Property Sheet" in the [TerraExplorer User Guide](#) for further information.

Cutting a Hole on Terrain

Cutting a hole on the terrain removes the terrain texture while maintaining the original contours of the terrain. The objects or feature layers created on the terrain in the area of the hole are drawn on the original contours of the terrain.

To cut a hole on the terrain:

1. On the **Tools** tab, in the **Objects and Layers** group, click **Hole on Terrain**. The Hole on Terrain property sheet is displayed.
2. With the cursor in the **3D Window**, click to add the first point of the hole.
3. Click again to add additional points of hole until you have marked the entire shape.
4. Right-click to finish the hole.

Note: It is recommended to keep the number of hole points to the minimum required to define the shape. Adding unnecessary points may adversely affect performance.

5. Continue to edit the hole or close the property sheet to finish the operation. **See** "Hole on Terrain Property Sheet" in the [TerraExplorer User Guide](#) for further information.

CHAPTER 20 ANALYSIS TOOLS

About Analysis Tools

The Tools tab provides you with a set of TerraExplorer analysis tools that help you validate photo parameters.

Tool	Description
Distance	Displays the aerial distance, elevation difference, horizontal distance and slope angle between two or more points in the 3D Window.
Area	Measure the following for a defined area on the terrain: ▪ Horizontal plane – Displays the measurement of the horizontal projection of an area in the 3D Window. ▪ Terrain surface – Displays the measurement of the marked area while taking into account terrain contours.
Query	Provides data about any point or object in the 3D World.

Distance Tool

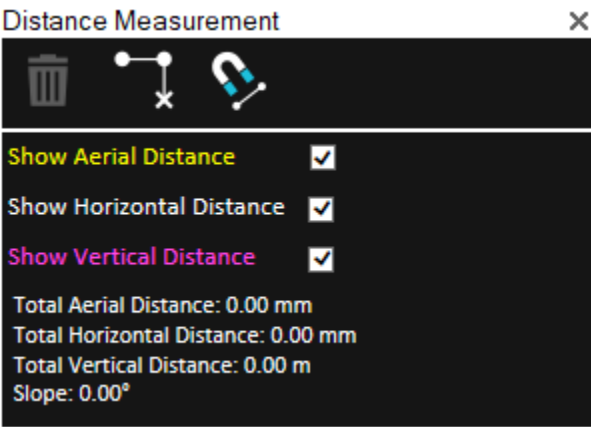
The Distance tool allows you to measure the aerial distance, elevation difference and slope between two or more points. The aerial distance is the actual distance between the points.

To use the Distance tool:

1. On the **Tools** tab, in the **Measure** group, click **Distance**. The Distance Measurement dialog is displayed with the following options regarding display of distance information in the **3D Window**:

Note: The aerial, horizontal, and vertical distance information are calculated and displayed in the Distance Measurement **dialog** even if the respective option was not selected

- **Show Aerial Distance** – Show a yellow line marking the aerial distance, i.e., the actual distance between points, in the 3D Window.
- **Show Horizontal Distance** – Show a white line marking the horizontal distance between points in the 3D Window.
- **Show Vertical Distance** – Show a purple line marking the vertical distance between points in the 3D Window.



Distance Measurement Dialog

- 2. If you want to snap the measurement points to an object's edge or vertex, click **Snap** . All edges, i.e., intersections of two plane faces, and vertices, i.e. intersections of two polylines are automatically detected.
- 3. In the **3D Window**, click any point on the terrain or on an object to define the start point of the measurement. A yellow line extends from the start point. Drag the cursor to the next point and click again. Repeat to add as many segments as required. If you want to delete the last point added, click **Delete Point** . Right-click to finish the measurement.

The Distance Measurement dialog displays the following measurements:

Measurement	Description
Total Aerial Distance	The aerial distance, i.e., the actual distance between the points. This measurement is the sum of the aerial distances between the points as you progress from the start point to the endpoint.
Total Horizontal Distance	The horizontal distance between the points. This measurement is the sum of the horizontal distances between the points as you progress from the start point to the endpoint.
Total Vertical Distance	The difference in elevation between the points. If you have marked more than two points, this measurement does not take into account the elevation values of the middle points.
Slope	The slope of the line between the first and last points. If you have marked more than two points, this measurement does not take into account anything other than the elevation at the start point and the elevation at the endpoint.

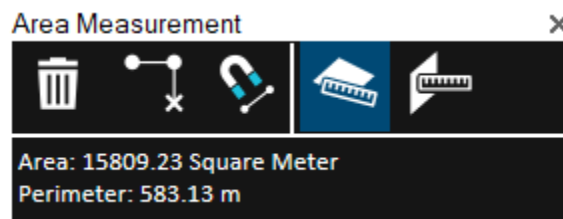
- 4. If you want to delete the measurement polyline in the 3D Window, click **Delete** .
- 5. To perform additional distance measurements, repeat steps 2-3 above.
- 6. To close this tool, click the X on the top right corner of the dialog.

Area Tool

Using the Area tool, you can calculate area on the terrain or on an arbitrary plane.

To use the Area tool:

1. On the **Tools** tab, in the **Measure** group, click **Area**. The Area Measurement dialog is displayed.



Area Measurement Dialog

2. Click either of the following commands:
 - **Horizontal plane area (2D measurement)**  - Calculate area on terrain. The area measured is the horizontal projection of the area you have outlined, even if some or all of your selected area encompasses mountainous terrain.
 - **3D plane area**  - Calculate area on an arbitrary plane. The polygon's orientation is determined by the first three points selected.
3. If you want to snap the measurement points to an object's edge or vertex, click **Snap** . All edges, i.e., intersections of two plane faces, and vertices, i.e. intersections of two polylines are automatically detected.
4. Draw a polygon in the 3D Window that defines the borders of the region being measured:
 - Place the polygon points in the **3D Window** by clicking in the desired locations. A red line extends from the start point. If a valid selection area was drawn, the line turns white after the third click, indicating that you can then right-click and obtain the required measurement. A minimum of three points must be selected. If you want to delete the last point added, click **Delete Point**.

Note: If the boundary line of the measured area crosses itself, it turns red, and a dialog is displayed informing that the polygon is invalid.
5. Right-click to finish the measurement. The Area dialog displays area and perimeter measurements.
6. If you want to delete the area polygon in the 3D Window, click **Delete** .
7. To perform additional area measurements, repeat steps 2-5 above.
8. To close this tool, click the X on the top right corner of the dialog.

Information Query Tool

The Query tool provides you with data about any point or object in the 3D World. For a point on the terrain, location and elevation information is provided, while for an object, the query also displays the object's perimeter and area. If the object selected was loaded from a layer with attribute information, the object's attributes are also displayed.

To use the information query tool:

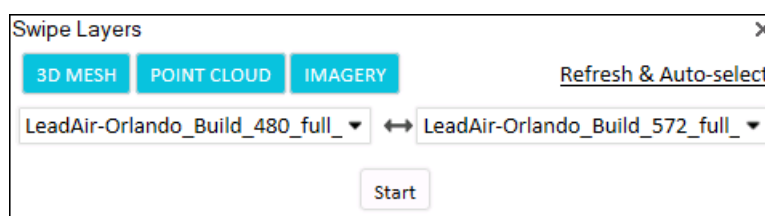
1. On the **Tools** tab, in the **Measure** group, click **Query**.
2. Move the cursor over any point in the **3D Window**. When you move the cursor over an object, the object is highlighted.
3. Click the left mouse button on the point or object for which you require information.

- In the Message Bar, the longitude, latitude and altitude are displayed.
 - The Query Results dialog is displayed, showing the longitude, latitude, and elevation. When an object is selected, general information about the object and attribute information, if available, is displayed.
4. With the Query tool selected, you can click again on any point in the terrain to find its coordinates. Repeat for as many points as you wish.

On the **Tools** tab, in the **Measure** group, click **Query** again to close the Information Query.

Swipe Mesh, Point Cloud, and Imagery Layers

The Swipe Layers tool enables you to compare two mesh, point cloud, or imagery layers in the same location, or between one of these layer types and the base terrain. Using the tool, you can horizontally swipe between them, dynamically revealing and hiding parts of each layer.



Swipe Layers Tool

To use the Swipe Layers tool:

1. Zoom in on an area with two visible layers, or with a single layer that you want to compare to the base terrain.
2. On the **Tools** tab, in the **Objects and Layers** group, click **Swipe Layers**. The Swipe Layers controls are displayed.
3. Toggle on/off any of the following: 3D Mesh, Point Cloud, and Imagery to filter the selection to the layer type(s) you want. If you want to refresh the dropdown layers, and automatically detect and select the layers in current view, click **Refresh & Auto-Select**.
4. Select the layers from the two dropdown lists, and click **Start**.
5. If comparing mesh model layers, select the mesh model display style:
 - Texture
 - Texture + wireframe
 - Solid color
 - Solid color + wireframe
 - X-ray - Semi-transparent model, enabling you to see through all model walls
6. Slide the slider all the way to the right to completely show the first layer, and all the way to the left to completely show the second one. As the slider moves between these extremes, the first layer is gradually clipped to reveal the second one.
7. Click **Auto-Repeat**  to automatically loop the slider value between 0-100% to dynamically change the top layer clipping. Click **Freeze** at any point to freeze the slider at a specific value.

CHAPTER 21 PHOTOMESH AUTOMATION

About PhotoMesh Automation

PhotoMesh provides you with REST APIs for automating PhotoMesh project and queue tasks.

- **PhotoMesh API** - REST API to automate the production phases, e.g., load photo files or list, set AT and reconstruction areas, begin and manage a build, and create new build versions.
- **ProjectQueue API** - REST API to create a project queue, and then activate, monitor, and manage the queue processing.

Creating a Projects Queue

The project queue lists the projects selected for processing, detailing their associated tasks and specifying the order in which they should be executed. The queue is created using the PhotoMesh Queue Editor, where you can either import a JSON file containing information on the projects and their tasks, in which case the relevant fields are automatically populated, or input the information manually. The Editor allows for editing this data and adding new projects to the queue. Once the queue is finalized, activate the processing to initiate PhotoMesh's build tasks in the specified sequence. The progress and management of these tasks are handled through the Projects Queue Manager. See "[Managing the Queue](#)" in this chapter for more information.

To create a queue:

1. From the **Start** page, open the **Automation Dashboard** and in the **Project Queue** section, click **Project Queue**. The Projects Queue Manager is displayed. Click **Back to Editor** to open the Projects Queue Editor. See "[Automation Dashboard](#)" in the "Basic Concepts" chapter for information.
2. Click **Load from File**, and browse to the projects queue JSON file you created. See "[Creating a Projects Queue JSON File](#)" in this chapter for information. The file is loaded, and the relevant fields are automatically populated based on the file information. If you want to enter all the values manually, click **New projects template**.

Note: When loading a projects queue JSON file from the Projects Queue Editor, the JSON array must begin with the "projects" string. E.g.,

```
{ "projects":
[
  {json proj1},
  {json proj2}
]
}
```

3. For each project you want processed, enter/modify the required information for your project file type:

Field	Description
Name	Free text describing the project.
Action	The action you want to perform in PhotoMesh. The following are the possible options: ▪ Create (and Build) New Project ▪ New Build ▪ New Build – Copy AT

	▪ Rebuild ▪ Pause ▪ Script (Python or JavaScript)
Project Path	Type the full path to: ▪ The PhotoMeshXML project file that should be loaded if performing a new build or rebuilding, ▪ The PhotoMeshXML project file that should be created if creating and building a new project. ▪ A script file if the action selected is script.
Build From	The step from which you want to start the build. Select "Auto" to have PhotoMesh automatically determine what processing must be performed to complete the project, avoiding unnecessary rebuilding.
Build Until	The step until which you want to build.
Inherit from Build	The name of the build version from which to inherit the project's base build parameters. This property is applied only when performing a New Build or New Build-Copy AT.
Preset	If the preset file is saved under the application's AppData or in the application files' Presets folder, enter the name of the preset files to be loaded by PhotoMesh, separated by a comma. Otherwise enter the full path of each preset, separated by a comma. Click Select to select one or more presets from a list, and then click OK , or type the full path to the file. The available preset files are in the Presets folder located under the folder from where PhotoMesh is running. Only the file name should be typed, without the file extension, e.g., "Draft_mode" or "Standard_GPS".
Max Pool Fusers	Maximum number of pool fusers to run.
Max AWS Fusers	Maximum number of Amazon Web Services (AWS) fuser instances to run.
AWS Startup Script	The startup script PhotoMesh should use when launching AWS fuser instances. This field is only displayed if Max AWS Fusers was set > 0.
AWS Configuration Name	The name of the configuration JSON file PhotoMesh should use to determine which AWS instances to launch. Only the file name should be passed, without the file extension. This field is only displayed if Max AWS Fusers was set > 0.
AWS Configuration JSON Path	The path to the folder where the AWS Configuration Name JSON file is saved (default is ./PhotoMesh/AWSConfigurations). This field is only displayed if Max AWS Fusers was set > 0.
Collections	The source of the project's photo information. This property is only displayed when Action was set to Create New Project. The following options are available: ▪ Load Folder ▪ Load Video

	▪ Load List (XML, Excel)
Collection XXX	This property is only displayed when Action was set to Create New Project: ▪ If Collections was set to Load Folder – Enter the name and full path to each of the project's collections. ▪ If Collections was set to Load Video – Enter the name and full path to each of the project's collections (containing the video file).
Excel	Full path to the photo list.
Properties	Full path to the collectionProperties.xml file (optional). There are no restrictions on the naming of the collection properties XML file, or on the location where this file is saved. This property is only displayed when Action was set to Create New Project and Collections was set to Load Folder.
Focal35	Equivalent focal length in millimeters for a 35 mm film camera. This property is only displayed when Action was set to Create New Project and Collections was set to Load Video.
Taken Every	Sampling interval, i.e., every how many seconds a frame should be captured. This property is applied only when loading a video. This property is only displayed when Action was set to Create New Project and Collections was set to Load Video.
Seconds Start	Type the start time from which video frame extraction should begin. This property is only displayed when Action was set to Create New Project and Collections was set to Load Video.
End	Type the end time for video frame extraction. This property is only displayed when Action was set to Create New Project and Collections was set to Load Video.

4. Click **Remove** to remove a project.
5. Click **Move up**  and **Move down**  to change the order of the projects in the queue.
6. Click Add to Queue.

Managing the Queue

The project queue is monitored and managed from the Projects Queue Manager. The Manager provides dynamic information on the status of the projects in the queue and enables you to start/stop the projects in the loaded queue, and change fuser allocation.

To manage the projects in your queue:

1. From the **Start** page, open the **Automation Dashboard** and in the **Project Queue** section, click **Project Queue**. The Projects Queue Manager is displayed. See "[Automation Dashboard](#)" in the "Basic Concepts" chapter for information.
2. Click **Queue Manager**. The Projects Queue Manager is displayed, with the left side displaying information on each of the projects in the queue. Click **Move up**  and **Move down**  to change the order of the projects in the queue.
3. If you want to remove any of the projects from the queue, click the **Remove** link above the project name. If you want to clear the entire queue, click **Remove All** at the top of the page.

4. Click **Build** to start building the projects in the queue.
5. If you want to abort the current build project, and all other builds in the queue, click **Abort & Stop**.
6. If you want to change the maximum number of AWS or pool fusers for the queue, enter the required number in the appropriate field, and click **Change**.

PhotoMesh API

PhotoMesh provides a REST API for PM Production to automate the production phases, e.g., load photo files or list, set AT and reconstruction areas, begin and manage a build, and create new build versions.

To view interactive documentation of the PhotoMesh REST API:

- From the **Start** page, open the **Automation Dashboard** and in the **PhotoMesh API** section, click **Reference Guide**. See "[Automation Dashboard](#)" in the "Basic Concepts" chapter for information.
- Note:** You can also access it from Resources >PhotoMesh API. See "[PhotoMesh Resources](#)" in the "Introduction" chapter for information.

Queue API

PhotoMesh provides a REST API to create a project queue and then activate, monitor, and manage queue processing. The project queue is created using the **add** API method, in which a project queue JSON file is passed that lists all the tasks that should be performed for each of the projects in the project queue. See "[Creating a Project Queue JSON File](#)" in this chapter for more information.

To view technical and interactive documentation of the Queue REST API:

- From the **Start** page, open the **Automation Dashboard** and in the **Project Queue** section, click **Reference Guide**. See "[Automation Dashboard](#)" in the "Basic Concepts" chapter for information.

To use PhotoMesh Queue API:

- From the **Start** page, open the **Automation Dashboard** and in the **Project Queue** section, click **Project Queue**. See "[Automation Dashboard](#)" in the "Basic Concepts" chapter for information.

Creating a Projects Queue JSON File

The projects queue JSON file lists all the tasks that should be performed for each of the projects in the project queue. It also includes photo information and build and fuser settings for each of the queue's projects. The steps below describe the file structure and syntax.

To create a projects queue JSON file:

1. Create a JSON object that holds an array whose objects are each of the projects to be included in the queue.

```
[
  {json proj1},
  {json proj2}
]
```

Note: When loading a projects queue JSON file from the PhotoMesh Queue Editor, the JSON array must begin with the "projects" string. E.g.,

```
{ "projects":
  [
```

```
{json proj1},
{json proj2}
]
}
```

2. Create project objects that are composed of name/value pairs separated by commas, which set the build tasks to perform, build and fuser settings, and photo information. The following name/value pairs should be included:

Field	Description
Mandatory Parameters	
action	Determines what action is performed. The following are the possible values: 0 – Create New Project (Create project, load photos and photo information, and then build the project), 1 – New Build, 2 – New Build-Copy AT, 3 – Rebuild, 4 – Pause, 5 – Script (Python or JavaScript).
projectPath	The full path to: ▪ The PhotoMeshXML project file that should be or created if creating and building a new project (action = 0). ▪ The PhotoMeshXML project file that should be loaded if performing a new build or rebuilding (action = 1, 2, or 3). ▪ A script file if the step is script (action = 5)
buildFrom	The step from which to start the build. The following are the possible values: 0 – Auto, 1 – Data Preparation, 2 – AT, 3 – Point Cloud, 4 – Model, 5 – Texturing, 6 – Output.
buildUntil	The step until which to perform the build. The following are the possible values: 0 – Auto, 1 – Data Preparation, 2 – AT, 3 – Point Cloud, 4 – Model, 5 – Texturing, 6 – Output.
preset	The name of the preset file to be loaded by PhotoMesh. Click select to select a preset or type the full path to the file. The available preset files are in the Presets folder located under the folder from where PhotoMesh is running. Only the file name should be passed, without the file extension, e.g., "Draft_mode" or "Standard_GPS".
Specific Parameters	
sourceType	The source type of the project's photo information. This parameter is applied only when creating a new project (action = 0). The following are the possible values: 0 – Load Folder, 1 – Load Video, 2 – Load List (Excel or XML file).
sourcePath	This parameter is applied only when creating a new project (action = 0): ▪ If creating a new project by loading a folder of photos (sourceType = 0), the sourcePath array contains the name and full path to each of the project's collections, and an optional "properties" value with the full path to

	the collectionProperties.xml file. When using the API, there are no restrictions on the naming of the collection properties XML file, or on the location where this file is saved. ▪ If creating a new project by loading a video (sourceType = 1), the sourcePath array contains the name and full path to each of the project's collections (containing the video file), focal length 35, Sampling interval (frameEvery), and start and end times for video frame extraction. ▪ If creating a new project by loading an Excel or XML photo list file (sourceType = 2), the sourcePath array contains the full path to the photo list.
inheritBuild	The name of the build version from which to inherit the project's base build parameters. This property is applied only when performing a New Build or – New Build-Copy AT (action = 1 or 2).
Fusers Parameters	
MaxLocalFusers	The maximum number of pool fusers to use for the build.
MaxAWSFusers	The maximum number of Amazon Web Services (AWS) fuser instances to launch for the build.
AWSFuserStartupScript	The startup script PhotoMesh should use when launching AWS fuser instances. This parameter is required if MaxAWSFusers was set > 0.
AWSBuildConfigurationName	The name of the configuration JSON file PhotoMesh should use to determine which AWS instances to launch. Only the file name should be passed, without the file extension. This parameter is required if MaxAWSFusers was set > 0.
AWSBuildConfiguration.JsonPath	The path to the folder where the AWSBuildConfigurationName JSON file is saved (default is ./PhotoMesh/AWSConfigurations). This parameter is required if MaxAWSFusers was set > 0.

Example 1: New Project from Folder

```
{ "projects":
[ {
  "comment": "New project from folder",
  "action": 0,
  "projectPath": "N:\\PM
Projects\\ProjectFolder\\ProjectName.PhotoMeshXML",
  "buildFrom": 1,
  "buildUntil": 6,
  "inheritBuild": "",
  "preset": "PhotoMesh Default",
```

```

    "workingFolder": "N:\\WorkingFolder",
    "MaxLocalFusers": 10,
    "MaxAWSFusers": 7,
    "AWSFuserStartupScript": "script",
    "AWSBuildConfigurationName": "AT_MultiCoreCPU",
    "AWSBuildConfigurationJsonPath": "",
    "sourceType": 0,
    "sourcePath": [
        {
            "name": "RGB",
            "path": "X:\\RGB",
            "properties": "X:\\RGB.xml"
        },
        {
            "name": "NIR",
            "path": "X:\\NIR",
            "properties": "X:\\NIR.xml"
        },
        {
            "name": "Thermal",
            "path": "X:\\THERMAL",
            "properties": "X:\\Thermal.xml"
        }
    ]
}
]
}

```

Example 2: New Project from Video

```

{"projects":
[ {
    "comment": "New project from video",
    "action": 0,
    "projectPath": "N:\\PM
Projects\\ProjectFolder\\ProjectName.PhotoMeshXML",
    "buildFrom": 1,
    "buildUntil": 6,
    "inheritBuild": "",
    "preset": "PhotoMesh Default",
    "workingFolder": "N:\\WorkingFolder",

```

```

    "MaxLocalFusers": 10,
    "MaxAWSFusers": 7,
    "AWSFuserStartupScript": "script",
    "AWSBuildConfigurationName": "AT_MultiCoreCPU",
    "AWSBuildConfigurationJsonPath": "",
    "sourceType": 1,
    "sourcePath": [
      {
        "name": "Collection 1",
        "path": "X:\\PMProjects\\Washington\\PMCollections",
        "focal35": 24,
        "frameEvery": "00:00:05",
        "startTime": "00:00:00",
        "endTime": "01:00:00"
      }
    ]
  }
]
}

```

Example 3: Rebuild Project

```

{"projects":
[ {
  "comment": "Rebuild project",
  "action": 3,
  "projectPath": "N:\\PM
Projects\\ProjectFolder\\ProjectName.PhotoMeshXML",
  "buildFrom": 1,
  "buildUntil": 6,
  "preset": "PhotoMesh Default",
  "workingFolder": "N:\\WorkingFolder",
  "MaxLocalFusers": 10,
  "MaxAWSFusers": 7,
  "AWSFuserStartupScript": "script",
  "AWSBuildConfigurationName": "AT_MultiCoreCPU",
  "AWSBuildConfigurationJsonPath": "" }
]
}

```

Troubleshooting

If you cannot access the PhotoMesh, PhotoMesh Fuser, or Project Queue API, this is generally due to a firewall blocking the required port or missing URL reservations. In this case, you can manually add from the command line a rule to allow the required port through the firewall and configure a URL registration. Below are the steps for each of the API's:

- [PhotoMesh API](#)
- [Fuser API](#)
- [Project Queue API](#)

PhotoMesh API

To open the PhotoMesh API port 8086:

1. Click the **Start**  button, type "cmd", and then select **Command Prompt** from the list.
2. In the command prompt window type the following and click **Enter**:

```
netsh advfirewall firewall add rule name="PM API WebAPI TCP Port 8086"
dir=in action=allow protocol=TCP localport=8086
```

3. Then type the following and click **Enter**:

```
netsh http add urlacl url=http://*:8086/ SDDL=D:(A;;GX;;;WD)
```

The first line adds a firewall rule named "PM API WebAPI TCP Port 8086" allowing it to bypass the firewall, and the second line assigns the rights for the port.

If you want to change the port of the PhotoMesh API, do the following:

1. In a text editor, open the *PhotoMesh.exe.config* file that is located in the application files folder, and replace the value of the "API.API URL" key.

```
<appSettings>
  <add key="API.API URL" value="http://XXXX" />
</appSettings></appSettings>
```

2. Open in a text editor the *.env* file that is located in the application files' **PhotoMesh.js** folder, and replace the value of *PHOTOMESH_EXE_SERVER_PORT*. E.g.,

```
PHOTOMESH_EXE_SERVER_PORT=XXXX
```

PhotoMesh Fuser API

To open the PhotoMesh Fuser API port 8085:

1. Click the **Start**  button, type "cmd", and then select **Command Prompt** from the list.
2. In the command prompt window, type the following and click **Enter**:

```
netsh advfirewall firewall add rule name="PM Fuser API WebAPI TCP
Port 8085" dir=in action=allow protocol=TCP localport=8085
```

3. Then type the following and click **Enter**:

```
netsh http add urlacl url=http://*:8085/ SDDL=D:(A;;GX;;;WD)
```

The first line adds a firewall rule named "PM Fuser API WebAPI TCP Port 8085" allowing it to bypass the firewall, and the second line assigns the rights for the port.

If you want to change the port of the Fuser API, do the following:

- In a text editor, open the *PMFuserService.exe.config* file located in the application files' **Fuser** folder, and replace the value of *webApiUrl*. E.g.,

```
<add key="WebApiUrl" value="http://*:XXXX/" />
```

Project Queue API

To open the Project Queue API fuser port 8087:

1. Click the **Start**  button, type "cmd", and then select **Command Prompt** from the list.
2. In the command prompt window, type the following and click **Enter**:

```
netsh advfirewall firewall add rule name="PM ProjectQueue WebAPI TCP
Port 8087" dir=in action=allow protocol=TCP localport=8087
```

3. Then type the following and click **Enter**:

```
netsh http add urlacl url=http://*:8087/ SDDL=D:(A;;GX;;;WD)
```

The first line adds a firewall rule named "PM ProjectQueue WebAPI TCP Port 8087" allowing it to bypass the firewall, and the second line assigns the rights for the port.

If you want to change the port of the Project Queue API, do the following:

- In a text editor, open the *.env* file located in the application files' **PhotoMesh.js** folder, and replace the *port* number.
E.g., PORT=8087 > PORT=XXXX